DENTAI:AI-POWERED DENTAL ASSESSMENT

AMIR KEIZRUL BIN MUDZIR

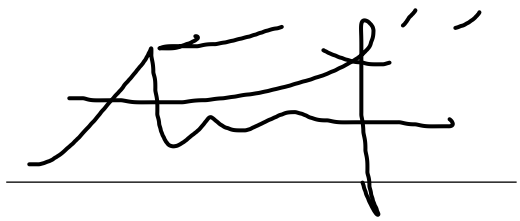
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For the Bachelor of Information Technology (Hons) in Software Engineering
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DECLARATION

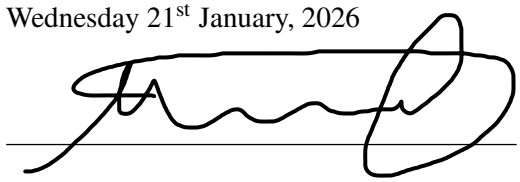
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LIST OF ABBREVIATIONS

SDLC Software Development Life Cycle

SRS Software Requirements Specification

AI Artificial Intelligence

API Application Programming Interface

CNN Convolutional Neural Network

IDE Integrated Development Environment

JAD Joint Application Development

RAD Rapid Application Development

UAT User Acceptance Testing

YOLOv11 Ultralytics Yolov11

ABSTRACT

The current project proposes a web application that allows helping in early dental health evaluation using artificial intelligence. The users of the system can upload intraoral images, which are evaluated with the help of a trained YOLOv11 deep learning model with the most widespread conditions in place, including gingivitis, ulcers, caries (cavities), and discoloured teeth. The backend that implements the Flask framework supports the use of the AI model to analyse images in real-time, whereas the frontend that is based on the React framework provides the user with an intuitive experience of uploading the images, viewing the results, and scheduling the appointments. Firebase stands as a backend database that handles user profiles and bookings. The underlying premise of the system is to fill the divide to reach early dental diagnostics, especially in underserved areas due to an efficient and economical solution that may quickly be adopted. The results of the evaluation indicate high accuracy of the models and mostly positive feedback on the user experience, which proves the prospects of the system to raise oral health awareness and provide early treatment by proactive involvement in digital interaction.

ABSTRAK

Projek ini mencadangkan sebuah aplikasi web yang membantu dalam penilaian awal kesihatan pergigian menggunakan kecerdasan buatan. Pengguna sistem ini boleh memuat naik imej intraoral yang akan dianalisis oleh model pembelajaran mendalam YOLOv11 yang telah dilatih, untuk mengesan beberapa kondisi pergigian yang biasa berlaku seperti gingivitis, ulser, karies (gigi berlubang), dan perubahan warna gigi. Bahagian belakang sistem dibangunkan menggunakan rangka kerja Flask untuk menyokong analisis imej secara masa nyata oleh model AI, manakala bahagian hadapan berasaskan React memberikan pengalaman pengguna yang intuitif untuk memuat naik imej, melihat keputusan, dan menjadualkan temu janji. Firebase digunakan sebagai pangkalan data belakang untuk mengurus profil pengguna dan tempahan. Tujuan utama sistem ini adalah untuk merapatkan jurang dalam mendapatkan diagnosis awal pergigian, terutamanya di kawasan yang kurang mendapat perkhidmatan, melalui penyelesaian yang cekap dan menjimatkan serta mudah digunakan. Hasil penilaian menunjukkan ketepatan model yang tinggi dan maklum balas pengguna yang kebanyakannya positif, membuktikan potensi sistem ini dalam meningkatkan kesedaran kesihatan mulut dan menyediakan rawatan awal melalui penglibatan digital yang proaktif.

Chapter 1

Introduction

1.1 Introduction

This project presents DentAI, a web-based dental health assessment system that integrates artificial intelligence with clinical and administrative management features. The system employs deep learning and object detection techniques to analyse intraoral dental images uploaded by patients, enabling the automated detection of common dental conditions such as gingivitis, ulcers, dental caries (cavities), and tooth discoloration Makarim et al. (2025).

DentAI is designed as a multi-role platform that supports patients, clinic administrators, and system administrators. Patients can register accounts, upload dental images for AI analysis, view assessment results, track their dental progress history, receive reminders, and book dental appointments. This allows users to monitor their oral health conveniently and encourages early detection before conditions worsen.

In addition to patient-facing features, the system provides tools for clinic administrators to manage appointments, maintain clinic information, and access patient dental records. System administrators are responsible for managing users, registering clinic hosts, overseeing feedback content, and analysing system trends to ensure efficient and reliable system operation.

By combining AI-powered dental image assessment with appointment scheduling and administrative functionalities, DentAI aims to bridge the gap between individuals and professional dental services. The system promotes proactive oral healthcare, improves accessibility to preliminary dental assessments, and offers a scalable and cost-effective solution, particularly beneficial for communities with limited access to dental professionals Tan (2023).

1.2 Problem Statement

Many dental conditions such as cavities, gum disease, and oral ulcers often develop without noticeable symptoms during their early stages. When left undetected, these conditions may progress into more severe problems that require complex, time-consuming, and costly treatments. Early identification of dental issues is therefore crucial in preventing long-term oral health complications.

Despite the importance of regular dental checkups, many individuals avoid visiting dental clinics due to high treatment costs, fear of dental procedures, or inconvenience Datey (2023). This avoidance leads to delayed diagnosis and increases the likelihood of serious dental conditions developing over time Steinvik (2023).

Furthermore, there is a lack of accessible and user-friendly tools that allow individuals to monitor their oral health independently. Without an affordable and convenient method for preliminary dental self-assessment, many people remain unaware of potential dental problems until symptoms become severe or painful. This highlights the need for a simple, AI-assisted solution that supports early detection and encourages proactive dental care.

1.3 Research Objectives

This research aims to achieve the following objectives:

Research Objective 1:

To study existing dental-related applications and object detection models suitable for analysing intraoral dental images.

Research Objective 2:

To develop a web based Dental assessment platform that utilizes a trained AI model to perform dental condition assessment.

Research Objective 3:

To evaluate the effectiveness and the accuracy of the DentAI platform by going through user acceptance testing.

1.4 Research Questions

While conducting the research, the following are the questions that seek answers from the researcher:

Research Question 1: How effective is the YOLOv11 model in detecting common dental conditions such as gingivitis, ulcers, caries, and tooth discoloration from intraoral images?

Research Question 2: How can an AI-based system be integrated into a web application to provide real-time dental assessments and appointment scheduling?

Research Question 3: What is the user perception and usability level of a web-based AI dental assessment system in terms of accuracy, convenience, and accessibility?

1.5 Research Methodology

The research methodology adopted for this project is based on a prototype-driven iterative development approach. This methodology emphasizes continuous improvement through repeated cycles of designing, building, demonstrating, and refining the system. By following this approach, the project can adapt to real-world challenges—especially in AI

model training and clinical feedback—while ensuring that each stage of development is aligned with the overall project objectives.

This methodology is suitable for AI-driven healthcare applications because it allows regular testing, user evaluation, and technical adjustments, ensuring that both the model and user interface evolve effectively throughout the project life cycle. The key characteristics of this methodology include:

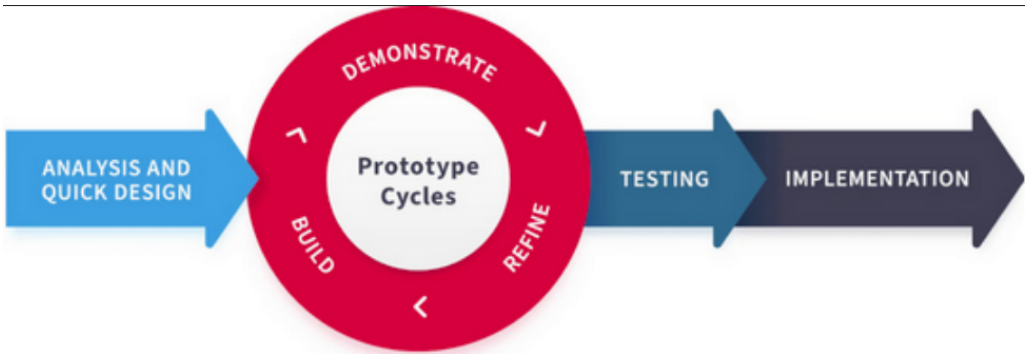


Figure 1.1: RAD Methodology

- **Iterative Development:** Breaks the project into several smaller and easier to digest stages, allowing continuous refinement of AI algorithms and user interface based on clinical feedback.
- **Enhanced Management:** Easy to manage and plan what goals need to be finished before proceeding to the next stage, particularly important for AI model training and validation dependencies.
- **Sequential Implementation:** Allows planning each feature implementation sequentially and reduces design ambiguity through prototype-driven development.

1.6 Research Scope and Limitation of the Study

The scope of the study is as follows:

(a) The scope focuses on the detection of four common dental conditions: gingivitis, ulcers, caries, and tooth discoloration using intraoral image analysis.

(b) The scope is limited to image-based diagnosis and appointment management; it does not replace professional dental consultation or X-ray-based diagnostics.

Limitations include the reliance on image quality, a limited range of detectable conditions, and the availability of ethically sourced dental image datasets for model training.

1.7 Contribution of the Research

(a) Developing an AI-powered diagnostic tool that supports early detection of dental issues via web-based access.

(b) Enhancing accessibility to dental assessments, especially for users in remote or underserved areas.

(c) Providing a scalable and integrated platform that combines diagnosis with appointment booking.

(d) Offering insights into the use of YOLOv11 in the dental diagnostic domain and demonstrating its feasibility through a real-world application.

1.8 Significance of the Research

The significance of this research lies in its potential to transform how individuals approach oral healthcare. By introducing a digital tool that enables self-assessment and facilitates early diagnosis, the system encourages proactive oral hygiene practices. It addresses common barriers to dental visits, such as cost, fear, and inconvenience, and promotes timely intervention Datey (2023). Furthermore, the integration of AI in dental diagnostics may inspire future innovations in teledentistry and personalized healthcare Batra (2022).

1.9 Thesis Structure

This thesis is organised as follows: Chapter 2 critically reviews the relevant literature in the research area. Chapter 3 provides the details of the research methodology in the research. Chapter 4 elaborates on the implementation of the system. Chapter 5 discusses the experimental result and the overall findings of the research. Finally, Chapter 6 summarises the research implications and provides recommendations for future research.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews existing literature and technologies relevant to the development of a web-based dental assessment platform that leverages artificial intelligence to detect oral conditions such as gingivitis, ulcers, caries (cavities), and tooth discoloration. The system includes a dental appointment booking feature and is developed using a React frontend, Flask backend, and Firebase database.

The review serves to identify technological gaps, examine best practices in medical imaging and web-based healthcare platforms, and justify the selection of tools and methods for this project. Key areas include computer vision for dental diagnostics, web technologies for healthcare, and the integration of real-time AI inference with cloud databases Khattak (2025).

2.2 Key Concepts and Terminologies

2.2.1 AI-Based Dental Condition Detection

Analysis of dental images with Artificial Intelligence implies separation and localization of visual motives in intraoral pictures. Object detection using Convolutional Neural Networks (CNNs), and in particular because of their efficiency and accuracy, object detectors such as YOLOv5 are popular in medical imaging fields of applications Makarim et al. (2025).

In this project, a YOLO model is trained to detect four dental conditions:

- **Gingivitis:** Inflammation of the gums.
- **Ulcers:** Open sores in the mouth.
- **Caries (Cavities):** Tooth decay.
- **Tooth Discoloration:** Abnormal changes in tooth colour.

The model is deployed and accessed through a Flask API, enabling predictions at real-time through the uploading of images on the Web Interface. Recent studies confirm that YOLO architectures outperform traditional classification models in detecting dental conditions from intraoral images due to their speed and localization capabilities Zhang et al. (2025).

2.2.2 Web Application In Health Services

The widespread nature, portability, and integration with a database and APIs make web apps a favourite form of providing health resources. Frontend development is possible with React, and Flask is the backend capable of providing such an AI model. The system is scalable and secure, which allows Firebase to work with data of users, schedule appointments, and authenticate users. Recent architectures for AI-based health monitoring systems have successfully validated this "React-Flask-Firebase" stack for ensuring modularity and secure data management in healthcare environments Tank (2025).

2.2.3 Importance of Early Dental Diagnosis

Research shows that early intervention of most teeth problems can significantly decrease the number of aggressive processes and enhance human long-term oral health conditions. Nevertheless, most people put off or neglect dentists because it is expensive, time wasting or they are unaware. An AI screening tool can be offered online to enable users to conduct prompt screening and make a wise decision to seek treatment. Studies indicate that delay in seeking care is significantly associated with lower oral health self-efficacy, a gap that accessible digital screening tools aim to bridge Steinvik (2023).

2.3 Current Issues In The Domain

2.3.1 Accessibility Of Dental Care

Poor dental accessibility to the rural and underserved population in countries such as Malaysia is common. Geographic constraints and the disparity between urban and rural healthcare infrastructure contribute to this inequality. Preliminary screenings and facilitation of the clinic visits through appointment scheduling can be provided well by web-based tools. Evidence from the National Health and Morbidity Survey indicates that rural populations in Malaysia face distinct barriers in utilizing oral healthcare, necessitating digital interventions Tan (2023).

2.3.2 Fear And Delay In Treatment

Fear of the dentist usually causes the neglect of essential examinations that increase the severity of minor disorders with time. An AI tool, which proposes confidential non-invasive evaluation, can ease the anxiousness of users and prompt them to do so sooner. Dental anxiety is a well-documented barrier, with approximately 50% of patients reporting moderate to severe anxiety which directly leads to avoidance of treatment until pain becomes unbearable Datey (2023).

2.3.3 Fragmentation In Health Tech Solutions

Most of the available systems serve only one purpose—either diagnosis or appointment. Integrated schemes are not available to bring together the detection of a condition alongside real-time decision support and clinic scheduling. While teledentistry has grown, current platforms often lack fully automated integration where a diagnostic result immediately triggers a scheduling workflow Batra (2022).

2.4 Review Of Related Works

2.4.1 Dental AI Research Platforms

Numerous studies have demonstrated the effectiveness of CNNs in identifying caries and periodontal disease from X-rays and intraoral photos. However, most are academic or clinical tools with no web interface or public accessibility. For instance, while high accuracy has been achieved in detecting caries using deep learning on bitewing radiographs, these models typically remain confined to research settings or desktop applications for radiologists rather than patient-facing web tools Görürgöz (2025).

2.4.2 Teledentistry Tools

Commercial solutions like Toothpic and Dental Monitoring provide virtual consultations, but they often require dentist review and do not support self-assessment or automated booking features. Their models are also not open-source or locally deployable. A review of these platforms highlights that while they facilitate remote monitoring, they heavily rely on asynchronous human interaction rather than real-time AI inference for immediate user feedback Batra (2022).

2.4.3 AI-Powered Diagnostic Apps

AI apps in dermatology and ophthalmology (e.g., SkinVision, EyeQue) offer real-time image classification for early detection. These serve as strong precedents for using AI via web interfaces in medical diagnostics. For example, SkinVision has demonstrated high sensitivity (up to 95%) in detecting skin conditions using smartphone-captured images, proving the viability of the consumer-grade camera approach for medical triage Sangers et al. (2022).

2.5 Review On Existing Solution

2.5.1 Comparison

Table 2.1: Comparison of Dental Service Features

Feature	**DentAi**	**Government Clinic**	**Toothpick.com**
Online Appointment Booking	✓	• **X** Phone-based booking only; long waiting time	✓
Real-Time Availability	✓	• **X** Limited slots; no live availability system	✓
AI Dental Pre-Assessment	✓	• **X** No AI; assessment only during physical visit	• **X** Platform focuses on booking, not diagnosis
Instant Initial Assessment	✓	• **X** 30–60 minutes waiting time even with appointment	• **X** Assessment only after seeing dentist
24/7 Accessibility	✓	• **X** Limited clinic hours (office hours only)	✓
Digital Records & History	✓	• **X** Mostly paper-based; records can be lost	• **X** No centralized dental assessment records
History Tracking	✓	• **X** No patient-accessible progress tracking	• **X** Booking history only, no health progress tracking
End-to-End Care System	✓	• **X** Manual, fragmented process	• **X** Acts as a marketplace, not a full care system

Table 2.1 shows the comparison between the proposed system and existing solutions.

2.6 Role Of Artificial Intelligence In Dental Diagnosis

Schwendicke et al. (2019) highlighted that traditional diagnostic methods can produce inconsistent outcomes due to individual judgment and manual interpretation of radiographs.

This variability is supported by findings from Kondori (2009), who discovered that diagnostic accuracy ranged from 65% to 80% depending on the dentist's experience and the case complexity. Such variability can result in missed diagnoses or unnecessary treatments.

To address these inconsistencies, AI-powered systems have emerged as promising tools in the dental field. Lin et al. (2021) demonstrated that AI models achieved over 90% accuracy when detecting common dental issues like caries, gingivitis, and root canal infections, with results validated against laboratory tissue analysis. These AI systems reduce reliance on visual guesswork by identifying patterns and features that are difficult for the human eye to detect consistently.

2.7 Summary

This literature review highlights the need for an integrated, user-friendly dental health platform accessible through the web. Existing solutions often lack public access, real-time detection, or region-specific support. The proposed web application bridges these gaps by combining AI-driven dental assessments with a booking system in a single React-Flask-Firebase architecture. This design not only promotes early dental intervention but also improves access to dental care in a scalable, affordable way.

Chapter 3

Research Methodology

3.1 Introduction

This chapter outlines the methods, tools, techniques, and software development life-cycle models utilized in this research project. It covers the technologies used, system architecture, and methodology models applied to the development of AI-powered dental assessment system. The Rapid Application Development (RAD) methodology was selected for its iterative nature, user-centric focus, and ability to quickly deliver functional prototypes through continuous user involvement and rapid prototyping cycles. This approach ensures quick delivery of working prototypes, continuous user feedback integration, and flexible adaptation to changing requirements.

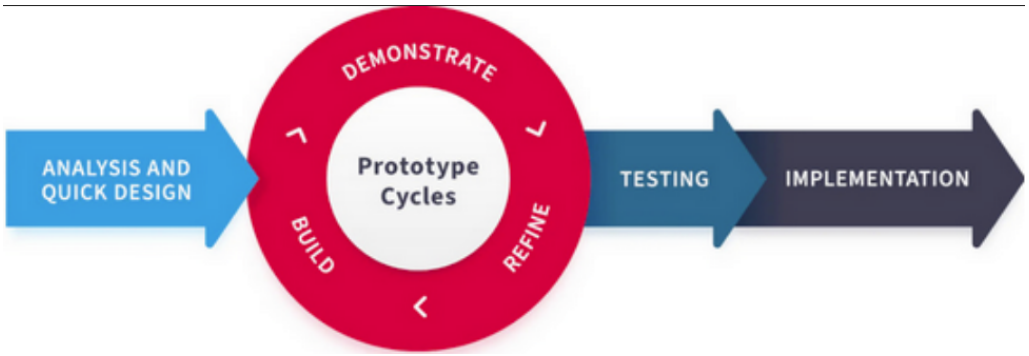


Figure 3.1: RAD Methodology

3.1.1 PHASE 1: ANALYSIS AND QUICK DESIGN

This initial phase combined requirement gathering with preliminary system design to establish the project scope rapidly. Instead of a lengthy planning phase, functional and non-functional requirements were identified through Joint Application Development (JAD) sessions involving stakeholders, potential users, and healthcare professionals.

Key Activities:

- **JAD Sessions:** Conducted collaborative workshops to define the scope of the dental assessment and appointment features.
- **Quick Design Modeling:** Created rapid paper prototypes and low-fidelity wireframes to visualize the user flow for image uploading and AI analysis.
- **Feasibility Analysis:** Prioritized requirements based on the technical feasibility of integrating the YOLOv11 model with the React frontend.
- **Technology Selection:** Finalized the tech stack (React, Flask, Firebase) to ensure rapid development capabilities.

3.1.2 PHASE 2: PROTOTYPE CYCLES (Build – Demonstrate – Refine)

This phase represents the core of the RAD methodology. Unlike traditional waterfall models, the "User Design" and "Construction" phases were merged into iterative Prototype Cycles. Development followed a continuous loop of Build, Demonstrate, and Refine.

Cycle 1: Core Architecture & Database

- **Build:** Setup of Firebase configuration and Flask backend skeleton.
- **Demonstrate:** Verified database connectivity and API endpoints.
- **Refine:** Optimized database structure for user authentication based on initial latency checks.

Cycle 2: AI Integration & Basic UI

- **Build:** Training the YOLOv11 model for Gingivitis and Caries detection and integrating it with a basic React interface.
- **Demonstrate:** Showcased the image upload and raw prediction results to stakeholders.
- **Refine:** Improved the "Build" code to handle image compression and adjusted AI confidence thresholds based on feedback.

Cycle 3: Full Feature & UI Polish

- **Build:** Developed the appointment booking interface and final styling using CSS.
- **Demonstrate:** Usability testing of the complete flow from diagnosis to booking.
- **Refine:** Enhanced visual hierarchy and error handling to ensure the system is user-friendly for non-technical patients.

3.1.3 PHASE 3: TESTING

Once the prototype cycles produced a feature-complete system, the project moved into a dedicated testing phase. This phase focused on validating the system logic and AI performance before the final rollout.

Key Activities:

- **AI Validation:** Rigorous testing of the YOLOv11 model against a validation dataset to ensure an accuracy target of $> 85\%$ for all four dental conditions.
- **System Testing:** End-to-end testing of the React frontend and Flask backend to ensure data consistency during appointment booking.
- **Cross-Platform Compatibility:** Verifying system responsiveness on mobile and desktop browsers to ensure accessibility.

3.1.4 PHASE 4: IMPLEMENTATION

The final phase focused on the "Cutover" activities—moving the system from a development environment to a live production environment.

Key Activities:

- **Deployment:** Hosting the React frontend and deploying the Flask API to the production server.
- **User Acceptance Testing (UAT):** Final validation by target users to confirm the system meets real-world needs.
- **Documentation & Handover:** Finalizing user guides and system documentation for future maintenance.

3.2 Development Tool

The development of AI-powered dental assessment system focuses on a unified web-based platform with integrated AI capabilities:

- **Frontend Development (React.js):**
  - The system is developed using React.js, a modern JavaScript framework for building dynamic and responsive user interfaces.
  - Component-based architecture ensures reusable UI elements and maintainable code structure.
  - Tailwind CSS for responsive design and consistent styling across devices.
- **Backend Development (Flask + AI Integration):**
  - Flask framework provides RESTful API architecture for system backend.
  - YOLOv11 deep learning model integrated for real-time dental condition detection.
  - Microservices architecture allows independent AI model updates and scaling.

- **Database and Authentication (Firebase):**
  - Firebase provides real-time database (Cloud Firestore) for user data and analysis history.
  - Authentication services for secure user registration and login.
  - Cloud storage for dental image uploads and processing.
- **Development Tools:**
  - Visual Studio Code as the primary development environment for its flexibility and multi-language support.
  - Git version control for collaborative development and code management.
  - Postman for API testing and validation during development cycles.
  - Firebase Console for database management and user analytics.
- **AI Development Tools:**
  - Python environment for YOLOv11 model training and optimization.
  - LabelImg for dental image annotation and dataset preparation.
  - Google Colab for model training with GPU acceleration.
- **Testing and Deployment:**
  - Jest for frontend component testing and validation.
  - Flask testing framework for backend API testing.
  - Cross-browser compatibility testing across multiple devices and platforms.
  - Firebase Hosting for production deployment and continuous integration.

3.3 Flowchart

3.3.1 Patient Flowchart

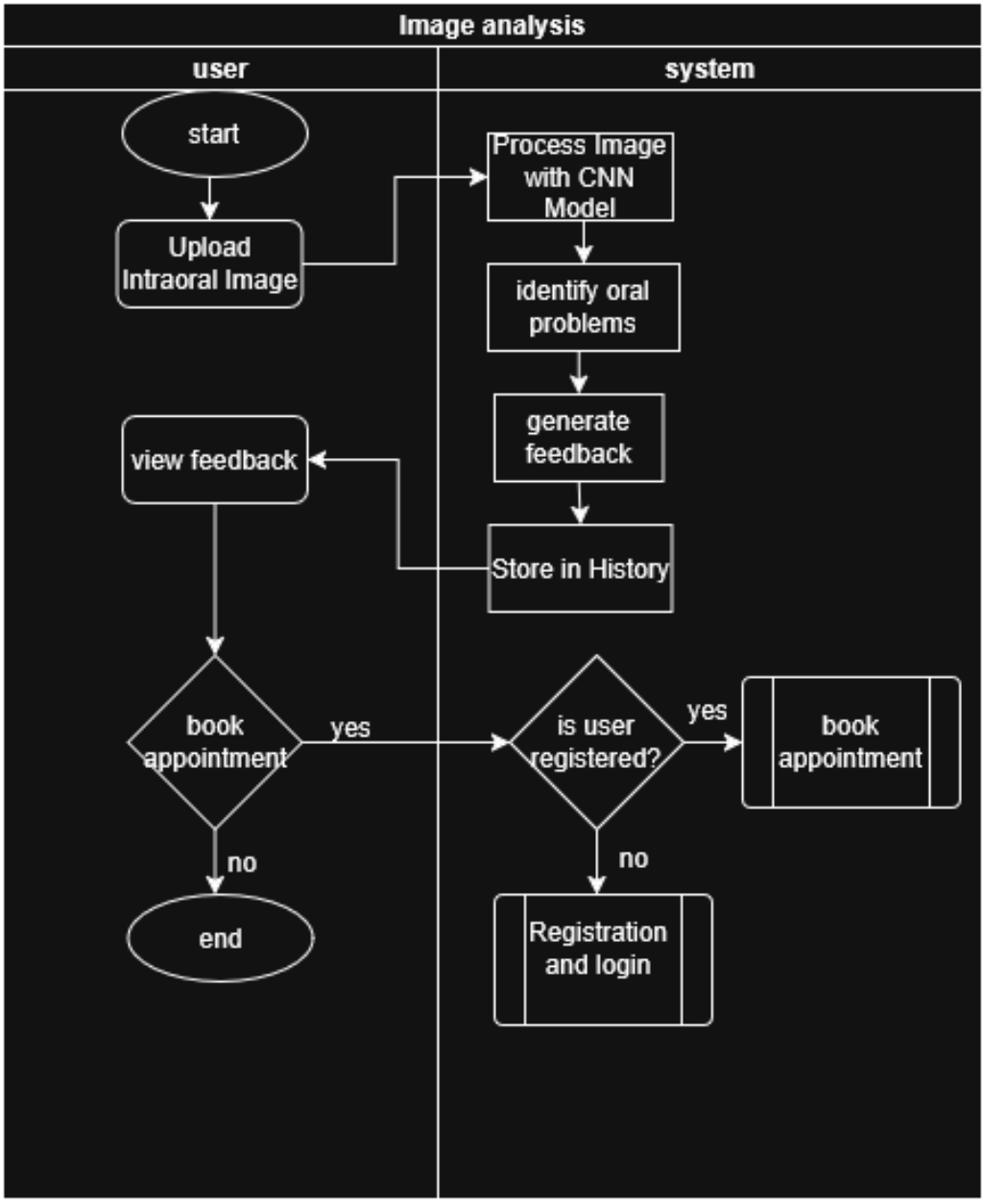


Figure 3.2: Image Analysis Flow Chart

This flowchart illustrates the image analysis process where the patient uploads a dental image. The system processes the image using an AI model to detect dental conditions and displays the assessment results to the patient.

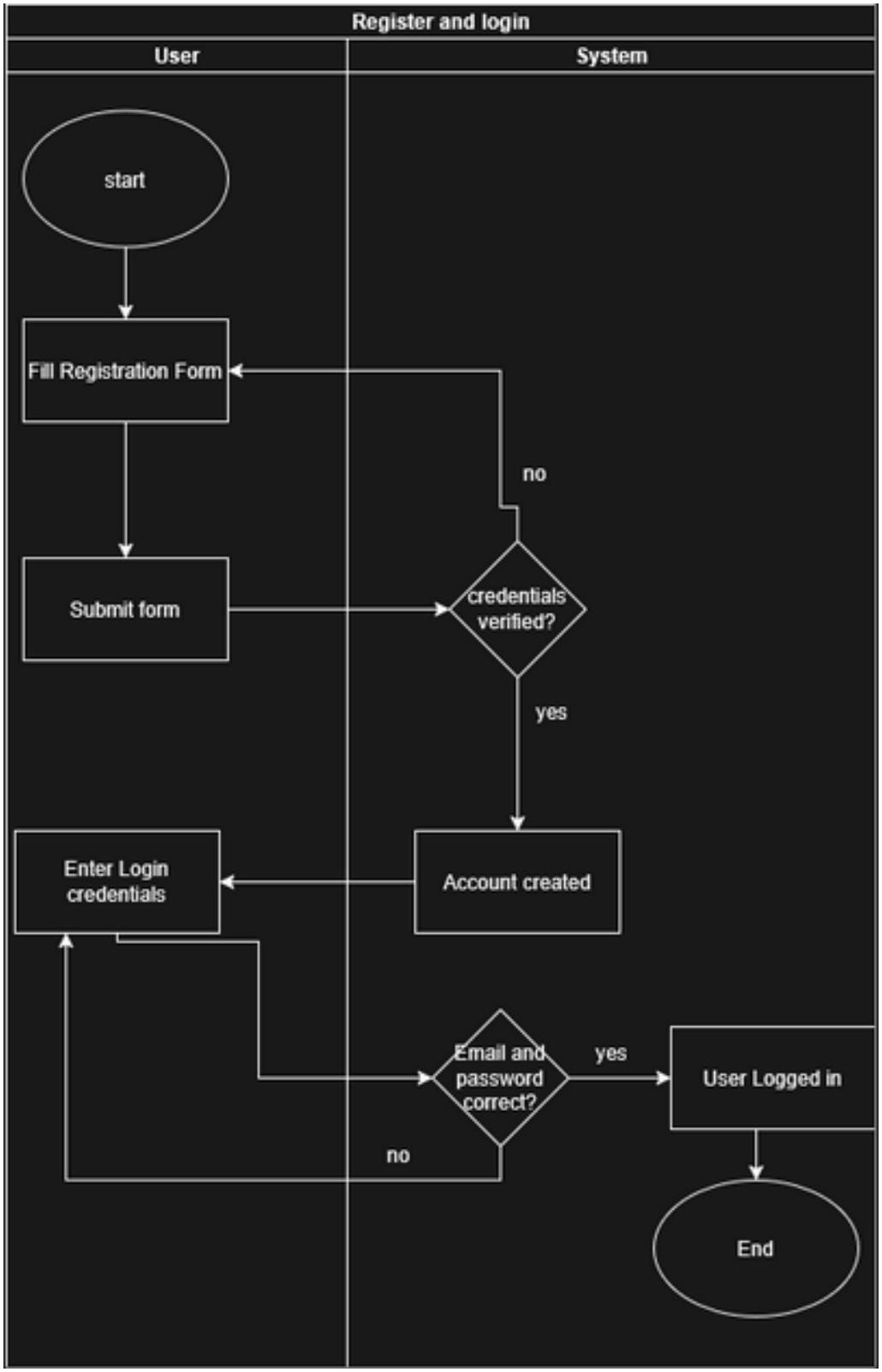


Figure 3.3: Register and Login Flow Chart

This flowchart shows the registration and login process for patients. New users register

by providing personal details, while existing users log in using their credentials to access the system.

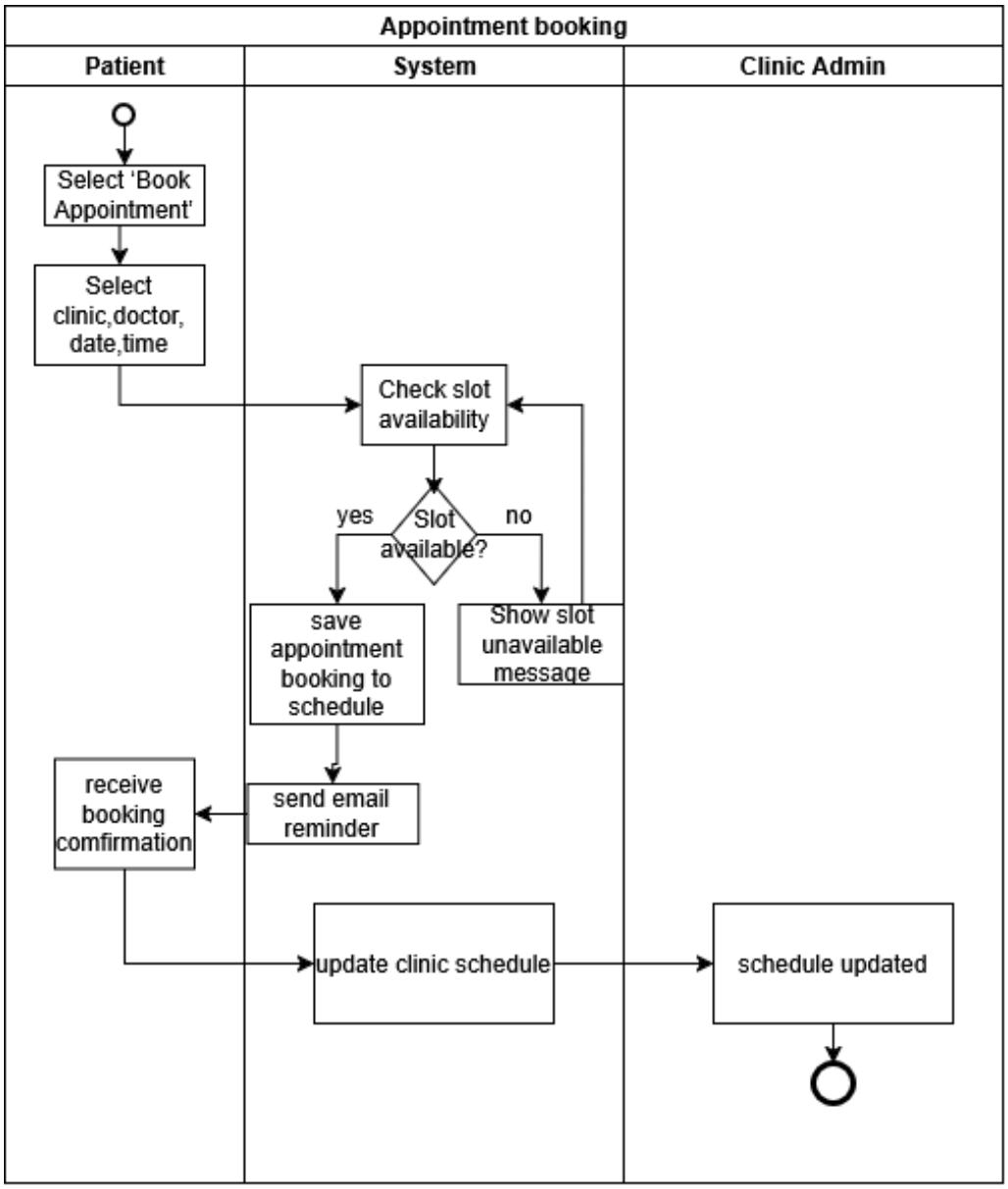


Figure 3.4: Appointment Booking Flow Chart

This flowchart describes the appointment booking process. Patients select a clinic, choose an available date and time, confirm the appointment details, and submit the booking request.

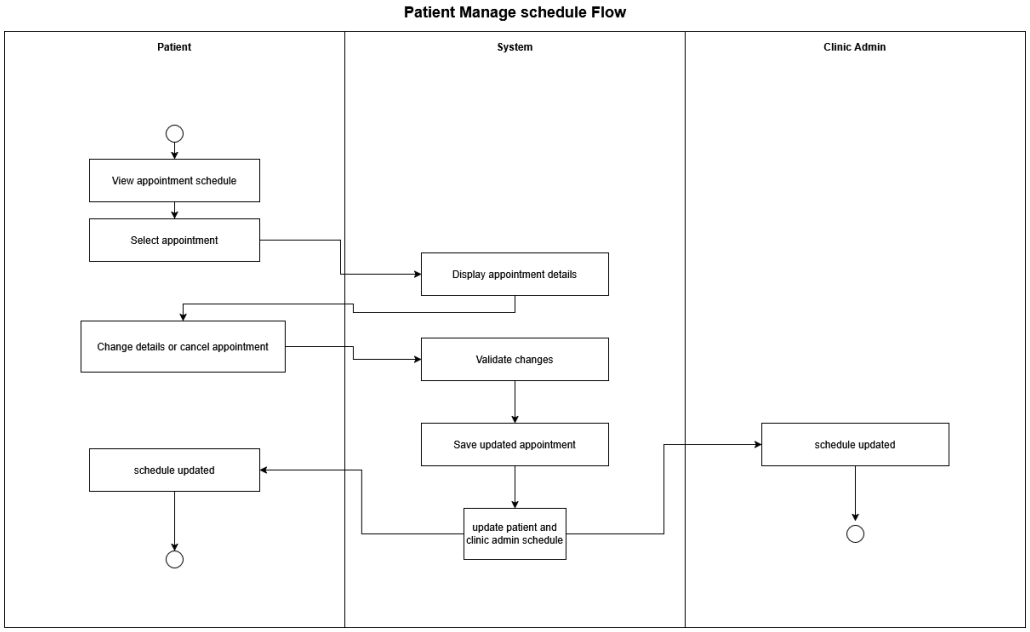


Figure 3.5: Patient Manage Schedule Flow Chart

This flowchart explains how patients manage their schedules. Patients can view, reschedule, or cancel their appointments through the system.

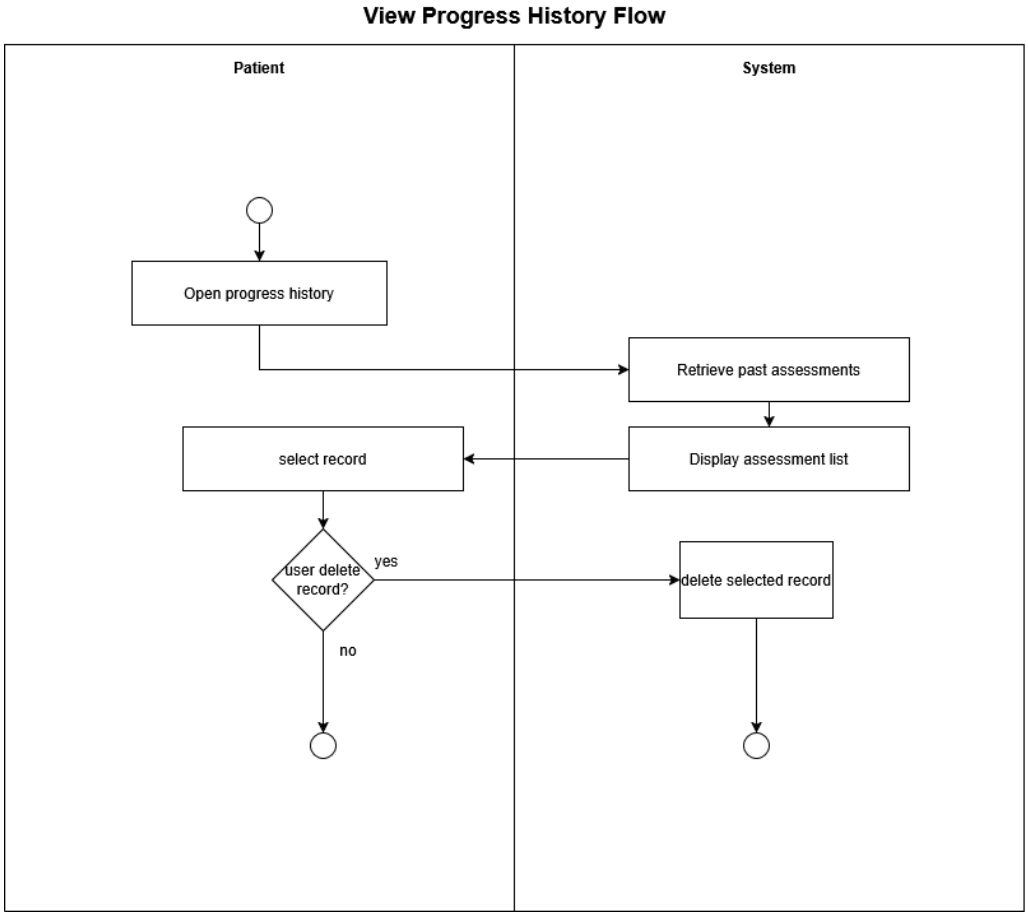


Figure 3.6: View Progress Flow Chart

This flowchart shows how patients view their dental progress history. The system displays previous assessment results and records, allowing patients to monitor their dental condition over time.

3.3.2 Admin Flowchart

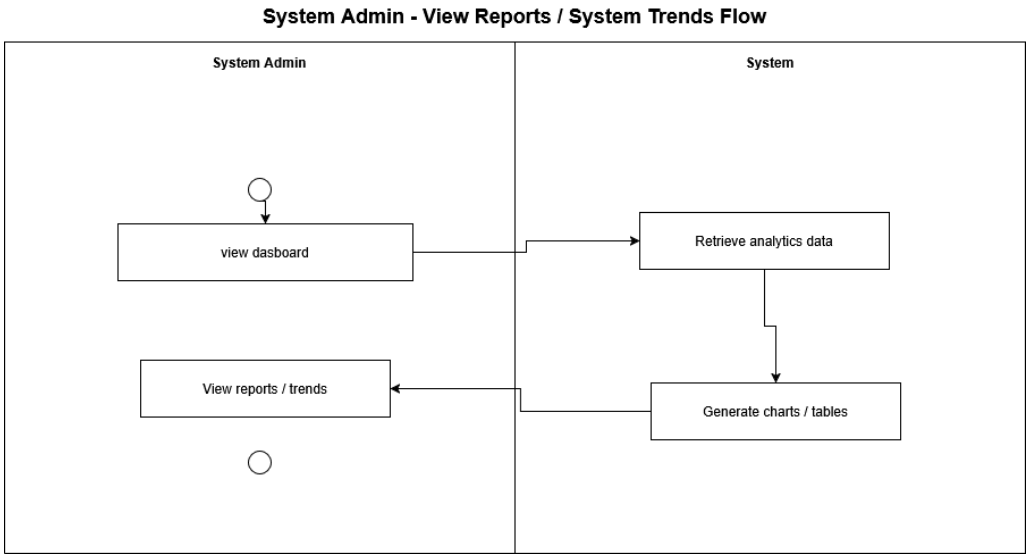


Figure 3.7: Admin View Reports/System Trends Flow Chart

This flowchart illustrates how the administrator views system reports and trends, including user activity, appointment statistics, and overall system performance.

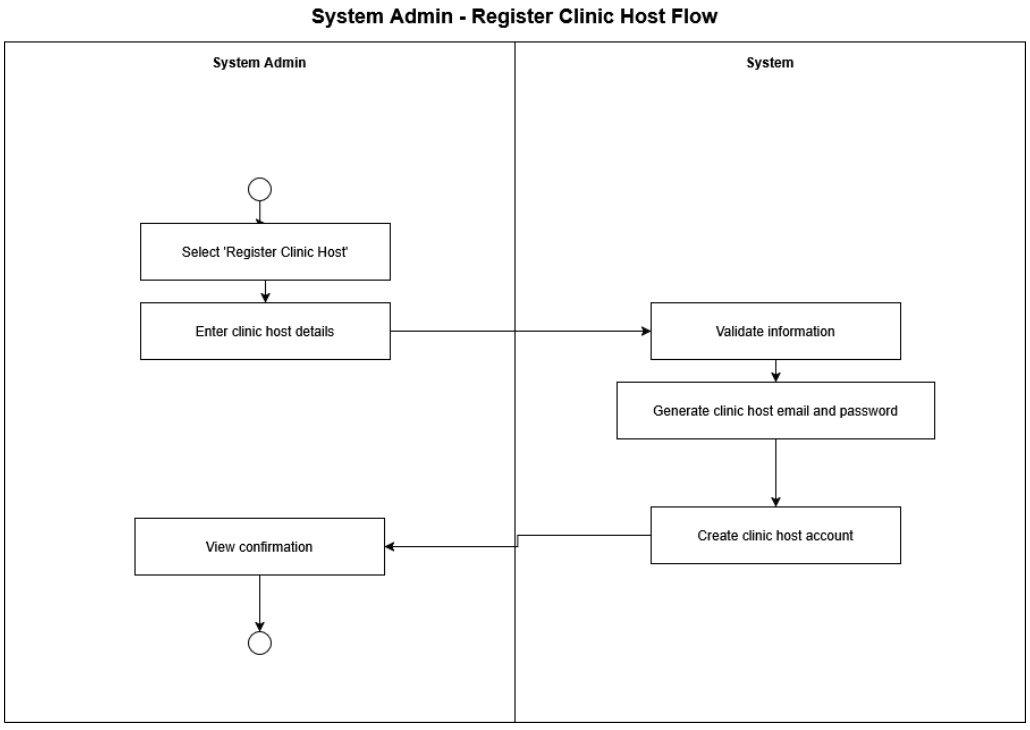


Figure 3.8: Admin Register Clinic Host Flow Chart

This flowchart describes the process of registering a clinic host. The administrator enters clinic details, verifies the information, and approves the clinic for system access.

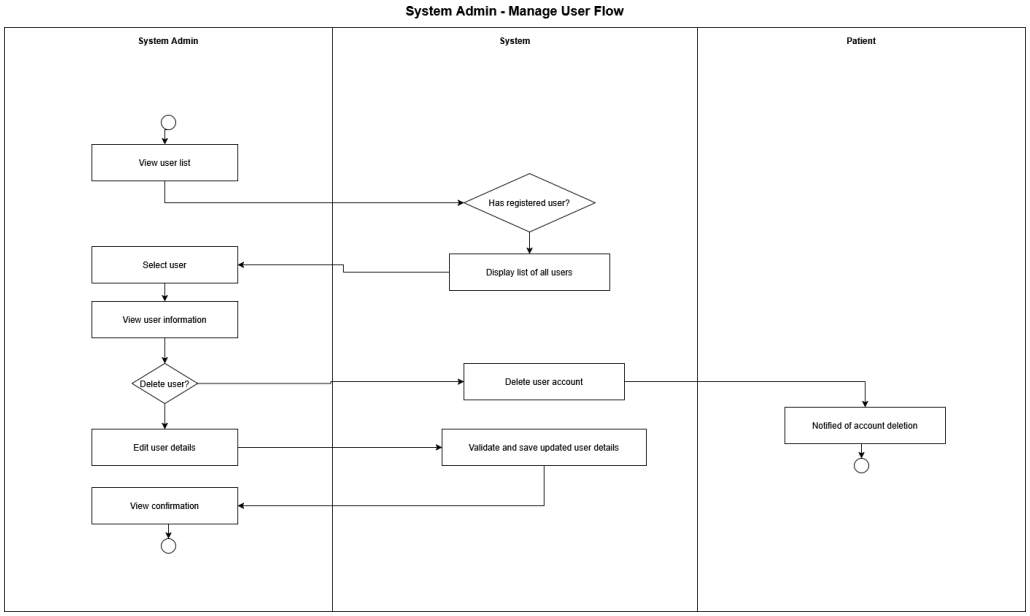


Figure 3.9: Admin Manage User Flow Chart

This flowchart shows how the administrator manages user accounts, including viewing user details, updating information, and deactivating accounts when necessary.

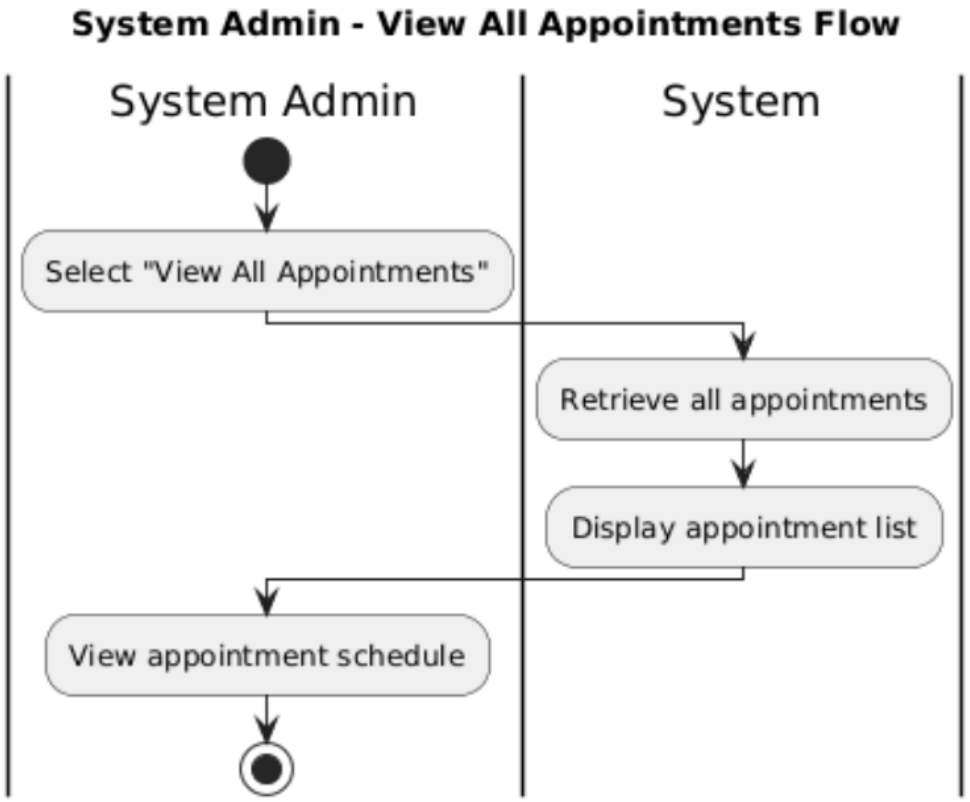


Figure 3.10: Admin View All Appointments Flow Chart

This flowchart represents the process of viewing all appointments across the system, allowing the administrator to monitor and manage appointment records.

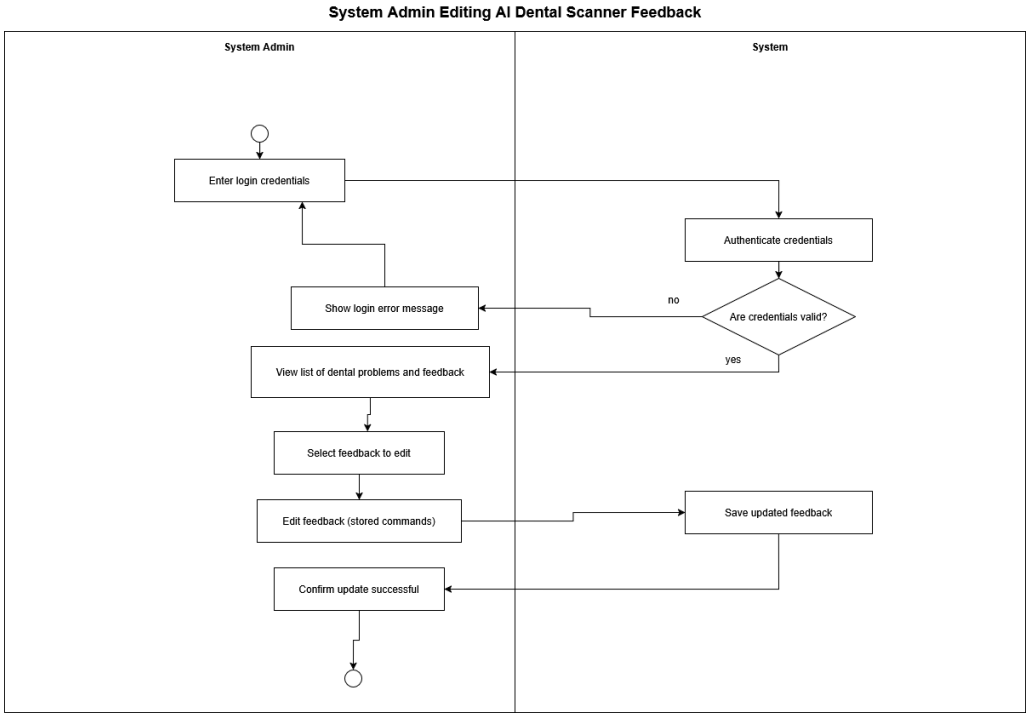


Figure 3.11: Admin Edit AI Dental Assessment Feedback Flow Chart

This flowchart explains how the administrator edits and manages AI-generated dental assessment feedback to improve clarity and accuracy of the system outputs.

3.3.3 Clinic Admin Flowchart

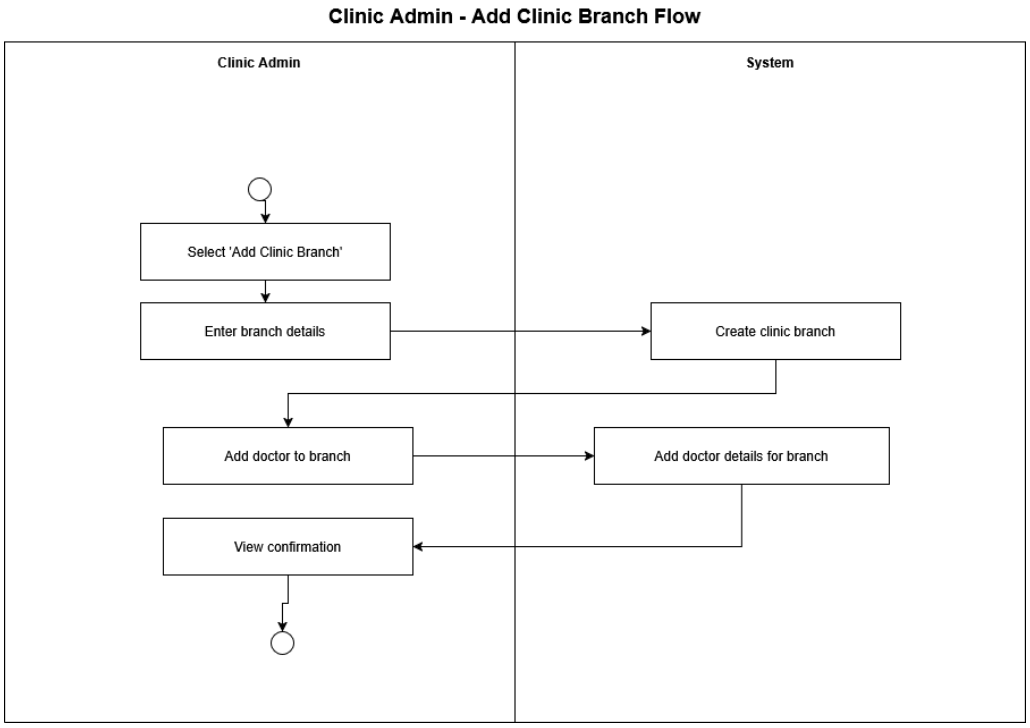


Figure 3.12: Clinic Add Branch Flow Chart

This flowchart shows the process of adding a new clinic branch by entering branch details such as location and operating hours.

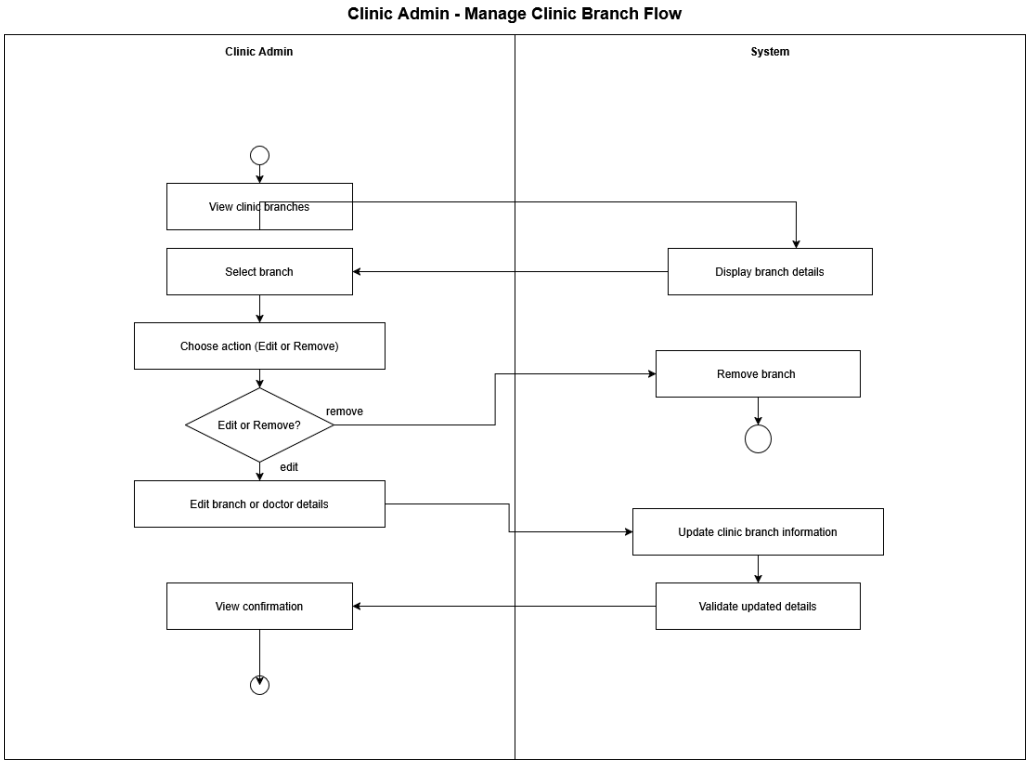


Figure 3.13: Clinic Manage Branch Flow Chart

This flowchart illustrates how clinic administrators manage existing branches, including updating or removing branch information.

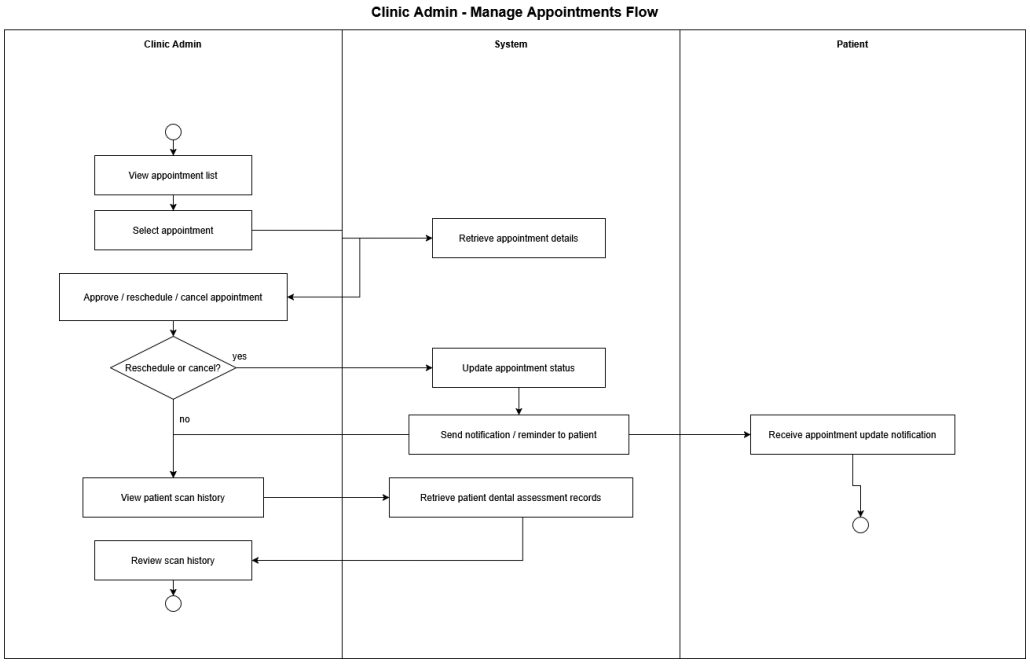


Figure 3.14: Clinic Manage Appointments Flow Chart

This flowchart represents the clinic-level appointment management process, where appointments can be viewed, approved, rescheduled, or cancelled.

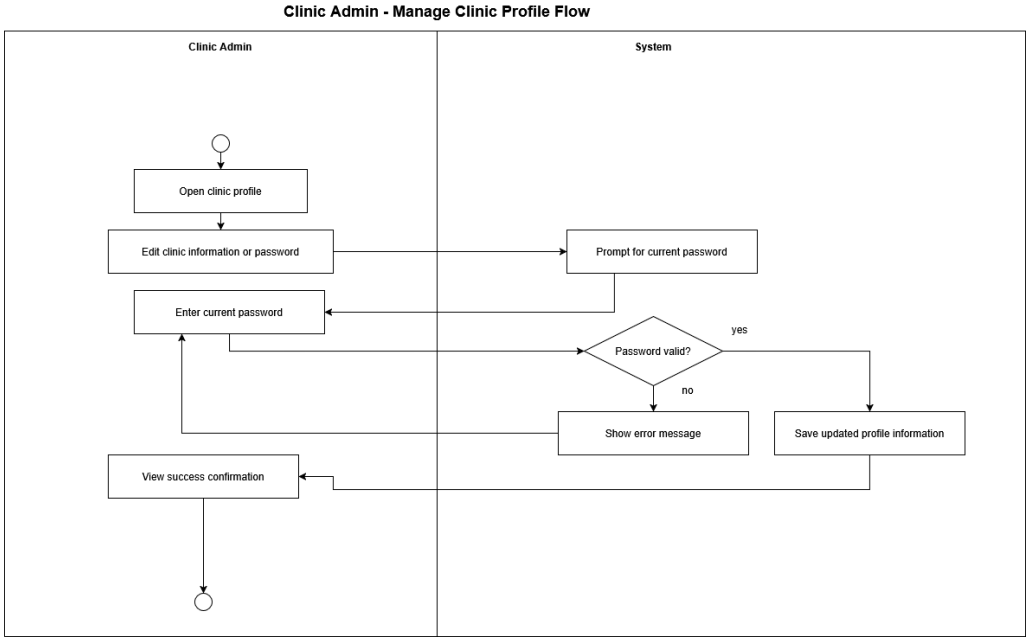


Figure 3.15: Clinic Manage Profile Flow Chart

This flowchart explains how clinic administrators manage and update clinic profile information such as contact details and services offered.

3.3.4 Use Case Diagrams

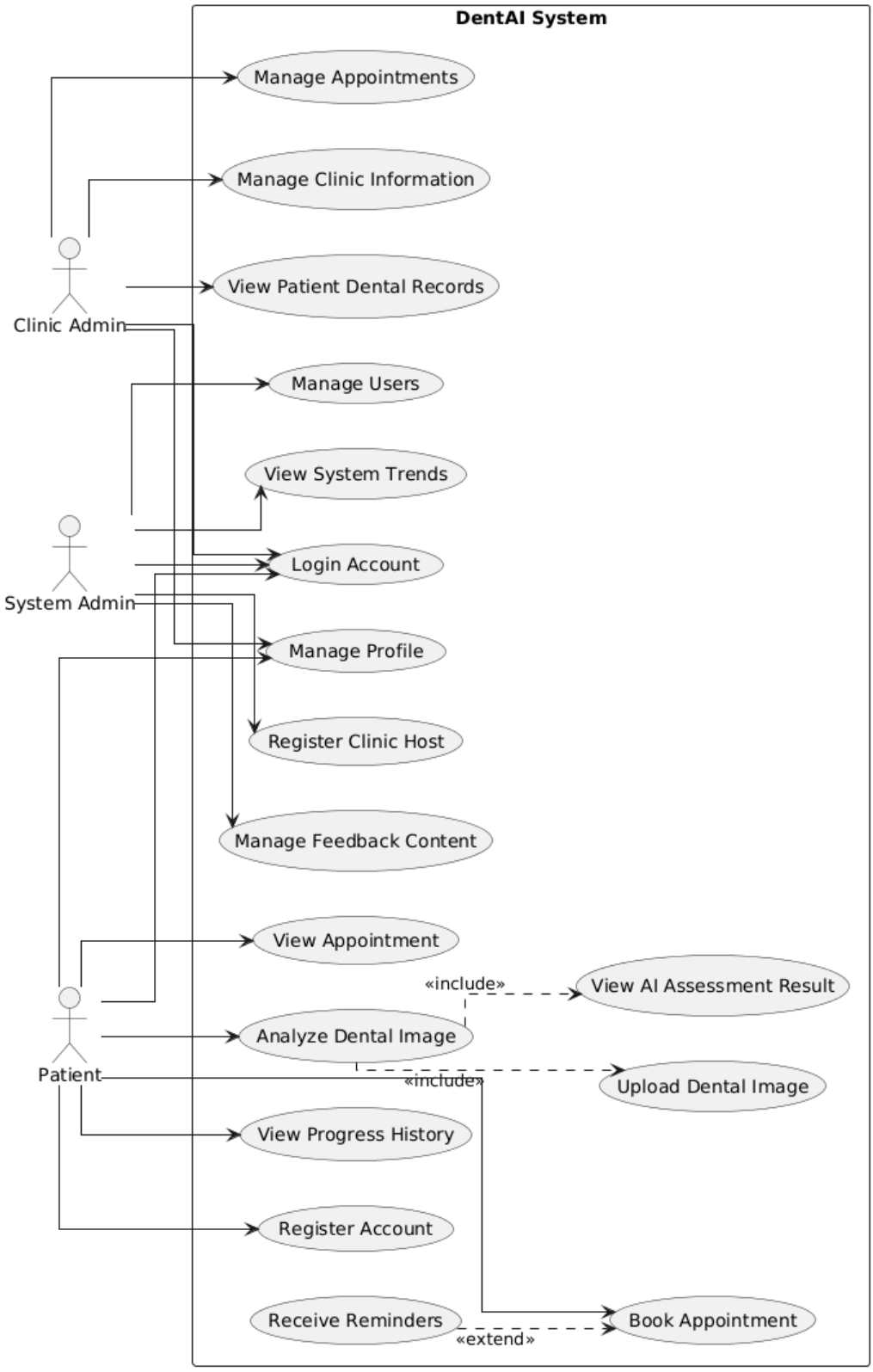


Figure 3.16: DENTAI: AI-Powered Dental Assessment Use Case Diagram

This use case diagram provides an overview of interactions between users and the DENTAI system, summarizing key functionalities such as dental assessment, appointment booking, and system management.

Table 3.1: Summary of System Use Cases

Use Case ID	**Use Case Name**
UC_01	Register Account
UC_02	Login Account
UC_03	Manage Profile
UC_04	Upload Dental Image
UC_05	Analyze Dental Image
UC_06	View AI Assessment Result
UC_07	View Progress History
UC_08	Book Appointment
UC_09	View Appointment
UC_10	Receive Reminders
UC_11	Manage Appointments
UC_12	Manage Clinic Information
UC_13	View Patient Dental Records
UC_14	Manage Users
UC_15	View System Trends
UC_16	Register Clinic Host
UC_17	Manage Feedback Content

3.3.5 Class Diagram

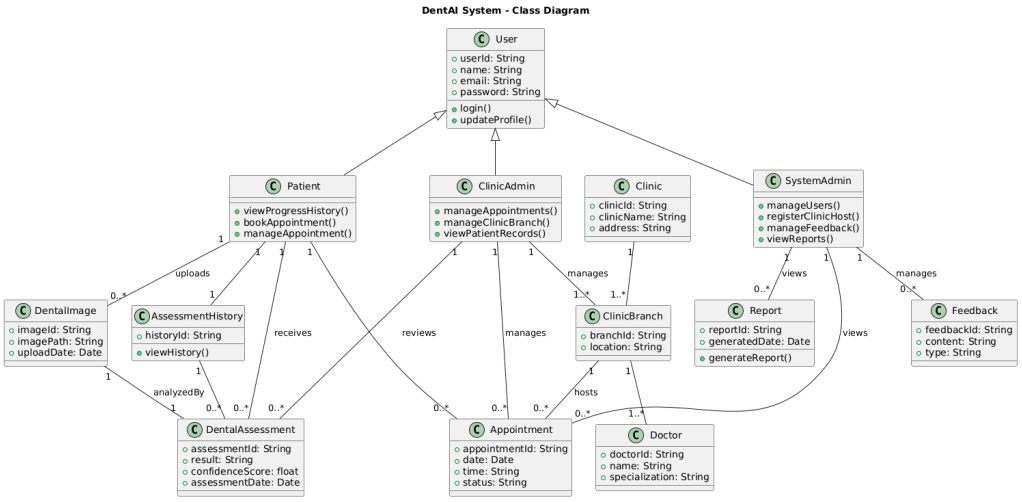


Figure 3.17: DENTAI:AI-Powered Dental Assessment Class Diagram

3.3.6 System Architecture

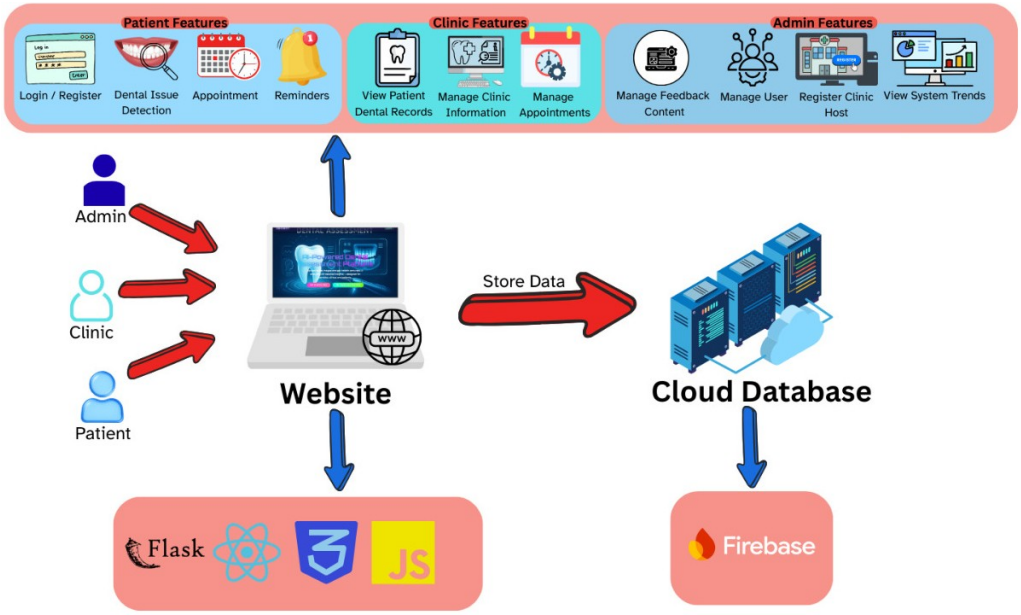


Figure 3.18: DENTAI:AI-Powered Dental Assessment System Architecture

Chapter 4

Implementation Of AI Powered Dental Assessment

4.1 Introduction

This chapter is an overview of the system implementation throughout the development of the DENTAI:AI-Powered Dental Assessment. It covers the system modules, user interface design, database implementation, and the integration of artificial intelligence functionalities. The chapter also explains how system features vary according to different user roles, including patients, clinics, and administrators.

4.2 Implementation Of The System

The DENTAI:AI-Powered Dental Assessment is developed to help users check and manage their dental health in one platform. The system uses artificial intelligence to detect dental problems and allows users to manage their dental activities easily.

The system includes several main modules such as Login and Register, Appointment, Progress History, Reminders, and Dental Issue Detection. It also provides features for

Clinic Admin to view patient dental records, manage clinic information, and manage appointments. For system administration, the Admin can manage users, register clinic hosts, manage feedback content, and view system trends. More details about each module will be explained in this chapter.

4.3 Development Tools

4.3.1 Visual Studio Code(IDE)

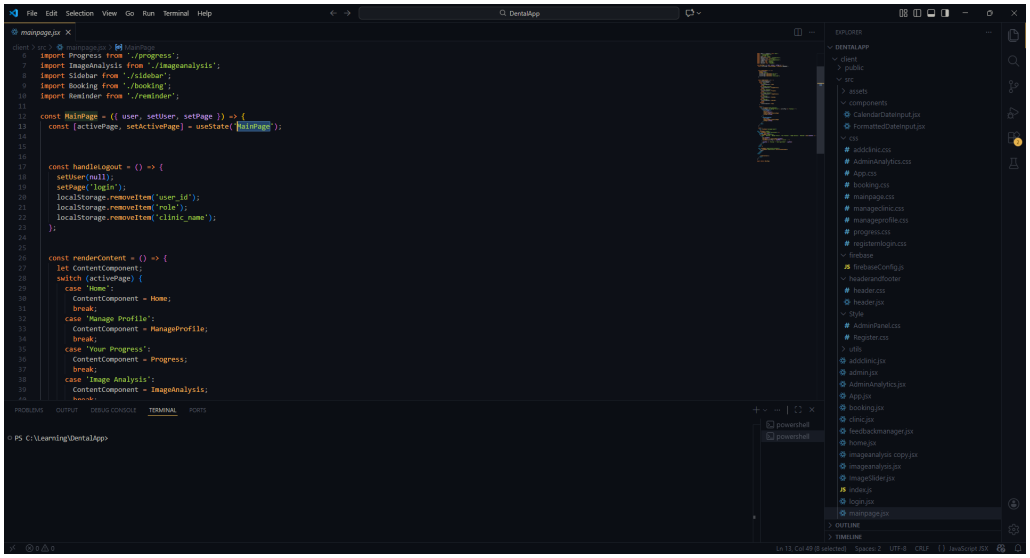


Figure 4.1: DENTAI:AI-Powered Dental Assessment Code In Visual Studio Code

Visual Studio Code (VS Code) is a software tool used to write and manage program code. It can run on different operating systems and is suitable for web application development. VS Code provides helpful features such as code suggestions, error checking, and extension support, which make development easier and faster.

In this project, VS Code is used to develop the DENTAI:AI-Powered Dental Assessment System. It is used to build the Flask backend, create the React frontend using JSX, design the interface with CSS, and connect the system with Firebase. Using VS Code allows the developer to organize the project files properly and debug the system during development.

4.3.2 Framework (Flask & React)

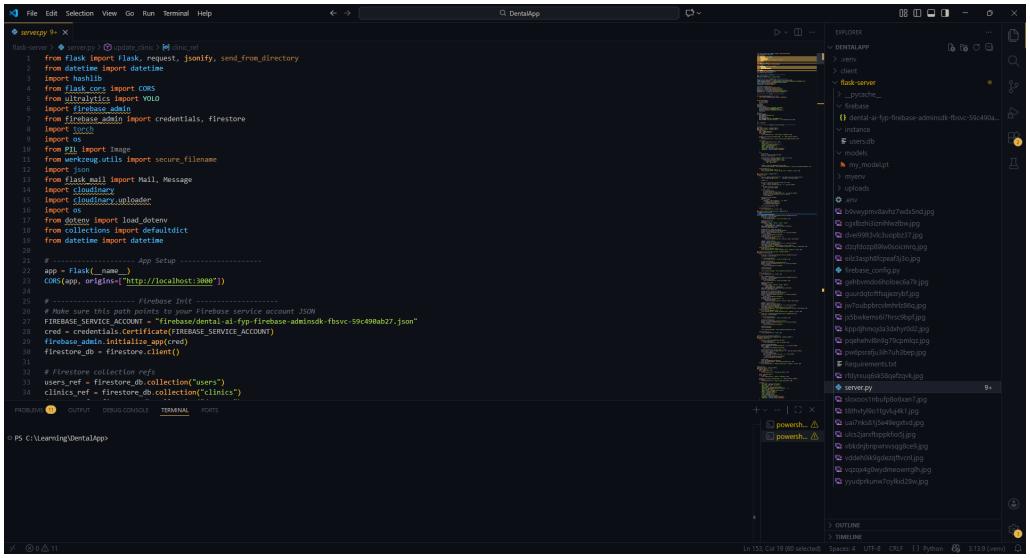


Figure 4.2: Flask in Visual Studio Code (VSCODE)

Flask is an open-source web framework written in Python. It follows a simple and flexible design that helps developers build web applications and APIs easily. Flask supports important features such as routing, request handling, user authentication, and database connection, which are suitable for developing backend systems.

In this project, Flask is used as the backend framework for the DENTAI:AI-Powered Dental Assessment System. It handles system logic, user login and registration, appointment processing, dental assessment data, and communication with the database and AI model.

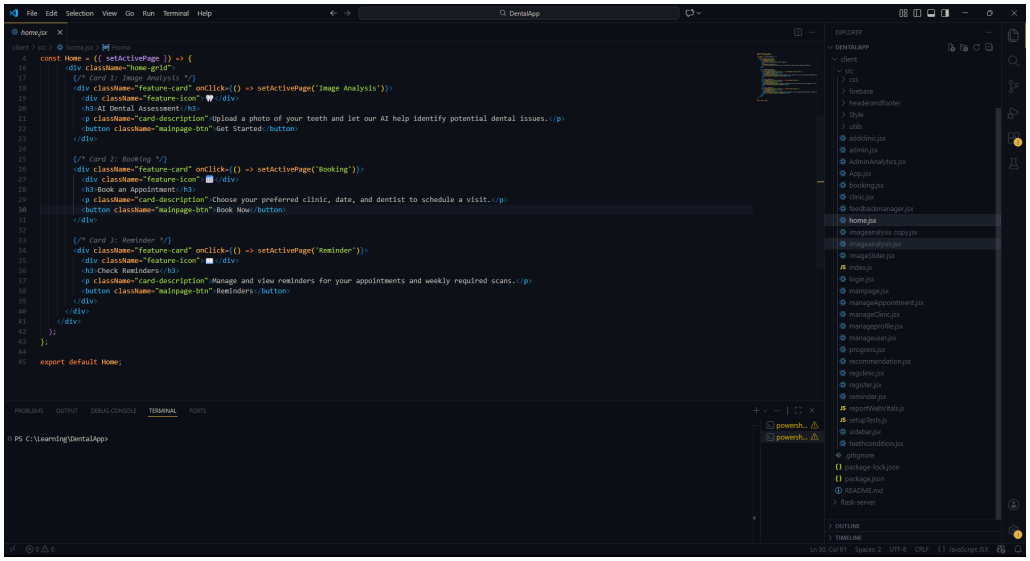


Figure 4.3: React in Visual Studio Code (VSCODE)

React is a JavaScript-based framework used to build user interfaces. It allows developers to create reusable components and manage page updates efficiently. React helps improve system performance and user experience.

For this system, React is used as the frontend framework to develop the user interface, manage user interactions, display dental assessment results, and communicate with the Flask backend.

4.3.3 Database (Firebase)

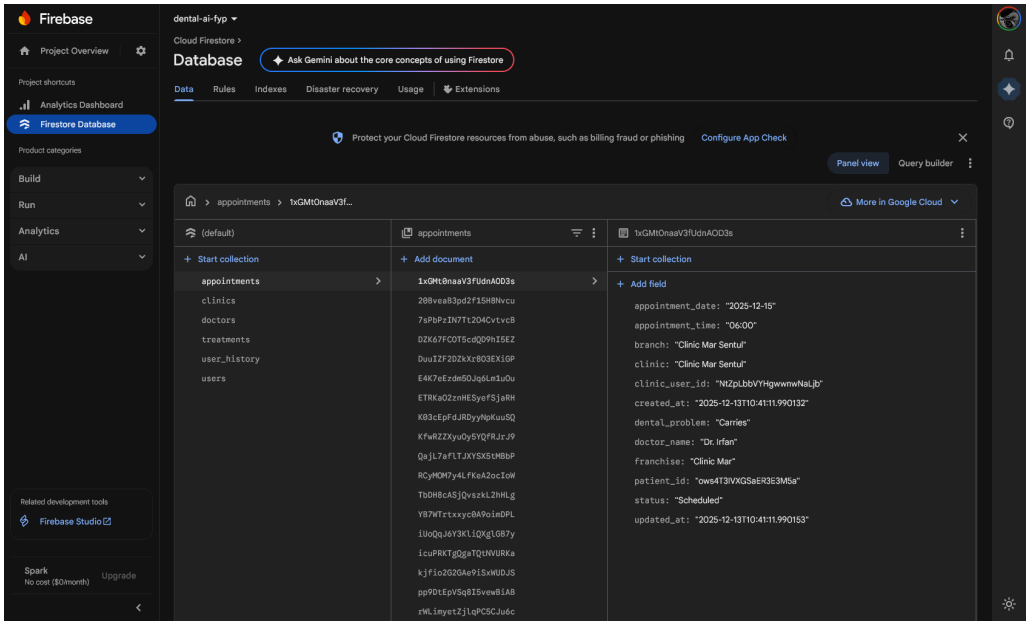


Figure 4.4: Firebase Database Console with the data for DENTAI:AI-Powered Dental Assessment

Firebase is a cloud-based database platform provided by Google. It is commonly used in web applications because it allows data to be stored and accessed in real time. Firebase supports secure data storage and easy integration with web systems, making it suitable for applications that require fast and reliable data access.

In the DENTAI:AI-Powered Dental Assessment System, Firebase is used as the main database to store system data such as user information, dental assessment records, appointment details, progress history, reminders, clinic information, and feedback data. It helps the system manage data efficiently and supports smooth communication between the frontend and backend.

4.4 Modules Implementation

The implementation focuses on the DENTAI:AI-Powered Dental Assessment system. The system has 12 main modules that work together to make dental health management easier

and more organized. These modules help the system run smoothly, improve efficiency, and make it simple for both patients and clinic staff to use. Overall, the system aims to provide a complete and easy-to-understand solution for monitoring and managing dental health.

4.4.1 Login & Register Module

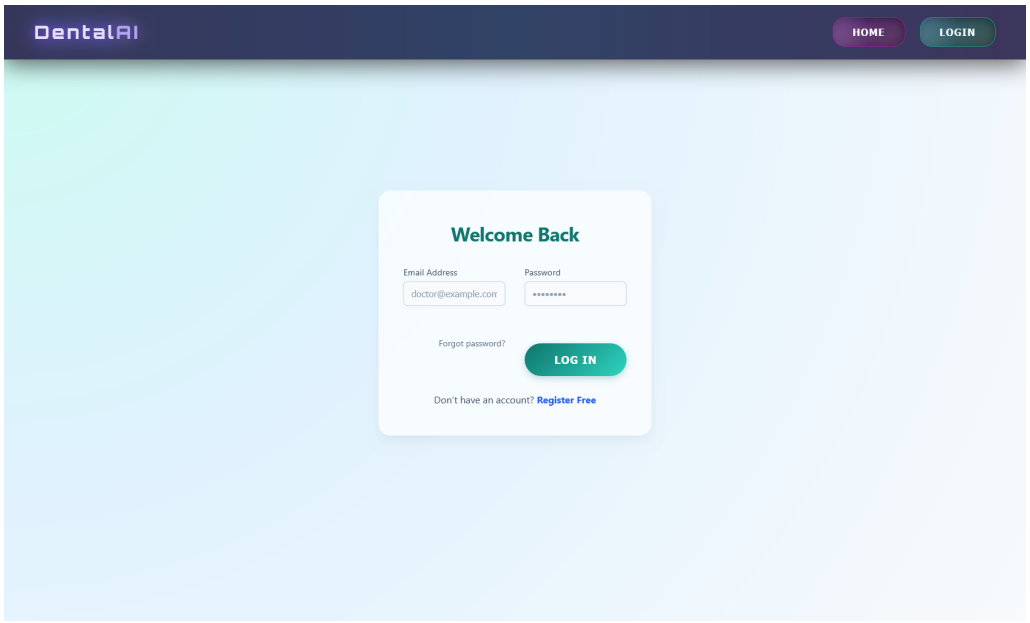


Figure 4.5: Login page

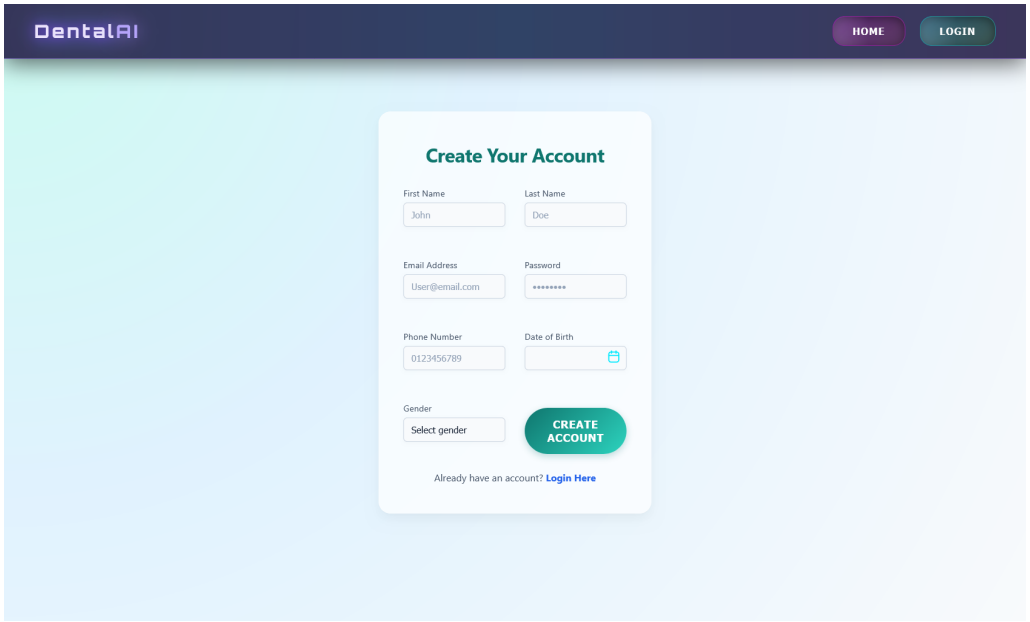


Figure 4.6: Register page

The login page of the DENTAI:AI-Powered Dental Assessment web-based system requires users to enter a username and password to access the system. This ensures that only authorized users, including patients, clinics, and system administrators, can access sensitive dental information and system features.

Upon successful login, users are redirected to their respective dashboards based on their assigned roles.

If incorrect login credentials are entered, the system displays an error message, locks the account, and prompts the user to re-enter the correct details.

4.4.2 Appointment Module

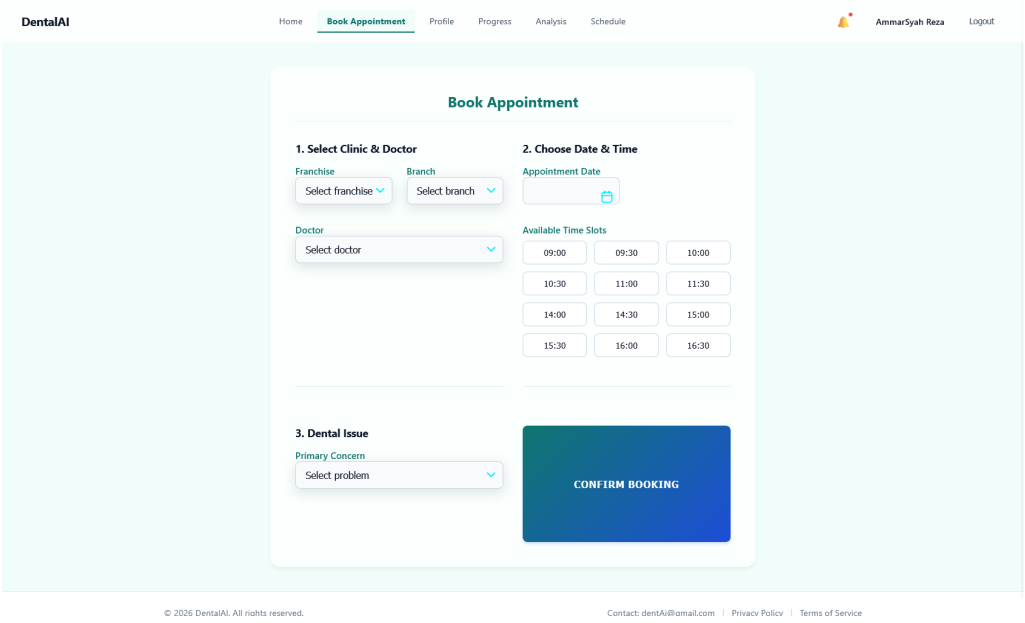


Figure 4.7: Appointment page

The Patient Appointment page helps patients book visits easily. First, they choose a clinic branch and a dentist. Then, they pick an available date and time to avoid scheduling problems.

Patients also select their dental issues so the dentist can prepare for the visit. Finally, they check their details and confirm the booking. The system saves the information to keep everything organized.

4.4.3 Progress History Module

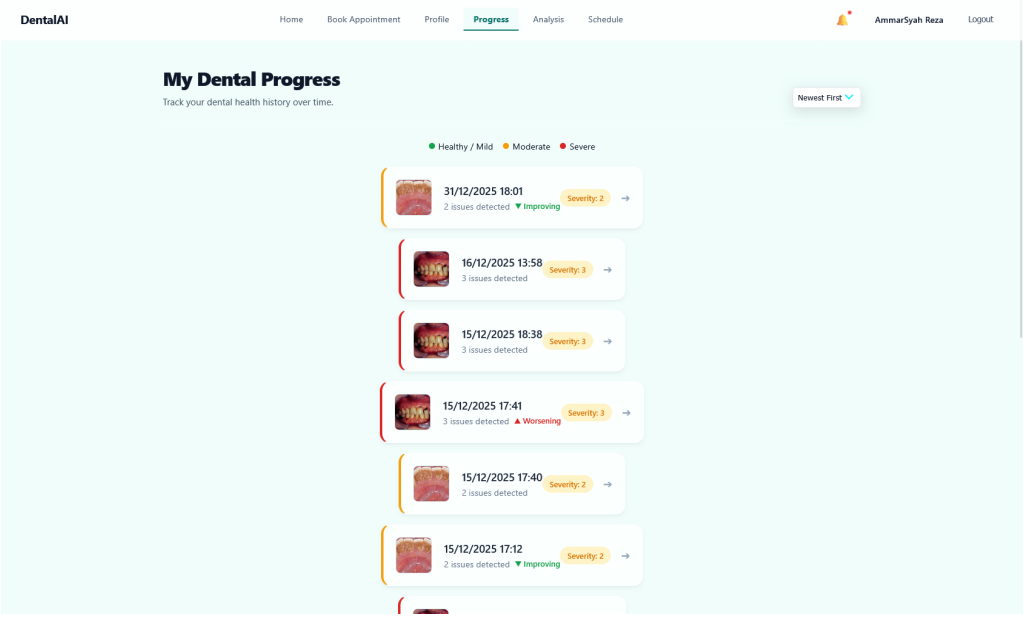


Figure 4.8: Progress page

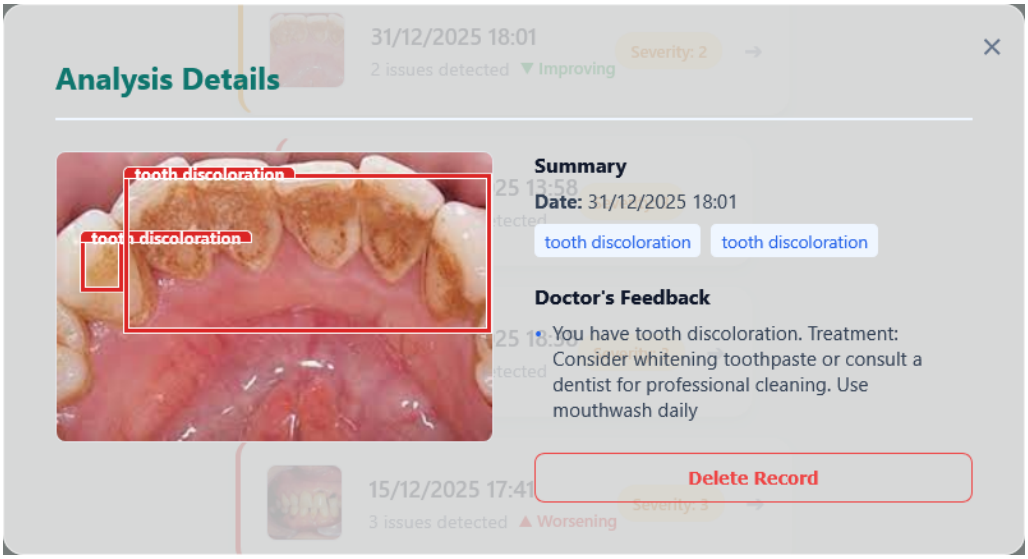


Figure 4.9: Progress details popup

The Progress page allows you to track your dental health history over time. Each entry displays a photo of your teeth, the date of the assessment, and the number of issues detected.

The system uses a color-coded system to show the severity of your condition: green for

Healthy, orange for Moderate, and red for Severe. You can also see if your condition is Improving or Worsening compared to previous checks.

By clicking on an entry, you can view the Analysis Details popup, which provides a summary of detected issues like tooth discoloration and specific feedback from a doctor. This popup also offers treatment suggestions, such as professional cleaning or daily mouthwash, and gives you the option to delete the record if necessary.

4.4.4 Reminders Module

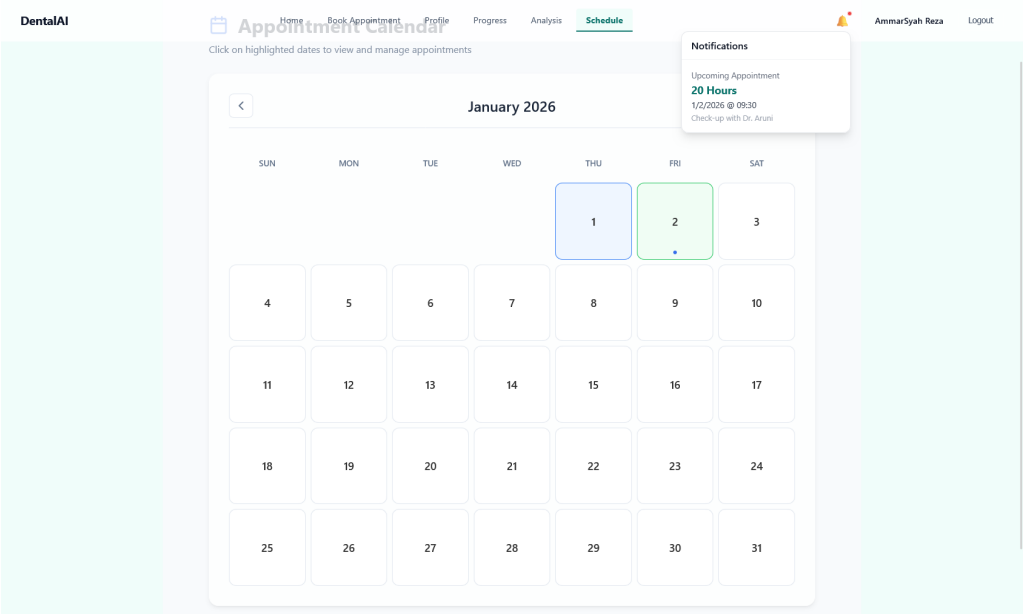


Figure 4.10: Reminders page (calendar form)

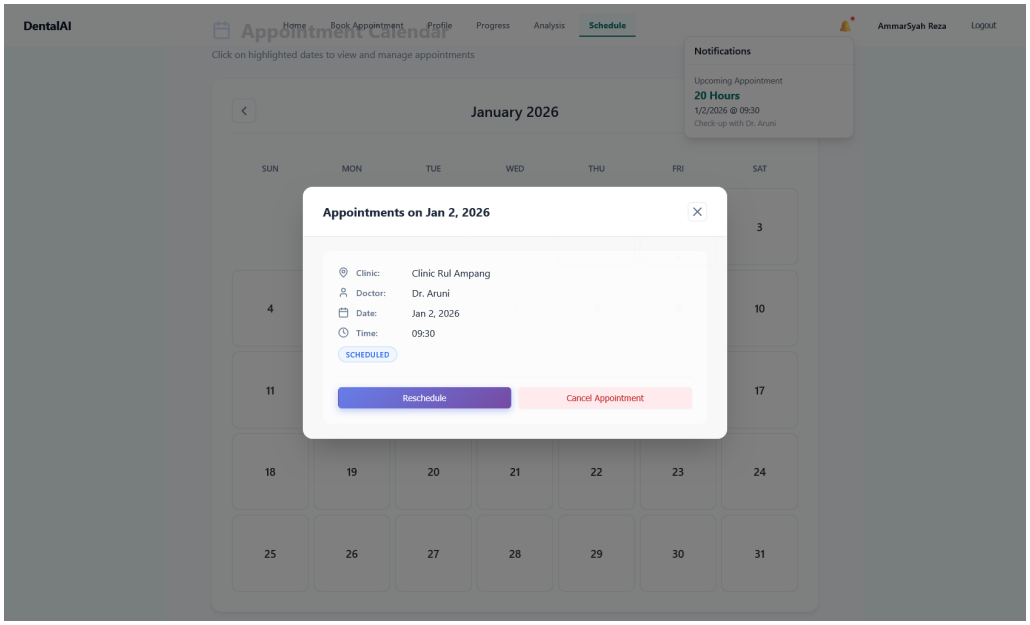


Figure 4.11: Reminder details popup

The Schedule page features an Appointment Calendar that allows you to view and manage your dental visits in a clear monthly format. Dates with scheduled appointments are highlighted, and a Notifications panel reminds you of upcoming visits, showing the countdown and specific time details.

By clicking on a highlighted date, an Appointments popup appears with full details, including the clinic name, doctor, and exact time. This popup provides convenient options to either Reschedule or Cancel Appointment, giving you full control over your dental care schedule.

4.4.5 Dental Issue Detection Module

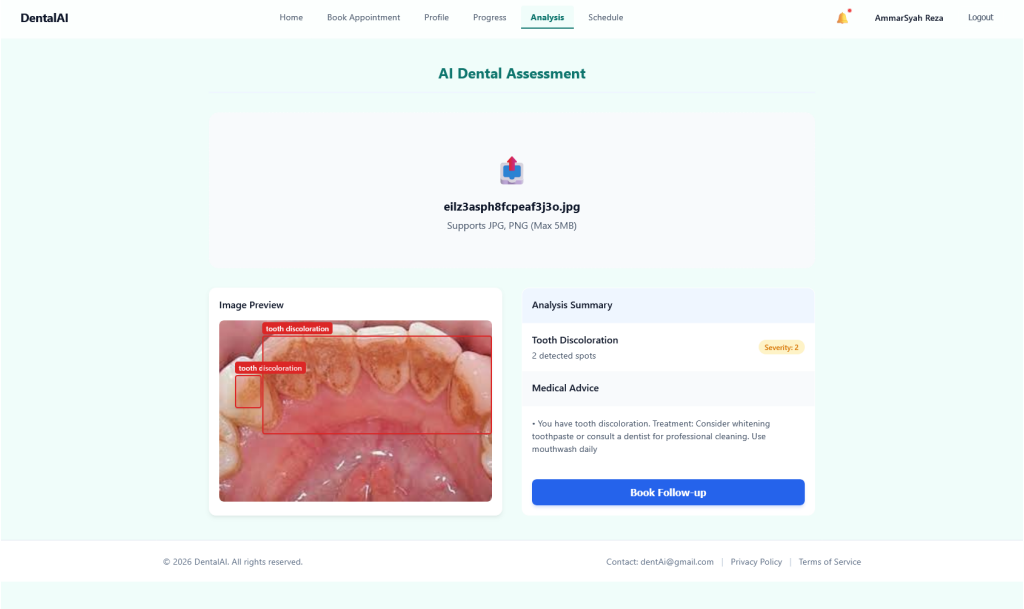


Figure 4.12: Dental analysis page

The Analysis page features the DENTAI:AI-Powered Dental Assessment tool, which allows users to receive an instant check-up by uploading a photo of your teeth. Once the image is processed, a detection box appears directly on the photo to highlight specific dental issues, such as tooth discoloration,carries, gingivitis,ulcer and calculus.

The system then provides an Analysis Summary showing the severity level and detailed Medical Advice for your condition. All results from this assessment are automatically saved to your Progress page so you can track your dental health history over time.

4.4.6 View Patient Dental Records Module

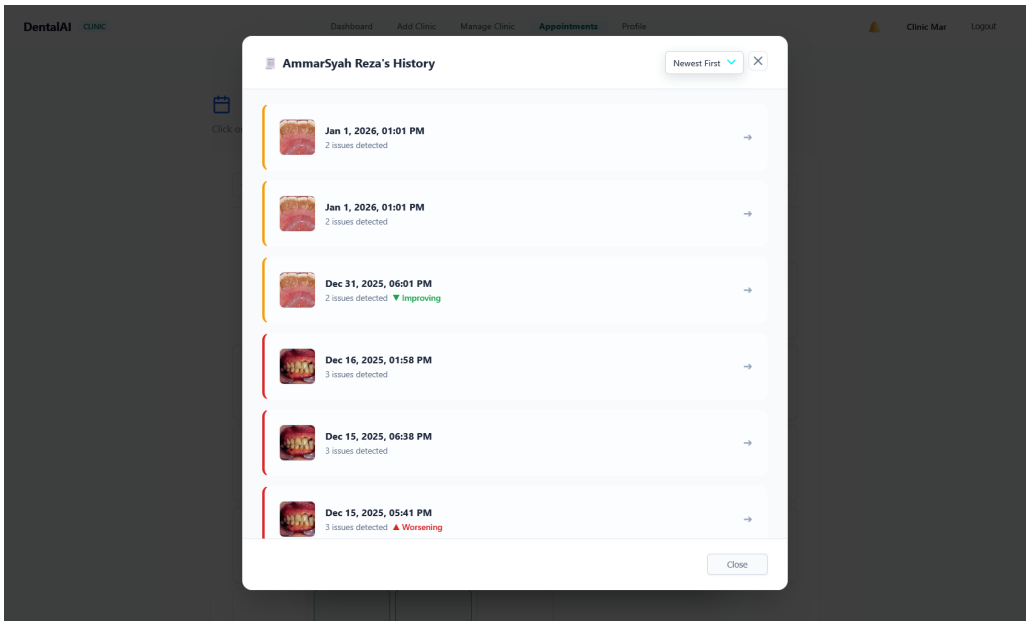


Figure 4.13: Patient dental records popup

The Patient History popup allows clinic staff to review a patient's complete dental scan records in one place. Each entry in the list shows a thumbnail of the dental photo, the exact date and time of the assessment, and the number of dental issues detected during that session.

The system uses a similar style from the progress page ,this will help the clinic to understand the patient's progress over time.

4.4.7 Manage Clinic Information Module

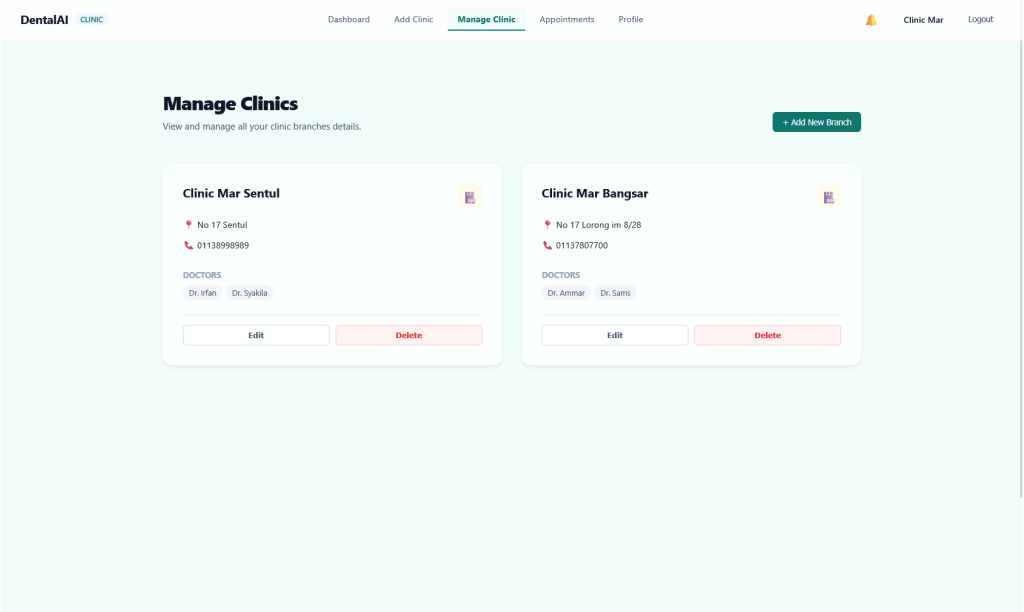


Figure 4.14: Manage clinic (edit/delete)

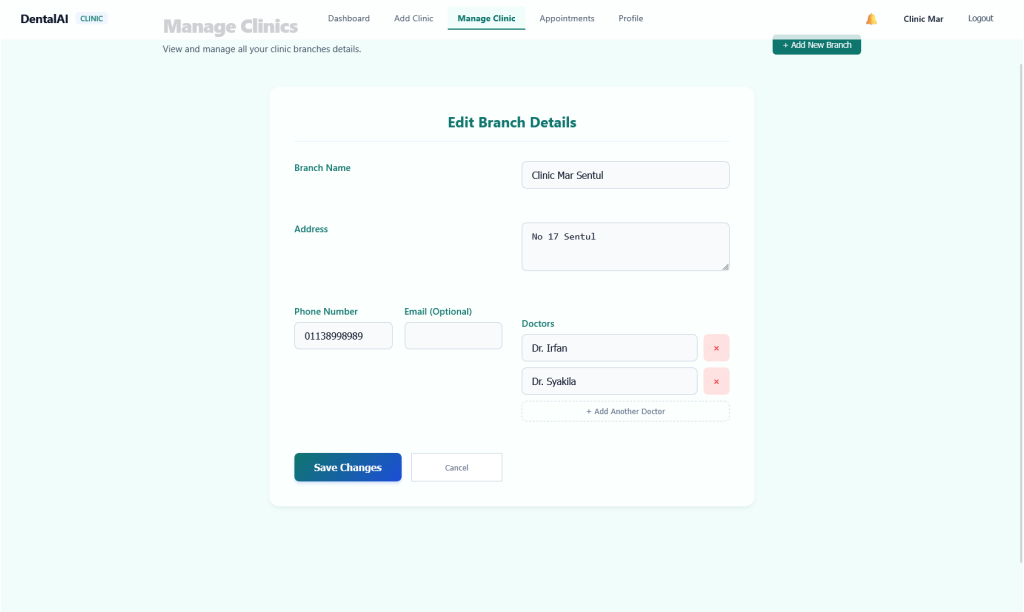


Figure 4.15: Manage clinic (edit page)

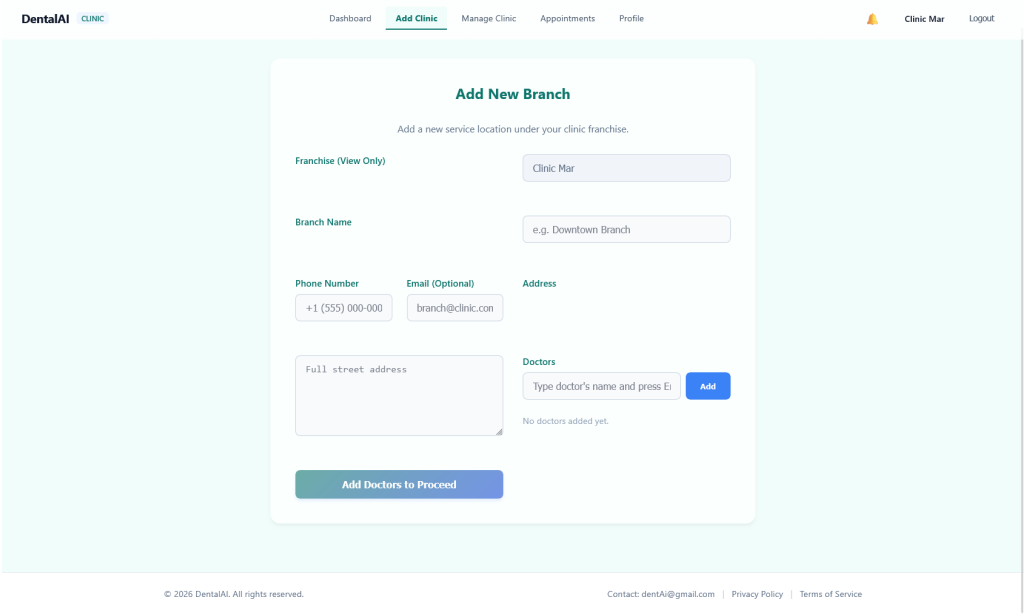


Figure 4.16: Manage clinic (add new branch)

The Manage Clinic page allows clinic admin to view and organize all their branch details in one place. Each branch is displayed as a card showing its address, contact number, and the list of doctors assigned to that location. From this main management page, you can easily update your clinic's information by clicking the Edit button or remove a branch entirely using the Delete button.

The Edit Branch Details page allows you to modify the branch name, address, and phone number for an existing location. You can also manage the staff roster by adding or removing doctors from the branch. Once all updates are finished, clicking Save Changes ensures your clinic's public information remains accurate for patients.

To grow your network, you can use the Add Clinic page to register a new service location under your franchise. This page requires you to enter specific details such as the branch name, phone number, address, and email. You can also add doctors to the new branch to finalize the setup.

4.4.8 Manage Appointments Module

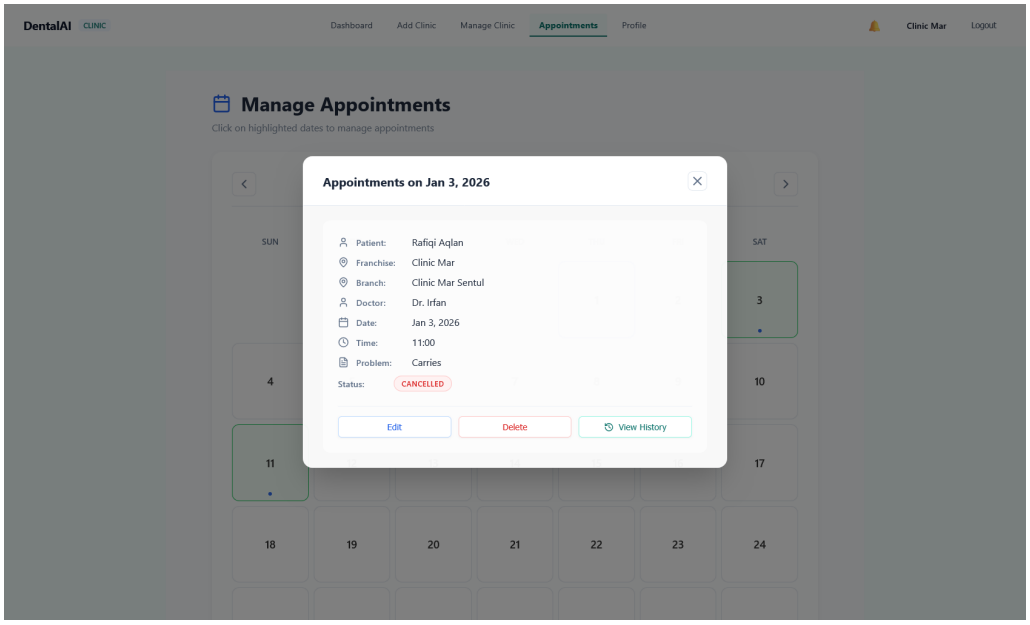


Figure 4.17: Manage appointments (edit/delete/view user history)

The Manage Appointments page provides a centralized calendar view for clinic staff to oversee all patient bookings. By clicking on a specific date, an Appointments popup appears, displaying a comprehensive list of scheduled visits, including patient names, assigned doctors, and the specific dental problems reported.

From this popup, staff can maintain the schedule by using the Edit button to update appointment details or the Delete button to remove a booking. Additionally, the View History button allows clinic users to access a patient's full dental record, enabling them to review past AI assessments and track health trends before the patient arrives.

4.4.9 Manage Feedback Content Module

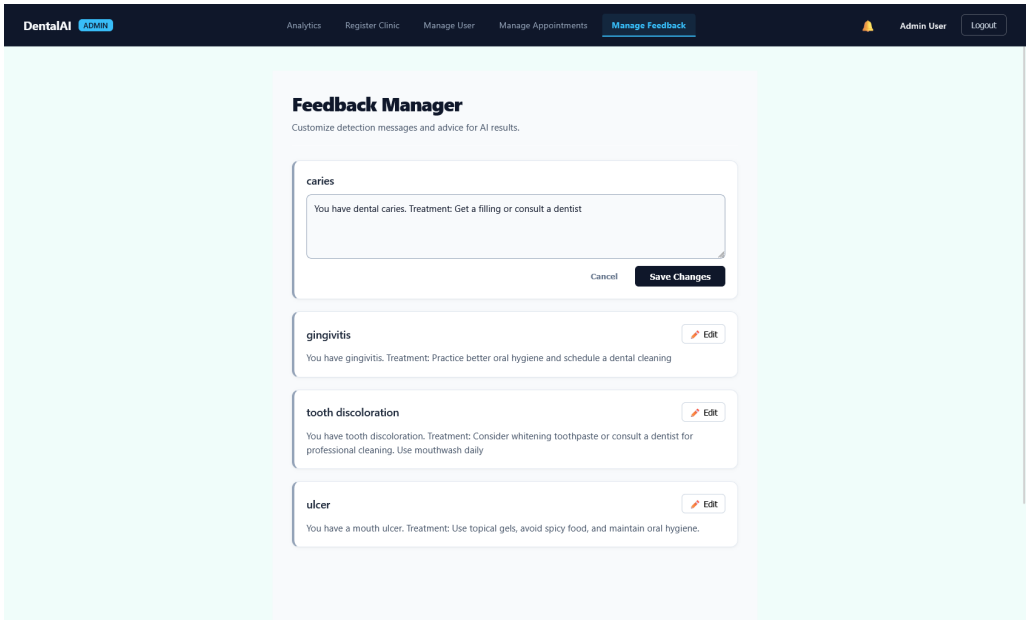


Figure 4.18: Manage feedback page

The Feedback Manager page allows administrators to customize the automated medical advice and detection messages for different dental conditions. Each specific issue, such as caries, gingivitis, tooth discoloration, or ulcers, has its own section where the recommended treatment and advice are stored. By clicking the Edit button, an admin can update the text to provide more accurate guidance, such as suggesting specific toothpastes or professional cleanings. Once the updates are finished, clicking Save Changes ensures that these new instructions will be automatically shown to patients in their Analysis Summary after an AI assessment.

4.4.10 Manage User Module

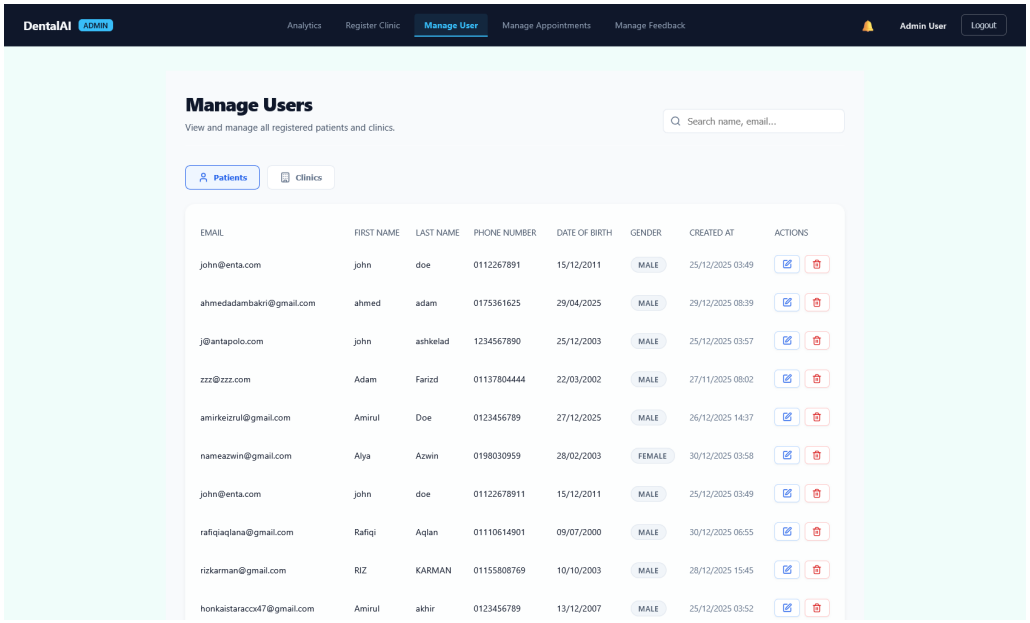


Figure 4.19: Manage User (Patient) page

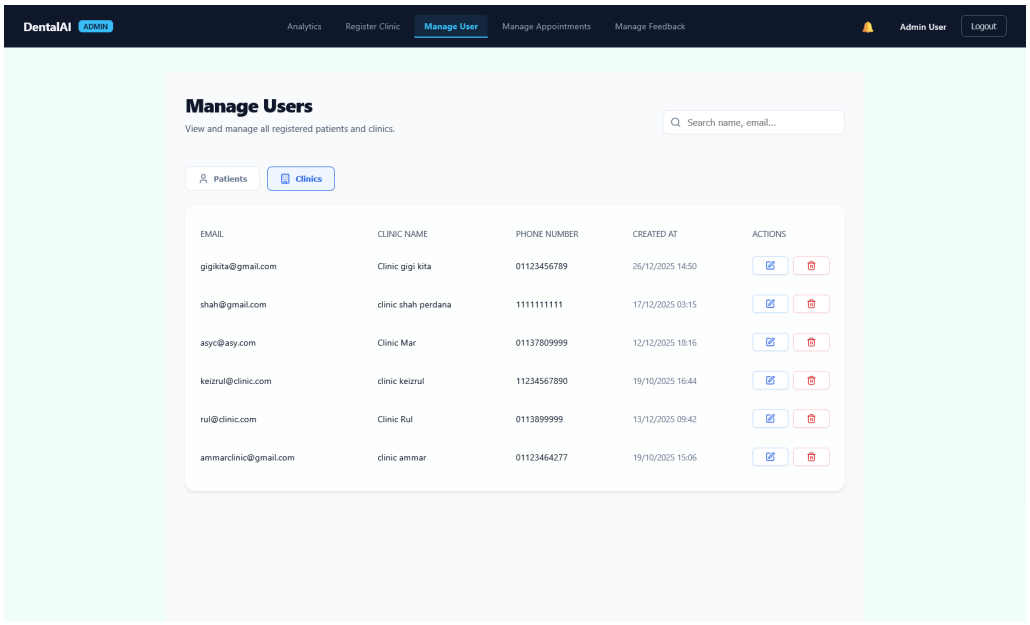


Figure 4.20: Manage User (Clinic) page

The Manage User (Patients) page allows administrators to oversee all registered patient accounts in a comprehensive table view. Each entry displays the patient's email, full name, phone number, date of birth, and gender, along with the date their account was created. Administrators can maintain the accuracy of the user database by using the Edit

action to update patient profile information or the Delete action to remove an account entirely.

The Manage User (Clinics) page provides a dedicated view for administrators to manage all clinic accounts registered on the platform. The system lists each clinic by its primary email, clinic name, and contact phone number, showing exactly when the account was first created. To ensure high-quality service, administrators have the authority to Edit clinic account details or use the Delete button to remove a clinic franchise from the system if it is no longer active.

4.4.11 Register Clinic Host Module

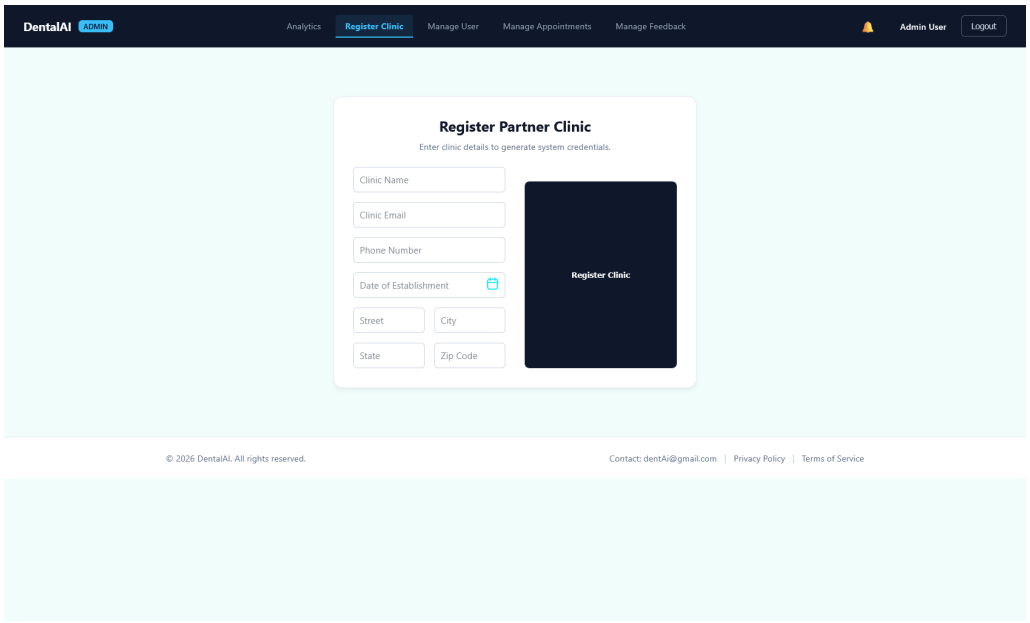


Figure 4.21: Register clinic (admin) for franchise

The Register Partner Clinic page allows administrators to enroll new clinic franchises into the system by entering their official details. This form requires the clinic's name, primary email address, and phone number to generate necessary system credentials. To complete the registration, administrators must also provide the clinic's date of establishment and a full physical address, including the street, city, state, and zip code. Once these details are submitted, the new franchise is officially added to the platform's database.

4.4.12 View System Trends Module

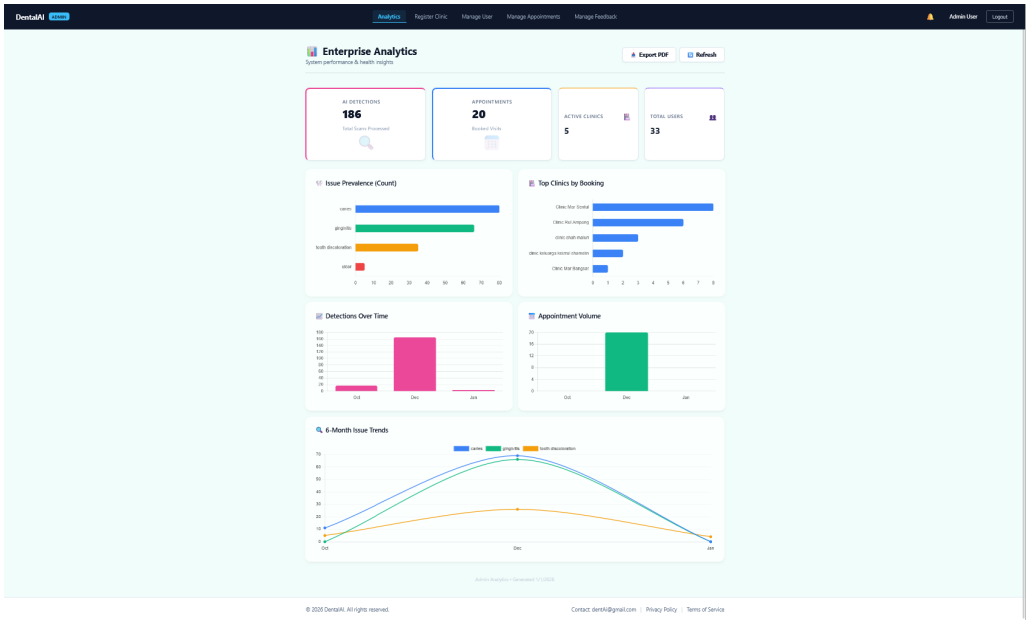


Figure 4.22: View System Trends(analytics) page

The Enterprise Analytics page provides administrators with a high-level overview of system performance and dental health trends through detailed visual reports. At the top of the page, key performance indicators show the total number of AI scans processed, the volume of booked appointments, and the current count of active clinics and users. To help with long-term planning, the page features bar charts that rank the most common dental issues, such as caries and gingivitis, and identify the top-performing clinics by booking volume. Additionally, administrators can track growth using the Detections Over Time and Appointment Volume graphs, or monitor shifting health patterns over a half-year period with the 6-Month Issue Trends line chart. For external reporting, the system includes an Export PDF button to quickly generate a shareable version of these insights.

4.5 Summary

This chapter describes the functional modules of the developed system, designed to support the distinct needs of Patients, Clinics, and Administrators. The Patient module

features AI-based self-assessment tools for immediate health analysis and a comprehensive scheduling system for booking appointments. The Clinic module allows dental providers to manage branch operations, organize doctor schedules, and access detailed patient history records for better diagnosis. Furthermore, the Administrator module ensures effective platform management through centralized

Chapter 5

Results and Discussion

5.1 Introduction

This chapter outlines the testing phase conducted for the AI Powered Dental Assessment web application. Testing plays a vital role in ensuring that all modules from Image analysis to appointment booking and admin functions are functioning properly and meet the defined requirements. The testing included both functional and user acceptance testing (UAT), carried out during various stages of development. Detecting and resolving bugs early in the process significantly reduces post-deployment issues and enhances user satisfaction.

5.1.1 Advantages Of Testing

- **Improves System Quality:** Through iterative testing cycles, defects were identified and fixed early, ensuring a smoother 83 final product.
- **Boosts User Confidence:** Verified stability and responsiveness increased trust among both users and administrators.
- **Reduces Future Maintenance Cost:** Catching issues during development avoids costly revisions post-launch.

- **Enhances Usability:** User-centric testing helped validate that the interfaces and flows were intuitive and met user expectations.

5.1.2 Testing Principles (Based On ISTQB)

a. **Presence of Defects:** Testing revealed the existence of bugs but not their complete absence.

b. **Exhaustive Testing is Impossible:** Testing focused on high priority features like the chatbot, search, and navigation.

c. **Early Testing:** Testing was initiated as early as UI design mock-ups using static review.

d. **Defect Clustering:** Bugs were mostly concentrated in map routing and chatbot responses.

e. **Pesticide Paradox:** Test cases were revised after major UI updates.

f. **Context Dependent:** As a dining app, testing focused heavily on recommendation accuracy and usability rather than intensive security checks.

g. **Absence of Errors Fallacy:** The app must not only be bug-free but also fulfil its core purpose of assisting in smart dining decisions.

5.1.3 Test Mission & Objectives

The mission and objective of this testing process for the AI Powered Dental Assessment application are to deliver the software product at its finest quality. This involves ensuring that both functional and quality attributes are satisfactory, the presence of defects is minimal, and testing activities are conducted systematically.

5.2 Test Strategy Used

The testing approach used for the AI-Powered Dental Assessment System (DentAI) is User Acceptance Testing (UAT), supported by Functional Testing using a Black Box Testing approach. Testing is conducted by the system developer to ensure that the application behaves according to user expectations, particularly for patients and dental practitioners.

Each feature and user role is tested based on functional requirements defined during the planning phase. Black box testing is applied to verify system functionality by validating inputs and outputs without considering internal implementation details. Additionally, UAT feedback is collected using Google Forms questionnaires to evaluate usability, accuracy, and overall user satisfaction.

5.2.1 Software Testing Plan Document (STP)

Use Case ID	UC-001	
Use Case Name	User Registration	
Description	This use case allows a new user to create an account by providing their full name, email, username, password, and password confirmation.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	The user is not already registered.	
Post-condition	A new user account is created and stored in the system database.	
Main Flow	Step	Action
	Pre-Condition	The user is not already registered.
	1.1	User navigates to the registration page.
	1.2	The system displays the registration form.
	1.3	User enters full name, email, username, password, and confirms password.
	1.4	The system validates the entered data (e.g., unique email/username, matching passwords).
	1.5	If valid, the system stores the user information in the database.
	1.6	The system displays a success message and redirects to the login page.
	Post-condition	A new user account is created and stored in the system database.
Alternate Flow	Pre-Condition	The student already has an account
	2.1	The student can navigate to login page
	2.2	The student can log in using his account
	2.3	The student will automatic login to student page
	Post-condition	Th student doesn't need to register to login if already got account

Figure 5.1: Use Case Description DENTAI:AI-Powered Dental Assessment For User Registration

Use Case ID	UC-002	
Use Case Name	Login	
Description	To allow users (User/Admin) to securely log into the system and access its features based on their roles.	
Primary Actor	User, Admin	
Include use cases	Access Main Page	
Pre-Condition	The user has already registered an account.	
Post-condition	The user is authenticated and redirected to the appropriate main page based on their role (User or Admin).	
Main Flow	Step	Action
	Pre-Condition	The user has already registered an account.
	1 User/Admin navigates to the login page. 2 User/Admin enters username and password. 3 System validates credentials. 4 If valid, the system determines the user's role. 5 System grants access and redirects to the appropriate main page.	
	Post-condition	The user is authenticated and redirected to the appropriate main page based on their role (User or Admin).
Alternate Flow	Pre-Condition	User/Admin enters invalid credentials.
	1 User/Admin enters incorrect username or password. 2 System displays an error message indicating invalid credentials. 3 User/Admin is prompted to re-enter login information. 4 User/Admin retries login or navigates to "Forgot Password".	
	Post-condition	The system displays an error message and prompts retry.
Robust Flow	Pre-Condition	User/Admin clicks on "Forgot Password" link.
	1 User/Admin clicks on the "Forgot Password" link on the login page. 2 System prompts user to enter registered email address. 3 User enters an email and submits the request. 4 System sends a password reset link to the provided email. 5 User/Admin accesses the link and sets a new password.	

Figure 5.2: Use Case Description DENTAI:AI-Powered Dental Assessment For Login

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC001.2		
Test Case Name	UI Validation Test Case for Required Fields		
Requirement Traceability	REQ_UR101		
Test Case Description	To ensure the registration form does not allow submission if any mandatory field is left blank.		
Item(s) to be tested			
1	Required field validation		
2	Error message display on incomplete submission		
Pre-condition	1. User is on the registration page 2. User has not filled all required fields		
Test Procedure Specification			

Figure 5.3: STP DENTAI:AI-Powered Dental Assessment For UI Validation

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing	
Test Type	Black Box Testing – Functional Testing			
Test Design Technique	Use Case Testing			
Risk Number	1 (Lowest)			
Test Case Number	TC002.1			
Test Case Name	Functionality Test Case for User/Admin Login			
Requirement Traceability	REQ_UR201			
Test Case Description	To verify that users and admins can log in using valid credentials and are redirected to the appropriate main page based on their role.			
Item(s) to be tested				
1	Login form fields (username and password			
2	Credential validation logic			
3	Role based redirection (User/Admin)			
4	Error handling for invalid login			
5	Password recovery process			
Pre-condition	1.User does already have an account. 2.User is on the login page.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

Figure 5.4: STP DENTAI:AI-Powered Dental Assessment For Login

Features	Likelihood (L)	Impact (I)	Total Risk (R = L * I)	Priority
User Authentication				
Failed User Registration	6	500	3000	3
Failed Login	6	500	3000	3
Failed Profile Update	5	300	1500	2
Image Analysis				
Failed Image Upload (JPEG/PNG)	7	1000	7000	4
Failed Image Preprocessing (Resizing)	6	500	3000	3
AI Model Misdiagnosis	9	1000	9000	5
Failed Bounding Box Highlighting	8	1000	8000	4
Condition Classification				
Failed Condition Classification (Health Status)	7	1000	7000	4
Failure to Support New Dental Conditions	8	1000	8000	4
Feedback				
Failed Generate Feedback Report	6	500	3000	3
Failed Actionable Feedback (Suggestions)	7	800	5600	4
Failed Visual Feedback (Highlight Issues)	6	500	3000	3
Progress Tracking				
Failed Save Health Reports	6	500	3000	3
Failed Visual Progress Charts	6	500	3000	3

Figure 5.5: Risk Assessment Matrix For DENTAI:AI-Powered Dental Assessment

The Software Testing Plan (STP) for the DentAI system was designed to ensure that all system modules function correctly and meet the specified requirements. The testing strategy focuses on validating key user workflows through use case–based test cases, risk analysis, and black-box testing.

Based on the STP tables shown in Figure 5.3 and Figure 5.4, test cases were derived directly from DentAI's core modules. Examples include the User Registration and Login modules, where testing verifies correct input handling, validation rules, error messages, and successful redirection upon completion. For the registration module, test cases ensure that users can create an account using valid credentials while preventing duplicate or invalid entries. For the login module, testing confirms that only registered users with correct credentials can access the system.

Black-box testing was applied to evaluate the system from an end-user perspective without considering internal implementation details. This approach ensures that DentAI behaves as expected when interacting with different user inputs, including valid, invalid, and missing data.

In addition, a risk analysis table is included in the STP to identify potential issues such as authentication failure, incorrect data submission, and unauthorized access. Corresponding test cases were designed to reduce these risks and improve system reliability.

Overall, the STP demonstrates a structured and systematic testing approach for DentAI, ensuring that critical modules such as registration and login operate reliably before users proceed to advanced features like image analysis and appointment booking.

5.2.2 User Acceptance Testing

User acceptance testing was conducted by giving survey to users in a suitable period.

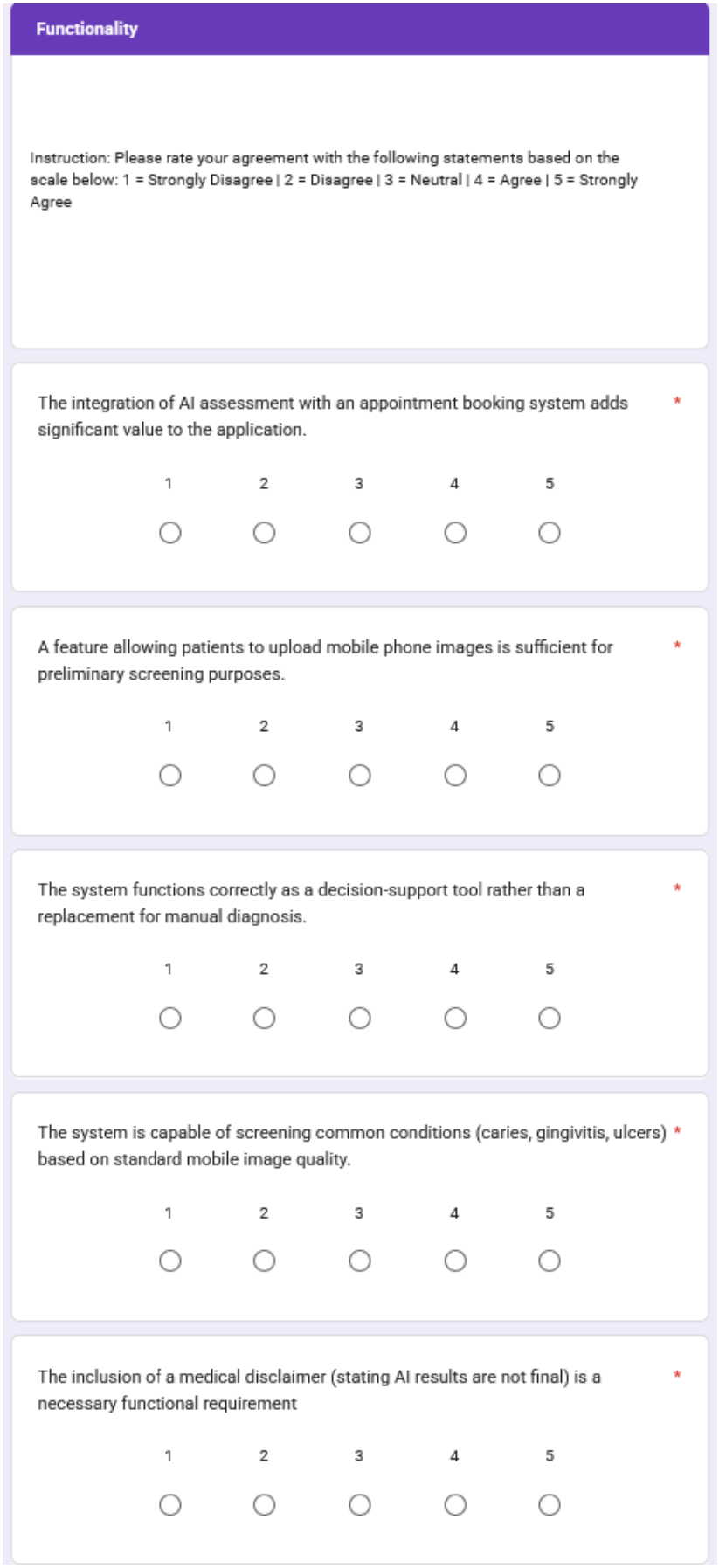


Figure 5.6: Survey Question For Functionality

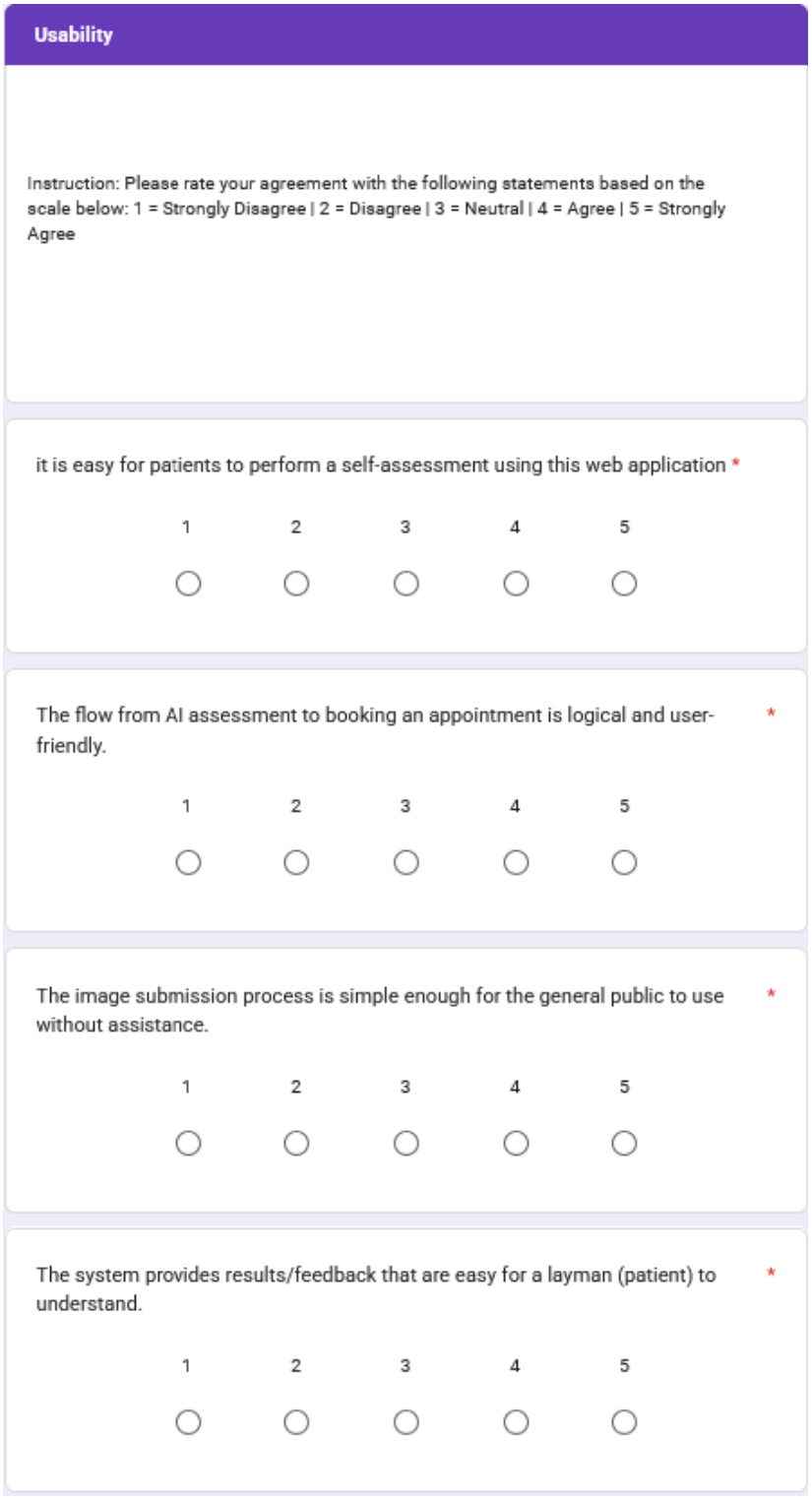


Figure 5.7: Survey Question For Usability

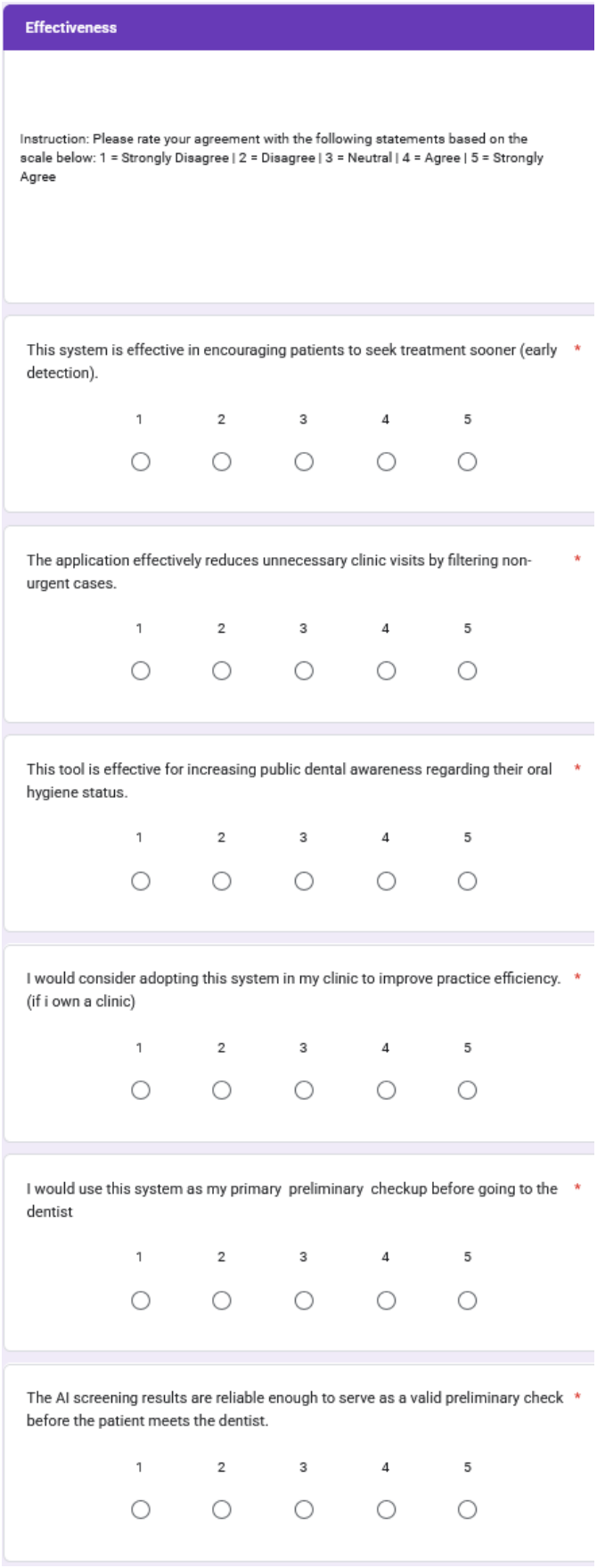


Figure 5.8: Survey Question For Effectiveness

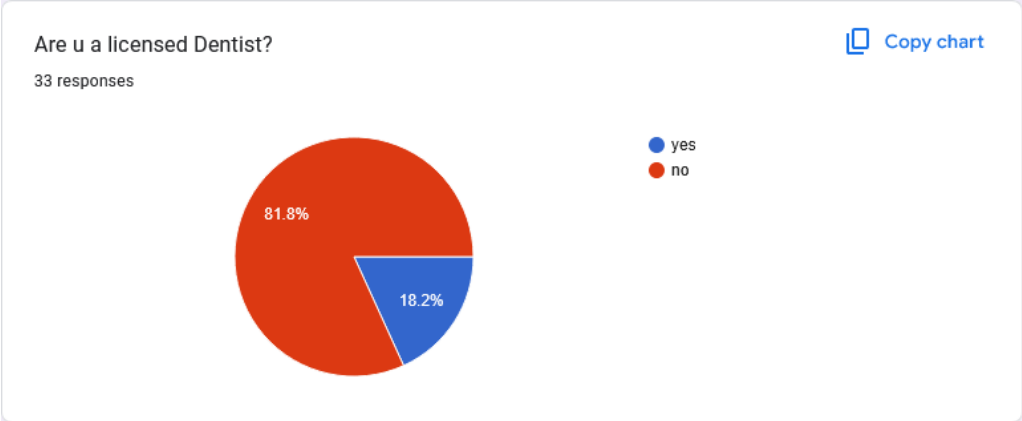


Figure 5.9: Result gathered from UAT with a sample size of 32 respondents.

Based on Figure 5.9 out of the 32 respondents 84% is a layman and 15.6% is a licensed Dentist. This distribution is important as it indicates that the system was evaluated by both general users and professionals with clinical experience. Feedback from laypersons reflects usability and accessibility for the public, while input from licensed dentists provides insights grounded in professional dental practice. The inclusion of both groups strengthens the evaluation by ensuring the system is assessed from both user-centric and expert perspectives.

5.2.3 Functionality

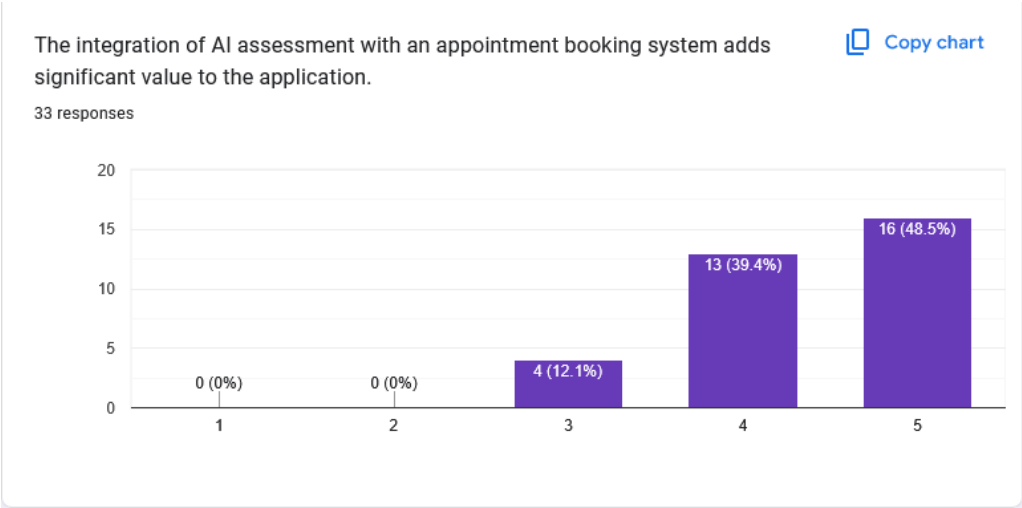


Figure 5.10: Result from Question 1 of Functionality Survey.

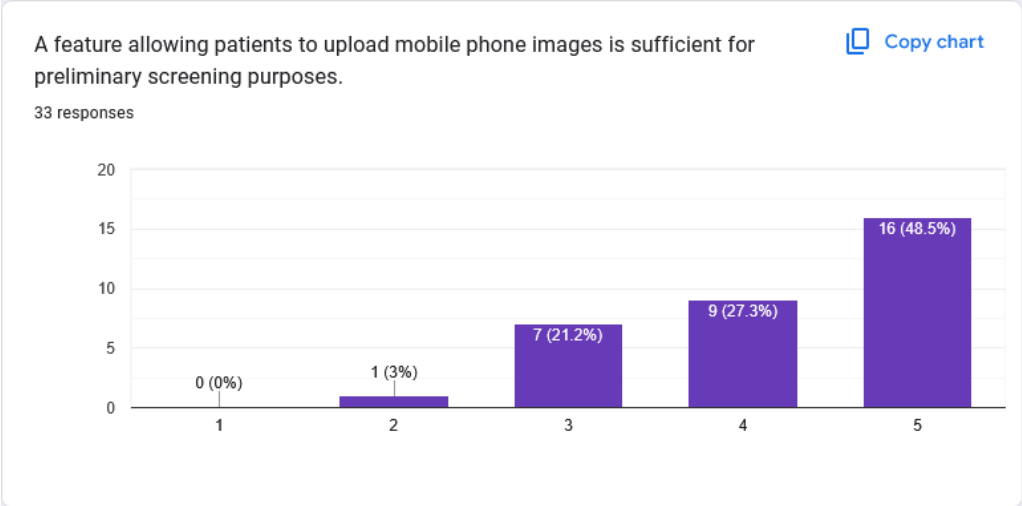


Figure 5.11: Result from Question 2 of Functionality Survey.

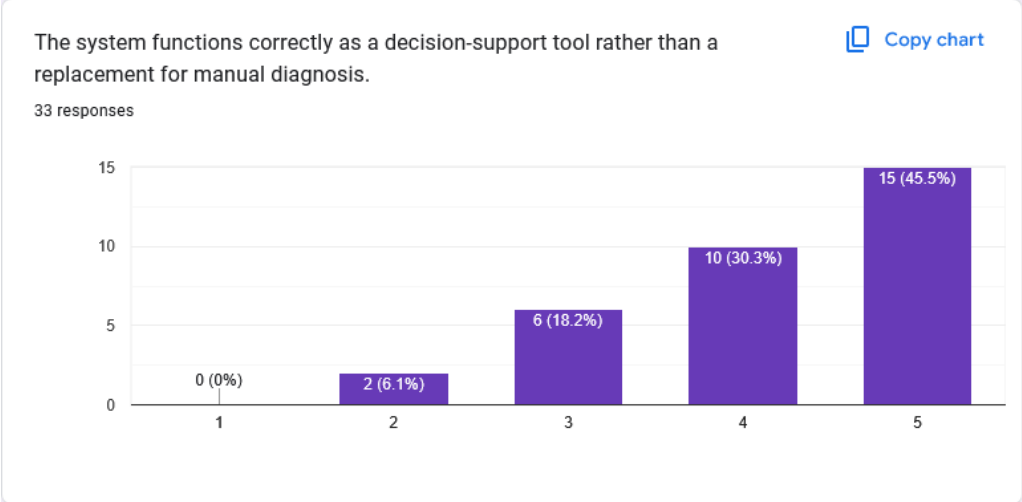


Figure 5.12: Result from Question 3 of Functionality Survey.

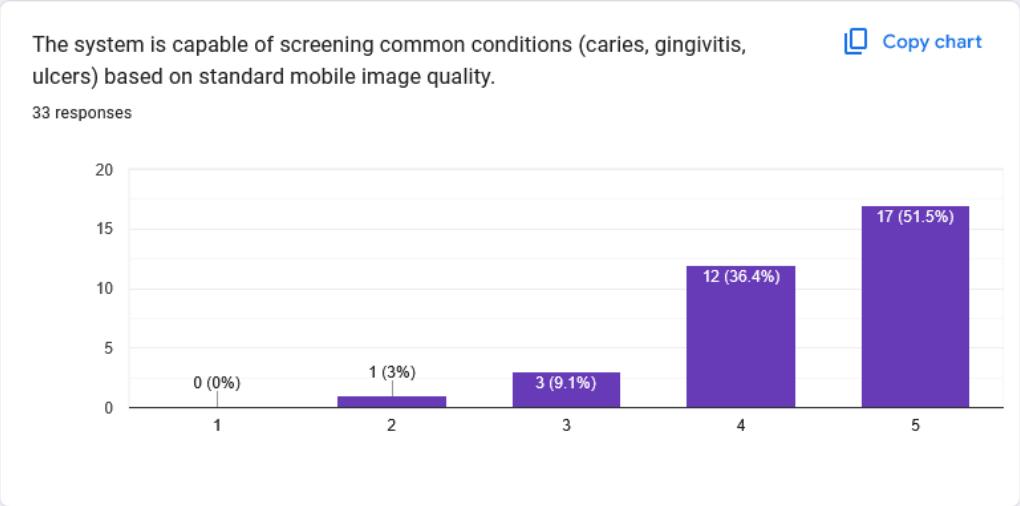


Figure 5.13: Result from Question 4 of Functionality Survey.

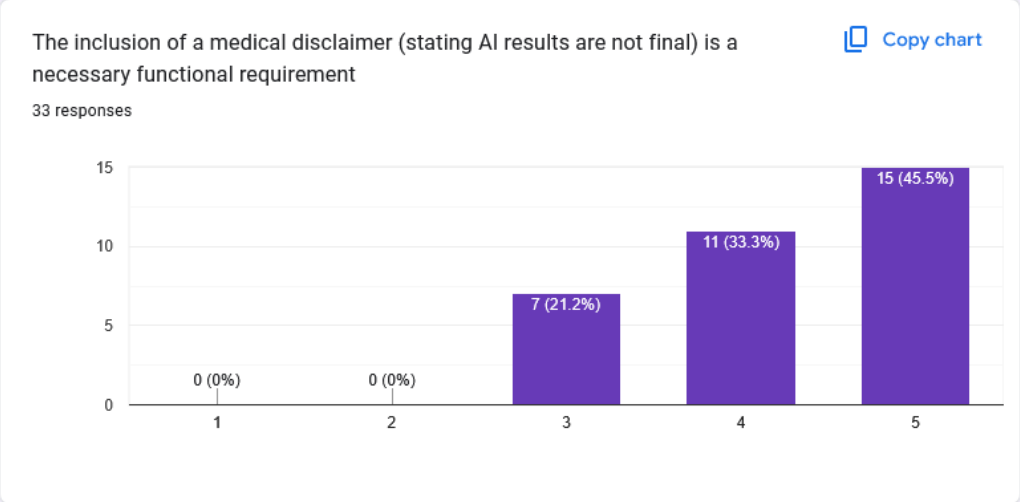


Figure 5.14: Result from Question 5 of Functionality Survey.

To assess the system's functionality, the evaluation was structured into five key functional categories: clinical safety, decision-support behaviour, input adequacy, screening capability, and system integration. Responses were measured using a 5-point Likert scale, ranging from Strongly Disagree to Strongly Agree.

As illustrated in Figure 5.14, 45.5% of respondents strongly agreed and 33.3% agreed that the inclusion of a medical disclaimer stating that AI results are not final is a necessary functional requirement, while 21.2% provided a neutral response. This indicates strong user awareness of ethical and clinical limitations associated with AI-based assessments.

For the statement regarding the integration of AI assessment with an appointment booking system in Figure 5.10, 48.5% of respondents strongly agreed and 39.4% agreed that this feature adds significant value to the application, with 12.1% responding neutrally. This highlights the importance of seamless clinical follow-up within the system.

In terms of the system functioning as a decision-support tool rather than a replacement for manual diagnosis in Figure 5.12, 45.5% of respondents strongly agreed and 30.3% agreed, while 18.2% selected neutral and 6.1% disagreed. These findings reinforce the system's intended role as a supportive aid rather than an autonomous diagnostic solution.

Regarding image acquisition in Figure 5.11, 48.5% of respondents strongly agreed and 27.3% agreed that allowing patients to upload mobile phone images is sufficient for preliminary screening purposes. Meanwhile, 21.2% provided a neutral response and 3.0% disagreed, indicating general acceptance of mobile image usage for early-stage assessment.

Finally, in Figure 5.13 when evaluating the system's ability to screen common dental conditions such as caries, gingivitis, and ulcers using standard mobile image quality, 51.5% of respondents strongly agreed and 36.4% agreed. Only 9.1% responded neutrally and 3.0% disagreed, demonstrating strong confidence in the system's screening capability.

Overall, the consistently high agreement percentages across all statements indicate that the system has successfully met its functional and non-functional requirements, particularly in terms of clinical relevance, usability, and practical deployment. Collectively, these categories confirm that the system meets its intended functional requirements for safe, supportive, and practical AI-assisted dental screening.

5.2.4 Effectiveness

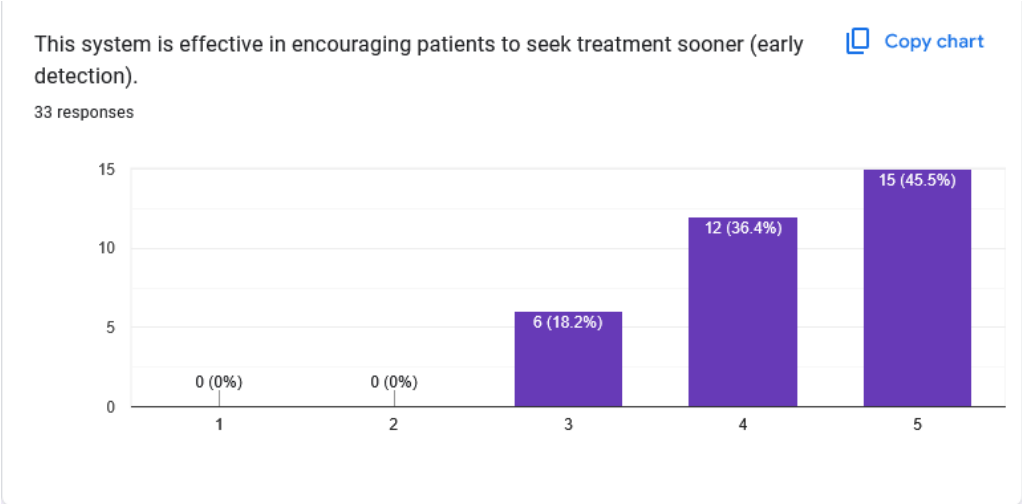


Figure 5.15: Result from Question 1 of Effectiveness Survey.

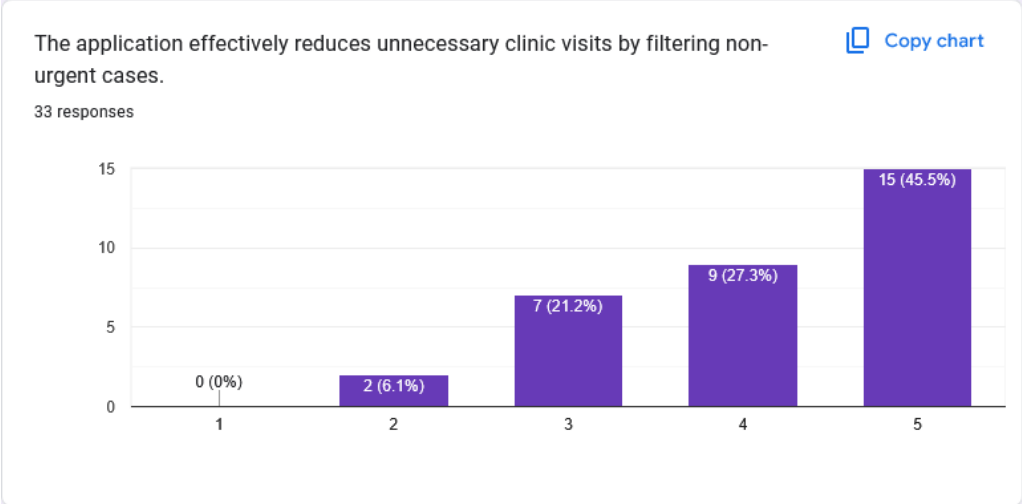


Figure 5.16: Result from Question 2 of Effectiveness Survey.

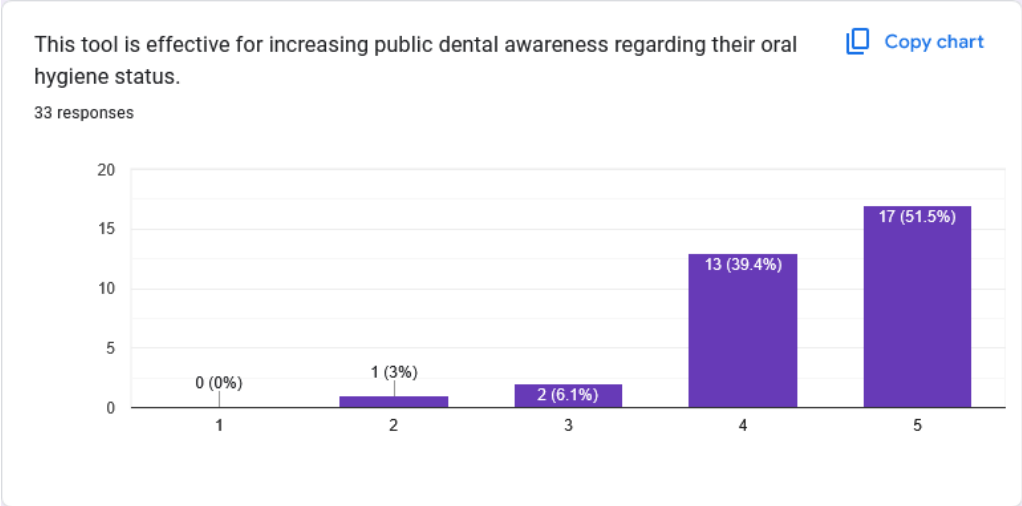


Figure 5.17: Result from Question 3 of Effectiveness Survey.

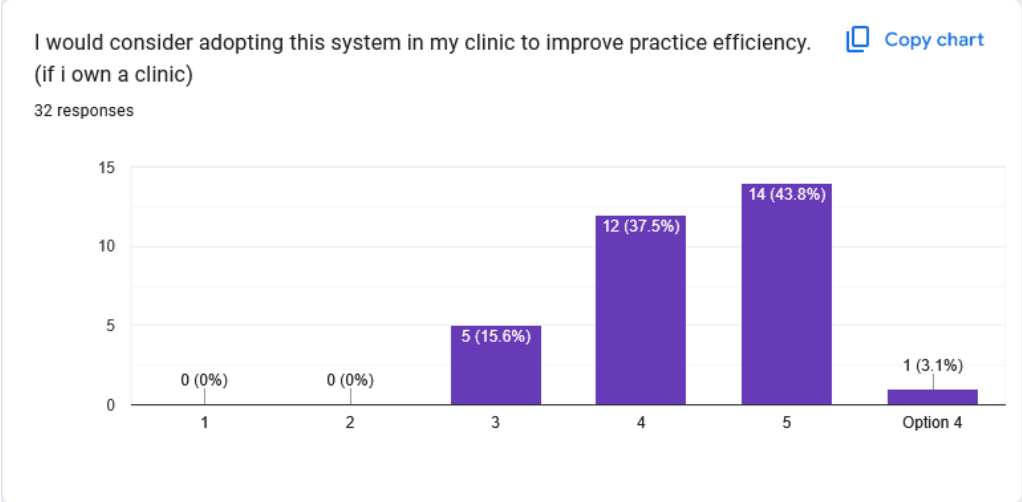


Figure 5.18: Result from Question 4 of Effectiveness Survey.

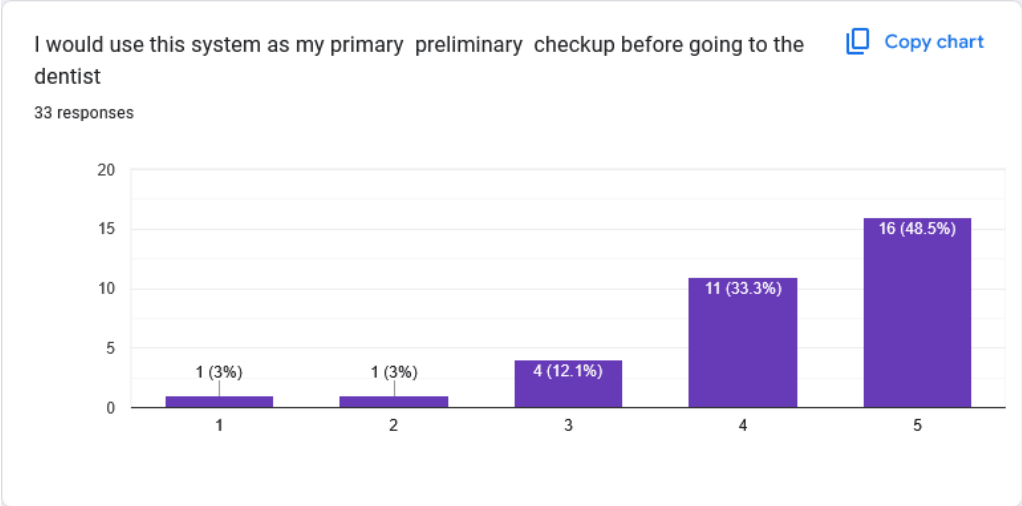


Figure 5.19: Result from Question 5 of Effectiveness Survey.

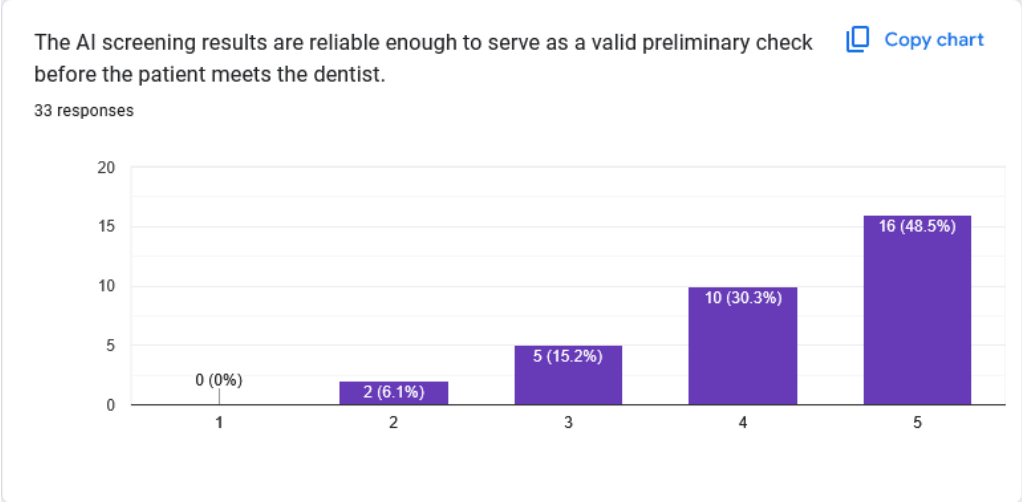


Figure 5.20: Result from Question 6 of Effectiveness Survey.

To evaluate the effectiveness of the dental AI system, respondents were asked to rate several statements related to early detection, clinical efficiency, public awareness, adoption intention, and reliability of AI screening outcomes. Responses were measured using a 5-point Likert scale, ranging from Strongly Disagree to Strongly Agree.

As shown in Figure 5.15, 45.5% of respondents strongly agreed and 36.4% agreed that the system is effective in encouraging patients to seek treatment earlier, while 18.2% selected a neutral response. This suggests that the system has strong potential to support early detection and timely dental intervention.

For the statement in Figure 5.16 concerning the reduction of unnecessary clinic visits by filtering non-urgent cases, 45.5% of respondents strongly agreed and 27.3% agreed, while 21.2% responded neutrally and 6.1% disagreed. These findings indicate that the system is perceived as effective in improving clinical efficiency and patient triage.

In terms of increasing public dental awareness in Figure 5.17, 51.5% of respondents strongly agreed and 39.4% agreed that the tool effectively improves understanding of oral hygiene status. Only 6.1% responded neutrally and 3.0% disagreed, demonstrating strong agreement on the system's educational value.

Regarding professional adoption, among respondents working in a clinical setting in Figure 5.18, 43.8% strongly agreed and 37.5% agreed that they would consider adopting the system in their clinic to improve practice efficiency, while 15.6% provided a neutral response. This reflects positive acceptance of the system within professional environments.

For patient-oriented usage in Figure 5.19, 48.5% of respondents strongly agreed and 33.3% agreed that they would use the system as a primary preliminary checkup before visiting a dentist, with 12.1% responding neutrally and 6.0% disagreeing. This indicates strong user confidence in the system as a first-line screening tool.

Finally, when assessing the reliability of AI screening results as a valid preliminary check prior to a dental consultation in Figure 5.19, 48.5% of respondents strongly agreed and 30.3% agreed. Meanwhile, 15.2% responded neutrally and 6.1% disagreed, suggesting overall trust in the system's screening reliability while still acknowledging the need for professional confirmation.

Overall, the high levels of agreement across all effectiveness-related statements demonstrate that the system has successfully met its effectiveness requirements, particularly in supporting early detection, clinical workflow optimization, public awareness, and user adoption. These results confirm that the system is effective as a preliminary AI-assisted dental screening solution that complements professional dental care rather than replacing it.

5.2.5 Usability

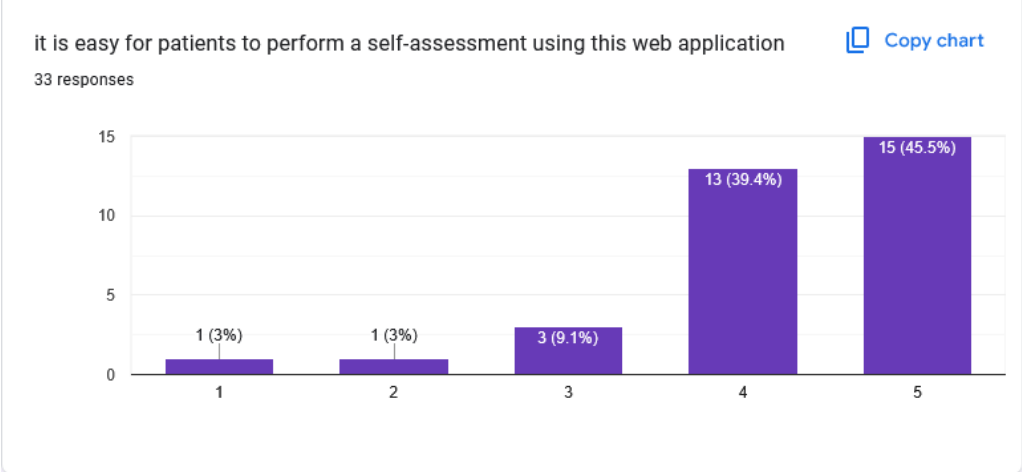


Figure 5.21: Result from Question 1 of Usability Survey.

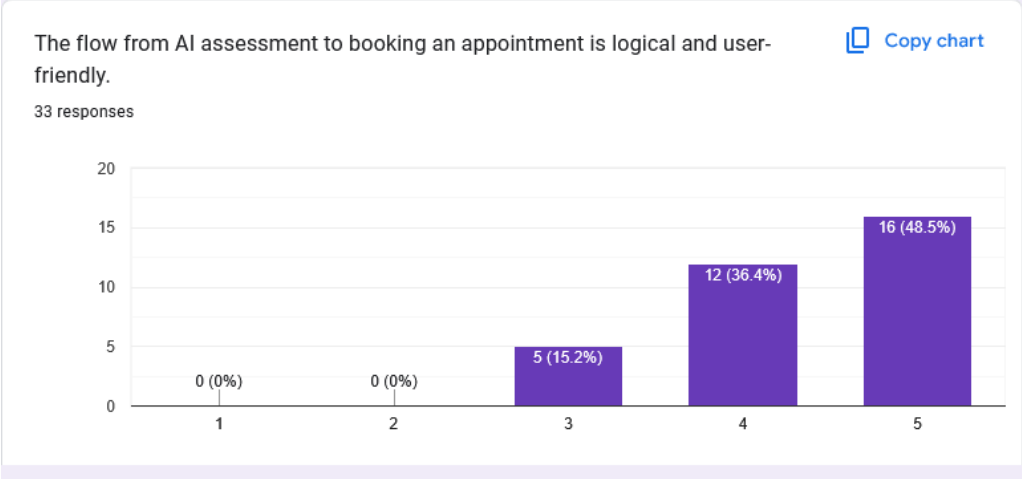


Figure 5.22: Result from Question 2 of Usability Survey.

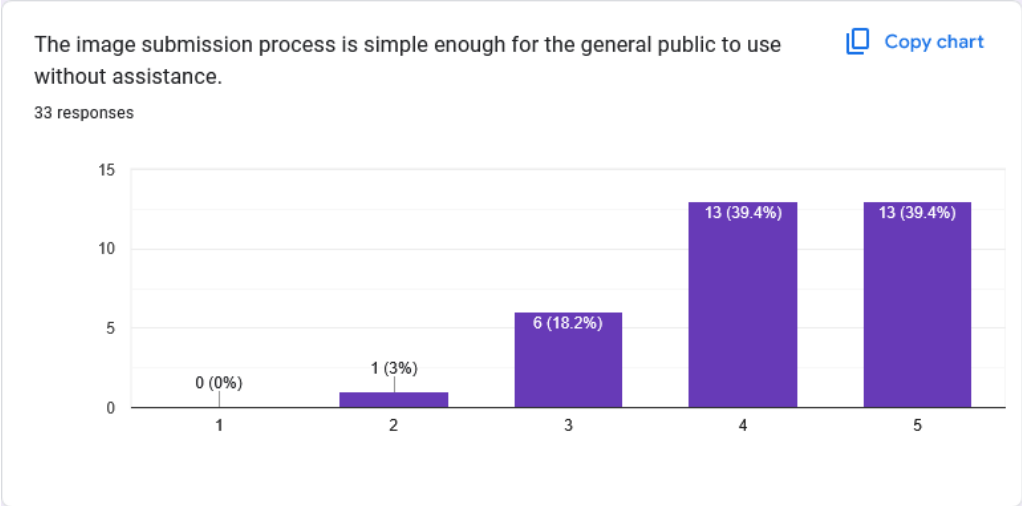


Figure 5.23: Result from Question 3 of Usability Survey.

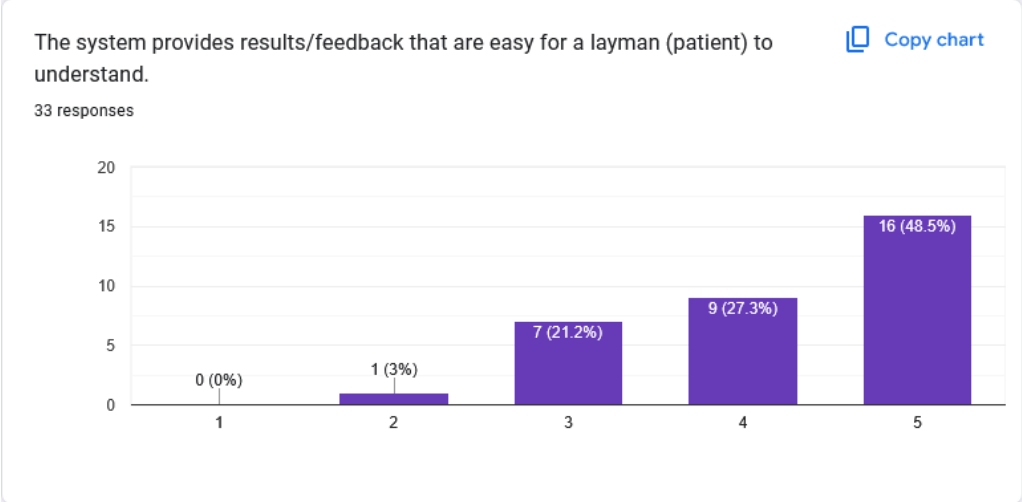


Figure 5.24: Result from Question 4 of Usability Survey.

To evaluate the usability of the dental AI web application, respondents were asked to rate several statements related to ease of use, clarity of system flow, simplicity of image submission, and understandability of system feedback. Responses were measured using a 5-point Likert scale, ranging from Strongly Disagree to Strongly Agree.

As illustrated in Figure 5.21, 45.5% of respondents strongly agreed and 39.4% agreed that it is easy for patients to perform a self-assessment using the web application. Meanwhile, 9.1% responded neutrally, and a small proportion of 6.0% disagreed. This indicates that the majority of users found the system intuitive and easy to operate independently.

Regarding the flow from AI assessment to booking an appointment in Figure 5.22, 48.5% of respondents strongly agreed and 36.4% agreed that the process is logical and user-friendly, while 15.2% selected a neutral response. No respondents expressed disagreement, suggesting a well-designed and coherent user journey.

For the image submission process in Figure 5.23, 39.4% of respondents strongly agreed and 39.4% agreed that submitting images is simple enough for the general public to use without assistance. Additionally, 18.2% responded neutrally and 3.0% disagreed, indicating strong usability with minor room for improvement.

In terms of result and feedback clarity in Figure 5.24, 48.5% of respondents strongly agreed and 27.3% agreed that the system provides results that are easy for laypersons to understand. Meanwhile, 21.2% responded neutrally and 3.0% disagreed, suggesting that while most users find the feedback clear, further simplification could enhance comprehension for all users.

Overall, the high levels of agreement across all usability-related statements indicate that the system has successfully met its usability requirements, particularly in terms of ease of navigation, clarity of interaction flow, simplicity of input, and understandability of outputs. These findings confirm that the web application is accessible and user-friendly for both general users and patients without technical or medical backgrounds.

5.3 Result Analysis

5.3.1 Discussion of UAT Result

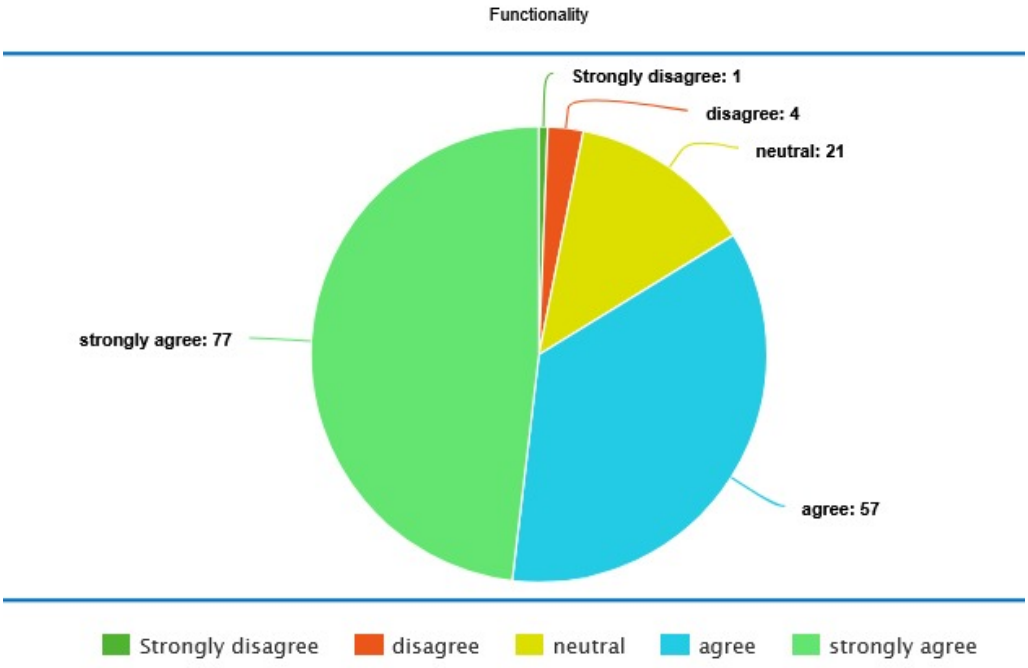


Figure 5.25: Result gathered from 32 respondents regarding functionality.

The functionality evaluation results show a strongly positive overall perception of the system. Many responses fall within the agreement spectrum, with 77 respondents selecting “strongly agree” and 57 selecting “agree”, indicating that most users believe the system functions as intended and meets its functional requirements. A smaller portion of respondents (21) selected a neutral response, suggesting minor uncertainty or limited exposure to certain features. Only 4 respondents disagreed and 1 strongly disagreed, representing a very small minority. Overall, this distribution demonstrates a high level of user confidence in the system’s functionality, confirming that the implemented features are reliable, appropriate, and well-aligned with user expectations, while indicating minimal functional issues.

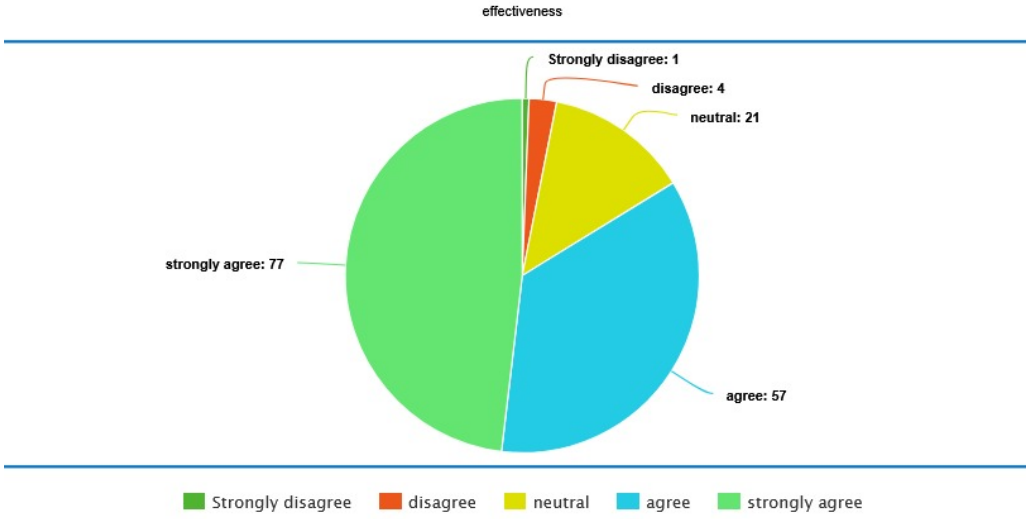


Figure 5.26: Result gathered from 32 respondents regarding effectiveness.

The effectiveness evaluation results indicate a high level of user confidence in the system's ability to achieve its intended outcomes. A substantial majority of respondents selected "strongly agree" (77) and "agree" (57), demonstrating that users perceive the system as effective in supporting early detection, improving decision-making, and enhancing overall dental care processes. A moderate number of respondents (21) chose a neutral response, which may reflect cautious optimism or limited experience with certain system features. Only a very small proportion of respondents expressed negative views, with 4 selecting "disagree" and 1 selecting "strongly disagree." Overall, this distribution confirms that the system effectively meets its effectiveness objectives, with strong acceptance among users and minimal concerns regarding its impact or reliability.

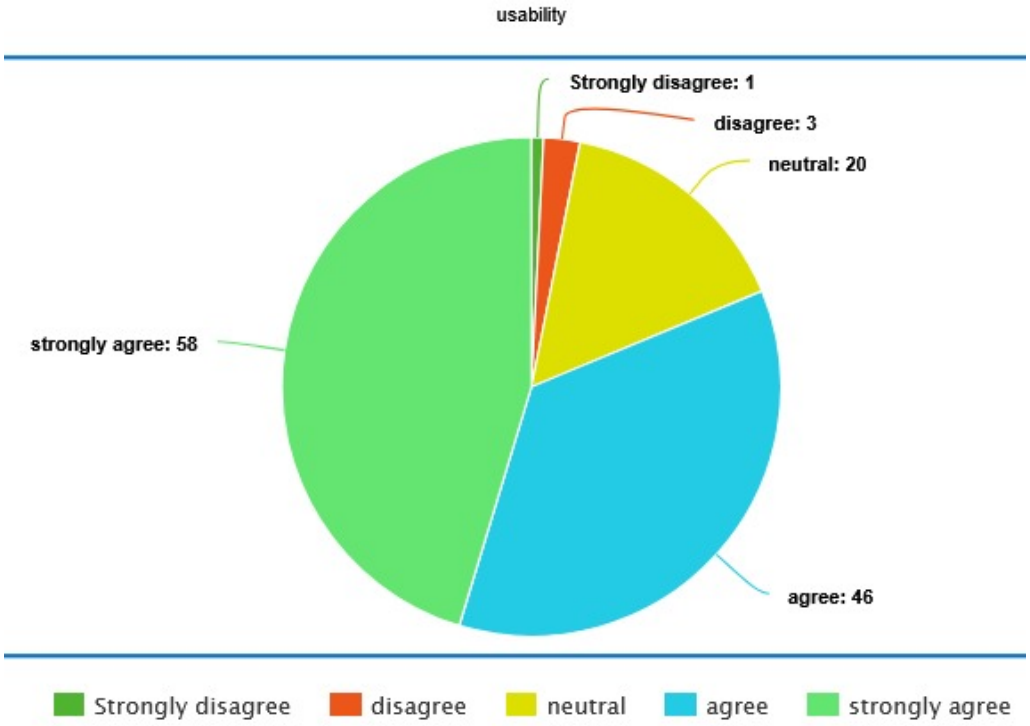


Figure 5.27: Result gathered from 32 respondents regarding usability.

The usability evaluation results demonstrate a strongly positive user experience with the system. A large majority of respondents selected “strongly agree” (58) and “agree” (46), indicating that users found the system easy to use, intuitive, and well-structured for performing dental self-assessments. A portion of respondents (20) chose a neutral response, which may reflect varying levels of familiarity or confidence when using certain features. Only a very small number of respondents expressed negative perceptions, with 3 selecting “disagree” and 1 selecting “strongly disagree.” Overall, these results confirm that the system successfully meets its usability objectives, providing a user-friendly interface and clear interaction flow suitable for both laypersons and clinical users.

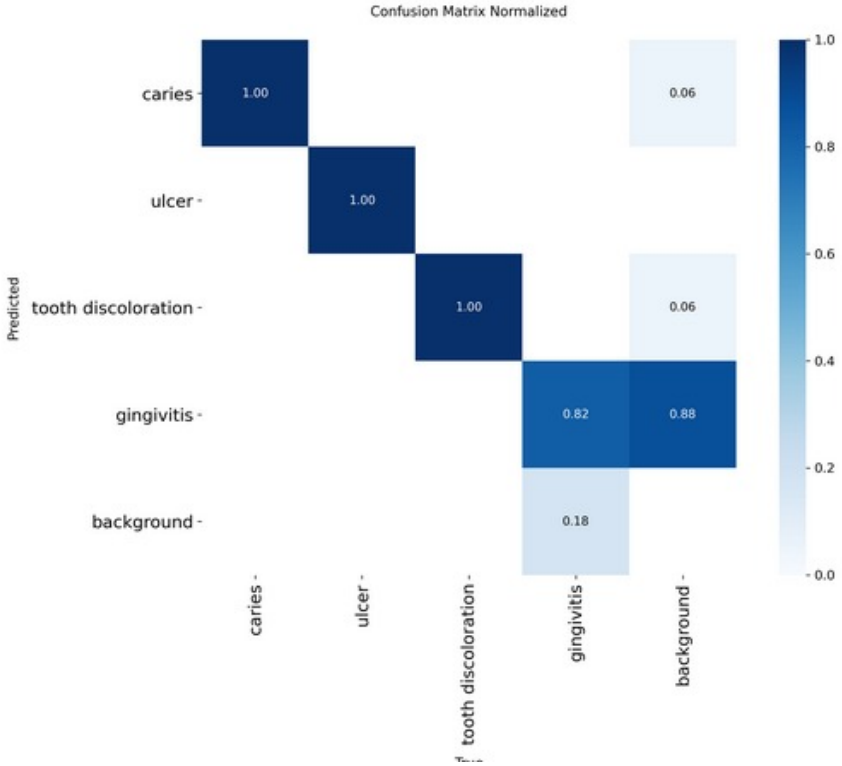


Figure 5.28: Confusion Matrix

Figure 5.28 shows the normalized confusion matrix for the DentAI model across five classes: caries, ulcer, tooth discoloration, gingivitis, and background. The model achieved perfect classification (1.00) for caries, ulcer, and tooth discoloration, indicating strong detection performance for these conditions.

Gingivitis was correctly classified in 82% of cases, with 18% misclassified as background, suggesting difficulty in distinguishing mild gum inflammation from normal gum appearance. A small proportion of background samples (6%) were incorrectly predicted as caries or tooth discoloration, indicating minor false positives.

Overall, the results demonstrate that DentAI performs reliably for most dental conditions, with misclassifications mainly occurring between gingivitis and background, highlighting an area for future improvement.

5.4 Summary

This chapter presented the results and discussion of the testing phase conducted for the AI-Powered Dental Assessment web application. Both Functional Testing and User Acceptance Testing (UAT) were employed to evaluate the system across three main quality categories: functionality, effectiveness, and usability, using a structured questionnaire with responses from 32 participants, including both laypersons and licensed dentists The inclusion of these two user groups ensured that the system was assessed from both user-centric and professional clinical perspectives.

Overall, the findings demonstrate strong positive acceptance of the system. The functionality evaluation confirmed that the system meets its intended purpose as a decision-support tool, with high agreement on clinical safety measures, screening capability, and system integration. The effectiveness results indicate that the system is perceived as valuable in encouraging early treatment, reducing unnecessary clinic visits, and improving public dental awareness, while also gaining acceptance for both patient use and potential clinical adoption. Additionally, the usability evaluation showed that the application is easy to use, intuitive, and accessible, particularly for lay users performing self-assessments and interpreting results.

In conclusion, the consistently high levels of agreement across all three evaluation categories confirm that the system successfully satisfies its functional, effectiveness, and usability requirements. These results validate the system's design and implementation as a reliable, effective, and user-friendly AI-assisted dental screening solution that complements professional dental care rather than replacing it, thereby fulfilling the objectives set for this project .

Chapter 6

Conclusion and Future Works

6.1 Introduction

This chapter presents the summary and general conclusion of the research work conducted throughout the development of the DentAI system. The chapter begins with a brief review of each preceding chapter, highlighting the key problems addressed and the major contributions achieved in this project. The DentAI system, an AI-powered web-based dental assessment platform, has been successfully designed, developed, and implemented.

The system was developed to assist users and dental practitioners by providing automated dental image analysis, early detection of common dental conditions, and supportive decision-making insights. Through the integration of artificial intelligence and web technologies, the system enables users to upload dental images for analysis and receive assessment results efficiently, while also supporting appointment booking with dental clinics.

During the User Acceptance Testing (UAT) phase, the system was evaluated by multiple users, and positive feedback was obtained regarding its usability, functionality, and overall user experience. These results demonstrate that the system meets user expectations and performs according to the defined requirements. The effectiveness of the system is further supported by the literature review presented in Chapter 2, the methodology and system

architecture discussed in Chapters 3 and 4, and the testing and verification processes described in Chapter 5.

Overall, the DentAI system has successfully fulfilled most of its proposed functional and non-functional requirements, achieved its primary objectives, and demonstrated the practicality and potential of applying AI-powered solutions in the dental healthcare domain. This project highlights the feasibility of leveraging artificial intelligence to enhance dental assessment, improve accessibility to preliminary dental evaluations, and support modern digital healthcare services.

6.2 Conclusion

Table 6.1: Summary of Objectives, Methodology, and Output

Objective	**Methodology**	**Output**
To study existing dental-related applications and object detection models suitable for analysing intraoral dental images.	Conduct a literature review on dental systems and object detection techniques.	Chapter 2
To design and develop a web-based application that utilises a trained AI model to perform dental condition assessment based on image analysis.	Implement and integrate the AI model into a web-based system for dental image assessment.	Chapter 3, 4
To evaluate the developed system in terms of accuracy, reliability, and user experience through performance testing and validation.	Perform system testing, validation, and user evaluation.	Chapter 5

6.3 Future Works

There are several areas that future researchers can explore to further enhance the AI-based dental detection system.

- **Future Work 1:** Additional dental disease classes can be included in the AI model, such as periodontitis, plaque buildup, tooth fractures, and abscesses. Expanding

the number of detectable conditions would allow the system to provide more comprehensive dental assessments.

- **Future Work 2:** The model can be trained using larger and more diverse datasets collected from multiple dental clinics. A bigger dataset with varied lighting conditions, image angles, and patient demographics can help improve model accuracy, reduce prediction errors, and enhance generalization.

- **Future Work 3:** The overall system can be enhanced by improving user interface design, system performance, and scalability. Future work may include better appointment management features, real-time notification systems, and cloud optimization to support a larger number of users and clinics efficiently.

In conclusion, the DentAI system has successfully achieved its objective of developing a smart, web-based platform for automated dental image analysis and preliminary dental condition assessment. Through a structured development process and comprehensive testing, the system demonstrates reliable AI-driven results while maintaining ease of use for users and dental practitioners. Positive user feedback further confirms its usability and practical value in supporting early dental assessment and informed decision-making. With future enhancements such as expanded detection classes, improved model accuracy using larger datasets, and deeper clinical integration, DentAI has strong potential to evolve into a more robust and impactful AI-powered dental healthcare solution.

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APPENDIX

Appendix A

Gantt Chart

	Project Schedule AI powered dental assessment 24 weeks	24 weeks
	Activity	Weeks needed
1	**Requirement Phase**	
	Tasks	
1.1	Write Proposal	6
1.2	Defense proposal	
1.3	Gather requirement	
1.4	Prioritise Requirement	
1.5	Document srs	
3	**Design Phase**	
	Tasks	
3.1	Design Architecture	4
3.2	Design Database	
3.3	Design Component	
3.4	Design interface	
3.5	Design storyboard	
4	**Implementation Phase**	
	Tasks	
4.1	Environment Setup	14
4.2	Frontend Development	
4.3	Backend/API Development	
4.4	Database Integration	
4.5	AI Model training and Integration	
4.6	App-Backend Connection	
4.7	End-to-end testing	
5	**Testing Phase**	
	Tasks	
5.1	Validate that the code meets the specified requirements	4
5.2	Identify and rectify bugs and issues	
5.3	Ensure that the system meets acceptance criteria	
5.4	Address and resolve any bugs or issues identified during testing	
5.5	Obtain user feedback on the system usability and functionality	
6	**Deployment Phase**	
	Tasks	
6.1	Deploy the system to a production environment for users to interact with	3
6.2	Monitor system performance during the initial user interactions	
6.3	Address any post-deployment issues promptly	
6.4	Provide ongoing support and maintenance for continuous improvement	
6.5	Gather user feedback for further enhancements and updates	

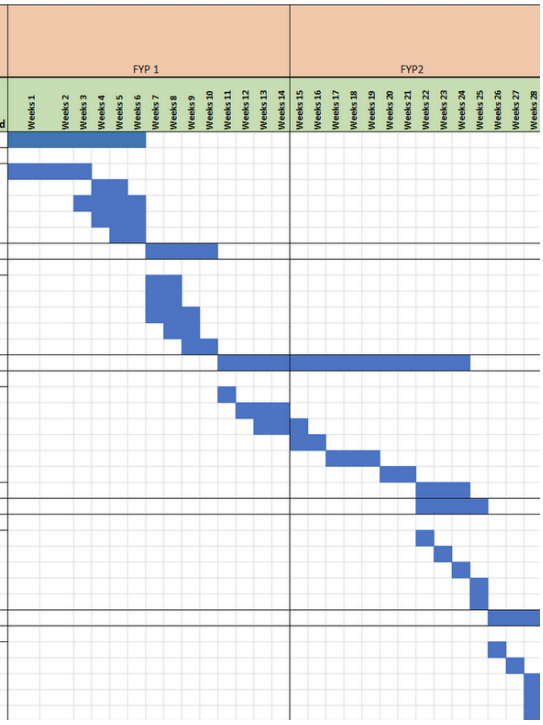


Figure A.1: Gantt Chart for DENTAI:AI-POWERED DENTAL ASSESSMENT

Appendix B

Questionnaire Sample

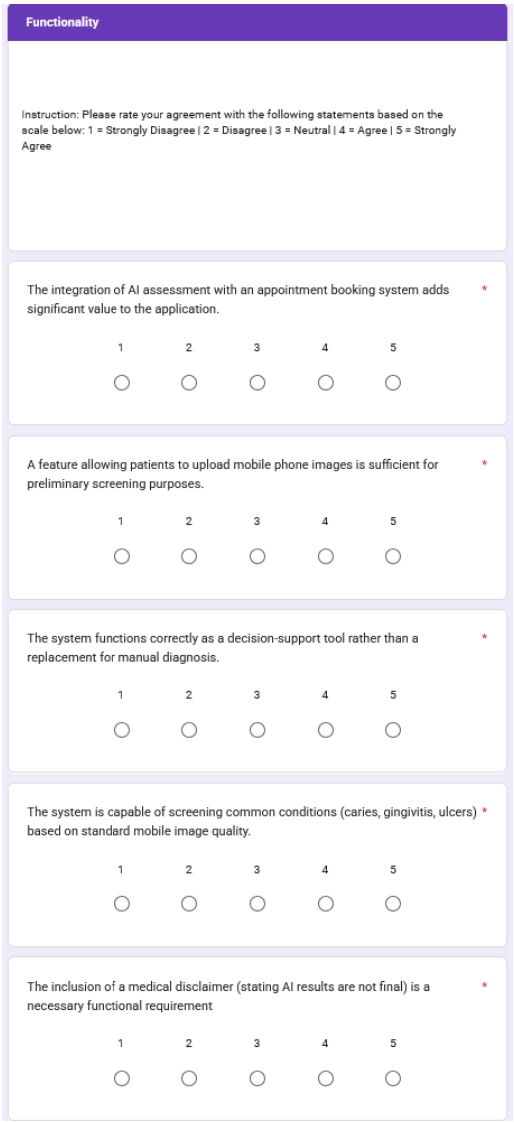


Figure B.1: The Questionnaire for this research. (Functionality)

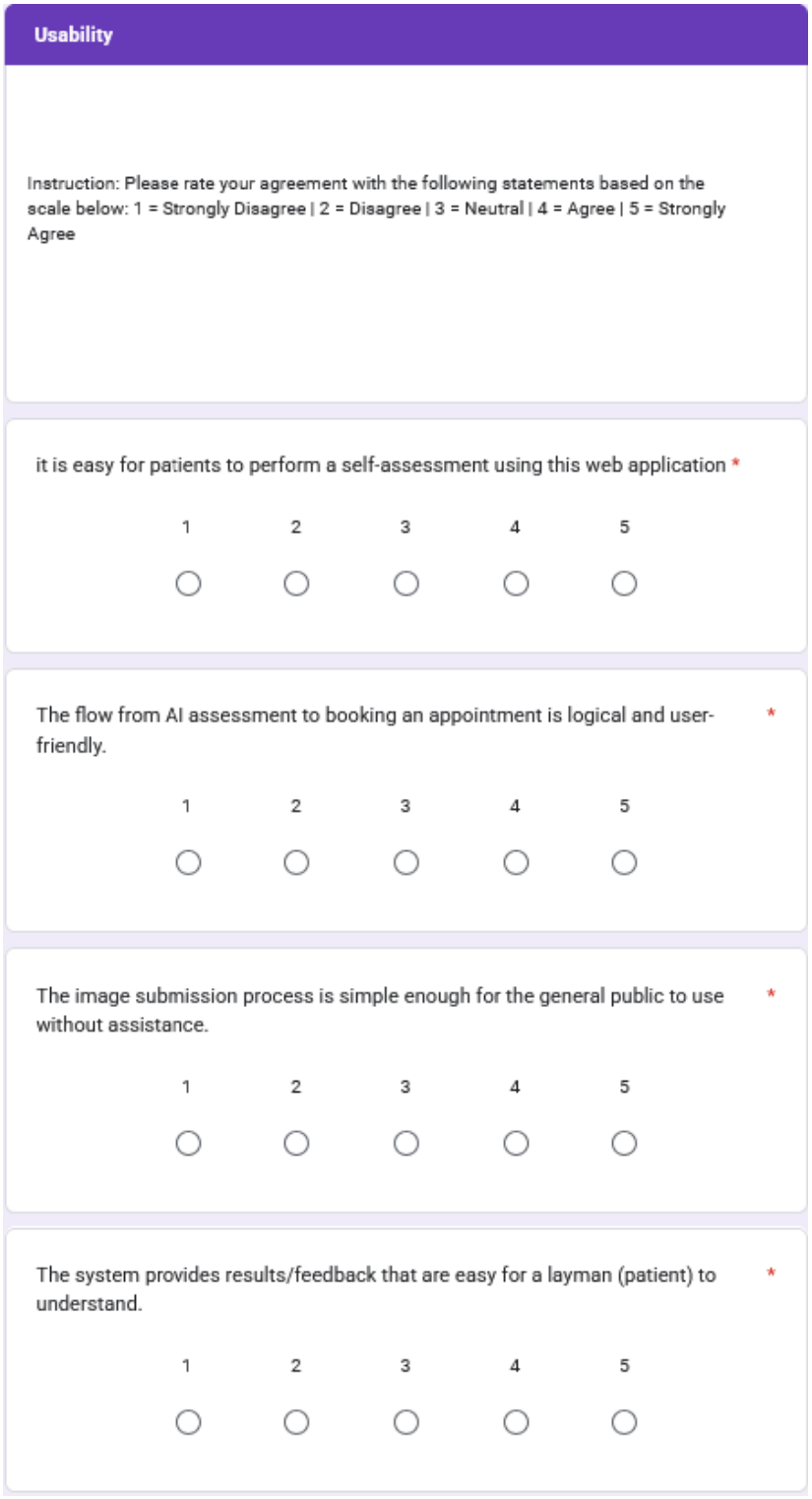


Figure B.2: The Questionnaire for this research. (Usability)

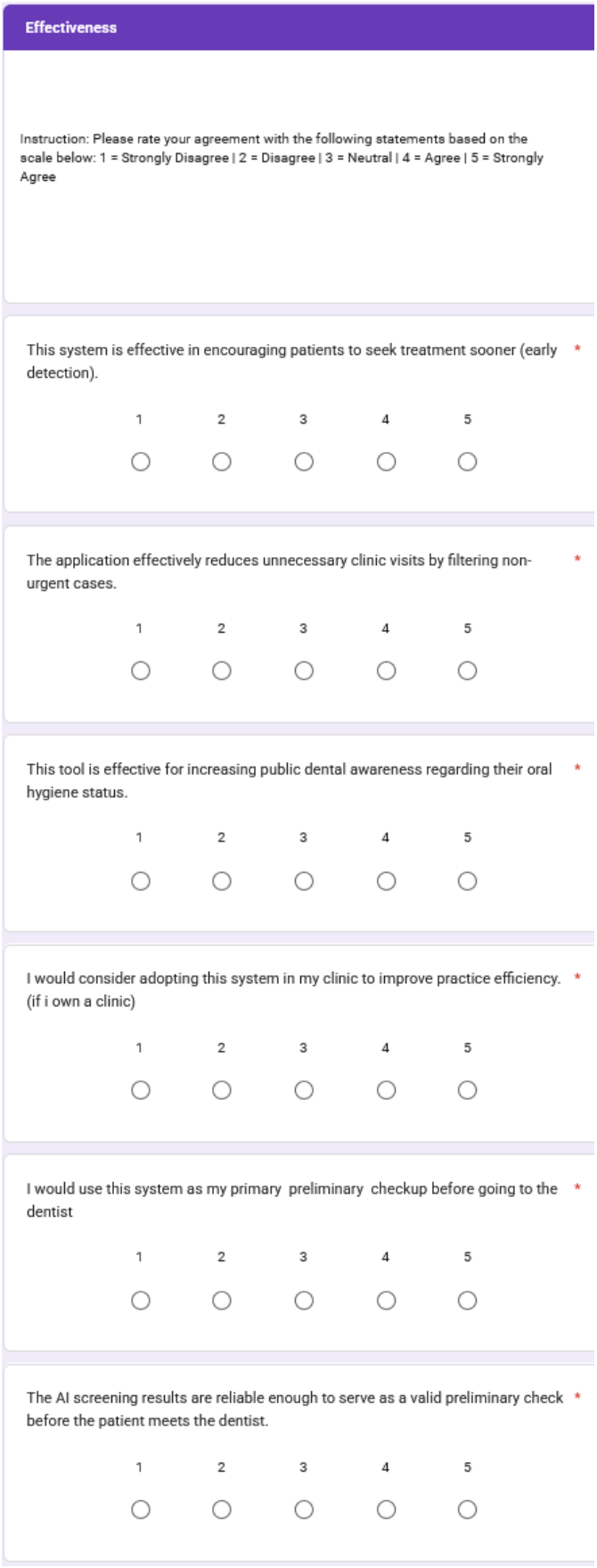


Figure B.3: The Questionnaire for this research. (Effectiveness)

Software Requirements Specification

for

<DentAI:Dental Powered Ai Assesment>

Version 1.0 approved

Prepared by Amir Keizrul Bin Mudzir, Ammar Syahansyah Bin Mohd Reza

<UniKL>

<25/5/2025>

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Revision History

Name	Date	Reason For Changes	Version
Amir Keizrul Bin Mudzir	22/6/2025	Change of stakeholder	1.1

1. Introduction

1.1 Purpose

This System Requirements Specification (SRS) document intends to specify the functional requirements and non-functional requirements of the AI-Powered Dental Assessment Web Application, Version 1.0. This project is specifically created for dental patients, general users who may want to have preliminary analysis of their dental conditions and dentist who will use the platform to aid in diagnosing patients at an early stage.

This project is targeted at the development of a centralized and web-based system that would use AI technology in identifying and classifying some common dental factors like gingivitis, caries (cavities), ulcers, and tooth discoloration based on pictures uploaded by the user. Its purpose is to provide quick, convenient, and easy-to-use tests, and give one an opportunity to make an appointment with registered dental clinics to get a further consultation and treatment..

This SRS outlines the scope and system requirements for the initial version release, including the following modules:

- User Account Management System
- AI Dental Image Assessment System
- Dental Condition Feedback & Result Display Module
- Appointment Booking System
- Clinic & Doctor Management Module
- Admin Dashboard System
- Feedback Management

These features are used to increase awareness among people about the dental health, support dental professionals in their work with outreach and preliminary diagnostics, as well as improve the process of connection between a patient and the dental service provider..

It will be used by developers, testers, and stakeholders of the project and people taking over its maintenance in the future. It constitutes a description of the proposed functionalities/constraints and the behavior/system and inter-factor of interaction with different user roles like General Users (Patients), Dental Clinic Staff, Dentists and System Administrators.

1.2 Document Conventions

This System Requirements Specification (SRS) document follows a standard document formatting and structure to ensure easy reading and consistency throughout the document. The following are used:

Typographical conventions	Description
Bold text	used for section headings and to highlight key terms or modules.
Numbered headings	Numbered headings (e.g., 1.0, 1.1, 2.0) follow the IEEE 830-1998 SRS documentation standard.
FR	Functional Requirement
NFR	Non-Functional Requirement
High	High Priority (Must have)
Medium	Medium Priority (Should have)
Low	Low Priority (Nice to have)

Each requirement in this document is assigned its own priority. Priorities are not assumed to be inherited by sub-requirements; all requirements are individually labeled.

1.3 Intended Audience and Reading Suggestions

The target audiences for this document are developers, project managers, programmers, designers, users and testers. The purpose of this document is to describe the system requirements specification for AI-Powered Dental Assessment web app. This document can be used as a reference for new user of staff to see the flow of the system. Several modellings have been described in the documentation including Use Case Diagram, UML Class Diagram and Module's swim lane flowchart Diagram.

When reading this SRS document, it is suggested that readers begin with Section 1 of the document to understand the purpose, scope, and context of the project. For Developers and designers, it is suggested that they should focus more on Sections 2 and 3, where the core features and system interactions are described in detail. Testers are recommended to focus on Section 3 that covers the requirements, to derive test cases. End users and stakeholders may find Section 2.3 (User Classes and Characteristics) and the summary of system functions useful for understanding how they will use the platform.

1.4 Project Scope

The AI-Powered Dental Assessment Web Application is a diagnostic-support tool designed to assist users in identifying early signs of common dental conditions through automated image analysis. By integrating artificial intelligence with web-based technologies, the system enables users to upload images of their teeth for automated assessment, receive preliminary feedback on detected dental conditions, and book appointments with nearby clinics for further consultation and treatment.

The software is developed to address the gap between early dental issue detection and timely professional care. It empowers users, especially those with limited access to dental check-ups to become more aware of their oral health status. At the same time, it supports dentists and clinics in outreach and patient engagement, reducing the burden of walk-in diagnostics and focusing more on treatment planning.

Key goals and benefits include:

Providing fast, accessible dental assessments powered by AI image analysis.

Increasing awareness of oral health issues among the public.

Improving user access to dental care services through integrated appointment booking.

Enhancing clinic-patient interaction through a digital platform.

Reducing workload on dental professionals for basic diagnostics.

This version (1.0) of the platform includes core functionalities such as image upload and analysis, condition detection for gingivitis, ulcers, caries, and tooth discoloration, results display, user account system, clinic/doctor management, and admin functions. This SRS defines the scope and requirements for the initial release, which will serve as a foundation for future enhancements such as progress tracking, treatment history, and multilingual support.

1.5 References

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2. Overall Description

2.1 Product Perspective

This is a web-based system that is named AI-Powered Dental Assessment Platform. The system's purpose is to assist users in conducting a preliminary analysis of their oral health by uploading images of their teeth, which are then analyzed using artificial intelligence. The platform identifies potential dental conditions such as gingivitis, ulcers, caries (cavities), and tooth discoloration. Users can then book an appointment with a registered dental clinic for further diagnosis and treatment based on the AI's results.

Currently, many people rely on traditional in-person consultations even for minor dental concerns, which can lead to unnecessary visits or delayed action due to inconvenience. This system aims to bridge that gap by offering instant, AI-based dental assessments through a centralized, accessible web platform. It is a new, self-contained system and not a follow-up version of any existing application. It is developed as an initiative to increase awareness, accessibility, and engagement with oral health, especially for users who may delay professional consultations due to time or cost concerns.

The AI-Powered Dental Assessment System operates in a web environment with the following functionalities and entities interacting with the system:

Entity	Functionality	Usage context
User (Patient)	Register, login, upload teeth image, receive AI result, book appointment, view history	Web-based application
Clinic Admin	Manage clinic details, dentist assignments, view appointments, manage stored feedback	Web-based application
System Admin	Manage users, clinics, dentists and post system announcements	Web-based application

User (Patient): After registration and login, users can upload an image of their teeth for AI assessment. The AI system will analyze the image and provide feedback on any detected dental conditions. Based on the results, users can book an appointment with a dentist from the listed clinics. Users can also view their appointment history and condition records for personal tracking.

Clinic Admin: Clinic administrators can manage their clinic's profile and details, Manage stored feedback and oversee all appointment-related activities within their clinic.

System Admin: The system admin oversees the entire platform. They manage all user accounts (patients, clinics, and dentists), approve or remove clinic profiles, and post important announcements or news updates visible to all users.

2.2 Product Features

The AI-Powered Dental Assessment platform offers several features designed to provide quick, AI-driven dental analysis while connecting users to professional dental services. Below are the major functionalities of the platform:

Feature ID	Feature Name	Description	Accessible Role
F001	Register Account (User Account Management)	User shall be able to create a new account to access the platform's services.	Patient, Dentist, Clinic Admin
F002	Login Account (User Account Management)	User shall be able to log into the platform securely.	Patient, Dentist, Clinic Admin, System Admin
F003	Logout (User Account Management)	User shall be able to log out from the platform.	All Users
F004	Upload Dental Image (AI Assessment)	Patient can upload an image of their teeth for AI diagnosis.	Patient

F005	AI Condition Detection (AI Assessment)	AI will analyze uploaded image and identify potential conditions: gingivitis, ulcers, caries, and discoloration.	Patient
F006	View AI Result (AI Assessment)	Patient can view visual feedback and textual explanation of detected conditions.	Patient
F007	Book Appointment (Appointment Booking)	Patient shall be able to book an appointment with a selected clinic based on AI result.	Patient
F008	Manage Appointments (Appointment Booking)	Dentist and Clinic Admin can view, confirm, or update appointment status.	Dentist, Clinic Admin
F009	Manage Clinic Profile (Clinic Management)	Clinic Admin can add or edit clinic details, including available dentists.	Clinic Admin
F010	Admin Dashboard (System Administration)	System Admin can manage users, clinics, dentists, and appointments.	System Admin
F011	Manage stored feedback	Clinic admin can edit the stored feedback for each class of detection	Clinic admin

2.3 User Classes and Characteristics

The AI-Powered Dental Assessment platform will be used by several distinct user classes, each with unique responsibilities, privileges, and levels of interaction. The system's interface and access control are designed to accommodate the needs of each user group as shown below:

User	Guest	Patient	Clinic Admin	System Admin
Subset of Functions Used	View homepage and general dental health info Upload image for AI assessment (view-only result) Cannot book appointments	Register and login Upload dental image Receive full AI analysis Book appointments View appointment and assessment history	Manage clinic profile Assign or remove dentists View and manage clinic appointment records	Manage all users and clinics Manage AI feedback knowledgebase Post system announcements

Frequency of Use	Occasional	High Frequency	Medium Frequency	Low Frequency
Technical Expertise / Required Knowledge	Low	Low to Medium Technical Expertise	Medium Technical Expertise	Medium to High Technical Expertise
Educational / Professional Level	General public or first-time visitors	General public, patients, or students	Dental clinic administrative staff	System administrators or IT personnel
Security / Privilege Level	No Privilege	Standard Privilege	Elevated Privilege	Full Administrative Privilege
User Importance	Low Importance	High Importance	Medium Importance	High Importance
Characteristic	Can only browse public content and perform limited AI image assessments with no booking access	Can perform full AI assessments, view detailed results, and book appointments with clinics	Oversee the clinic's operation including managing bookings and profile data	Responsible for platform-wide user, clinic, and system data management

2.4 Operating Environment

Category	Description
Operating system	Windows
Browser	Firefox
Ram	16gb
Chip	Ryzen 5 5600 Intel i5 9th gen
System type	64 bit os
storage	250gb free space

2.5 Design and Implementation Constraints

The design and implementation of the AI-Powered Dental Assessment Web Platform will be shaped by several technical and regulatory constraints to ensure the system meets its goals while maintaining security, performance, and compatibility standards. Below are the key constraints developers must adhere to:

2.5.1 Corporate and Regulatory Policies

- Data Privacy and Patient Confidentiality:
  All user data, including uploaded dental images, personal profiles, and AI diagnosis results, must be securely stored and handled in compliance with data protection guidelines. Privacy of patient data is a top priority and must adhere to ethical standards for medical-related applications.

2.5.2 Hardware Limitations

- Server Capacity:
  The platform will be hosted on a cloud-based server. However, during initial deployment, processing power, storage availability, and bandwidth will be limited and must be optimized for performance efficiency.

- Client Device Compatibility:
  The system must be responsive and usable on various devices including desktops, laptops, tablets, and smartphones with different screen sizes and hardware specifications.

- Concurrent User Sessions:
  The platform must be able to handle multiple concurrent users—especially during peak usage such as when multiple users are uploading images or booking appointments simultaneously.

2.5.3 Technologies, Tools, and Databases

- Development Stack:
  The platform will be developed using React.js for the frontend and Python Flask for the backend. AI processing will utilize TensorFlow/Keras models for dental image analysis. Firebase will serve as the primary database for managing user, clinic, appointment, and AI result data.

- AI Model Constraints:
  AI model inference must be performed efficiently. Model size and response time will be limited by server memory and CPU/GPU resources, especially during simultaneous assessments.

- Image Handling:
  The system must limit the upload size and format (e.g., JPG, PNG) of dental images to ensure smooth processing and avoid storage overflow.

2.5.4 Communication Protocols

- HTTPS for Secure Communication:
  All data exchanges between the user and server must occur over HTTPS to protect sensitive health-related data from interception or manipulation.

2.5.5 Security Considerations

- Data Encryption:
  All sensitive information, including user credentials, dental assessment results, and appointment history, must be encrypted in both transit and storage to ensure maximum security and compliance with healthcare data protection standards.

2.6 User Documentation

The following user documentation materials will be provided alongside the AI-Powered Dental Assessment platform to assist users in understanding, navigating, and utilizing the system effectively:

1. User Manual Document

   Audience: All users (Guest, Patient, Clinic Admin, System Admin)

Purpose: This manual will serve as a comprehensive PDF guide that explains how to use the platform, covering tasks such as uploading dental images, viewing AI analysis results, booking appointments, and managing clinic information.

2. Video Tutorials

Audience: All users (Guest, Patient, Clinic Admin, System Admin)

Purpose: A set of short, step-by-step video guides will be made available to visually demonstrate how to perform key actions within the platform. These will help first-time users become familiar with the system.

2.7 Assumptions and Dependencies

Assumptions

- It is presumed that users of the system will have constant access to stable internet connection especially the patients as the system will be based entirely on the web.
- The assumption is that the users are to visit the platform using modern and updated web browsers like Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari to have the best service delivery and perform efficiently and compatibly.
- It is presupposed that the uploaded image should be clear and in the newest formats (e.g., JPG, PNG) to have correct AI analysis.
- Key stakeholders (e.g., clinic admins and system administrators) are supposed to participate actively in the system development and testing aspects to give a significant feedback and guarantee their proper functionality.
- It is implied: Such resources as user guides and tutorial videos will be provided to allow users to familiarize themselves with the platform and a method of technical support will be available once launched.

Dependencies

- User Acceptance and Training:
  The effectiveness of the system is pegged on the participation of the users. Depending on the readiness of patients and clinics to switch to the AI-assisted platform, the process of adoption involves moving away from the traditional approach (manual diagnosis or offline bookings)..

- Internet Connectivity and Device Compatibility:
  The performance of the platform relies on the good internet connection and the capability of the device utilized by the user. Image upload or appointment scheduling could be their detrimental functions in case of having low bandwidth or obsolete hardware.

- Data Security and Compliance:
  The practice of security best practices must be followed. The lack of data encryption, or safe handling of patient information, may result in possible instances of data breaches or data breach of trust.

3. System Features

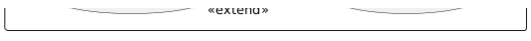
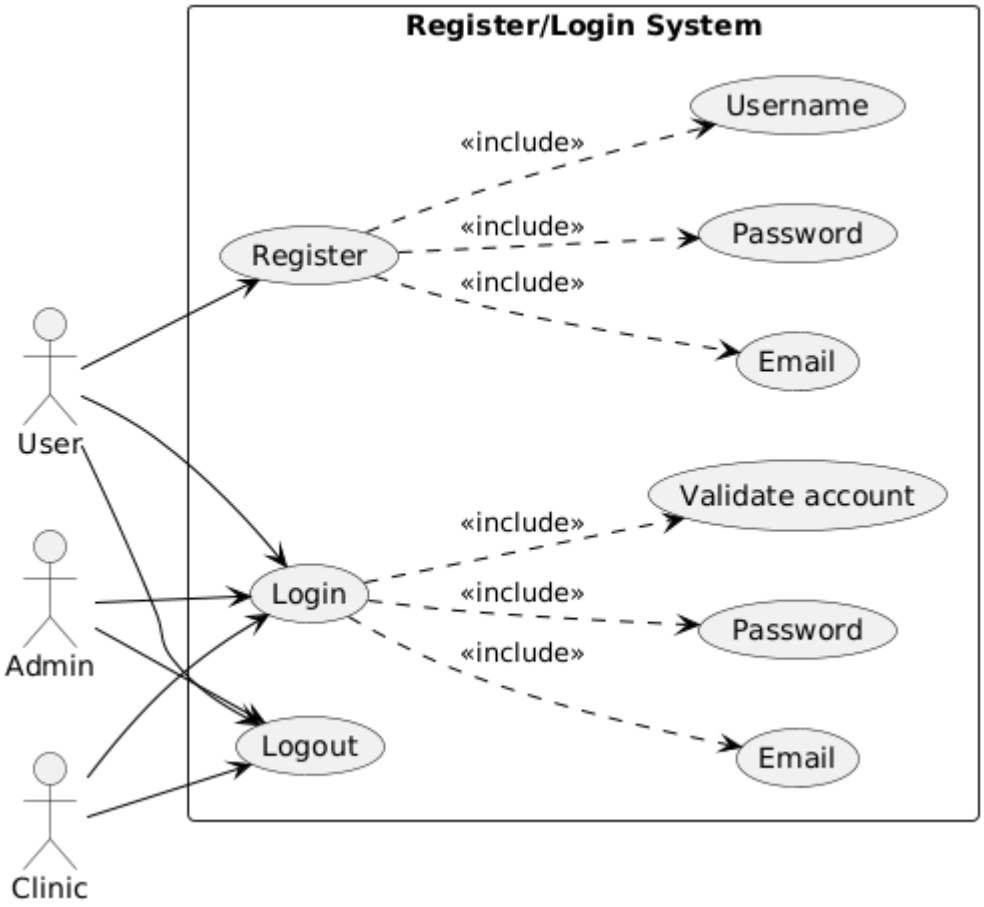


Figure 1 Use Case Diagram Ai powered dental assessment system- overall application

3.1 System Feature 1 : User Account Authentication System

3.1.1 Description and Priority

In this section, it will discuss the account signup, user login and user logout functionality. Priority will be discussed below.



3.1.2 Stimulus/Response Sequences

Figure 2 Use Case Diagram - Register Management

Use Case ID	UC-001	
Use Case Name	User Registration	
Description	This use case allows a new user to create an account by providing their full name, email, username, password, and password confirmation.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	The user is not already registered.	
Post-condition	A new user account is created and stored in the system database.	
Main Flow	Step	Action
	Pre-Condition	The user is not already registered.
	1.1	User navigates to the registration page.
	1.2	The system displays the registration form.
	1.3	User enters full name, email, username, password, and confirms password.
	1.4	The system validates the entered data (e.g., unique email/username, matching passwords).
	1.5	If valid, the system stores the user information in the database.
	1.6	The system displays a success message and redirects to the login page.
	Post-condition	A new user account is created and stored in the system database.
Alternate Flow	Pre-Condition	The student already has an account

	2.1	The student can navigate to login page
	2.2	The student can log in using his account
	2.3	The student will automatic login to student page
	Post-condition	Th student doesn't need to register to login if already got account
Robust Flow	Pre-Condition	User is not registered yet and is on the Registration page.
	3.1	The student register using email that already being
	3.2	The system displays an error message indicating the email
	3.3	The student required to use other email to register
	Post-condition	The system must gracefully handle invalid or duplicate inputs, ensuring no incorrect data is stored.

Table 1 Use case statement - user registration

Use Case ID	UC-002	
Use Case Name	Login	
Description	To allow users (User/Admin) to securely log into the system and access its features based on their roles.	
Primary Actor	User, Admin	
Include use cases	Access Main Page	
Pre-Condition	The user has already registered an account.	
Post-condition	The user is authenticated and redirected to the appropriate main page based on their role (User or Admin).	
Main Flow	Step	Action
	Pre-Condition	The user has already registered an account.
	1 User/Admin navigates to the login page. 2 User/Admin enters username and password. 3 System validates credentials. 4 If valid, the system determines the user's role. 5 System grants access and redirects to the appropriate main page.	
	Post-condition	The user is authenticated and redirected to the appropriate main page based on their role (User or Admin).
Alternate Flow	Pre-Condition	User/Admin enters invalid credentials.
	1 User/Admin enters incorrect username or password. 2 System displays an error message indicating invalid credentials. 3 User/Admin is prompted to re-enter login information. 4 User/Admin retries login or navigates to "Forgot Password".	
	Post-condition	The system displays an error message and prompts retry.
Robust Flow	Pre-Condition	User/Admin clicks on "Forgot Password" link.
	1 User/Admin clicks on the "Forgot Password" link on the login page. 2 System prompts user to enter registered email address. 3 User enters an email and submits the request. 4 System sends a password reset link to the provided email. 5 User/Admin accesses the link and sets a new password. 6 System confirms the password has been reset successfully.	
	Post-condition	System sends a password reset email to the user.

Table 2 Use case statement - Login

Use Case ID	UC-003	
Use Case Name	Logout	
Description	To allow users (User/Admin/clinic admin) to log out of the system	
Primary Actor	User, Admin,clinic admin	
Include use cases	-	
Pre-Condition	The User, Admin,clinic admin is currently logged in	
Post-condition	The User, Admin,clinic admin is logged out the system	
Main Flow	Step	Action
	Pre-Condition	The user has already registered an account.
	1. User clicks the logout button. 2. System terminates the current session. 3. System redirects user to the login page with a confirmation message.	
	Post-condition	The User, Admin,clinic admin is logged out the system
Alternate Flow	Pre-Condition	**Session already expired:**
	1. System automatically redirects to login page when a protected page is accessed.	
	Post-condition	User is logged out, and session is securely terminated.
Robust Flow	Pre-Condition	-
	-	
	Post-condition	-

Table 3 Use case statement - Logout

3.1.3 Functional Requirements

Requirement Traceability ID	Functional Requirement ID	Description	Priority	Author
REQ_F101	FR_001	Student shall be able to register an account in the system.	High	Amir Keizrul Bin Mudzir
REQ_F201	FR_002	Users shall be able to login an account in the system.	High	Amir Keizrul Bin Mudzir
REQ_F301	FR_003	Users shall be able to logout from their session.	High	Amir Keizrul Bin Mudzir

3.2 System Feature 2 : Image analysis

3.2.1 Description and Priority

In this section, it will show image analysis functionality. The priority will be discussed below.

3.2.2 Stimulus/Response Sequences

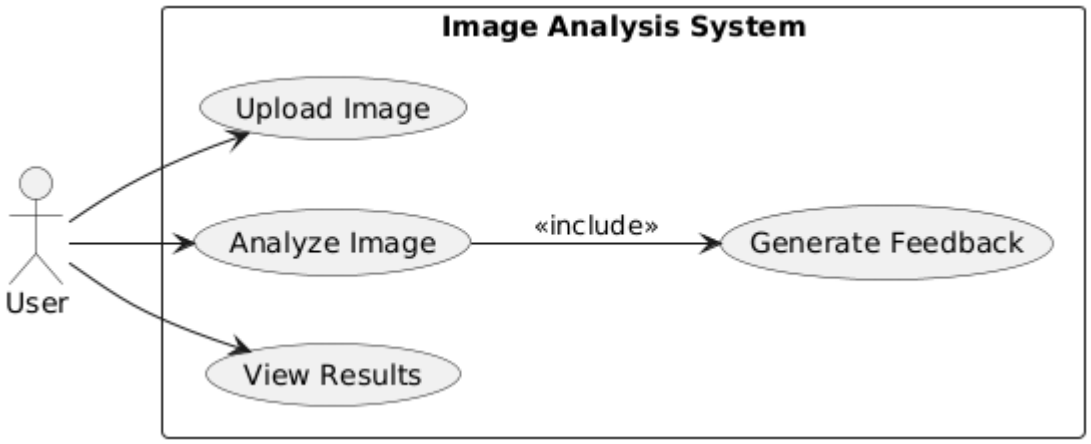


Figure 3 Analyze Image

Use Case ID	UC-004
Use Case Name	Upload image

Description	To allow users to upload an intraoral image to be analyzed by the AI model for dental condition detection.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	The user must be logged in and have access to the upload page.	
Post-condition	The image is stored in the system and sent for analysis. A result or feedback is generated.	
Main Flow	Step	Action
	Pre-Condition	The user must be logged in and have access to the upload page.
	1 User navigates to the “Upload Image” section. 2 System prompts the user to choose an image file. 3 User selects and submits an intraoral image. 4 System validates the file type and size. 5 System stores the image in the database or cloud storage. 6 Image is sent to the AI model for analysis. 7 System notifies the user that the image has been uploaded successfully.	
	Post-condition	The image is stored in the system and sent for analysis. A result or feedback is generated.
Alternate Flow	Pre-Condition	User submit an invalid file
	1. If the uploaded file is not a valid image format or too large. 3. System shows an error message and prompts re-upload. 5. User re-uploads a valid image.	
	Post-condition	The system displays an error message and prompts retry.
Robust Flow	Pre-Condition	Upload fails
	1. If storage/upload fails due to server or network error. 2. System displays an error message (e.g., “Upload Failed”). 3. User can retry uploading or contact support.	
	Post-condition	System sends an error message

Table 4 Use case statement - Upload image

Use Case ID	UC-005	
Use Case Name	Analyze image	
Description	To analyze the uploaded intraoral image using an AI model to detect dental conditions	
Primary Actor	System	
Include use cases	generate feedback	
Pre-Condition	A valid intraoral image has been uploaded by a logged-in user.	
Post-condition	The AI model returns detection results that are stored and displayed to the user.	
Main Flow	Step	Action
	Pre-Condition	A valid intraoral image has been uploaded by a logged-in user.
	1 System receives the uploaded image. 2 The AI model analyses the image for dental conditions. 3 System generates a result with identified conditions and confidence levels. 4 Results are stored in the database and linked to the user's account.	
	Post-condition	The AI model returns detection results that are stored and displayed to the user.
Alternate Flow	Pre-Condition	No Condition Detected
	1. If the AI model finds no significant dental condition. 2. System generates a message indicating no issues were found.	
	Post-condition	The system displays a message indicating no issues.
Robust Flow	Pre-Condition	Analysis Error
	1.If the AI model fails due to corrupted image or internal error. 2. System logs the error and notifies the user of failure.	
	Post-condition	User is prompted to re-upload the image or try again later.

Table 5 Use case statement - Analyze image

Use Case ID	UC-006	
Use Case Name	View result	
Description	Allows the user to view feedback from their image analysis	
Primary Actor	user	
Include use cases	-	
Pre-Condition	User submitted and system completed image analysis	
Post-condition	Feedback is displayed to the user and can be reviewed at any time.	
Main Flow	Step	Action
	Pre-Condition	User submitted and system completed image analysis
	1 User navigates to the Feedback section after logging in. 2 System fetches all related feedback records for the user. 3 Feedback is retrieved from the database. 4 Feedback (text, date, treatment suggestions, etc.) is displayed. 5 User reviews the feedback and may proceed to track progress or book appointment.	
	Post-condition	Feedback is displayed to the user and can be reviewed at any time.
Alternate Flow	Pre-Condition	No Feedback Available
	1.If the user has no completed analyses or feedback records. 2. System shows a message: "No feedback available yet."	
	Post-condition	User is prompted to upload an image or return to dashboard.
Robust Flow	Pre-Condition	System Error While Fetching Feedback
	1 If feedback retrieval fails due to a database error. 2 System logs the error and displays an error message.	
	Post-condition	User is given the option to retry
Use Case ID	UC-007	
Use Case Name	Generate Feedback	
Description	This use case allows the system to generate feedback based on image analysis results	
Primary Actor	System	
Include use cases	Image Analysis	
Pre-Condition	User has uploaded an image and the analysis has been successfully completed.	
Post-condition	Feedback is generated and stored in the database, associated with the user's profile.	
Main Flow	Step	Action
	Pre-Condition	User has uploaded an image and the analysis has been successfully completed.
	1 User uploads an image and initiates analysis.	

	2 System analyzes the image using an AI model. 3 System processes analysis results to create textual feedback. 4 Feedback includes diagnosis and recomendation 5 Feedback is saved in the database and becomes available to view.	
	Post-condition	Feedback is generated and stored in the database, associated with the user's profile.
Alternate Flow	Pre-Condition	
	-	
	Post-condition	
Robust Flow	Pre-Condition	Invalid Image or Model Error
	1 Image format is unsupported or unclear for analysis. 2 System throws an error and asks the user to upload a new image. 3 If the AI model fails internally, system logs the error and informs the user.	
	Post-condition	User is prompted to try again later

Table 6 Use case statement - View result

3.2.3 Functional Requirements

Requirement Traceability ID	Functional Requirement ID	Description	Priority	Author
REQ_F401	FR_004	User shall be able to upload an image for analysis.	High	Amir Keizrul Bin Mudzir
REQ_F402	FR_005	System shall analyze the uploaded image using AI-based object detection.	High	Amir Keizrul Bin Mudzir
REQ_F403	FR_006	System shall display the detected object(s) from the image analysis.	High	Amir Keizrul Bin Mudzir
REQ_F404	FR_007	System shall display relevant feedback based on the detected object(s).	High	Amir Keizrul Bin Mudzir

3.2.4 4.1.3 use case: Booking Appointment

3.3 System Feature 3: Appointment Booking

3.3.1 Description and Priority

In this section it will show the appointment booking functionality, where the user can select date,book appointments,cancel/reschedule and the clinic admin can approve or decline said schedule.The priority will be discussed as follows

3.3.2 Stimulus/Response Sequences

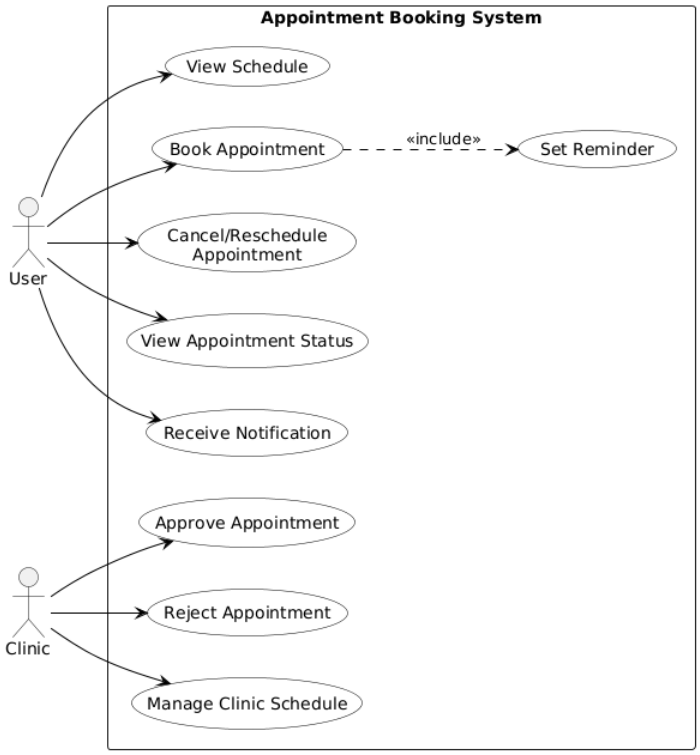


Figure 4 Use case: Booking Appointment

Use Case ID	UC-008	
Use Case Name	View Schedule	
Description	This use case allows the User to view available appointment slots provided by the Clinic.	
Primary Actor	User	
Include use cases	Login	
Pre-Condition	The User is logged in to the system..	
Post-condition	The available appointment schedule is displayed.	
Main Flow	Step	Action
	Pre-Condition	The User is logged in to the system..
	1 User selects "View Schedule". 2 System fetches available slots from the clinic's schedule. 3 System displays available appointment dates and times.	
	Post-condition	The available appointment schedule is displayed.
Alternate Flow	Pre-Condition	No available appointment
	1. If no slots are available, system shows a message: "No appointments available."	
	Post-condition	
Robust Flow	Pre-Condition	
	-	
	Post-condition	

Table 7 Use case statement - View Schedule

Use Case ID	UC-009	
Use Case Name	Book Appointment	
Description	This use case allows the User to request an appointment based on available slots.	
Primary Actor	User	
Include use cases	Login	
Pre-Condition	The User is logged in and has viewed the available schedule.	
Post-condition	The appointment request is submitted and pending approval by the clinic	
Main Flow	Step	Action
	Pre-Condition	The User is logged in to the system.
	1 User selects a preferred date and time from the schedule. 2 User submits the booking request. 3 System saves the request and notifies the clinic.	
	Post-condition	The appointment request is submitted and pending approval by the clinic
Alternate Flow	Pre-Condition	No available appointment
	1. If the selected time slot becomes unavailable during submission, system notifies the user and asks to select another time.	
	Post-condition	
Robust Flow	Pre-Condition	
	-	
	Post-condition	

Table 8 Use case statement - Book appointment

Use Case ID	UC-010	
Use Case Name	Set Reminder	
Description	This use case allows the User to set a calendar reminder for their approved appointment.	
Primary Actor	User	
Include use cases	Book Appointment	
Pre-Condition	The appointment has been approved by the Clinic	
Post-condition	A reminder is added to the User's calenda	
Main Flow	Step	Action
	Pre-Condition	The appointment has been approved by the Clinic
	1 User views the approved appointment. 2 User selects “Set Reminder”. 3 System asks for preferred reminder time (e.g., 1 day before). 4 Reminder is added to the user's system calendar or notified internally.	
	Post-condition	A reminder is added to the User's calendar.
Alternate Flow	Pre-Condition	No available appointment
	Post-condition	
Robust Flow	Pre-Condition	
	-	
	Post-condition	

Table 9 Use case statement - Set reminder

Use Case ID	UC-011	
Use Case Name	Cancel/Reschedule Appointment	
Description	This use case allows the User to cancel or reschedule an existing appointment.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	The User has an existing appointment.	
Post-condition	Appointment is canceled or rescheduled successfully	
Main Flow	Step	Action
	Pre-Condition	The User has an existing appointment.
	1 User selects “Cancel/Reschedule Appointment”. 2 System displays upcoming appointments. 3 User chooses to either cancel or pick a new slot. 4 If rescheduled, system sends new request to the Clinic. 5 If canceled, system removes the appointment.	
	Post-condition	Appointment is canceled or rescheduled successfully
Alternate Flow	Pre-Condition	No available appointment slot
	If rescheduling and new slot is unavailable, system notifies user to try again later.	
	Post-condition	Notify user to try later
Robust Flow	Pre-Condition	
	-	
	Post-condition	

Table 10 Use case statement - Cancel/reschedule Appointment

Use Case ID	UC-012	
Use Case Name	Approve Appointment	
Description	This use case allows the Clinic to view and approve or reject appointment requests made by users. Approval confirms the appointment and notifies the user.	
Primary Actor	Clinic	
Include use cases	-	
Pre-Condition	The Clinic is logged into the system and there are pending appointment requests.	
Post-condition	Appointment is either approved (confirmed) or rejected (removed), and the user is notified accordingly.	
Main Flow	Step	Action
	Pre-Condition	pending appointment requests.
	1 Clinic navigates to "Pending Appointments" section. 2 System displays all pending requests. 3 Clinic selects an appointment request. 4 Clinic approves the appointment. 5 System marks the appointment as confirmed and notifies the user.	
	Post-condition	Appointment approved
Alternate Flow	Pre-Condition	pending appointment requests
	1Clinic rejects the appointment request. 2 System deletes or archives the request and notifies the user of rejection.	
	Post-condition	Appointment rejected
Robust Flow	Pre-Condition	No pending request
	If no pending appointments are available, system displays “No pending requests.”	
	Post-condition	Display messages

Table 11 Use case statement - Approve Appointment

Use Case ID	UC-013	
Use Case Name	Manage Clinic Schedule	
Description	This use case allows the Clinic to view and approve or reject appointment requests made by users. Approval confirms the appointment and notifies the user.	
Primary Actor	Clinic	
Include use cases	-	
Pre-Condition	Clinic is authenticated and has access to the schedule management interface.	
Post-condition	Clinic schedule is updated and reflected in the user-facing booking system.	
Main Flow	Step	Action
	Pre-Condition	Clinic is authenticated and has access to the schedule management interface.
	1 Clinic navigates to the "Manage Schedule" section. 2 Clinic views the current schedule. 3 Clinic adds, updates, or deletes time slots. 4 System saves the changes and updates the available schedule for users.	
	Post-condition	Clinic schedule is updated and reflected in the user-facing booking system.
Alternate Flow	Pre-Condition	A pending appointment is schedule on the slot
	1 Clinic attempts to modify a time slot with an existing confirmed appointment. 2 System prevents the change and prompts the clinic to handle existing appointments before editing.	
	Post-condition	
Robust Flow	Pre-Condition	Server error
	If the system is unable to retrieve the schedule due to a server error, it displays an error message and logs the issue.	
	Post-condition	Display error message

Table 12 Use case statement - Manage clinic schedule

3.3.3 Functional Requirements

Requirement Traceability ID	Functional Requirement ID	Description	Priority	Author
REQ_F501	FR_008	User shall be able to view available appointment dates.	High	Amir Keizrul Bin Mudzir
REQ_F502	FR_009	User shall be able to book an appointment with a clinic.	High	Amir Keizrul Bin Mudzir
REQ_F503	FR_010	User shall be able to set a reminder for a booked appointment.	Medium	Amir Keizrul Bin Mudzir
REQ_F504	FR_011	Clinic admin shall be able to approve pre-booked appointments.	High	Amir Keizrul Bin Mudzir
REQ_F505	FR_012	Clinic admin shall be able to manage the clinic's available schedule.	High	Amir Keizrul Bin Mudzir
REQ_F506	FR_013	User shall be able to reschedule an existing appointment.	High	Amir Keizrul Bin Mudzir
REQ_F507	FR_014	User shall be able to cancel a booked appointment.	High	Amir Keizrul Bin Mudzir

3.4 System feature 4: User management

3.4.1 Description and Priority

In this section it will discuss about user management. Where the admin can manage user and register clinic.

3.4.2 Stimulus/Response Sequences

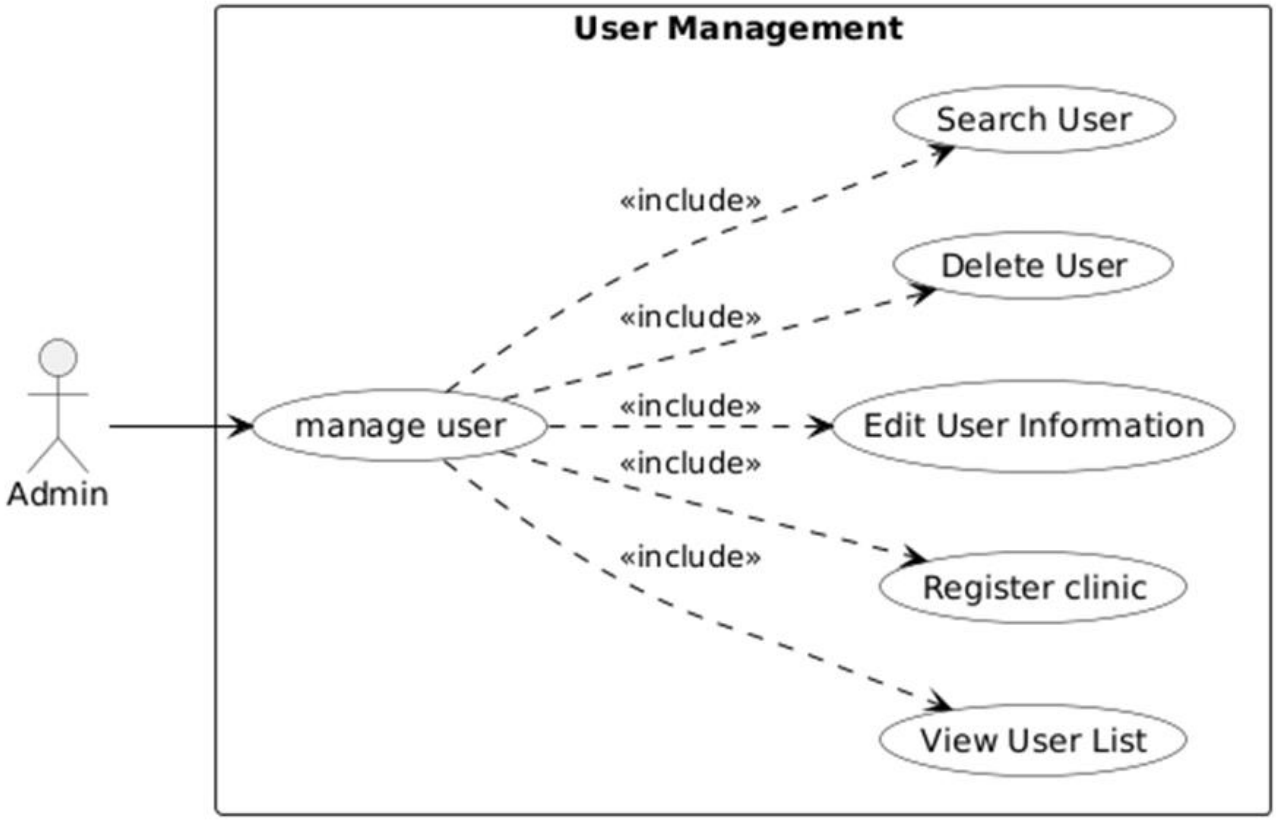


Figure 5 Use case: Manage user

Use Case ID	UC-014	
Use Case Name	Manage User	
Description	Allows the admin to manage registered users in the system, including viewing, deleting, or resetting their passwords.	
Primary Actor	Admin	
Include use cases	View User List, Delete User, search user, edit user information, register clinic.	
Pre-Condition	Admin is logged into the system.	
Post-condition	User information is accessed or updated based on admin action.	
Main Flow	Step	Action
	Pre-Condition	Admin is logged into the system.
	1 Admin navigates to the Manage Users section on the dashboard. 2 System displays available user management actions. 3 Admin chooses to view, delete, or reset password of a user.	

	4 Corresponding included use case is triggered.	
	Post-condition	User information is accessed or updated based on admin action
Alternate Flow	Pre-Condition	No users exist in the system
	1 System displays "No users found" message.	
	Post-condition	Admin is notified of the issue; no changes to user data occur.
Robust Flow	Pre-Condition	System fails to load the user list due to server error
	System displays an error and suggests retrying later.	
	Post-condition	No updates are performed; system state remains unchanged.access based on security protocol.

Table 13 Use case statement - Manage user

Use Case ID	UC-015	
Use Case Name	View User List	
Description	Allows the Admin to view the list of all registered users.	
Primary Actor	Admin	
Include use cases	View User List, Delete User, Reset User Password	
Pre-Condition	Admin is authenticated and authorized..	
Post-condition	System displays user list.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin clicks on “View User List.” 2 System fetches and displays all users in a table format.	
	Post-condition	System displays user list.
Alternate Flow	Pre-Condition	No users exist in the system
	1 System displays "No users found" message.	
	Post-condition	System still remains functional; no list is shown
Robust Flow	Pre-Condition	database is unreachable
	1 If database is unreachable, system shows error and retry option.	
	Post-condition	No data shown; admin remains on the same page.

Table 14 Use case statement - view user list

Use Case ID	UC-016	
Use Case Name	Delete User	
Description	Allows the Admin to remove a user account from the system.	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	Admin is authenticated and user exists in the system.	
Post-condition	User account is deleted and removed from user list.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin selects a user from the list. 2 Admin clicks "Delete." 3 System confirms action and deletes the user.	
	Post-condition	System displays user list.
Alternate Flow	Pre-Condition	Admin cancels the deletion
	1 Admin cancels the deletion at the confirmation prompt..	
	Post-condition	No action is taken; user remains in the system.
Robust Flow	Pre-Condition	database is unreachable
	1 If database is unreachable, system shows error and retry option.	
	Post-condition	No data shown; admin remains on the same page.

Table 15 Use case statement - Delete user

Use Case ID	UC-017	
Use Case Name	Edit user information	
Description	Allows the Admin to update user details such as name, contact information, or role.	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	User account exists and admin is authenticated	
Post-condition	The updated user information is saved and reflected in the system.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin selects a user from the user list. 2 Admin clicks "Edit." 3 System displays editable user information form. 4 Admin updates the desired fields. 5 Admin submits the changes. 6 System validates and saves the updated information.	
	Post-condition	System displays a success message and updated user list.
Alternate Flow	Pre-Condition	Admin cancels the update process.
	1 Admin clicks "Cancel" or navigates away before saving.	
	Post-condition	No changes are made; user information remains unchanged.
Robust Flow	Pre-Condition	System encounters a validation or database error.
	1 If there is a validation error (e.g., invalid email), system prompts for correction. 2 If database is unreachable, system shows an error and allows retry or cancel.	
	Post-condition	User data is not updated; admin remains on the edit page until resolved.

Table 16 Use case statement - Edit user information

Use Case ID	UC-018	
Use Case Name	Search User	
Description	Allows the Admin to search for a specific user in the system using keywords like name or email.	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	Admin is authenticated and the system has existing users.	
Post-condition	Search results are displayed based on the entered criteria.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin enters a search keyword (e.g., name, email) in the search bar. 2 Admin clicks "Search" or presses Enter. 3 System processes the request and queries the database. 4 System displays a list of matching users.	
	Post-condition	System displays the filtered user list based on the search criteria.
Alternate Flow	Pre-Condition	Search input does not match any user.
	1. System displays "No users found" message.	
	Post-condition	System returns to the search page; no users are displayed..
Robust Flow	Pre-Condition	System encounters input error or database issue.
	1 If search input is invalid (e.g., too short), system displays validation error. 2 If the database is unreachable, system shows error and retry option.	
	Post-condition	No results displayed; admin is notified and remains on the search page.

Table 17 Use case statement - Search user

Use Case ID	UC-019
Use Case Name	Register Clinic
Description	Allows the Admin to register a new clinic account in the system by entering clinic details and creating associated login credentials.
Primary Actor	Admin
Include use cases	-
Pre-Condition	Admin is authenticated and has permission to register clinics.

Post-condition	A new clinic account is successfully created and added to the system.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin clicks on "Register Clinic" button. 2 System displays clinic registration form. 3 Admin enters required clinic details (e.g., clinic name, address, email, phone number, login credentials). 4 Admin submits the form. 5 System validates the input and creates the clinic account. 6 System displays confirmation message and adds the clinic to the list.	
	Post-condition	Clinic is saved in the system and appears in the clinic list.
Alternate Flow	Pre-Condition	Admin cancels the registration process.
	1. Admin clicks "Cancel" or navigates away before submitting the form.	
	Post-condition	No clinic is registered; form is discarded
Robust Flow	Pre-Condition	System encounters input or database error.
	1 If required fields are missing or invalid, system prompts for correction. 2 If email already exists, system shows duplication error. 3 If the database is unreachable, system shows error and retry/cancel option.	
	Post-condition	Clinic account is not created until errors are resolved.

Table 18 Use case statement - Register Clinic

3.4.3 Functional Requirements

Requirement Traceability ID	Functional Requirement ID	Description	Priority	Author
REQ_F601	FR_015	Admin shall be able to view the list of registered users.	High	Amir Keizrul Bin Mudzir
REQ_F602	FR_016	Admin shall be able to search for a specific user by name or email.	High	Amir Keizrul Bin Mudzir

REQ_F603	FR_017	Admin shall be able to edit user information.	High	Amir Keizrul Bin Mudzir
REQ_F604	FR_018	Admin shall be able to delete a user from the system.	High	Amir Keizrul Bin Mudzir
REQ_F605	FR_019	Admin shall be able to register a clinic into the system.	High	Amir Keizrul Bin Mudzir

3.5 System Feature 5 : Profile Management

3.5.1 Description and Priority

In this section it will disscuss about the user profile management. Where they can upload phots,change password and update personal information.

3.5.2 Stimulus/Response Sequences

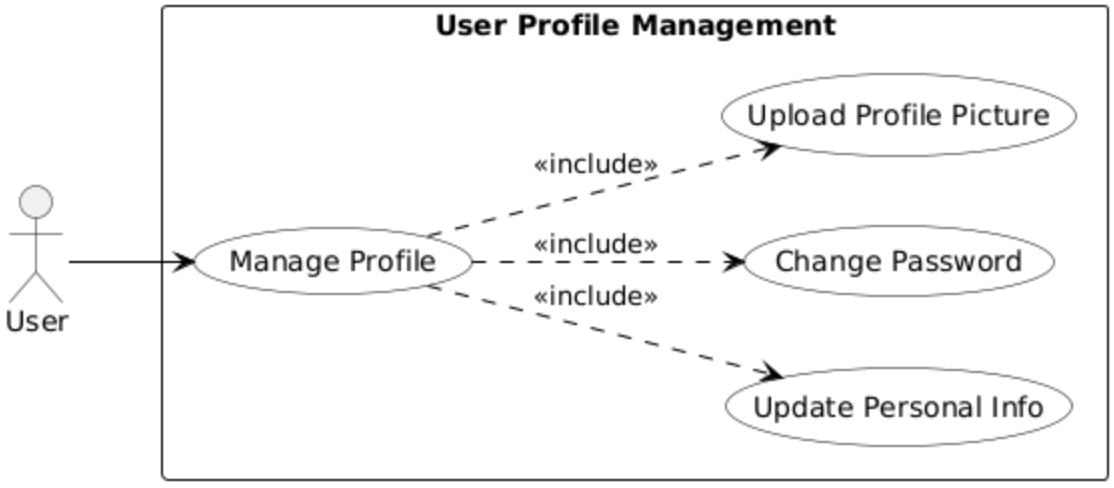


Figure 6 Use case: Manage Profile

Use Case ID	UC-020
Use Case Name	Update Personal Info
Description	Allows the user to edit their personal information such as full name, email, phone number, or address.
Primary Actor	User

Include use cases	-	
Pre-Condition	User is authenticated and has a profile.	
Post-condition	Updated personal info is saved and displayed in the user's profile.	
Main Flow	Step	Action
	Pre-Condition	User is logged into the system.
	1 User navigates to the profile section. 2 User clicks on “Edit Personal Info.” 3 System displays editable fields. 4 User updates the desired fields. 5 User clicks "Save." 6 System validates and saves the changes.	
	Post-condition	Updated information is shown in the user profile.
Alternate Flow	Pre-Condition	User cancels the update.
	1. User clicks "Cancel" before saving. 1.	
	Post-condition	No changes are made
Robust Flow	Pre-Condition	Invalid input or server/database issue.
	1 System displays error messages for invalid fields. 2 If the database is unreachable, system shows an error and retry option.	
	Post-condition	Changes are not saved until the issue is resolved.

Table 19 Use case statement - Update personal info

Use Case ID	UC-021	
Use Case Name	Change Password	
Description	This use case allows the user to securely change their password.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User is authenticated.	
Post-condition	Password is updated in the system.	
Main Flow	Step	Action
	Pre-Condition	User is logged into the system.
	1 User goes to the "Change Password" section. 2 User enters current password. 3 User enters and confirms new password.	

	4 System validates input. 5 System updates the password and shows confirmation.	
	Post-condition	New password is saved and used for future logins..
Alternate Flow	Pre-Condition	User cancels password change.
	1 User clicks "Cancel" before saving.	
	Post-condition	Password remains unchanged.
Robust Flow	Pre-Condition	Input errors or backend failure.
	1 If current password is incorrect, system displays an error. 2 If new password doesn't meet criteria, prompt user to fix. 3 If system error occurs, show retry/cancel options.	
	Post-condition	Password is not changed until errors are resolved.

Table 20 Use case statement - Change password

Use Case ID	UC-022	
Use Case Name	Upload Profile Picture	
Description	Allows the user to upload or change their profile picture.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User is authenticated.	
Post-condition	New profile picture is saved and shown in the user profile.	
Main Flow	Step	Action
	Pre-Condition	User is logged into the system.
	1 User clicks “Upload Profile Picture.” 2 System prompts user to choose an image file. 3 User selects an image and clicks “Upload.” 4 System validates file format and size. 5 System uploads the picture and updates the profile.	
	Post-condition	New profile picture is displayed in the system.
Alternate Flow	Pre-Condition	User cancels the upload
	1 User clicks "Cancel" before saving.	
	Post-condition	No image is uploaded or changed..
Robust Flow	Pre-Condition	Invalid file
	1 If file is not an image or too large, system shows error. 2 If upload fails due to network or server issue, system shows error and retry option.	
	Post-condition	No changes made until a valid image is successfully uploaded.

Table 21 Use case statement - upload profile picture

Requirement Traceability ID	Functional Requirement ID	Description	Priority	Author
REQ_F701	FR_020	User shall be able to edit their personal information.	High	Amir Keizrul Bin Mudzir
REQ_F702	FR_021	User shall be able to change their account password.	High	Amir Keizrul Bin Mudzir

REQ_F703	FR_022	User shall be able to upload or update their profile photo.	Medium	Amir Keizrul Bin Mudzir

3.6 System features 6 : manage feedback

3.6.1 Description and Priority

In this section the function of managing stored feedback will be discuss.

3.6.2 Stimulus/Response Sequences

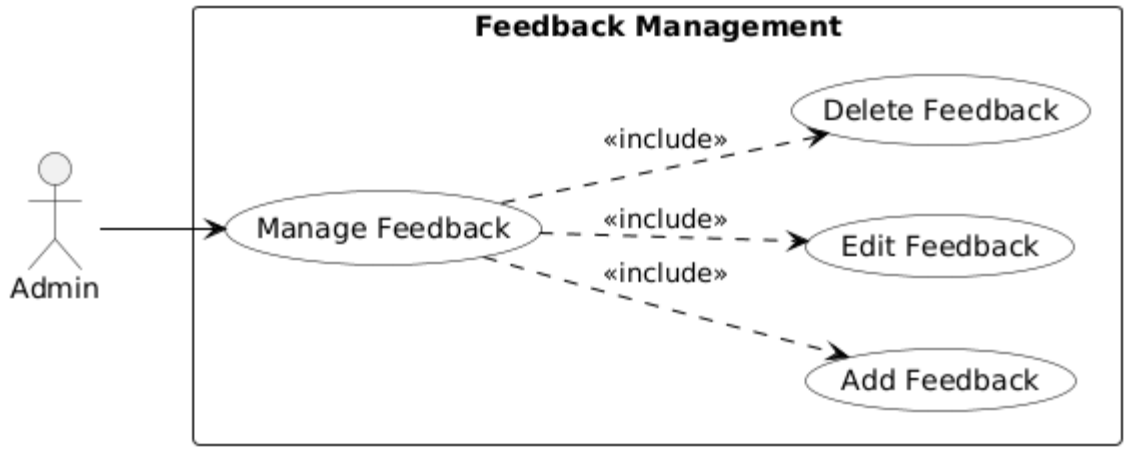


Figure 7 Use case: Manage Feedback

Use Case ID	UC-023	
Use Case Name	Add Feedback	
Description	Allows the clinic Admin to manually add feedback into the system	
Primary Actor	Clinic admin	
Include use cases	-	
Pre-Condition	Clinic Admin is authenticated.	
Post-condition	New feedback is stored in the system	
Main Flow	Step	Action
	Pre-Condition	User is logged into the system.
	1 Clinic Admin clicks "Add Feedback." 2 Clinic Admin enters details such as feedback text 3 Clinic Admin submits the feedback. 4 Clinic Admin validates and stores the data.	
	Post-condition	System displays success message and shows the added feedback.

Alternate Flow	Pre-Condition	Missing Fields
	1 Clinic Admin submits incomplete data. 2 System shows validation errors.	
	Post-condition	No feedback is stored.
Robust Flow	Pre-Condition	System Error
	1 If the system fails to store data, an error message and retry option are shown.	
	Post-condition	Admin remains on the form.

Table 22 Use case statement - Add feedback

Use Case ID	UC-024	
Use Case Name	Edit Feedback	
Description	Allows the Clinic Admin to edit existing feedback records linked to object detection results.	
Primary Actor	Clinic Admin	
Include use cases	-	
Pre-Condition	Clinic Admin is authenticated and feedback exists.	
Post-condition	Feedback is updated in the system.	
Main Flow	Step	Action
	Pre-Condition	Admin accesses the feedback list.
	1 Clinic Admin selects feedback to edit. 2 Clinic Admin updates fields (e.g., text, label, comment). 3 Clinic Admin saves the changes. 4 System updates the record.	
	Post-condition	Updated feedback is displayed in the list.
Alternate Flow	Pre-Condition	Cancel Editing
	1. Clinic Admin exit without saving	
	Post-condition	No changes made.
Robust Flow	Pre-Condition	Update Error
	1 If saving fails, system shows an error message.	
	Post-condition	Original data remains unchanged.

Table 23 Use case statement - edit feedback

Use Case ID	UC-025
Use Case Name	Delete Feedback
Description	Allows the Clinic Admin to remove stored feedback entries that are no longer needed.
Primary Actor	Admin

Include use cases	-	
Pre-Condition	Clinic Admin is authenticated and feedback exists.	
Post-condition	Feedback entry is deleted from the system.	
Main Flow	Step	Action
	Pre-Condition	Admin is logged in and viewing feedback list.
	1 Clinic Admin selects a feedback entry. 2 Clinic Admin clicks "Delete." 3 System asks for confirmation. 4 Clinic Admin confirms deletion. 5 System removes the feedback.	
	Post-condition	Feedback is no longer visible in the list.
Alternate Flow	Pre-Condition	Cancel Deletion
	1. Clinic Admin cancels the confirmation prompt.	
	Post-condition	Feedback remains unchanged.
Robust Flow	Pre-Condition	Deletion Failure
	1 If deletion fails, system displays an error.	
	Post-condition	Feedback remains in the system.

Table 24 Use case statement - Delete feedback

3.6.3 Functional Requirements

Requirement Traceability ID	Functional Requirement ID	Description	Priority	Author
REQ_F801	FR_023	Clinic admin shall be able to create new feedback for detected object classes.	High	Amir Keizrul Bin Mudzir
REQ_F802	FR_024	Clinic admin shall be able to edit existing feedback associated with detected object classes.	High	Amir Keizrul Bin Mudzir
REQ_F803	FR_025	Clinic admin shall be able to delete feedback associated with detected object classes.	High	Amir Keizrul Bin Mudzir

3.7 System Feature 7 : progress tracking

3.7.1 Description and Priority

In this section the functionality of progress tracking will be discuss. Where the user will view their past analyusis,view result and download their report.

3.7.2 Stimulus/Response Sequences

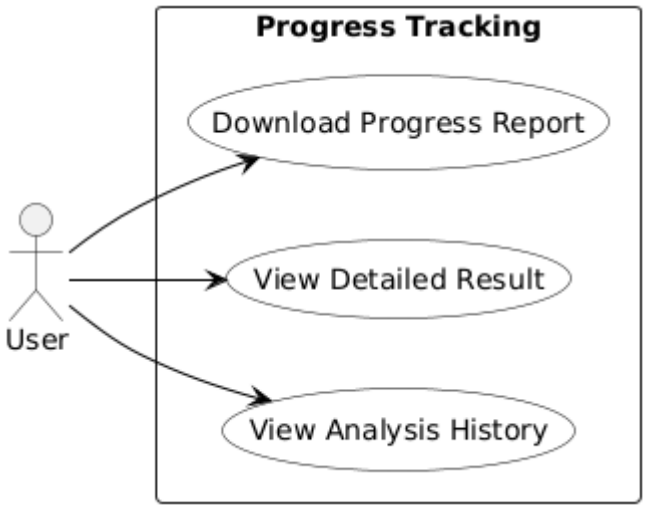


Table 25 Use case: Track Progress

Use Case ID	UC-026	
Use Case Name	View Analysis History	
Description	Allows the user to view a list of previously analyzed images along with the result summary.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User is authenticated and has at least one analysis history.	
Post-condition	A list of past analyses is displayed.	
Main Flow	Step	Action
	Pre-Condition	User is logged in.
	1 User navigates to "Progress" page. 2 System retrieves analysis history. 3 System displays each record (date, result, image thumbnail).	

	Post-condition	User sees a list of past analyses.
Alternate Flow	Pre-Condition	**No history exists**
	System shows “No history available” message.	
	Post-condition	User remains on same page.
Robust Flow	Pre-Condition	**Database is unreachable**
	System shows error message.	
	Post-condition	User stays on page with retry option .

Table 26 Use case statement - view analysis history

Use Case ID	UC-027	
Use Case Name	View Detailed Result	
Description	Allows the user to click on an individual analysis entry to view full detection results and feedback.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User has selected an analysis entry.	
Post-condition	Full result and image are shown.	
Main Flow	Step	Action
	Pre-Condition	User is logged in.
	1 User selects an entry from the history. 2 System fetches detailed result (condition, severity, feedback). 3 System displays detailed result and associated image.	
	Post-condition	Detailed data is shown.
Alternate Flow	Pre-Condition	**Selected entry is missing or corrupted**
	1 System notifies user of issue.	
	Post-condition	User remains on same page.
Robust Flow	Pre-Condition	

		Server error occurs
	1 System shows error with retry button.	
	Post-condition	User remains on previous page.

Table 27 Use case statement - View detailed history

Use Case ID	UC-028	
Use Case Name	Download Progress Report	
Description	Allows the user to download a PDF report of their past dental analyses.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User has at least one analysis history.	
Post-condition	File is downloaded successfully.	
Main Flow	Step	Action
	Pre-Condition	User has at least one analysis history.
	1 User clicks on "Download Report" button. 2 System generates a PDF summary of analysis results. 3 System downloads the file to user's device.	
	Post-condition	User receives the file.
Alternate Flow	Pre-Condition	**No history available**
	1 System disables button or shows "No data" alert.	
	Post-condition	User remains on same page.
Robust Flow	Pre-Condition	**PDF generation fails**

	1 System shows error with retry button.	
	Post-condition	User remains on page with retry option.

Table 28 Use case statement - Download Progress Report

3.7.3 Functional Requirements

Requirement Traceability ID	Functional Requirement ID	Description	Priority	Author
REQ_F901	FR_026	User shall be able to view their past image analysis history.	High	Amir Keizrul Bin Mudzir
REQ_F902	FR_027	User shall be able to view detailed results for each analysis session.	High	Amir Keizrul Bin Mudzir
REQ_F903	FR_028	User shall be able to download a report summarizing their dental health progress.	Medium	Amir Keizrul Bin Mudzir

4. External Interface Requirements

4.1 User Interfaces

4.1.1 Register and Login

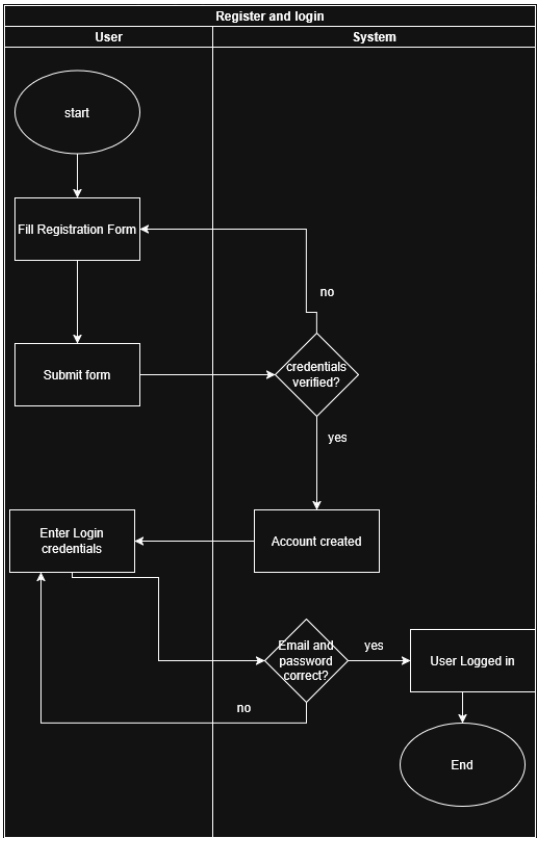


Figure 10 Register and Login Flowchart

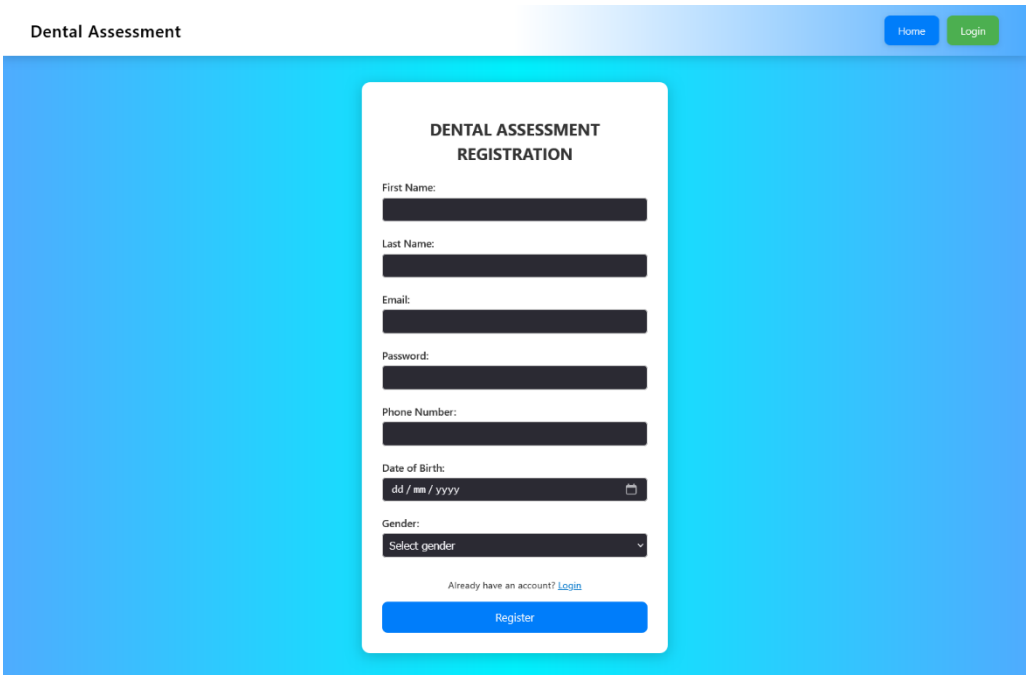


Figure 11 Register Screen

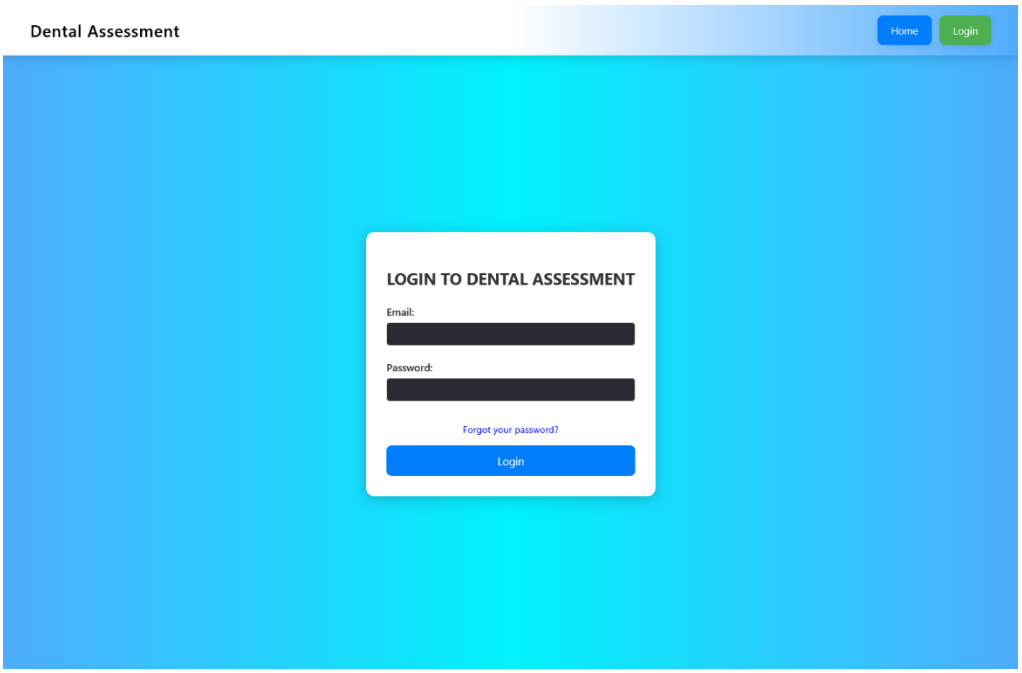


Figure 12 Login Screen

Field	**Details**
Requirement ID	REQ_F101
Version	1.0

Item	Register and Login
Description	User shall be able to register new accounts and login to access system functionality.
Purpose	To authenticate users and provide secure access to personalized features.
Format	• Username - String • Email - String • Password - String • Phone Number - String • User ID - Int (auto-generated)
Valid Range	Unique username/email, Password min 8 characters, Valid email format
Related I/O	Email verification, Session management
Author	Ammar Syahansyah Bin Mohd Reza

Table 35 Requirement - Register Account and login

Field	Details
Requirement ID	REQ_F102
Version	1.0
Item	Logout Account
Description	Users shall be able to logout from their session.
Purpose	To terminate the session and return to the login page.
Format	• Session ID – Int
Valid Range	Active session only
Related I/O	Logout button, redirect to login
Author	Ammar Syahansyah Bin Mohd Reza

Table 36 Requirement - Logout

4.1.2 Image Analysis

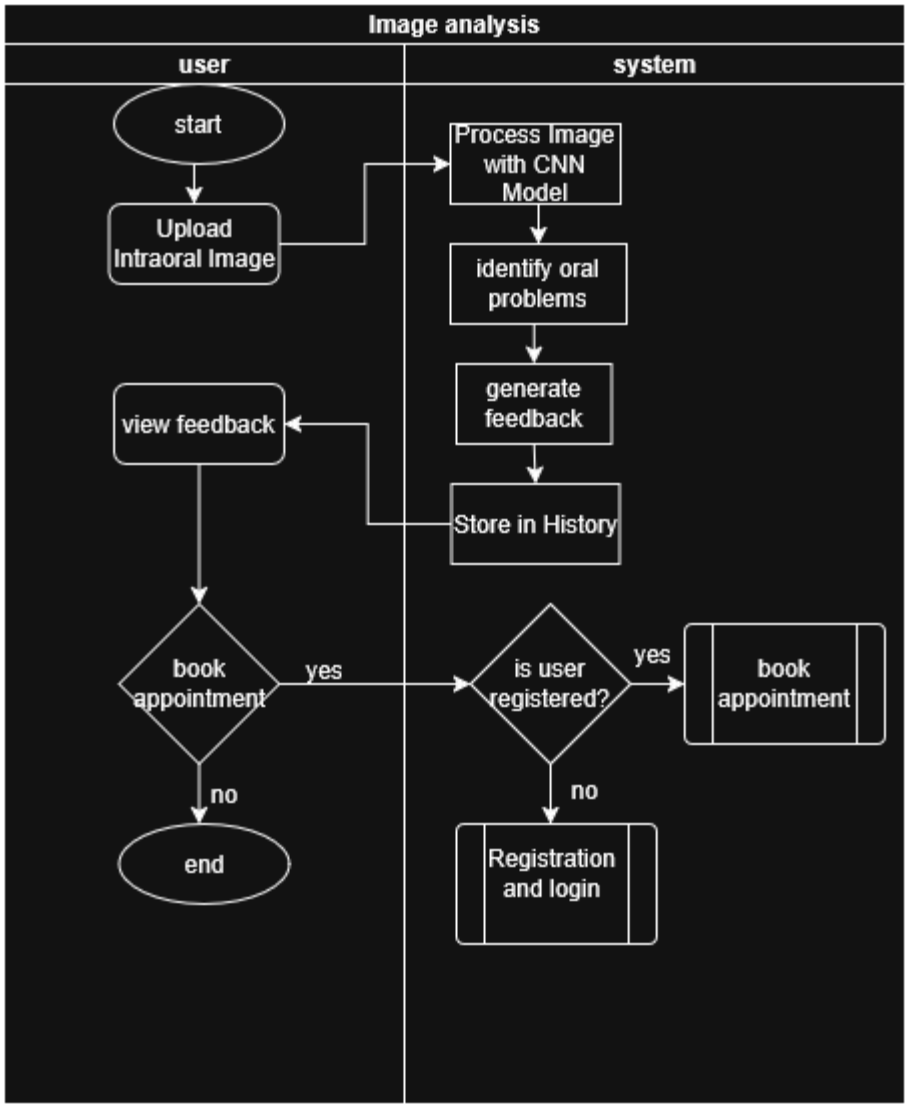


Figure 13 Image Analysis Flowchart

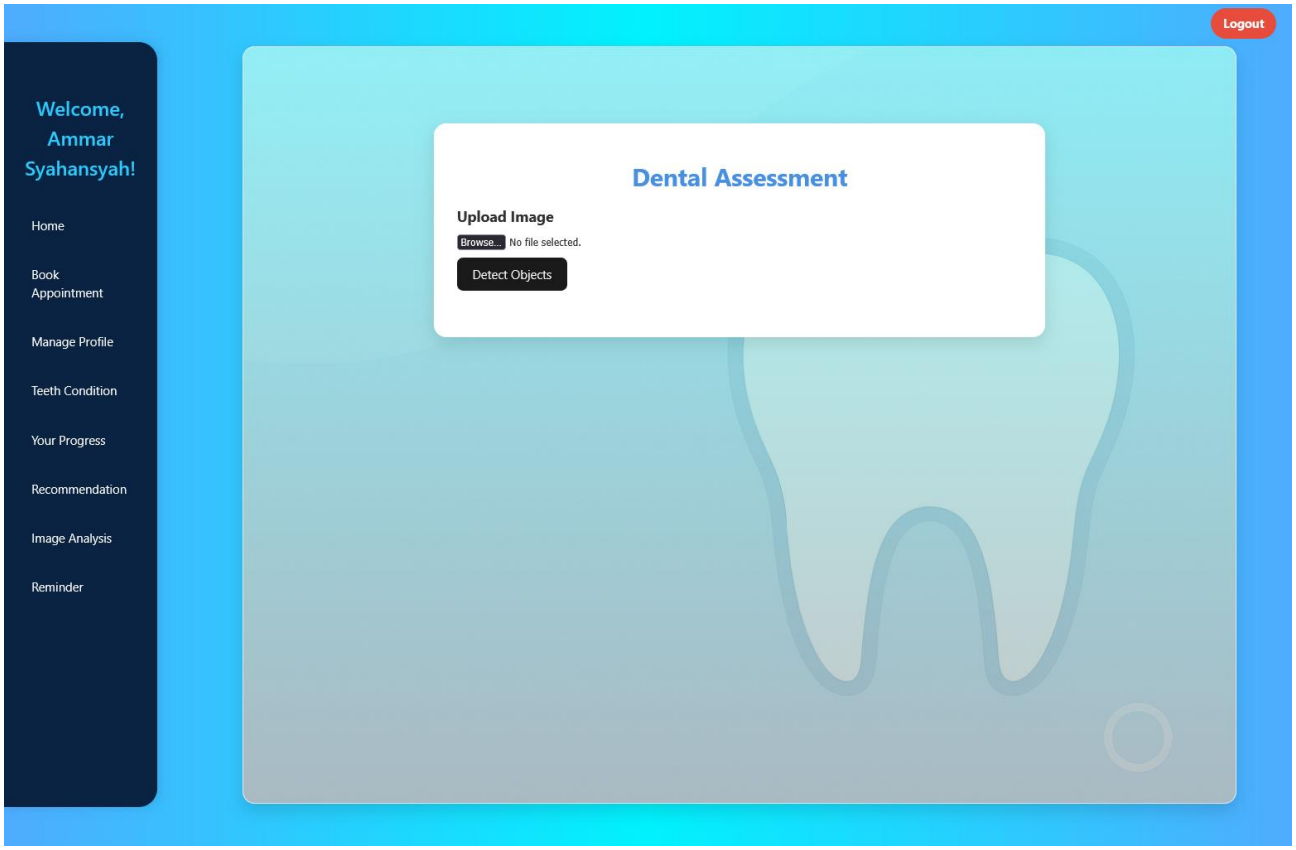


Figure 14 Upload Dental Image Screen

Field	**Details**
Requirement ID	REQ_F201
Version	1.0
Item	Upload Dental Image
Description	User shall be able to upload an image for analysis.
Purpose	To allow AI analysis of dental conditions.
Format	• Image – Binary (JPG, PNG) • User ID – Int
Valid Range	Image size < 10MB; Valid formats only
Related I/O	Upload form; success/error feedback
Author	Ammar Syahansyah Bin Mohd Reza

Table 37 Requirement – Upload Dental Image

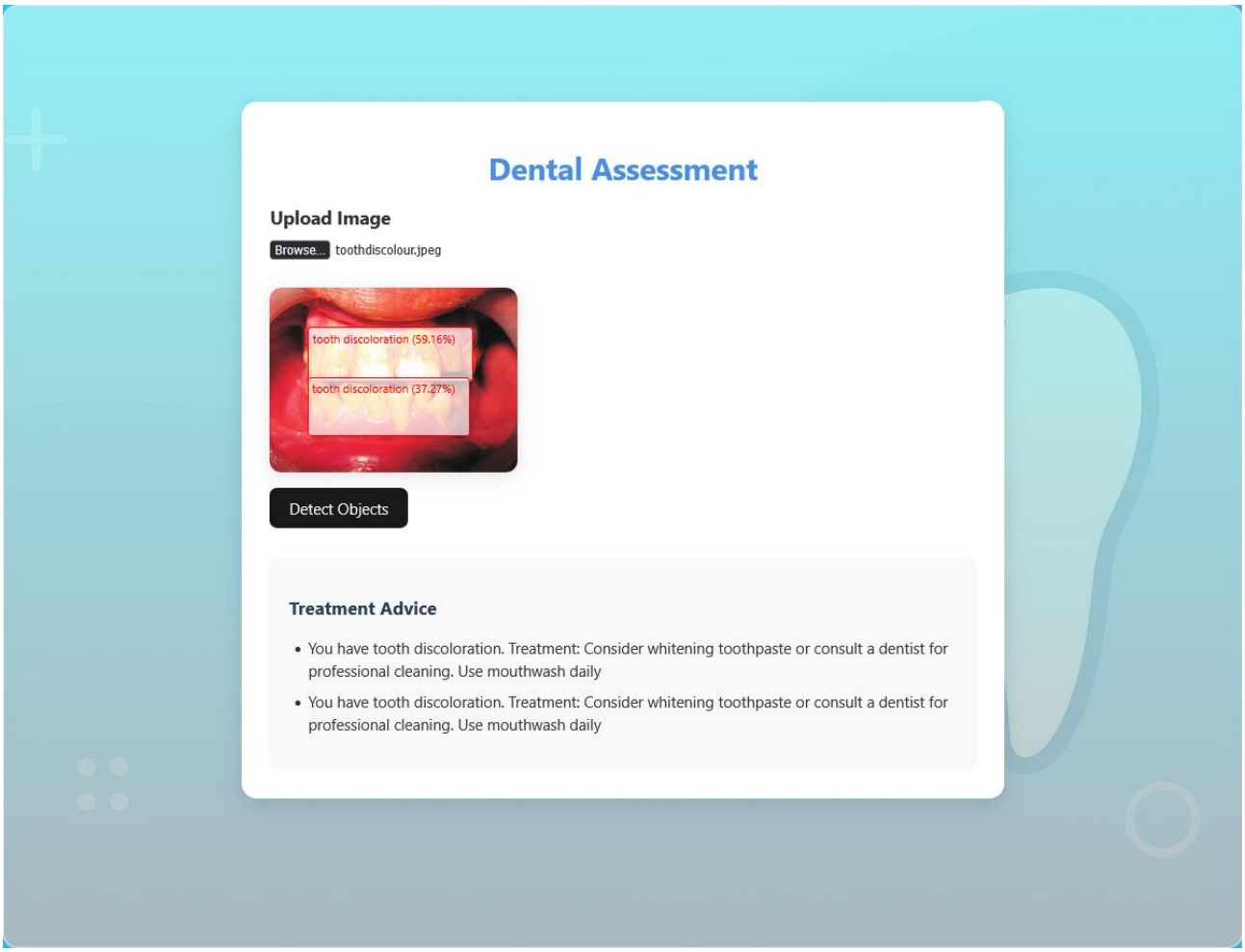


Figure 15 Analyze Image

Field	**Details**
Requirement ID	REQ_F202
Version	1.0
Item	Analyze Image
Description	System shall analyze the uploaded image using AI-based detection.
Purpose	To detect gingivitis, caries, ulcers, and tooth discoloration.
Format	• Image ID – Int • Result JSON – Object • Confidence Score – Float
Valid Range	Result must include class and probability score
Related I/O	AI Model, Database storage

Author	Ammar Syahansyah Bin Mohd Reza

Table 38 Requirement – Analyze Image

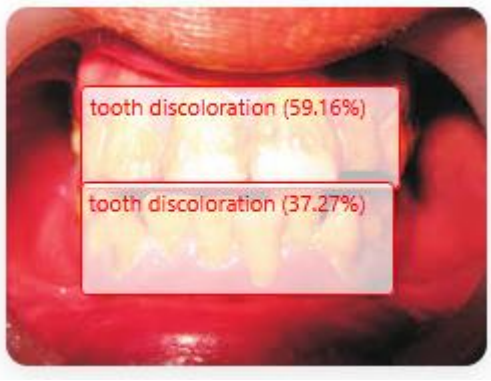


Figure 16 View Ai Result Screen

Field	**Details**
Requirement ID	REQ_F203
Version	1.0
Item	View AI Result
Description	User shall be able to view the results of the AI analysis.
Purpose	To inform user of their condition based on image scan.
Format	• Feedback Text – String • Timestamp – Date • Scan Image – Binary
Valid Range	N/A
Related I/O	Feedback viewer, database fetch
Author	Ammar Syahansyah Bin Mohd Reza

Table 39 Requirement – View Ai Result

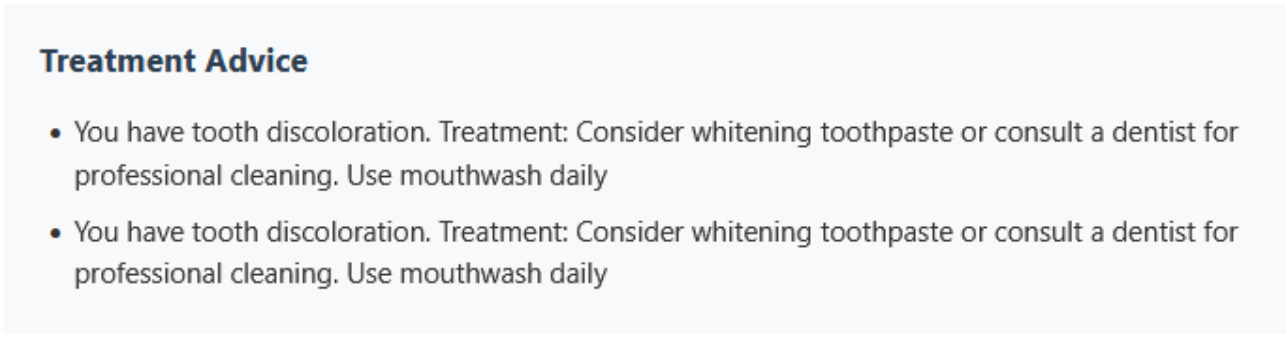


Figure 17 Feedback Screen

Field	**Details**
Requirement ID	REQ_F204
Version	1.0
Item	Generate Feedback
Description	System shall generate diagnostic feedback after analysis.
Purpose	To provide actionable results and recommendations.
Format	• Feedback Text – String • Feedback ID – Int • Result Label – String
Valid Range	N/A
Related I/O	Analysis Result; Feedback Module
Author	Ammar Syahansyah Bin Mohd Reza

Table 40 Requirement – Generate Feedback

4.1.3 Appointment Booking

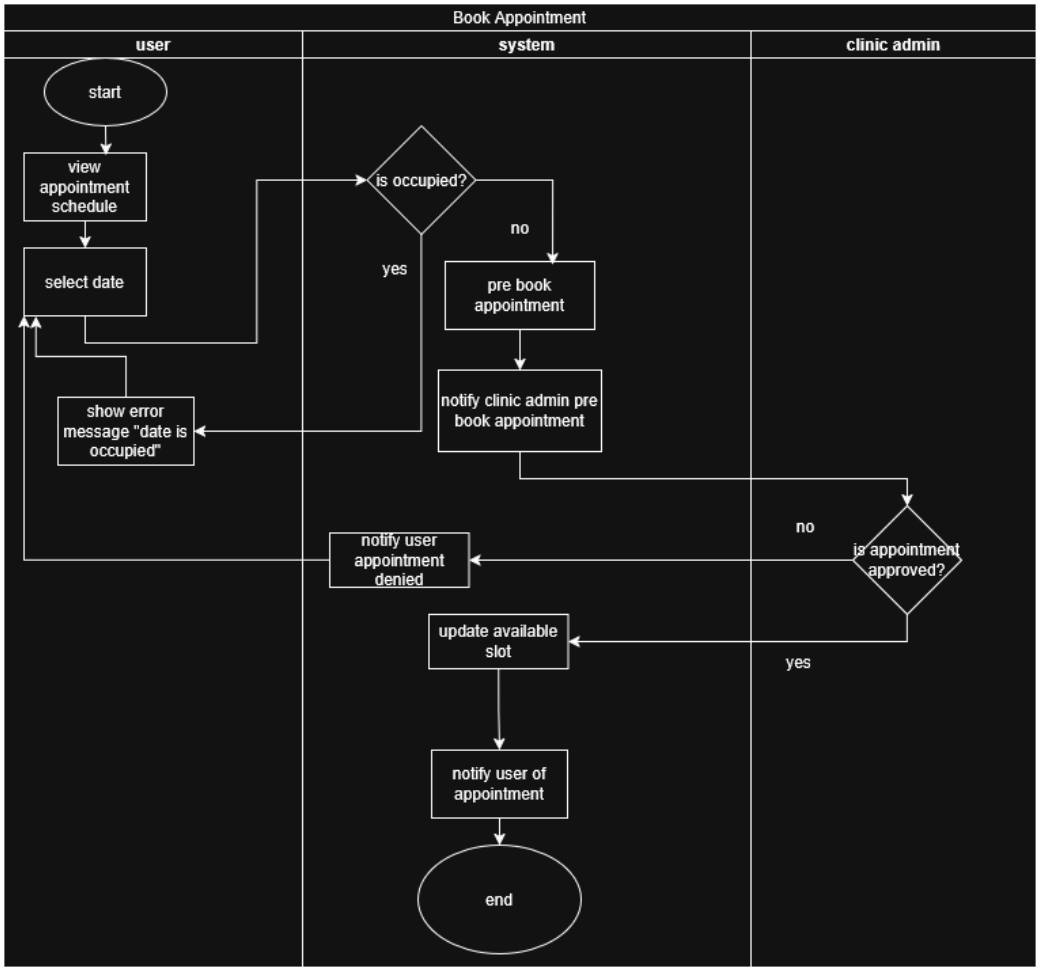


Figure 18 Appointment Booking Flowchart

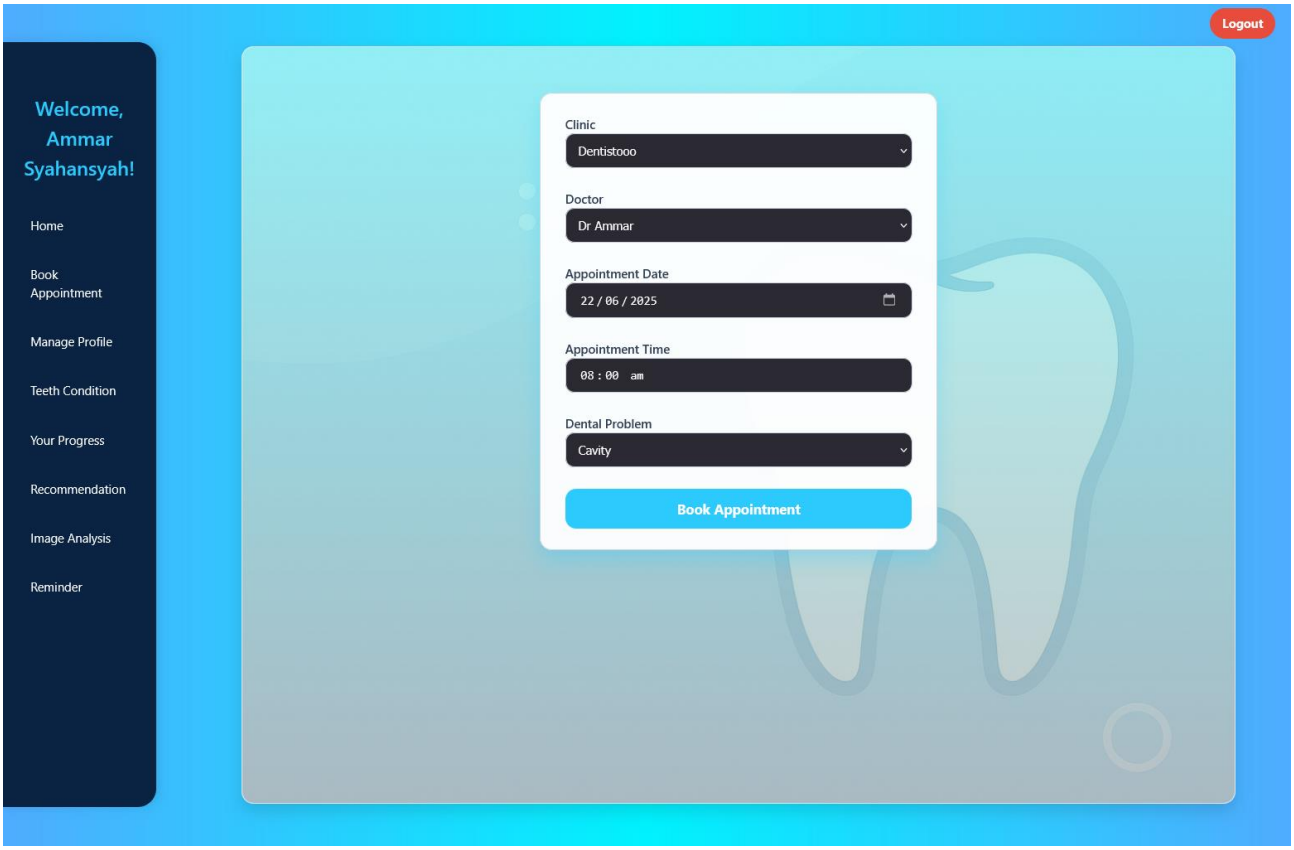


Figure 19 Book Appointment Screen

Field	**Details**
Requirement ID	REQ_F301
Version	1.0
Item	Book Appointment
Description	User shall be able to book dental appointments through the system with automatic conflict detection.
Purpose	To enable patients to schedule appointments while preventing double-booking.
Format	• User ID - Int • Date - Date • Time - Time • Appointment Type - String
Valid Range	Future dates only, Business hours only

Related I/O	Calendar system, Email notifications
Author	Ammar Syahansyah Bin Mohd Reza

Table 41 Requirement – Appointment Booking

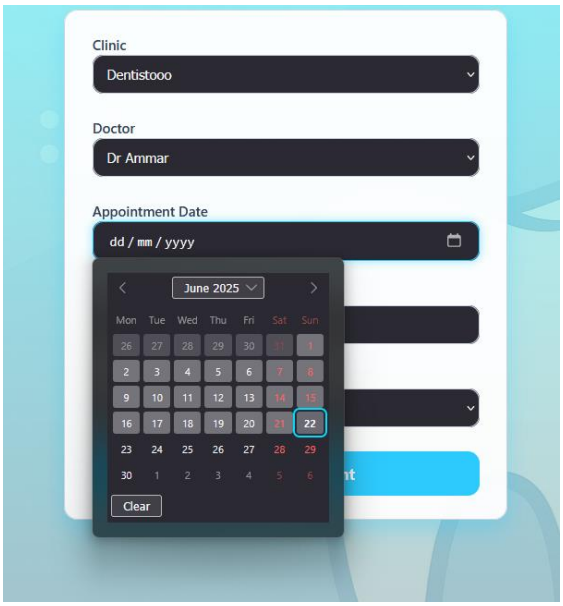


Figure 20 View Appointment Schedule Screen

Field	Details
Requirement ID	REQ_F302
Version	1.0
Item	View Appointment Schedule
Description	User shall be able to view available appointment dates.
Purpose	To let users choose appointment slots.
Format	• User ID – Int • Date – Date • Time – Time
Valid Range	Only available and future time slots
Related I/O	Calendar API; Schedule display UI
Author	Ammar Syahansyah Bin Mohd Reza

Table 42 Requirement – View Appointment Schedule

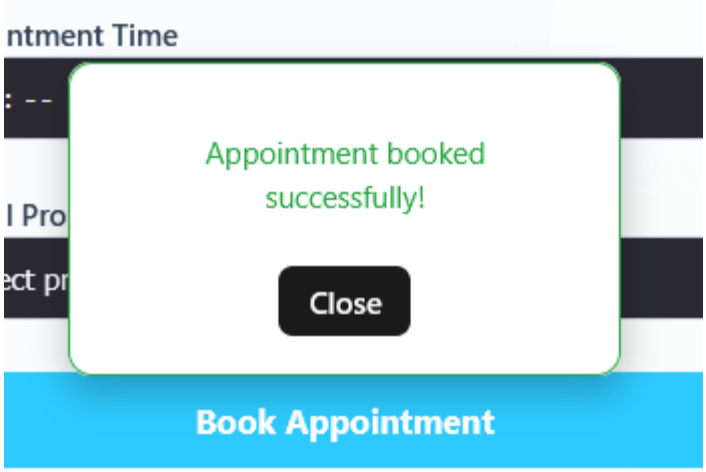


Figure 21 Book Appointment Screen

Field	**Details**
Requirement ID	REQ_F303
Version	1.0
Item	Book Appointment
Description	User shall be able to book an appointment with a clinic.
Purpose	To let users confirm an appointment for diagnosis/treatment.
Format	• Appointment ID – Int • Date – Date • Time – Time
Valid Range	Future available slots only
Related I/O	Booking form; confirmation message
Author	Ammar Syahansyah Bin Mohd Reza

Table 35 Requirement – Book Appointment

Field	**Details**
Requirement ID	REQ_F503
Version	1.0
Item	Appointment Reminder
Description	User shall be able to set a reminder for a booked appointment.

Purpose	To notify users before upcoming appointments.
Format	• Reminder Time – DateTime • Appointment ID – Int
Valid Range	Must be before appointment time
Related I/O	Calendar integration; Notification system
Author	Ammar Syahansyah Bin Mohd Reza

Table 43 Requirement – Appointment Reminder

Field	**Details**
Requirement ID	REQ_306
Version	1.0
Item	Cancel/Reschedule Appointment
Description	User shall be able to cancel or reschedule a booked appointment.
Purpose	To allow flexibility in appointment management.
Format	• Appointment ID – Int • New Date – Date
Valid Range	Reschedule must be to available future slots
Related I/O	Appointment management interface
Author	Ammar Syahansyah Bin Mohd Reza

Table 44 Requirement – Cancel/Reschedule Appointment

4.1.4 Manage User (Admin)

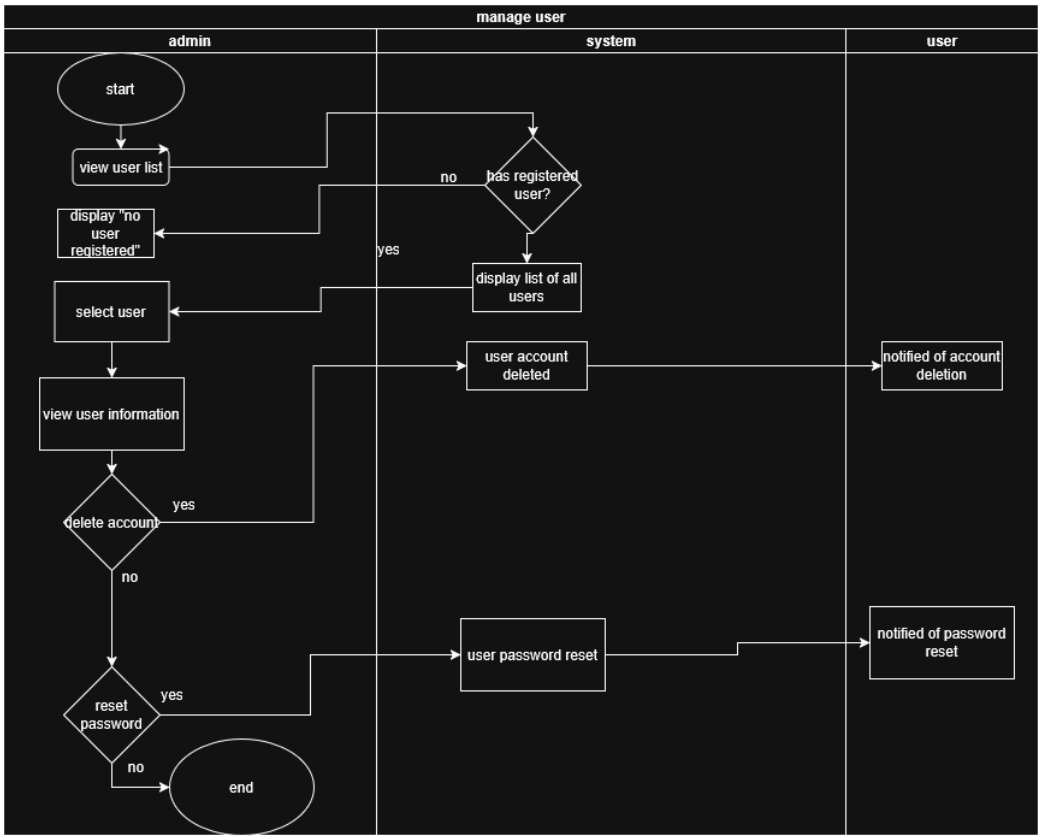


Figure 22 Manage User(Admin) Flowchart

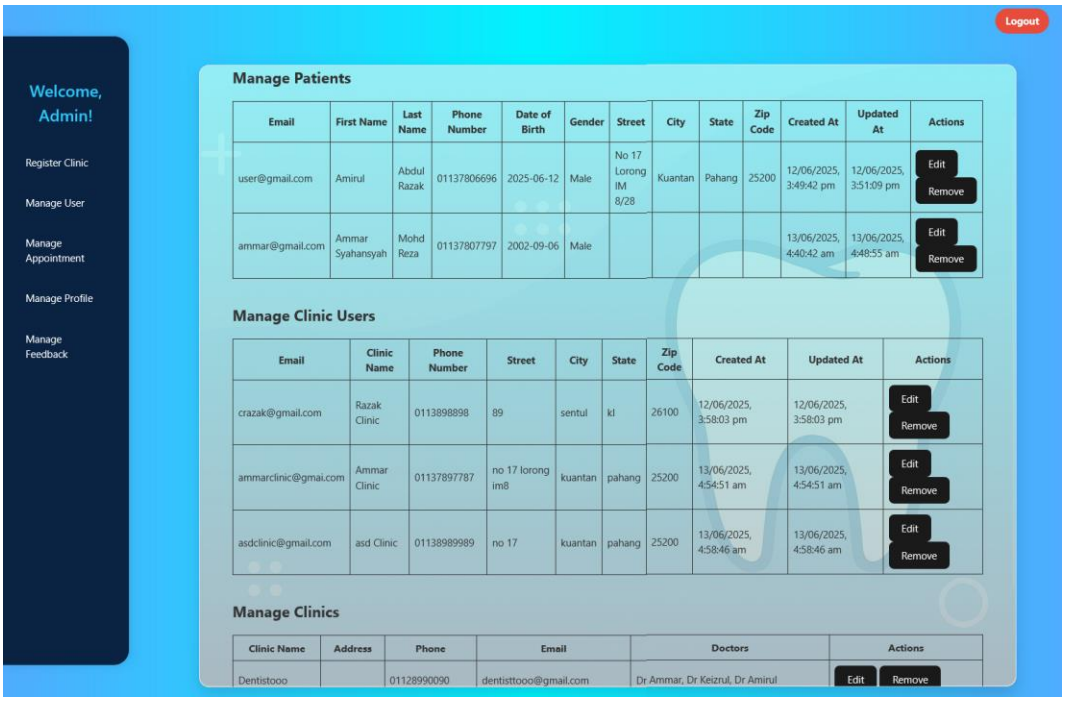


Figure 23 Admin View User Screen

Field	Details
Requirement ID	REQ_F401
Version	1.0
Item	View Registered Users
Description	Admin shall be able to view all registered users.
Purpose	To monitor and manage user accounts.
Format	• Admin ID – Int • User Table – List of user entries
Valid Range	Admin access required
Related I/O	Admin dashboard
Author	Ammar Syahansyah Bin Mohd Reza

Table 45 Requirement – Admin View User

Field	Details
Requirement ID	REQ_F402
Version	1.0
Item	Search User
Description	Admin shall be able to search for users using keywords.
Purpose	To locate and manage specific users.
Format	• Search Keyword – String • Search Result – Table
Valid Range	N/A
Related I/O	Admin panel search bar
Author	Ammar Syahansyah Bin Mohd Reza

Table 46 Requirement – Search User



Figure 24 Edit User Information

Field	Details
Requirement ID	REQ_F403
Version	1.0
Item	Edit User Information
Description	Admin shall be able to edit user profile data.
Purpose	To correct or update user details.
Format	
Valid Range	Valid formats required
Related I/O	Edit form; validation
Author	Ammar Syahansyah Bin Mohd Reza

Table 47 Requirement – Edit User Information



Figure 25 Delete User Account Screen

Field	Details
Requirement ID	REQ_F404
Version	1.0
Item	Delete User Account

Description	Admin shall be able to delete a user from the system.
Purpose	To remove invalid or inactive accounts.
Format	• User ID – Int
Valid Range	Valid existing user
Related I/O	Confirmation prompt
Author	Ammar Syahansyah Bin Mohd Reza

Table 48 Requirement – Delete User Account

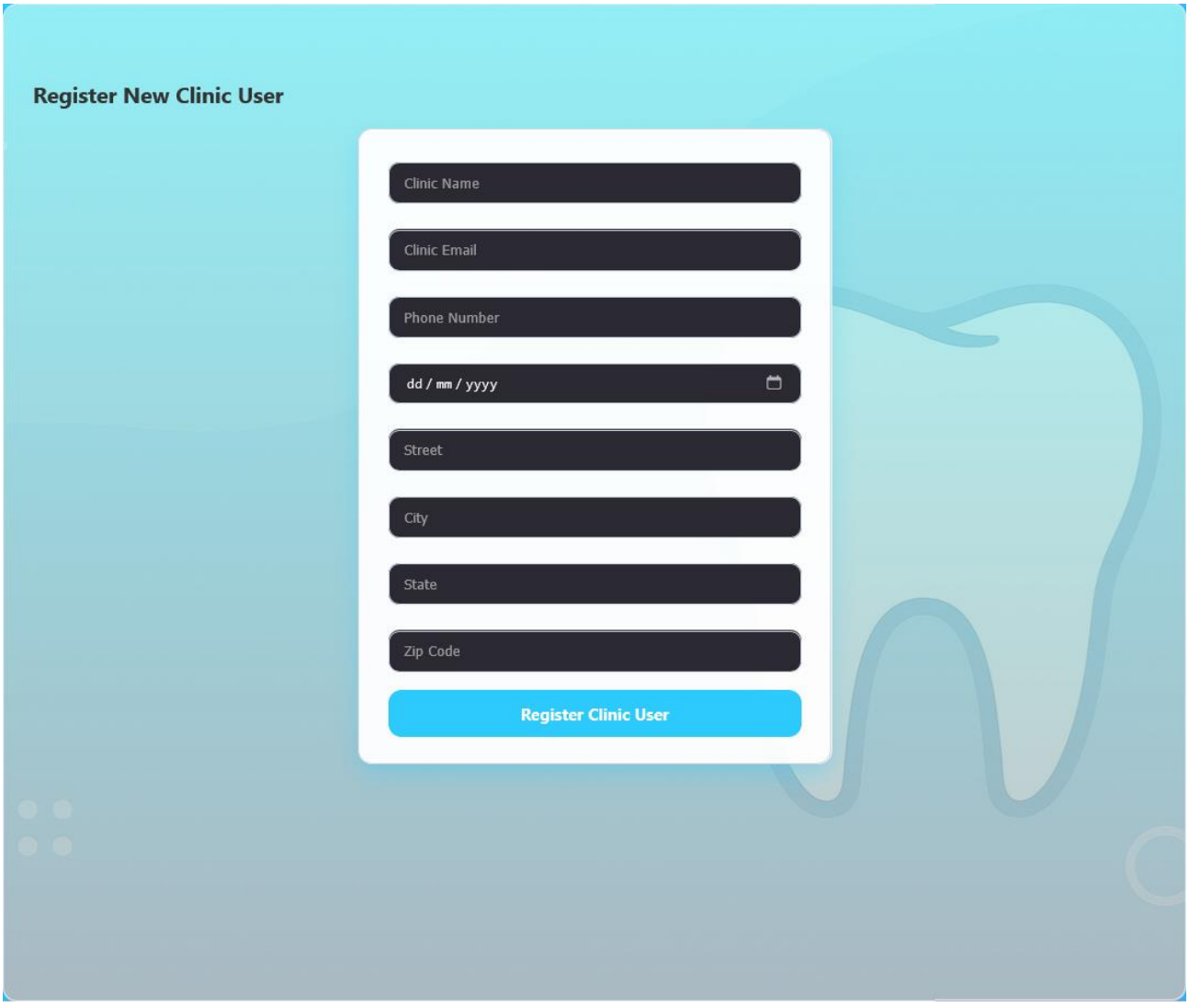


Figure 26 Register New Clinic Screen

Field	Details

Requirement ID	REQ_F405
Version	1.0
Item	Register Clinic
Description	Admin shall be able to register a new clinic.
Purpose	To onboard new clinics to the platform.
Format	• Clinic Name – String • Email – String • Address – String
Valid Range	Unique email and valid fields
Related I/O	Clinic registration form
Author	Ammar Syahansyah Bin Mohd Reza

Table 49 Requirement – Register New Clinic

4.1.5 User Profile Management

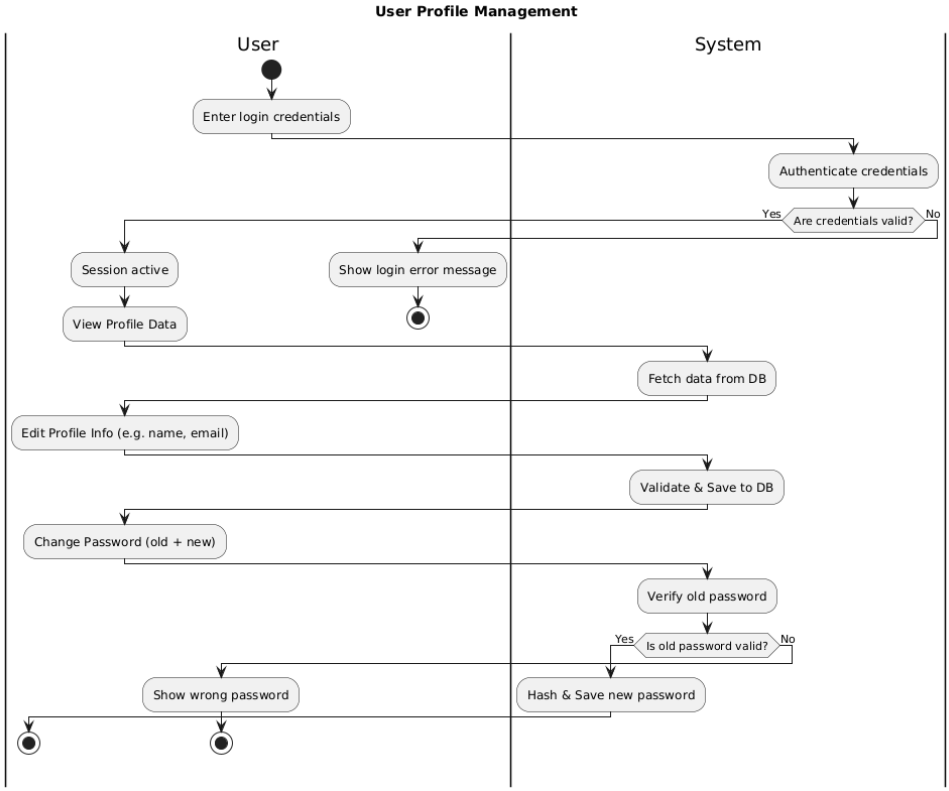


Figure 27 User Profile Management Flowchart

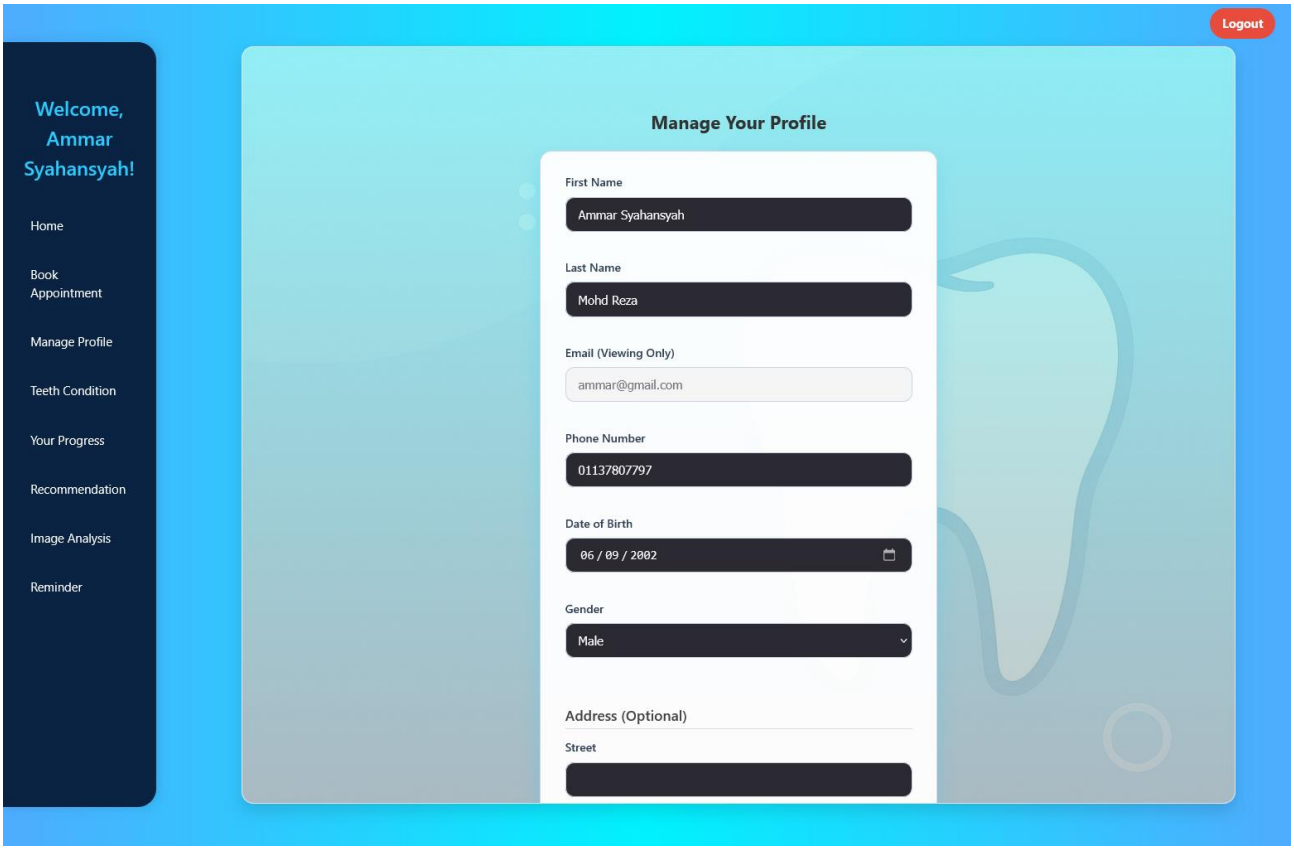


Figure 28 Edit Profile Screen

Field	**Details**
Requirement ID	REQ_F501
Version	1.0
Item	User Profile Management
Description	User shall be able to view and edit their profile information and change passwords.
Purpose	To allow users to maintain their personal information and account security.
Format	• • User ID – Int • Name – String • Email – String • Phone – String
Valid Range	Valid email format, Phone number format, Password min 8 characters

Related I/O	Database validation, Email verification
Author	Ammar Syahansyah Bin Mohd Reza

Table 50 Requirement – User Profile Management

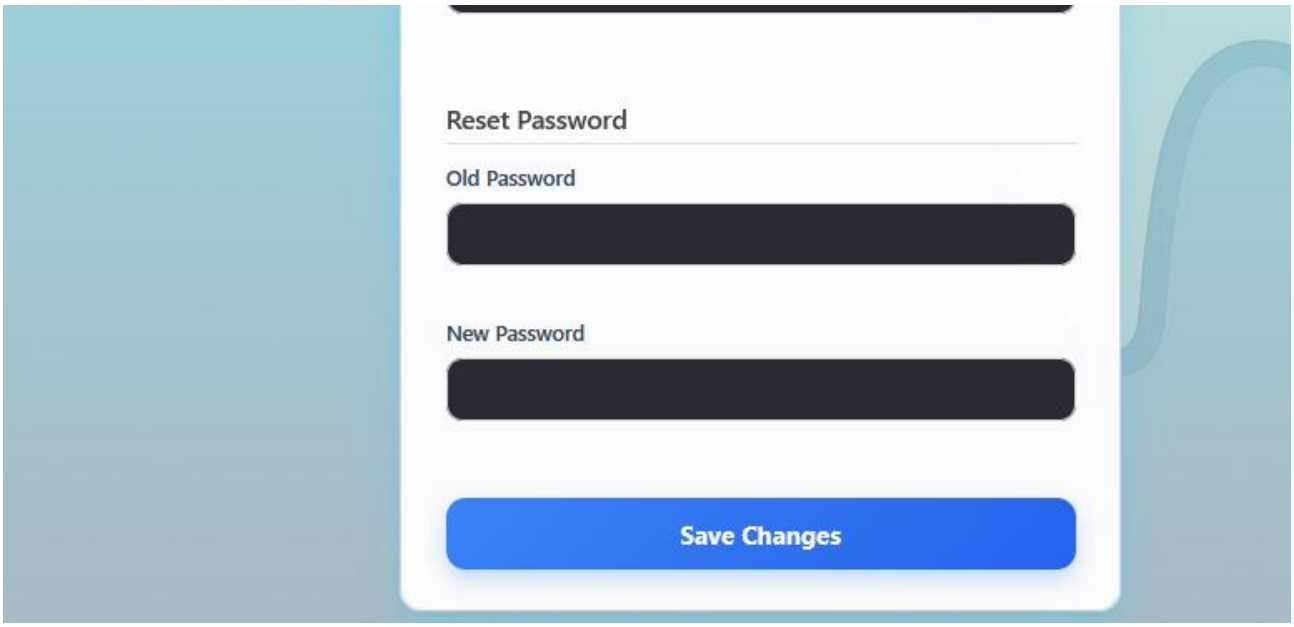


Figure 29 Change Password Screen

Field	Details
Requirement ID	REQ_F502
Version	1.0
Item	Change Password
Description	User shall be able to securely change their password.
Purpose	To maintain account security and allow credential updates.
Format	• Old Password – String • New Password – String
Valid Range	Password must be at least 8 characters
Related I/O	Change password form
Author	Ammar Syahansyah Bin Mohd Reza

Table 51 Requirement – Change Password

Field	Details

Requirement ID	REQ_F503
Version	1.0
Item	Upload Profile Picture
Description	User shall be able to upload or change their profile picture.
Purpose	To personalize user profile.
Format	• Image File – Binary • Supported: JPG, PNG
Valid Range	Image size < 5MB
Related I/O	Profile image upload interface
Author	Ammar Syahansyah Bin Mohd Reza

Table 52 Requirement – Upload Profile Picture

4.1.6 Admin Feedback Manager

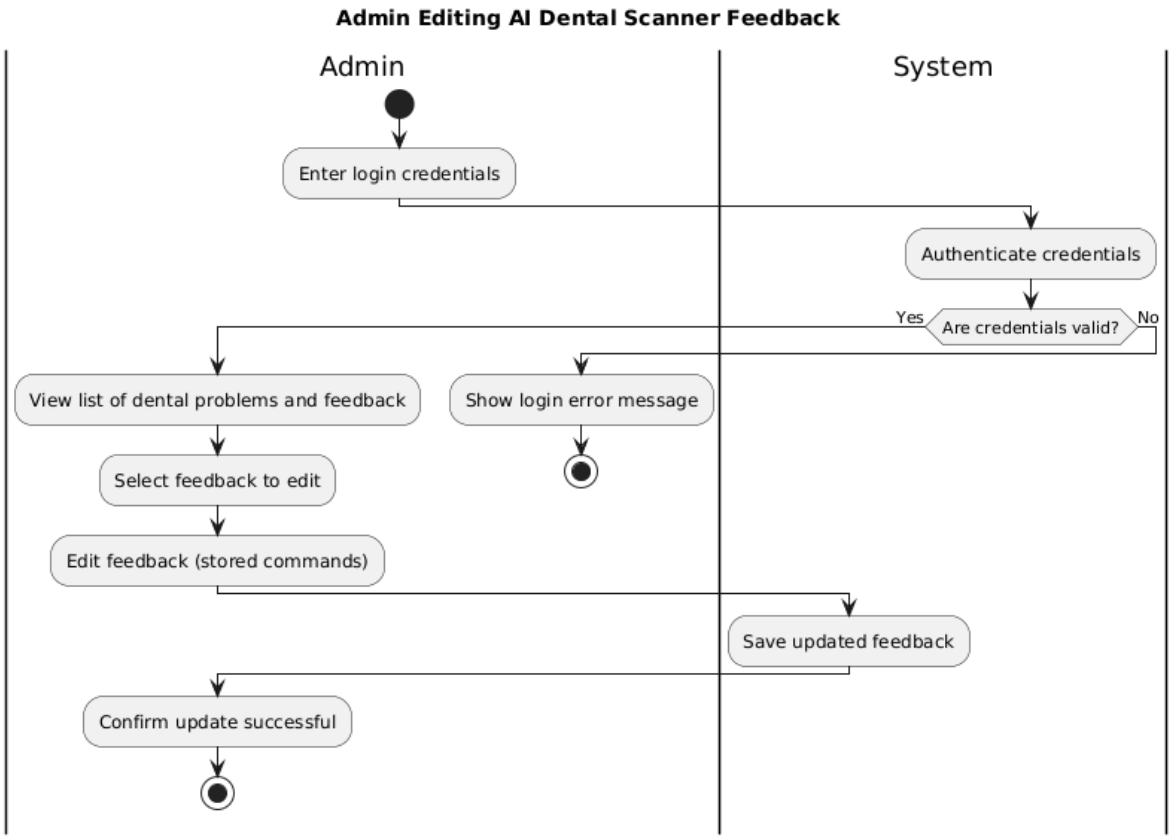


Figure 30 Admin Feedback Manager Flowchart

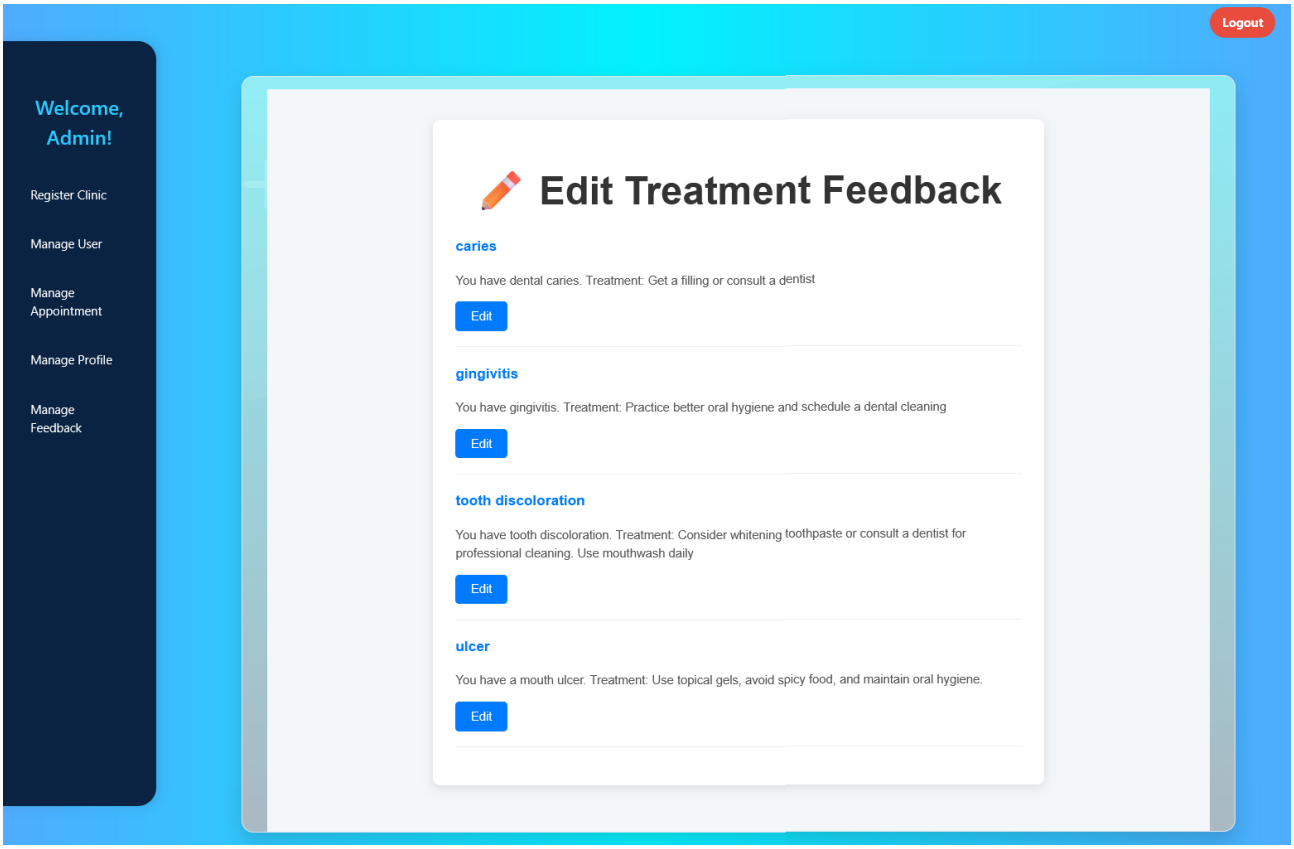


Figure 31 Manage Feedback Screen

Field	**Details**
Requirement ID	REQ_601
Version	1.0
Item	Admin feedback manager
Description	Admin shall be able to edit AI-generated dental scanner feedback and stored commands in the system.
Purpose	To allow administrators to modify and improve AI feedback for better accuracy and relevance.
Format	• Admin ID – Int • Feedback ID – Int • Command Text – String • Updated Feedback - String

Valid Range	Admin ID > 0, Feedback ID > 0
Related I/O	Database update operations
Author	Ammar Syahansyah Bin Mohd Reza

Table 53 Requirement – Admin Feedback Manager

Field	**Details**
Requirement ID	REQ_F602
Version	1.0
Item	Add Feedback
Description	Clinic Admin shall be able to create new feedback records.
Purpose	To associate dental conditions with expert recommendations.
Format	• Feedback Text – String • Condition Label – String
Valid Range	N/A
Related I/O	Feedback management module
Author	Ammar Syahansyah Bin Mohd Reza

Table 54 Requirement – Add Feedback

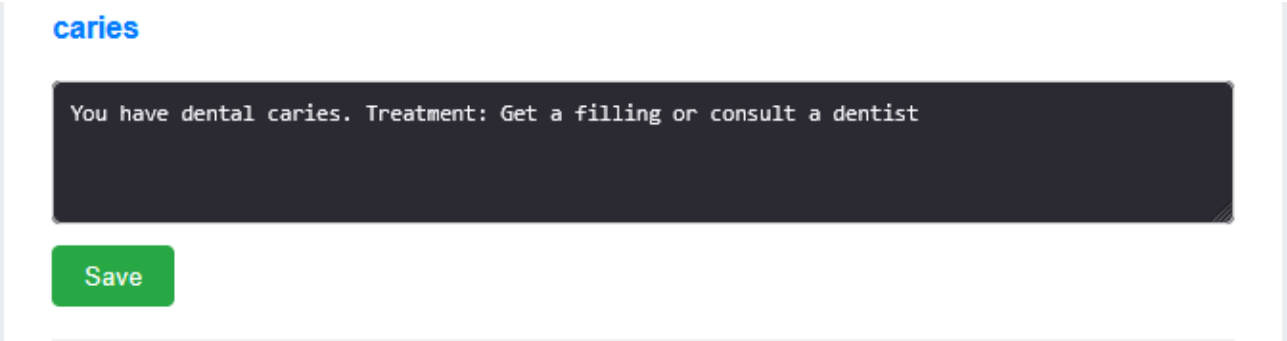


Figure 32 Edit Feedback Screen

Field	**Details**
Requirement ID	REQ_F603
Version	1.0
Item	Edit Feedback

Description	Clinic Admin shall be able to update feedback entries.
Purpose	To ensure accuracy of condition-specific guidance.
Format	• Feedback ID – Int• Updated Feedback – String
Valid Range	Valid input only
Related I/O	Feedback editor
Author	Ammar Syahansyah Bin Mohd Reza

Table 55 Requirement – Edit Feedback

Field	**Details**
Requirement ID	REQ_F604
Version	1.0
Item	Delete Feedback
Description	Clinic Admin shall be able to remove unnecessary feedback.
Purpose	To manage and clean outdated recommendations.
Format	• Feedback ID – Int
Valid Range	Feedback must exist
Related I/O	Delete confirmation module
Author	Ammar Syahansyah Bin Mohd Reza

Table 56 Requirement – Edit Feedback

4.1.7 Appointment Approval

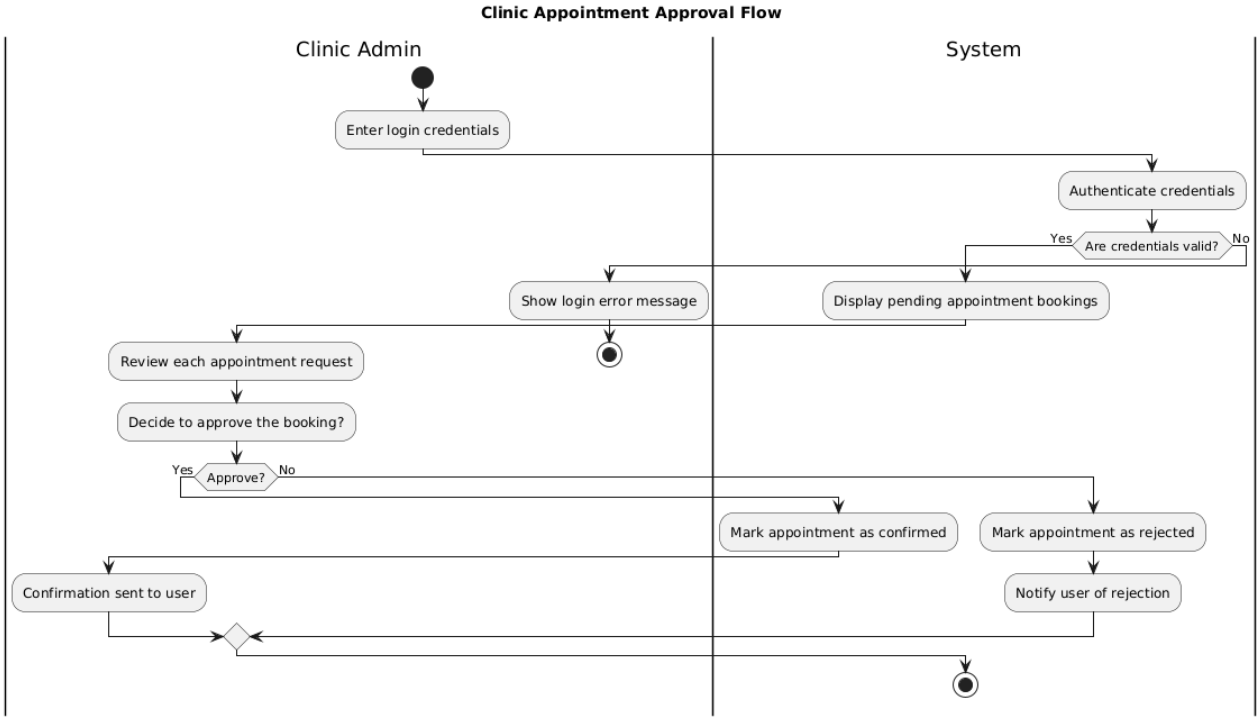


Figure 33 Clinic Appointment Approval Flowchart

Field	**Details**
Requirement ID	REQ_F701
Version	1.0
Item	Clinic Admin Appointment Approval
Description	Clinic admin shall be able to review and approve/reject pending appointment bookings.
Purpose	To provide administrative control over appointment scheduling and resource management.
Format	• Admin ID – Int • Appointment ID – Int • Approval Status – Boolean • Rejection Reason - String
Valid Range	Valid Admin credentials required

Related I/O	User notification system
Author	Ammar Syahansyah Bin Mohd Reza

Table 57 Requirement – Clinic Appointment Approval

4.1.8 Clinic Schedule Management

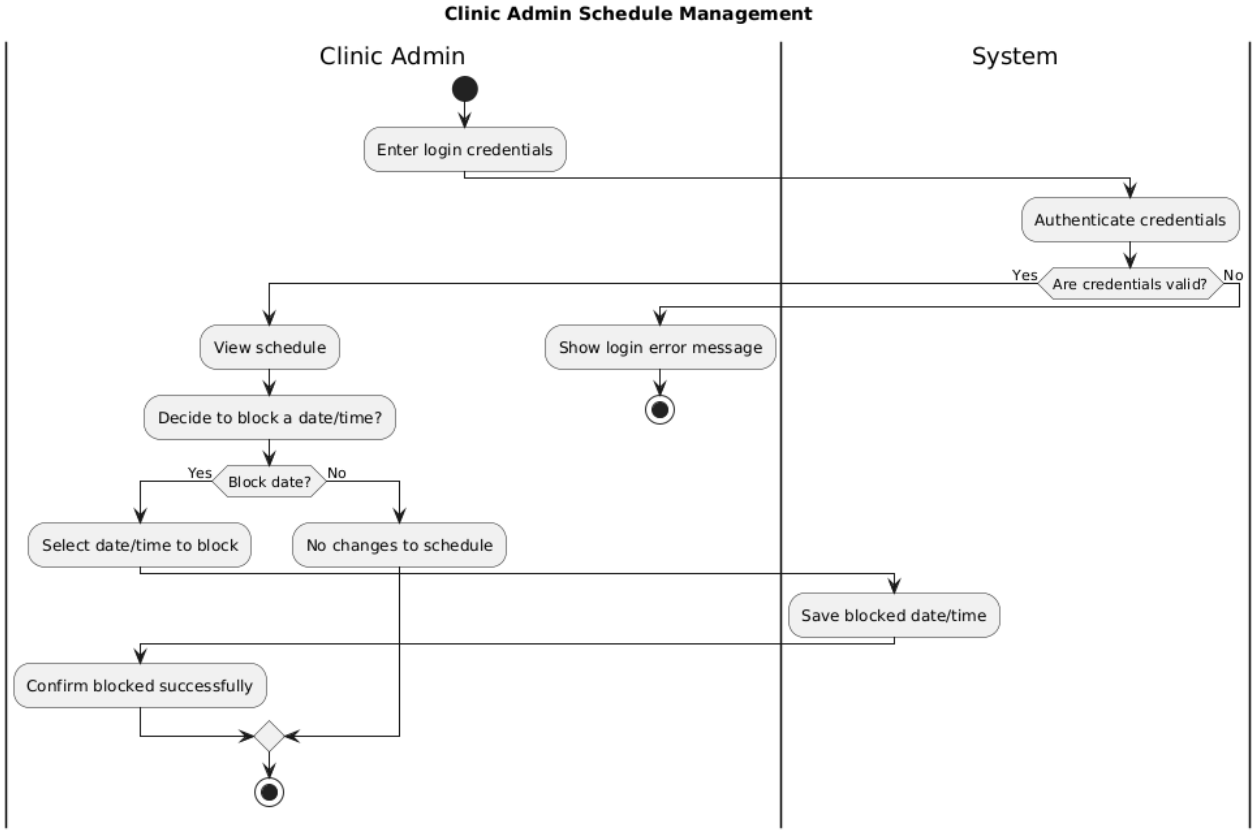


Figure 34 Clinic Schedule Management Flowchart

Field	Details
Requirement ID	REQ_F801
Version	1.0
Item	Schedule Management
Description	Clinic admin shall be able to view schedules and block specific dates/times from booking.
Purpose	To manage clinic availability and prevent bookings during unavailable periods.

Format	• Admin ID – Int • Block Date – Date • Block Time – Time • Block Reason - String
Valid Range	Future dates only
Related I/O	Calendar system
Author	Ammar Syahansyah Bin Mohd Reza

Table 58 Requirement – Clinic Schedule Management

4.1.9 Patient Viewing Ai Dental Feedback

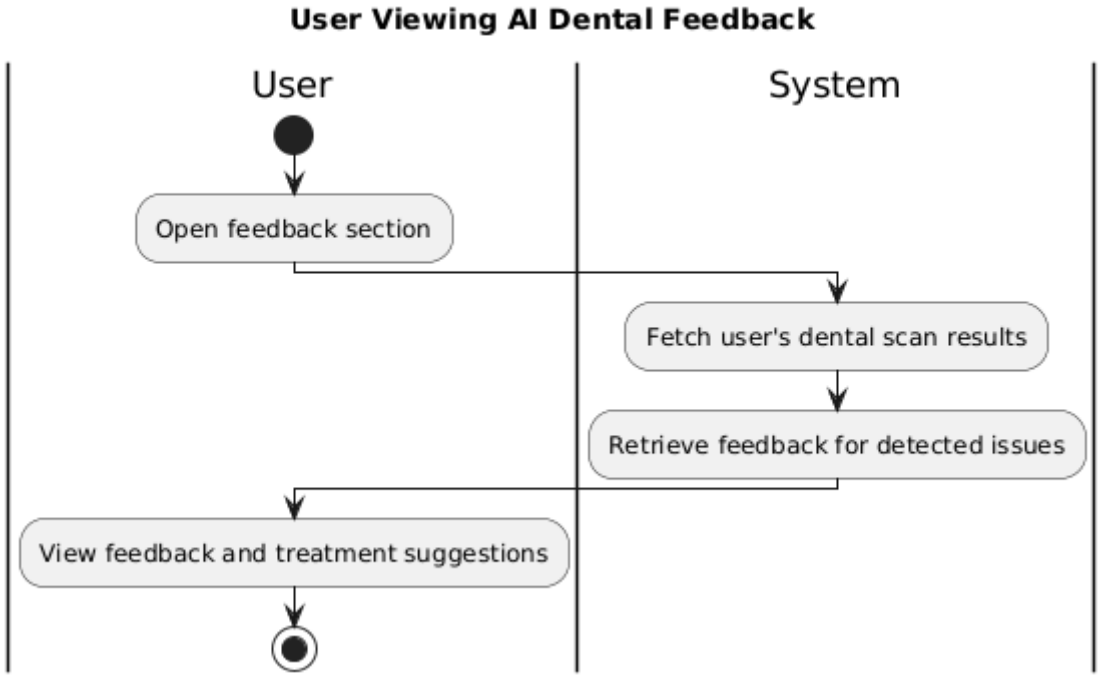


Figure 35 Patient Viewing Ai Dental Feedback Flowchart

Field	Details
Requirement ID	REQ_F901
Version	1.0
Item	View AI Dental Feedback

Description	User shall be able to view AI-generated dental feedback and treatment suggestions for their scans.
Purpose	To provide patients with accessible dental analysis and recommendations.
Format	• • User ID – Int • Scan ID – Int • Feedback Text – String • Treatment Suggestions
Valid Range	Valid user session required
Related I/O	Image processing system
Author	Ammar Syahansyah Bin Mohd Reza

Table 59 Requirement – Patient Viewing Ai Dental Feedback

4.1.10 View Progress

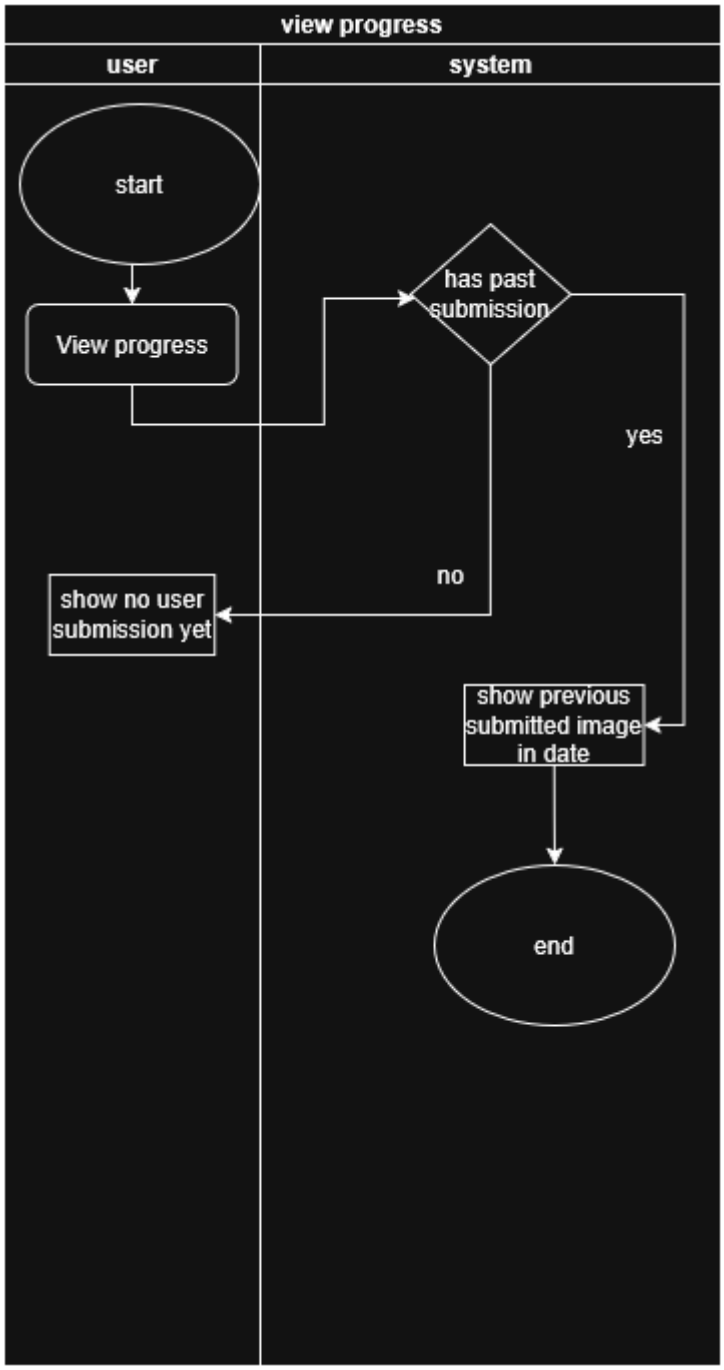


Figure 36 View Progress Flowchart

Field	Details
Requirement ID	REQ_F901
Version	1.0

Item	View Analysis History
Description	User shall be able to view their past image analysis records.
Purpose	To help users track changes in their dental condition over time.
Format	• User ID – Int • Scan Records – List of entries (Date, Condition)
Valid Range	User must have at least one scan record
Related I/O	History viewer interface; database read
Author	Amir Keizrul Bin Mudzir

Table 60 Requirement – View Analysis History

Field	**Details**
Requirement ID	REQ_F1002
Version	1.0
Item	View Detailed Result
Description	User shall be able to see full AI result and feedback for each scan.
Purpose	To allow deeper understanding of each diagnosis.
Format	• Scan ID – Int • Full Result – JSON • Feedback – String
Valid Range	Valid scan entry required
Related I/O	Detailed result viewer
Author	Amir Keizrul Bin Mudzir

Table 61 Requirement – View Detailed Result

Field	**Details**
Requirement ID	REQ_F903
Version	1.0
Item	Download Progress Report
Description	User shall be able to export past scan data into a PDF report.

Purpose	To keep a record or share progress with dental professionals.
Format	• User ID – Int • Report – PDF File
Valid Range	User must have history to export
Related I/O	PDF generation tool; Download prompt
Author	Amir Keizrul Bin Mudzir

Table 62 Requirement – Download Progress Report

4.2 Hardware Interfaces

The system operates as a web-based solution and does not require specific hardware integration. It depends on:

Client Devices: Personal computers, smartphones, or tablets with camera access for image capture.
Server Infrastructure: Cloud-hosted servers capable of handling AI image inference, database operations, and API requests.
External Calendar Sync (optional): Integration with device calendars for appointment reminders.

No embedded devices or physical hardware modules are needed to run or interact with the system

4.3 Software Interfaces

Category	Technologies / Tools
Programming Language	• JavaScript • Python
Framework	• React.js • Flask
Development Tools	• Visual Studio Code • GitHub • Google Chrome
Database	• Firebase Realtime Database • Firebase Firestore
Authentication	• Firebase Authentication

Table 63 Software Requirements

Name	Description
JavaScript	• A versatile client-side scripting language used to create dynamic and interactive web content.
Python	• A high-level, general-purpose programming language used for backend logic, APIs, and AI integration.

Table 64 Programming Language

Name	Description
React.js	• A JavaScript library developed by Facebook for building fast, dynamic UIs using reusable components.
Flask	• A lightweight Python web framework for building RESTful APIs and backend services.

Table 65 Framework Description

Name	Description
Visual Studio Code	• A powerful code editor with built-in debugging, Git support, and extensions.
GitHub	• A web-based platform for version control and collaborative software development.
Postman	• A tool used for testing APIs by sending HTTP requests and inspecting responses.
Google Chrome	• A modern web browser with powerful developer tools used for testing and debugging web applications.

Table 66 Development Tools Description

Name	Description
Firebase Realtime Database	• A cloud-hosted NoSQL database that stores data in JSON format and syncs in real-time across clients.
Firebase Firestore	• A flexible, scalable NoSQL cloud database for storing and syncing data in real time and offline mode.

Table 67 Database Description

Name	Description
Firebase Authentication	• Provides backend services to authenticate users using email/password and tokens.

Table 68 Authentication Description

Name	Description
YOLOv5 API	• An object detection model served via Flask API to analyze dental images and return predictions (bounding boxes, labels, confidence).

Table 68 AI Model Description

4.4 Communications Interfaces

- The application uses secure protocols to ensure seamless and protected communication between components:
- HTTPS: All communication between the frontend, backend, and Firebase services is encrypted using SSL.
- JSON Format: RESTful APIs use JSON for request and response payloads.
- Email Services: Firebase triggers or SMTP integration for sending password reset links, appointment confirmations, and admin alert

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Performance Requirement ID	Description	Priority	Author
PFR001	The AI-powered dental web app should load dashboard pages within 3 seconds under normal internet conditions.	High Priority	Ammar Syahansyah Bin Mohd Reza
PFR002	AI analysis results for uploaded images should be returned within 2.5 seconds.	High Priority	Ammar Syahansyah Bin Mohd Reza
PFR003	The system should support at least 100 concurrent users without degradation in performance.	High Priority	Ammar Syahansyah Bin Mohd Reza
PFR004	Form submissions (e.g., registration, feedback, booking) must complete within 5 seconds.	High Priority	Ammar Syahansyah Bin Mohd Reza
PFR005	Appointment booking changes should reflect across user views within 3 seconds.	High Priority	Ammar Syahansyah Bin Mohd Reza
PFR006	The system must handle batch uploads (e.g., dental image sets) without freezing or timeout.	High Priority	Ammar Syahansyah Bin Mohd Reza
PFR007	The application must be accessible on all modern web browsers (Chrome, Firefox, Safari, Edge).	High Priority	Ammar Syahansyah Bin Mohd Reza
PFR008	The system should operate 24/7 with a monthly uptime of at least 99.5%.	High Priority	Ammar Syahansyah Bin Mohd Reza

Table 69 Performance Requirements

5.2 Safety Requirements

Safety Requirement ID	Description	Priority	Author
SFR001	Unauthorized access to user health data must be prevented using role-based authentication.	High Priority	Ammar Syahansyah Bin Mohd Reza
SFR002	All communications must be encrypted using HTTPS to protect data in transit.	High Priority	Ammar Syahansyah Bin Mohd Reza
SFR003	Users can only view or modify data based on their assigned roles and permissions.	High Priority	Ammar Syahansyah Bin Mohd Reza
SFR004	Regular backups of the Firebase database must be scheduled to prevent data loss.	High Priority	Ammar Syahansyah Bin Mohd Reza
SFR005	The image upload module must filter and reject unsafe or unsupported file types (e.g., EXE, ZIP).	High Priority	Ammar Syahansyah Bin Mohd Reza

Table 69 Safety Requirements

5.3 Security Requirement

Security Requirement ID	Description	Priority	Author
SCR001	All users must authenticate using a valid email and password to access the platform.	High Priority	Ammar Syahansyah Bin Mohd Reza

SCR002	Passwords must be securely hashed and stored using Firebase Authentication's best practices.	High Priority	Ammar Syahansyah Bin Mohd Reza
SCR003	Role-based access control (patient, clinic admin, system admin) must be enforced throughout the application.	High Priority	Ammar Syahansyah Bin Mohd Reza
SCR004	All API endpoints must be served over HTTPS to prevent interception or tampering.	High Priority	Ammar Syahansyah Bin Mohd Reza
SCR005	Personal and diagnostic data must be inaccessible to unauthorized users.	High Priority	Ammar Syahansyah Bin Mohd Reza
SCR006	The system must follow Malaysia's PDPA guidelines for user data handling and consent.	High Priority	Ammar Syahansyah Bin Mohd Reza
SCR007	URL manipulation to bypass login or access restricted content must be prevented.	High Priority	Ammar Syahansyah Bin Mohd Reza

Table 69 Security Requirements

5.4 Software Quality Attribute

- Usability
  - The system must be intuitive for both patients and administrators. New users should be able to perform key functions (upload image, book appointment) within 10 minutes of use. Interfaces include tooltips, icons, and visual feedback. The sidebar allows seamless role-based navigation.
- Availability
  - The system should be accessible 24/7, with brief scheduled maintenance communicated in advance. Uptime target is 99.5% per month.
- Reliability
  - The system should process requests correctly and consistently. It should recover gracefully from temporary disruptions and perform as expected under peak usage.
- Maintainability
  - Modular code design enables future enhancements, bug fixes, and AI model updates with minimal disruption. Code should follow clean and reusable practices.
- Compatibility
  - The platform must work on major web browsers and adapt responsively to mobile, tablet, and desktop devices.
- Security & Robustness
  - The system must resist common threats (e.g., SQL injection, XSS). Invalid or unexpected inputs must be gracefully handled without crashes.

6. Other Requirements

6.1 Legal and Institutional Compliance

The system must comply with the Personal Data Protection Act (PDPA) of Malaysia. A consent agreement (Terms & Privacy Policy) must be acknowledged by users during account registration.

6.2 Reuse Objectives of Project

The system is built with modularity in mind. Common modules such as authentication, image analysis, and appointment management can be reused in future dental or health diagnostic platforms.

6.3 Mobile Responsiveness

The platform must be mobile-responsive, ensuring that all core features (e.g., booking, results viewing, image upload) are accessible via smartphone or tablet with no loss of functionality.

6.4 Requirement Traceability

Requirement ID	Functional Requirement ID	Use Case ID	Feature Name	Description	Priority	Author
REQ001	FR_004	UC-004	Upload Image	Patient uploads intraoral image for analysis	High Priority	Ammar Syahansyah Bin Mohd Reza
REQ002	FR_005, FR_006, FR_007	UC-005, UC-006, UC-007	Get AI Result	System analyzes image and displays diagnostic result	High Priority	Ammar Syahansyah Bin Mohd Reza
REQ003	FR_009	UC-009	Book Appointment	Patient books appointment with clinic	High Priority	Ammar Syahansyah Bin Mohd Reza
REQ004	FR_011, FR_012	UC-011, UC-012, UC-013	Manage Appointment	Clinic Admin views and updates status of booking	High Priority	Ammar Syahansyah Bin Mohd Reza
REQ005	FR_012	UC-013	Manage Clinic Info	Clinic Admin edits clinic name, location, and schedule	Medium Priority	Ammar Syahansyah Bin Mohd Reza
REQ006	FR_015–FR_019	UC-014–	Manage User Roles	Admin manages user access and roles	Medium Priority	Ammar Syahansyah

		UC-019				Bin Mohd Reza
REQ007	FR_028	UC-028	Export AI Result	User exports result in PDF format	Medium Priority	Ammar Syahansyah Bin Mohd Reza

Table 70 Requirement Traceability

Appendix A: Glossary

Term	Definition
AI	Artificial Intelligence. Used to analyze dental images to detect oral conditions.
Appointment Booking	The module allowing users to schedule dental visits with clinics via the platform.
Clinic Admin	A user role responsible for managing clinic schedules, dentists, and approving appointments.
Feedback	Result or recommendation generated after AI dental image analysis.
Firebase	Backend platform used for database storage, authentication, and real-time operations.
Intraoral Image	An image of the inside of a patient's mouth used for dental analysis.
Patient	End-user who uploads dental images and books appointments.
System Admin	The administrator with the highest privilege, managing users, clinics, and platform settings.
YOLOv5	An object detection model used for analyzing dental images.
Use Case	A documented scenario describing how a user interacts with the system to achieve a goal.
SRS	Software Requirements Specification — the document outlining system functionalities and needs.
Role-Based Access Control	A security mechanism ensuring users access only data and features allowed for their roles.

Table 71 Glossary

Appendix B: Analysis Models

Use Case Diagram

System Architecture Diagram

UML Class Diagram

Appendix C: Issues List

Issue ID	Description	Status	Resolution Plan
ISS-001	Final confirmation needed on multilingual content (BM-English).	Pending	Translate UI elements and create language toggle in settings.

ISS-002	AI model currently only detects 4 dental conditions.	In Progress	Future model expansion to include root canal, plaque, etc.
ISS-003	Consent management for dataset reuse not fully implemented.	Pending	Add checkbox in upload page for research consent.
ISS-004	Admin panel lacks role audit logging.	In Progress	Implement action logging feature in backend.
ISS-005	PDF export of results is not yet integrated.	Pending	Plan to use jsPDF or React-to-PDF in next sprint.

Table 72 Issue List

Software Test Plan Document

AI POWERED DENTAL ASSESSMENT

Name: Amir Keizrul Bin Mudzir

Student ID:52213223121

Name: ,Ammar Syahansyah Bin MohdReza

Student ID: 52213223239

INFORMATION REGARDING DOCUMENT AND ITS APPROVAL

Version number	Date	Amend by	Role in Project	Amendmend reason
1.0	**8/5/2025**	**Amir Keizrul Bin Mudzir** **Ammar Syahansyah Bin Mohd Reza**		**First draft stp**
2.0	**1/6/2025**	**Amir Keizrul Bin Mudzir** **Ammar Syahansyah Bin Mohd Reza**		**Final software test plan**

I. Software Test Plan Document Purpose:

The purpose of this document is to ensure that the software developed will meet client satisfaction and expectations as per requirements defined clearly. To validate that the system has robustness, efficiency, and adaptability, this document intends to show that all the requirements of the clients are duly met. Besides, it ensures that the plan is practical enough to meet the standards and regulations to comply with. At the final level, the Software Testing Plan (STP) would also act as a framework and planning guide in executing and applying the tests.

II. Software Test Plan Contents:

A test plan provides testing execution strategy during the testing session. It gives an overall view of almost all the aspects, like objectives, scope, risks, strategy, approach/technique, participating teams, test objects, resources, entry and exit criteria, test prioritization, and tool support. Moreover, it also describes the modules as well as their functionalities.

III. Software Test Plan Document in term of topics to be covered:

This document intends to bring together several vital elements, including the testing strategy, scope,risks and contingencies, entry and exit criteria, and item pass or fail criteria, among others. Further, it will offer a description of the Test Specification, which would simplify and portray the existing test cases being developed. The Test Environment will show all the necessary facilities, hardware, the software necessary, and other requirements to ensure seamless execution during the tests. Finally, this will include the Test Schedule, Staff Responsibilities, and the Training needs focusing on the roles of those personnel in the overall schedule.

All items in this document are designed to align with and address the planning and design details outlined in the Software Requirement Specification (SRS), ensuring the project meets its defined objectives.

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Test policy

This section explains major philosophy of the organization of (organization) regarding testing and its terms, meaning and practices. However, most of these policy aspects are widely applied in any other organization even UniKL MIIT may have this in their test policy since this is based on ISTQB (worldwide standard).

1.1 Software Testing Term Definition

Testing is a development activity that is all about process of finding and identifying defects by performing test and checking the actual result match the expected results or not. It also known as process of verification and validation which is based on ISO-9000 standard as shown below:

- Validation - Confirmation by examination and through the provision of objective evidence that specified requirements have been fulfilled. To summarize ISO statement, validation confirms that product build with correct requirement that satisfy the customer requirement.
- Verification - Confirmation by examination and through provision of objective evidence that the requirements for a specific intended use or application have been fulfilled. To summarize the ISO statement, verification confirms that software has been created and fulfilled.

Testing also defines what will happen once a defect occurs (failure). Testing is performed by the testing team which then the report/result will be forwarded to the developer team so that they can start debugging.

1.2 Advantages of Testing

There are several significant advantages of testing that can produce magnificent software application with supreme quality. These are numerous advantage that can be extract from testing:

- Testing increases quality in terms of the software attribute indirectly in which defect can be detected and corrected earlier prior to delivery. In addition to that, testing did increase process quality as well since defects are documented which means it can be analyzed, processed and solution to that error can be reused again to prevent same error again.

- Testing boosts confidence of all stakeholders which affects perception of team management regarding the software quality as well its robustness due to very few or no failure are found.

- Testing is like liability insurance where it costs big money, but it ensures all the stakeholders to be at ease as it cover and mitigate all the possible risks.

- Testing that is successfully conducted leads to major reductions in cost as defects can cause huge costs.

1.3. Testing Principles

There are seven testing principles that this organization applies which are based on ISTQB standard.

Principle 1: Testing shows presence of defects. Testing can determine and reveal defects are present but unable to verify and identify that there are no defects at all are presents. According to Edger W. Dijkstra, 1970 “Program testing can be used to show presence of bugs, but never to show their absence!”

Principle 2: Exhaustive testing is impossible. Full complete/exhaustive testing is truly not feasible except for trivial situations. Risk-based testing will be used to produce risk analysis to determine priorities of features module that need to be carried out.

Principle 3: Early testing Testing is not a phase that happens at the end of software development, it should start as early as possible. Early testing (like static testing; review) takes place parallel to development activities, defects are detected earlier thus cost less to fix.

Principle 4: Defect clustering Testing effort shall be focused proportionally to the expected and observed defect density of modules. A small number of modules often contain most of the defects discovered during pre-release testing/or responsible for most of operational failures.

Principle 5: Repetitions have no effects – Pesticide Paradox Repeating test brings no new insight thus test cases need to be regularly reviewed, revised, and modified. So that we won't execute the same test and test cases repeatedly, there is a need for changes.

Principle 6: Testing is context dependent. Testing depends on the context usage of that software. For example, safety-critical software systems are TESTED DIFFERENTLY (more intensive in technique) from commercial applications.

Principle 7: Absence of errors fallacy System without failures does not imply system will meet user expectations. Sometimes there are no errors and failures, but the functionalities given are not what the user wants and expected.

1.4 Test Missions & Objectives

Delivering the software product at its finest quality is this organization's goal where the functional and quality attributes are satisfactory, defect presence is almost minimal, testing activities are conducted systematically.

1.5 Communication Among Stakeholders

It is important for all related stakeholders to participate in the testing process and give insight/feedback regarding the software product. Open minded, professional, analytical and straight forward attribute is a must for the testing to perform smoothly according to what being planned.

1.6 SDLC Identification

The Rapid Application Development (RAD) model, as the methodology in the Systems Development Life Cycle (SDLC), is adopted by Perfect Wave Academy (PWA) in developing Swimming Class Management System.

Rapid Application Development chosen mainly because of the nature of the project in terms of quick prototyping and making it iterative through all features and modules like Student Module, Coach Module, Notification Module, payment plan, self-learning module. RAD is able to change its demands as new requirements emerge so that a project may progress even when all major details are not entirely known at the beginning.

RAD is supposed to be faster in development and will give early feedback from stakeholders. The only thing is that this approach fits perfectly into the schemes of Perfect Wave Academy because the system contains modules that have a lot of user interaction such as student session booking and shopping. This is achieved through early involvement of the users in the testing of the modules.

In addition, RAD's empirical approach to development and testing offers early integration of activities like preparation, planning, execution, and so on in testing into the very first development stages. Rapid prototyping and allowing user evaluation during each iteration will increase opportunities for developer-stakeholder collaboration with relevant insights for system improvement.

Through the RAD, the Self-Learning Module or Payment Plan Module functionality may be prototyped, tested, and improved iteratively to identify possible risks, bugs, and usability issues that may arise. This

minimizes the likelihood of mistakes and permits the final systems be as robust and closely aligned to what the end-users really need.

Finally, the reason why RAD is great for Swimming Class Management System is because of its high interactivity, encourages end-user feedback, and provides for Constant change, which means delivery of the whole user-friendly system will be timely.

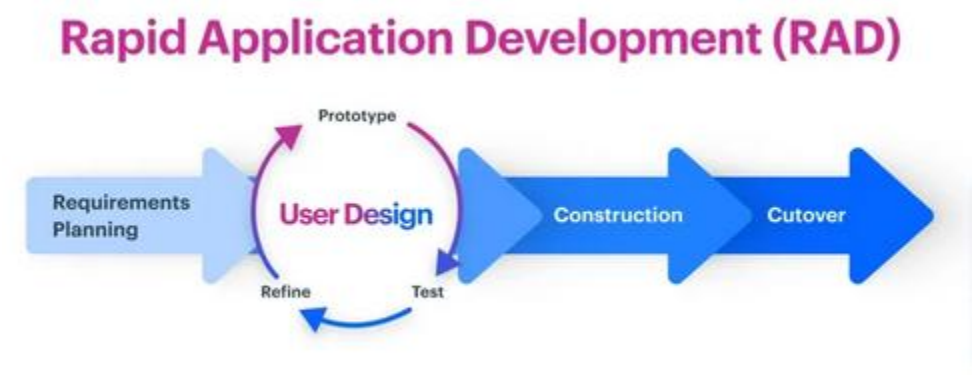


Figure 1Rapid Application Development (RAD) Model Diagram

Phases	Description
Requirement Planning	This phase focuses on gathering the high-level requirements of the project. Research is conducted to understand the system's objectives, scope, and resources required for development. For your Swimming Class Management System, this is where you identify the key system features such as student booking, payment plans, coach schedules, and reporting. In the context of testing, this phase also includes the creation of a test plan at a high level, outlining the strategies and scope of testing to ensure all functional requirements will be validated.
User Design (Prototype, Test, Refine)	This phase is focused on creating a prototype of the system that users can interact with. For the Swimming Class Management System, prototypes for modules like Student Session Booking or Payment Plan Module would be developed rapidly. The prototype helps visualize how the final system will function and allows for early user feedback.

	• Prototype: A basic version of the system is created, showcasing core functionalities like session booking and payment tracking. • Test: The prototype undergoes User Acceptance Testing (UAT), where the client (e.g., students or coaches) evaluates the functionality and provides feedback. Functional testing is also done to ensure the system works as per the Software Requirements Specification (SRS). • Refine: Based on feedback from testing, the prototype is refined. Any issues found during UAT or functional testing are addressed, and the system is improved. The looping process continues as the prototype is iteratively updated based on client feedback until it meets their expectations.
Construction	Once the prototype is validated and refined, the system goes into full development. In this phase, all the modules of the system—say, Coach Scheduling, Notification Module, Payment Plan—are coded and integrated to have the whole system up and running. The developers ensure that all the features and requirements are implemented as designed initially, while internal testing activities such as unit testing and integration testing are carried out to identify and fix technical issues. In this stage, the prototype is tested extensively using two most important techniques: Functional Testing and User Acceptance Testing (UAT). • Functional Testing: This testing is carried out to find bugs, errors, and defects in a system; that all functionalities are working properly according

	to the Software Test Plan and software Requirements Specification. Both documents serve as a reference for the execution of tests. • UAT: The system is exposed to the client in order to take note of their general opinion and comments. In case it doesn't satisfy the customer's requirements, the details will be noted down, analyzed, and the prototype updated with the findings. This may cause a new cycle of analysis, design, implementation, and testing as the system is developed based on the recommendations given by the client or due to new requirements. The cycles continue until the client is completely satisfied with the system. Since this is evolutionary prototyping, the improved prototype is not thrown away; it is incrementally developed into the final working system. The deliverables from this phase include a stable, functioning system with frequent evaluation and client feedback to ensure the final product is as expected and ready for deployment.
Cutover	The final phase involves the deployment of the system into production. In the case of the Swimming Class Management System, this means rolling out the final working version of the system to all users, such as students, coaches, and administrators. During this phase, further testing may be performed to ensure everything works in a live environment. After deployment, maintenance routines are set up to monitor system performance, address issues, and provide ongoing support. The system is now in active use, and the feedback gathered during this phase could lead to future updates or improvements.

Table 1RAD Phases Description

INTRODUCTION TO TEST PLAN

The document outlines an Acceptance Test Plan to test the developed AI powered dental assessment system. ATP will be used by the acceptance test team to plan, conduct and document the acceptance testing. In this context of acceptance level, black box testing is used where it is specification-based testing technique that refer SRS to conduct a functional test were the test case is derived through use case testing which is based on use case specification from SRS. Hence during the testing period, several references of documentation are used which is stated in the references part.

2.1 System Description

AI Powered Dental Assessment is a smart dental health app that helps users check their teeth using AI. By taking a photo of their mouth, users can get instant feedback on common dental problems like cavities, ulcer, gingivitis, and tooth discoloration. The app is easy to use and works on computers. It also gives tips to improve oral health and stores user data safely using Firebase. The goal is to make dental care more accessible and help people take better care of their teeth from home.

Modules list for user:

- **Register Account:**

  Users can create an account by providing details such as full name, email, username, password, and password confirmation to gain access to the system.

- **Login / Logout:**

  Allows users to securely log in using their registered credentials and log out after completing their session.

- **Image Upload & Analysis:**

Users can upload intraoral images (teeth photos) to the system, which will automatically analyze them using a CNN-based deep learning model.

- **Dental Condition Classification:**

Based on the image analysis, the system shows the problem in their teeth

- **Feedback Display:**

Provides users with clear and concise feedback regarding their dental condition after analysis.

- **Progress Tracking & Profile Management:**

Users can view a history of past image submissions and received feedback. The system also shows trends such as improvement or deterioration over time. Users can update personal details like name, age, and brushing habits.

- **Treatment Recommendations:**

The system suggests possible treatment options based on the diagnosis (e.g., "Visit a dentist for deep cleaning") and offers tips for oral hygiene and preventive care.

- **Reminder Notifications:**

Users can opt-in to receive reminders for future dental checkups or re-analysis. The system sends these reminders via push notifications or email based on user preference.

- **Appointment booking**

Users can book an appointment at the clinic on selected time

Modules list for admin:

- **Admin Login:**

Admins can securely log in using their username and password to access the admin panel.

- **User Management:**

  Admins can view the list of all registered users along with their profiles and image analysis reports. They have the authority to delete suspicious or inappropriate content submitted by users.

- **Feedback Management:**

  Admins can edit or delete auto-generated feedback shown to users and manage educational content related to proper dental care and hygiene practices.

Module list for clinic

- **Clinic Login:**

  Clinic staff can securely log in to access the appointment management system.

- **Appointment Review & Approval:**

  Clinics can view appointment requests submitted by users and approve or reject them based on availability and urgency.

- **Appointment Scheduling Management:**

  Allows clinics to manage available time slots, reschedule appointments, and notify users of changes.

2.2 Assumptions

To conduct and execute the test, it assumed that:

ref	Assumption
1	Functional requirements have been reviewed and acknowledged.
2	There is a Software Requirements Specification (SRS) as reference where it contains all the functional requirements.
3	Static test has been conducted via observation, reading and review of tested document (SRS)
4	The test resources and test environment described in section 6 is available
5	System, database, and server able to run at smooth performance
6	Identified defects and bugs are expected to be fixed/resolved before the testing
7	All changes to requirements will be notified and communicated to test team

Table 2 Assumption

2.3 References

SRS (Software Requirements Specification)	There is a Software Requirements Specification (SRS) to be referenced where it contains all the functional requirements.
IEEE Standard 829-1983	IEEE Standard for Software Test Plan.
Dishcovery Proposal	A proposal that defines Dishcovery overall during early planning.
ISTQB Slides	Numerous slides from chapter 1 until 7 are used as references. Obtained from Software Testing class.

ISO/IEC 25010:2011	Reference to define quality requirements and quality attributes.

Table 3 Reference

2.4. Document Overview

The STP document consists of six sections as brief below:

- **Section 1** contains Test Policy components which explain the major philosophy of the development organization regarding their testing and its terms, testing SOP (Standard of Procedure), meaning and practices.

- **Section 2** explains Introduction to Test Plan which gives an overview about the current acceptance test plan purpose, system description, references, and document structure overview.

- **Section 3** contains Test Strategy which describes the testing approach, test objective, test scope, test process, test level, risk and contingency, entry and exit criteria, item pass/fail criteria, suspension criteria and resumption requirements.

- **Section 4** contains Test Case Specification which describes the current test cases in development along with Use Case Diagram.

- **Section 5** contains Test Environment which describes all the required facilities, hardware, software, and other necessary requirements for test to be executed smoothly.

- **Section 6** contains Test Schedule, which discusses an overall schedule plan including testing activities.

TEST STRATEGY

A test strategy is a document that explains the overall plan for testing during the software development process. It outlines how testing will be done and is a key part of the test plan.

3.1 Test Objective

This test plan for the AI Powered Dental Assessment System aims to verify that the application operates smoothly, performs reliably, and delivers its core functionalities as intended. The testing efforts will concentrate on detecting and correcting any defects to ensure the system performs effectively for all user roles, including patients, dental professionals and administrators. Additionally, the testing will confirm that the system meets the functional and non-functional requirements defined in the Software Requirements Specification (SRS) and fulfills the expectations of key stakeholders.

Here are the main objectives:

1. Requirement Verification
    a. Ensure that all function and non-functional requirements outlined in the Software Requirements Specification are implemented and tested.
2. AI Model Accuracy
    a. Validate the deep learning model so that it correctly detects and classifies dental conditions using real world images.
3. Feature Testing
    a. Confirm that key features such as Image Upload, Condition Classification, Progress Tracking, Personalized Recommendations and Reminders are functioning correctly.
4. Usability
    a. Ensure that the system is user friendly, responsive and easy to navigate for user with varying technical backgrounds.

5. Security and Data Privacy
    a. Test security measures to protect user data.
6. Performance and Efficiency
    a. Evaluate how well the system processes images, return results and manages multiple users.
7. Accessibility
    a. Verify that the application compiles with accessibility standards and is useable by individuals with diverse backgrounds.
8. System Stability after Update
    a. Conduct regression testing after updates to confirm existing functionality remains unaffected.
9. Stakeholder Satisfaction
    a. Gather and incorporate feedback from user and supervisors to ensure the system aligns with expectations and is ready for deployment.
10. Preparing for Real World Use
    a. Confirm the system readiness for public use by simulating real world conditions and scenarios

3.2 Scope

The scope of this Acceptance Test Plan is to verify all functional requirements that has been listed in Software Requirements Specification for the AI Powered Dental Assessment System accurately implemented and tested.

3.3 Scope

The item to be tested is AI Powered Dental Assessment System

3.3.1 Feature Tested

There is a risk-based testing approach that includes a risk analysis conducted under the project's scope. The analysis evaluates key features such as Image Analysis, Condition Classification, and Treatment Recommendations based on their likelihood of failure, impact on user health outcomes, and overall priority. Features with high-risk attributes such as inaccuracies in dental condition detection or incorrect treatment suggestions will be prioritized for rigorous testing to ensure reliability and user safety. The testing strategy aligns with the project's use of deep learning (CNN/YOLO model) and web-based deployment (Next.js/Flask), focusing on critical workflows like image uploads, AI analysis, and feedback generation.

3.4 Test Process

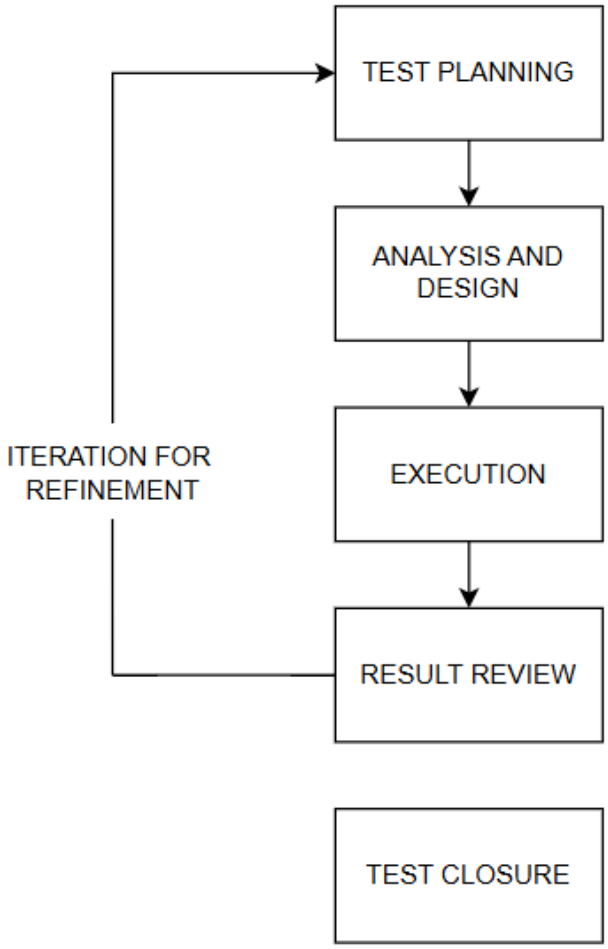


Figure 2 Test Process

Figure X Test Process

There are five main processes involved , the diagram above depicts the test process approach that will be taken.

3.4.1 Test Planning

1.System Analysis & Architecture Review

- Evaluate the system architecture and integration points to identify critical testing requirements.

2.Testing Objectives & Criteria Definition

- Establish testing goals aligned with project objectives and define entry/exit criteria.

3.Test Scenario Identification

- Derive test scenarios from functional and non-functional requirements.

4.Resource Planning

- Allocate personnel, tools, hardware, and timelines for test execution.

5.Test Deliverables Definition

- Document test plans, cases, scripts, and reports to ensure requirement traceability.

6.Test Schedule Development

- Define testing phases, milestones, and deadlines with risk mitigation buffers.

7.Review & Finalization

- Validate the test plan with stakeholders and refine based on feedback.

3.4.2 Analysis and Design

1.Analysis

- Conduct a detailed review of system requirements and specifications to identify testable components.
- Analyze functional and non-functional requirements to determine testing scope and coverage.
- Evaluate potential risks and challenges in the system to prioritize critical test areas.

2.Design

- Develop a structured test approach based on the analysis findings.
- Define test cases and scenarios to validate system functionality and performance.
- Outline test data requirements and environment setup for accurate test execution.
- Document the test design to ensure clarity and alignment with project objectives.

3.4.3 Execution

1.Test Execution

- Execute test cases as per the defined test plan and schedule.
- Log detailed results, including pass/fail status and observed defects.
- Validate system behavior against expected outcomes for each test scenario.

2.Defect Management

- Report identified defects with clear steps to reproduce, severity, and priority.
- Track defect resolution progress and verify fixes through retesting.

3.Progress Monitoring

- Maintain real-time documentation of test execution status.
- Update stakeholders on testing progress and any critical issues.

3.4.4 Result Review

1.Test Result Analysis

- Evaluate test outcomes against defined success criteria
- Identify patterns in test failures or system weaknesses

2.Defect Analysis

- Review all logged defects for completeness and accuracy

- Assess defect trends to identify systemic quality issues

3.Reporting

- Document findings and observations in test summary report
- Highlight critical issues and recommendations

5.Stakeholder Review

- Present results to project team and stakeholders
- Obtain formal acceptance of testing outcome

3.4.5 Test Closure

1.Verification

- Confirm completion of all test deliverables
- Validate resolution of critical defects

2.Artifact Management

- Archive test scripts, plans, and environments for future reuse

3.Process Evaluation

- Assess testing effectiveness and efficiency
- Document lessons learned and improvement opportunities

4.Knowledge Transfer

- Share insights with stakeholders and future teams

3.5 Item Pass/Fail Criteria

The test oracle used to determine whether a test item passes or fails is based on the functional requirements and use case specifications outlined in the AI Dental Assessment System Software Requirements Specification (SRS).

Each test conducted will be evaluated against these documented requirements. If the outcome of the test does not satisfy the expected behavior defined in the functional requirements or corresponding use cases, it will be considered a test failure. In such cases, a defect or incident report will be created. This report will be subject to review and analysis by the responsible tester and assigned developer to determine the root cause and appropriate resolution.

The system will only be deemed to pass a test when:

- The expected results, as per the SRS, are fully achieved without deviation.
- The AI component correctly identifies and classifies dental conditions with an accuracy rate that meets or exceeds the defined acceptance threshold.
- The system maintains usability and performance standards, including timely image processing and reliable result generation.
- All security, data integrity, and user interaction flows conform to the specified design.

3.6 Suspension Criteria and Resumption Requirements

3.6.1 Suspension Criteria

Testing will be temporarily suspended if the AI-powered dental system becomes inaccessible due to technical issues, or if testing is disrupted by physical factors such as power failure, network outage, or natural disasters affecting the test environment.

3.6.2 Resumption Requirement

Testing will resume once the system is fully operational again and the testing environment has returned to a stable and secure condition, ensuring that testing activities can continue safely and effectively.

3.7 Entry and Exit Criteria

This section will state when will the testing start and when will it end

3.7.1 Entry Criteria

- The tools in tern of hardware and software for the testing is ready
- The test environment has been setup
- Requirements in the SRS are documented , reviewed and approved by the stakeholders

3.7.2 Exit Criteria

- All the test cases executed successfully based on the test coverage of 80%. Test coverage defines how much testing is conducted should achieve 80% or above . The formula for calculating the test coverage is stated below

  C: total test cases executed / total number of test cases * 100%

  C: 40/50 * 100% = 80%
- Test results have been documented , reviewed and approved by stakeholders
- System functionallity meets the acceptance criteria outlined in the SRS.

3.8 Risk and Contingencies

No	Risk	Impact	Contingencies
1	The software is unavailable	• Testing cannot proceed • Delay occurs	Increase testing personnel or reduce the number of test cases to recover lost time.
2	Unavailability of test environment	• Inability to execute tests • Delay occurs	Prepare a checklist to ensure the environment is ready; assign more testers if delays occur.
3	Lack of knowledge or resources	• Inaccurate testing • Delay occurs	Consult with experts or conduct on the spot training
4	Time and cost limitations	• Insufficient time and budget to complete full testing	Prioritize and execute only high-priority
5	Large number of defects	• Test execution is blocked • Delay occurs	Prioritize fixing critical defects and retest before proceeding further.
6	Lack of test data	• Incomplete testing • Inaccurate testing	Perform experience-based testing or generate synthetic data if real data is unavailable.
7	Poor AI model performance	• Inaccurate assessments • False positives/negatives	Retrain the model or adjust thresholds, and consider testing with different datasets.
8	Environmental disruptions	• Physical disruptions like power failure or network outage	Have backup systems and alternate test plans ready to mitigate environmental issues.

Table 4 Risk and Contingencies

3.9 Test Level

To ensure the AI Powered Dental Assessment System functions as intended, multiple testing levels will be applied throughout the process. The primary focus will be on Acceptance Testing, which will confirm that the system meets all defined requirements and is ready for deployment.

Acceptance Testing

At this level, the AI Powered Dental Assessment System will be evaluated to ensure that it meets all functional requirements and performs as expected for end-users. This phase validates the system's readiness for real-world deployment. The testing will use a black-box testing approach, focusing solely on system functionality, ensuring that it works according to the specified requirements. The testing process will involve users (patients, dentists, and administrators) with the support of developers to ensure the system's viability.

3.9.1 Black Box Testing

The black-box testing approach will assess the system's input and output, focusing purely on the functional requirements defined in the SRS. This method, also known as Specification-Based Testing, ensures that the system performs as expected based on the requirements, without delving into the underlying code.

3.9.1.1 Functional Testing

Functional Testing will verify that all features of the system work as intended. Key functionalities to be tested include:

- User Authentication
    - Verifying that users can register, log in, and manage their profiles securely,
- Image Analysis

  - Ensuring users can upload dental images in common formats and preprocess those images

- Condition Classification
  - Confirming that the system classifies dental conditions
- Feedback Generation
  - Verifying that the system generates reports
- Progress Tracking
  - Ensuring the system can store and display users' dental health reports
- Treatment Recommendations
  - Verifying that the system provides tailored treatment suggestions, preventive measures, and recommends specific dental products
- Reminders and Alerts
  - Ensuring that the system allows users to set reminder
- Admin Dashboard
  - Verifying that the admin dashboard functions correctly
- Feedback and Support
  - Ensuring the system includes a feedback form for users to report issues or suggest improvements.

3.9.2 Usability Testing

Usability testing focuses on making sure the system is easy to use and meets user expectations. Testers observe how users interact with the app, especially when uploading images, reviewing dental reports, and navigating the app. The goal is to ensure key features, like treatment recommendations, progress tracking, and feedback, are clear and easy to understand. The app should help users easily understand their dental conditions and suggested actions without confusion. Usability testing also ensures the app is responsive,

accessible, and flows smoothly from one task to the next. User feedback will help improve the system, making it both functional and engaging.

3.9.3 Risk Based Testing

Risk-based testing prioritizes areas with the highest potential impact on the system. This approach focuses on critical components, such as the AI image analysis model and user data security. For example, errors in detecting dental conditions like cavities could lead to incorrect treatment recommendations, while security vulnerabilities could expose sensitive health information. Testing will emphasize these high-risk areas to ensure they perform correctly and can handle failure scenarios, ensuring the system's reliability and security before deployment.

3.9.3.1Risk Analysis

Risk-Based Testing focuses on testing parts of the system with the highest risk of failure and impact. This ensures that key features, such as **image analysis** and **user data security**, are thoroughly tested to prevent major issues. Risk is assessed based on the likelihood of failure and the impact it would have. The risk analysis formula is:

- **Likelihood (L)**: From 1 (low) to 9 (high).
- **Impact (I)**: A scale based on cost (e.g., 1, 10, 50, 100, 500, 1000).
- **Total Risk = L * I**

Testing is focused on high-risk areas, ensuring critical functions are reliable and reducing the chance of major failures.

Total Risk	Priority Number
R < 3000	1 (Lowest)
R >= 3000	2 (Low)

R >= 4000	3 (Medium)
R >= 8000	4 (High)
R >= 9000	5 (Highest)

Table 5 Risk Classification for Priotization

Features	**Likelihood (L)**	**Impact (I)**	**Total Risk (R = L * I)**	**Priority**
User Authentication				
Failed User Registration	6	500	3000	3
Failed Login	6	500	3000	3
Failed Profile Update	5	300	1500	2
Image Analysis				
Failed Image Upload (JPEG/PNG)	7	1000	7000	4
Failed Image Preprocessing (Resizing)	6	500	3000	3
AI Model Misdiagnosis	9	1000	9000	5
Failed Bounding Box Highlighting	8	1000	8000	4
Condition Classification				
Failed Condition Classification (Health Status)	7	1000	7000	4
Failure to Support New Dental Conditions	8	1000	8000	4
Feedback				
Failed Generate Feedback Report	6	500	3000	3
Failed Actionable Feedback (Suggestions)	7	800	5600	4
Failed Visual Feedback (Highlight Issues)	6	500	3000	3
Progress Tracking				
Failed Save Health Reports	6	500	3000	3
Failed Visual Progress Charts	6	500	3000	3

Failed Compare Current vs Previous Reports	5	400	2000	2
Treatment Recommendations				
Failed Generate Treatment Recommendations	8	1000	8000	4
Failed Suggest Preventive Measures	7	800	5600	4
Failed Recommend Specific Dental Products	6	500	3000	3
Reminders and Alerts				
Failed Set Reminders	6	500	3000	3
Failed Push Notifications	7	600	4200	4
Failed Urgent Alerts (e.g., Severe Cavity)	8	1000	8000	4
Admin Dashboard				
Failed Admin Dashboard Updates	8	1000	8000	4
AI Model Accuracy Monitoring Failure	9	1000	9000	5
Feedback and Support				
Failed Feedback Form Submission	5	300	1500	2

Table 6 Risk Analysis for Risk Based Testing

4. TEST CASES

Test cases contain a step-by-step flow of action where it will be performed by testers. Each test case contains tested by whom, test level, test type, test design technique, risk number, test case number, test case name, requirement traceability, test case description, preconditions, post conditions, item(s) to be tested, test procedure specification, input step(s), test data, expected result, actual result, pass/fail, and stakeholder (user).

4.1 use case AI powered dental assessment system – overall application.

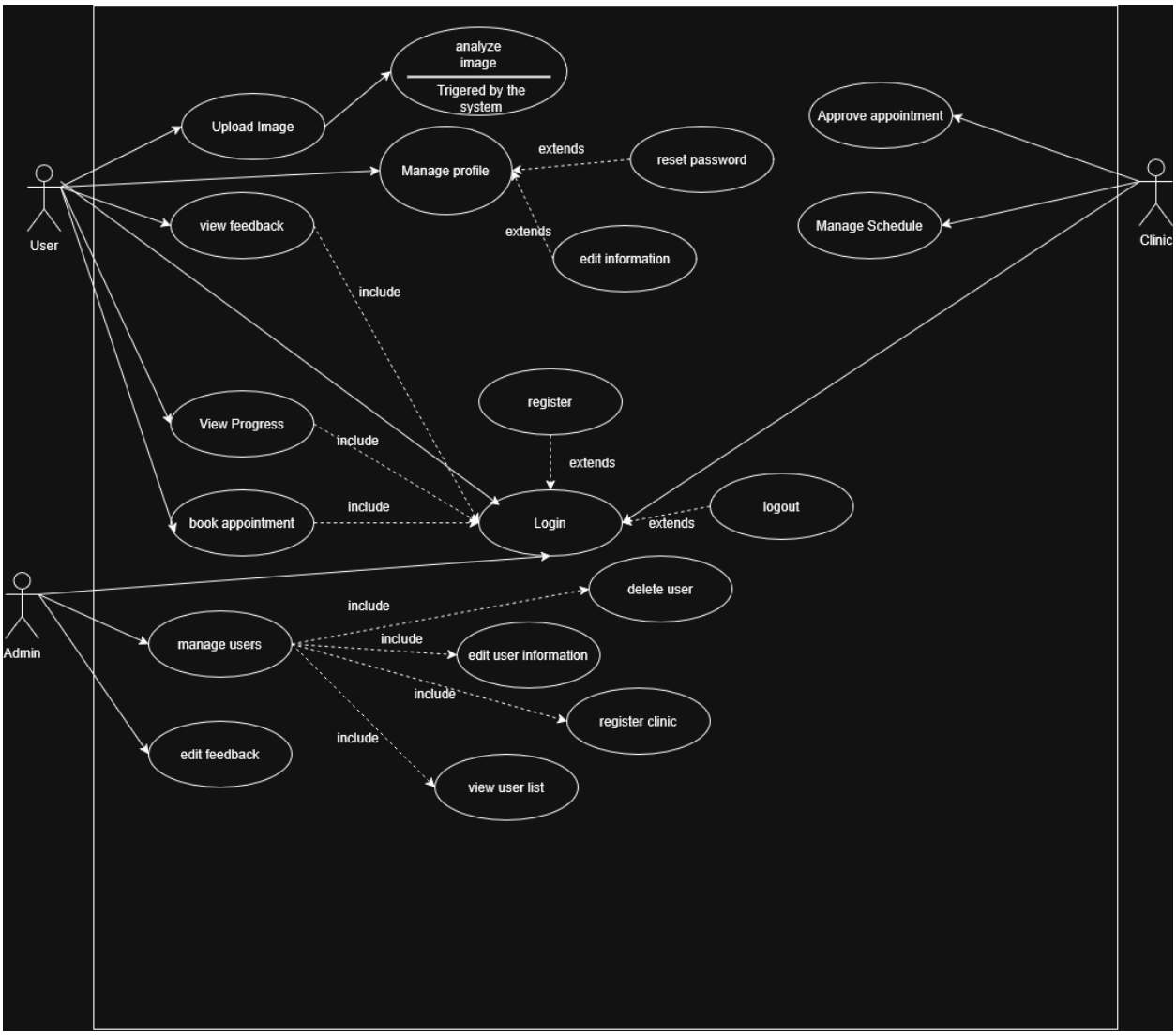


Figure 3 Use Case Diagram Ai powered dental assessment system- overall application

4.1.1 Use Case: Register management

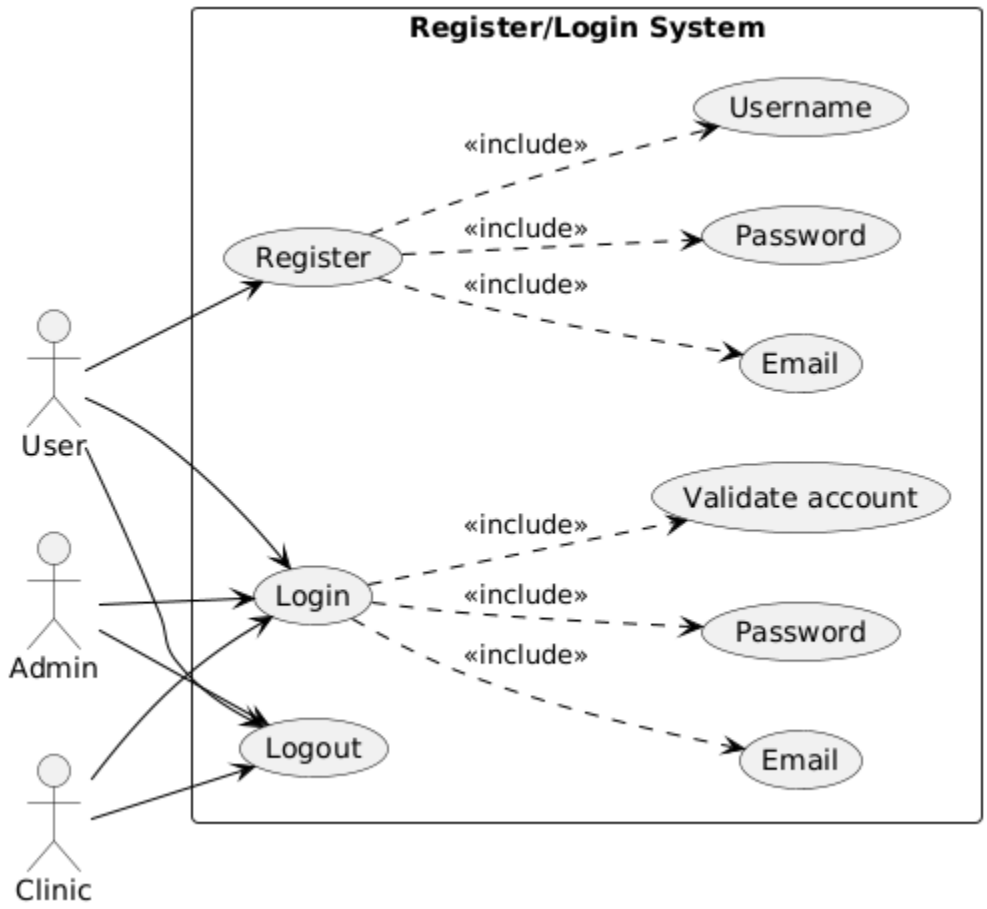


Figure 4 Use Case Diagram - Register Management

Use Case ID	UC-001	
Use Case Name	User Registration	
Description	This use case allows a new user to create an account by providing their full name, email, username, password, and password confirmation.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	The user is not already registered.	
Post-condition	A new user account is created and stored in the system database.	
Main Flow	Step	Action
	Pre-Condition	The user is not already registered.
	1.1	User navigates to the registration page.
	1.2	The system displays the registration form.
	1.3	User enters full name, email, username, password, and confirms password.
	1.4	The system validates the entered data (e.g., unique email/username, matching passwords).
	1.5	If valid, the system stores the user information in the database.
	1.6	The system displays a success message and redirects to the login page.
	Post-condition	A new user account is created and stored in the system database.
Alternate Flow	Pre-Condition	The student already has an account
	2.1	The student can navigate to login page
	2.2	The student can log in using his account
	2.3	The student will automatic login to student page
	Post-condition	Th student doesn't need to register to login if already got account

Robust Flow	Pre-Condition	User is not registered yet and is on the Registration page.
	3.1	The student register using email that already being
	3.2	The system displays an error message indicating the email
	3.3	The student required to use other email to register
	Post-condition	The system must gracefully handle invalid or duplicate inputs, ensuring no incorrect data is stored.

Table 7 Use case statement - user registration

Use Case ID	UC-002	
Use Case Name	Login	
Description	To allow users (User/Admin) to securely log into the system and access its features based on their roles.	
Primary Actor	User, Admin	
Include use cases	Access Main Page	
Pre-Condition	The user has already registered an account.	
Post-condition	The user is authenticated and redirected to the appropriate main page based on their role (User or Admin).	
Main Flow	Step	Action
	Pre-Condition	The user has already registered an account.
	1 User/Admin navigates to the login page. 2 User/Admin enters username and password. 3 System validates credentials. 4 If valid, the system determines the user's role. 5 System grants access and redirects to the appropriate main page.	
	Post-condition	The user is authenticated and redirected to the appropriate main page based on their role (User or Admin).
Alternate Flow	Pre-Condition	User/Admin enters invalid credentials.
	1 User/Admin enters incorrect username or password. 2 System displays an error message indicating invalid credentials. 3 User/Admin is prompted to re-enter login information. 4 User/Admin retries login or navigates to "Forgot Password".	
	Post-condition	The system displays an error message and prompts retry.
Robust Flow	Pre-Condition	User/Admin clicks on "Forgot Password" link.
	1 User/Admin clicks on the "Forgot Password" link on the login page. 2 System prompts user to enter registered email address. 3 User enters an email and submits the request. 4 System sends a password reset link to the provided email. 5 User/Admin accesses the link and sets a new password.	

	6 System confirms the password has been reset successfully.	
	Post-condition	System sends a password reset email to the user.

Table 8 Use case statement - Login

Use Case ID	UC-003	
Use Case Name	Logout	
Description	To allow users (User/Admin/clinic admin) to log out of the system	
Primary Actor	User, Admin,clinic admin	
Include use cases	-	
Pre-Condition	The User, Admin,clinic admin is currently logged in	
Post-condition	The User, Admin,clinic admin is logged out the system	
Main Flow	Step	Action
	Pre-Condition	The user has already registered an account.

	1. User clicks the logout button. 2. System terminates the current session. 3. System redirects user to the login page with a confirmation message.	
	Post-condition	The User, Admin,clinic admin is logged out the system
Alternate Flow	Pre-Condition	**Session already expired:**
	1. System automatically redirects to login page when a protected page is accessed.	
	Post-condition	User is logged out, and session is securely terminated.
Robust Flow	Pre-Condition	-
	-	
	Post-condition	-

Table 9 Use case statement - Logout

4.1.2 Use Case: Analyze Image

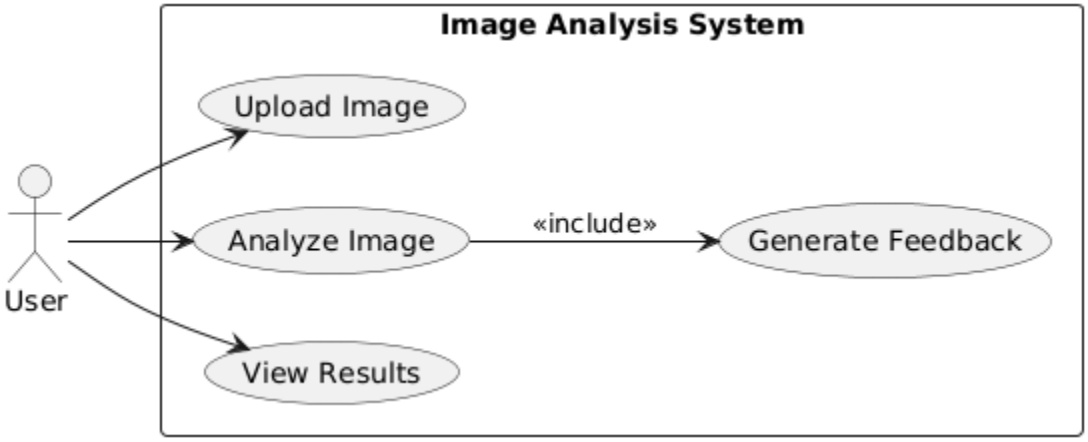


Figure 5 Analyze Image

Use Case ID	UC-004	
Use Case Name	Upload image	
Description	To allow users to upload an intraoral image to be analyzed by the AI model for dental condition detection.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	The user must be logged in and have access to the upload page.	
Post-condition	The image is stored in the system and sent for analysis. A result or feedback is generated.	
Main Flow	Step	Action
	Pre-Condition	The user must be logged in and have access to the upload page.
	1 User navigates to the “Upload Image” section. 2 System prompts the user to choose an image file. 3 User selects and submits an intraoral image. 4 System validates the file type and size. 5 System stores the image in the database or cloud storage. 6 Image is sent to the AI model for analysis. 7 System notifies the user that the image has been uploaded successfully.	

	Post-condition	The image is stored in the system and sent for analysis. A result or feedback is generated.
Alternate Flow	Pre-Condition	User submit an invalid file
	1. If the uploaded file is not a valid image format or too large. 3. System shows an error message and prompts re-upload. 5. User re-uploads a valid image.	
	Post-condition	The system displays an error message and prompts retry.
Robust Flow	Pre-Condition	Upload fails
	1. If storage/upload fails due to server or network error. 2. System displays an error message (e.g., “Upload Failed”). 3. User can retry uploading or contact support.	
	Post-condition	System sends an error message

Table 10 Use case statement - Upload image

Use Case ID	UC-005	
Use Case Name	Analyze image	
Description	To analyze the uploaded intraoral image using an AI model to detect dental conditions	
Primary Actor	System	
Include use cases	generate feedback	
Pre-Condition	A valid intraoral image has been uploaded by a logged-in user.	
Post-condition	The AI model returns detection results that are stored and displayed to the user.	
Main Flow	Step	Action
	Pre-Condition	A valid intraoral image has been uploaded by a logged-in user.
	1 System receives the uploaded image. 2 The AI model analyses the image for dental conditions. 3 System generates a result with identified conditions and confidence levels. 4 Results are stored in the database and linked to the user's account.	
	Post-condition	The AI model returns detection results that are stored and displayed to the user.
Alternate Flow	Pre-Condition	No Condition Detected
	1. If the AI model finds no significant dental condition. 2. System generates a message indicating no issues were found.	
	Post-condition	The system displays a message indicating no issues.
Robust Flow	Pre-Condition	Analysis Error
	1.If the AI model fails due to corrupted image or internal error. 2. System logs the error and notifies the user of failure.	
	Post-condition	User is prompted to re-upload the image or try again later.

Table 11 Use case statement - Analyze image

Use Case ID	UC-006	
Use Case Name	View result	
Description	Allows the user to view feedback from their image analysis	
Primary Actor	user	
Include use cases	-	
Pre-Condition	User submitted and system completed image analysis	
Post-condition	Feedback is displayed to the user and can be reviewed at any time.	
Main Flow	Step	Action
	Pre-Condition	User submitted and system completed image analysis
	1 User navigates to the Feedback section after logging in. 2 System fetches all related feedback records for the user. 3 Feedback is retrieved from the database. 4 Feedback (text, date, treatment suggestions, etc.) is displayed. 5 User reviews the feedback and may proceed to track progress or book appointment.	
	Post-condition	Feedback is displayed to the user and can be reviewed at any time.
Alternate Flow	Pre-Condition	No Feedback Available
	1.If the user has no completed analyses or feedback records. 2. System shows a message: "No feedback available yet."	
	Post-condition	User is prompted to upload an image or return to dashboard.
Robust Flow	Pre-Condition	System Error While Fetching Feedback
	1 If feedback retrieval fails due to a database error. 2 System logs the error and displays an error message.	
	Post-condition	User is given the option to retry
Use Case ID	UC-007	
Use Case Name	Generate Feedback	
Description	This use case allows the system to generate feedback based on image analysis results	
Primary Actor	System	
Include use cases	Image Analysis	

Pre-Condition	User has uploaded an image and the analysis has been successfully completed.	
Post-condition	Feedback is generated and stored in the database, associated with the user's profile.	
Main Flow	Step	Action
	Pre-Condition	User has uploaded an image and the analysis has been successfully completed.
	1 User uploads an image and initiates analysis. 2 System analyzes the image using an AI model. 3 System processes analysis results to create textual feedback. 4 Feedback includes diagnosis and recomendation 5 Feedback is saved in the database and becomes available to view.	
	Post-condition	Feedback is generated and stored in the database, associated with the user's profile.
Alternate Flow	Pre-Condition	
	-	
	Post-condition	
Robust Flow	Pre-Condition	Invalid Image or Model Error
	1 Image format is unsupported or unclear for analysis. 2 System throws an error and asks the user to upload a new image. 3 If the AI model fails internally, system logs the error and informs the user.	
	Post-condition	User is prompted to try again later

Table 12 Use case statement - View result

4.1.3 use case: Booking Appointment

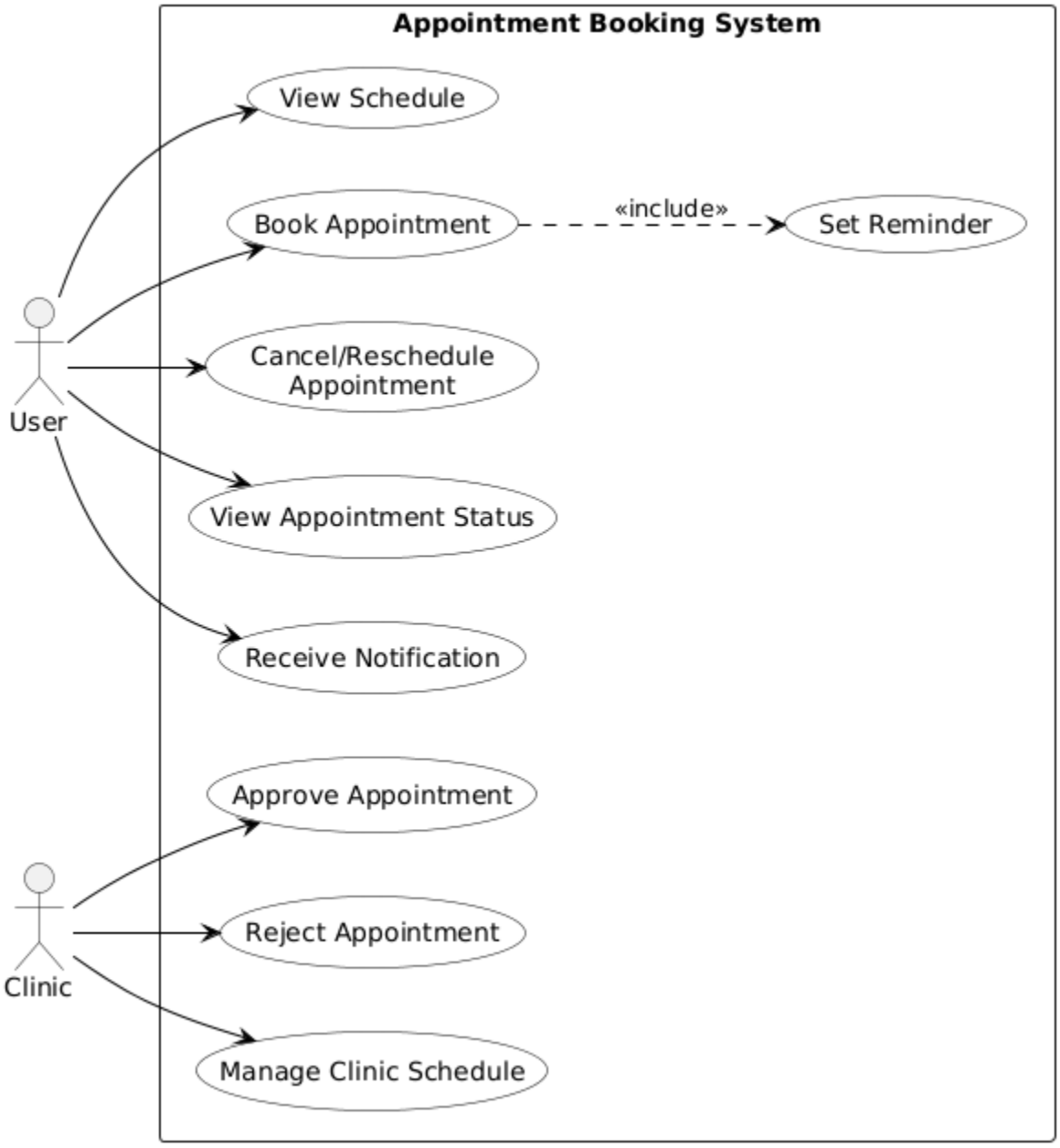


Figure 6 Use case: Booking Appointment

Use Case ID	UC-008	
Use Case Name	View Schedule	
Description	This use case allows the User to view available appointment slots provided by the Clinic.	
Primary Actor	User	
Include use cases	Login	
Pre-Condition	The User is logged in to the system..	
Post-condition	The available appointment schedule is displayed.	
Main Flow	Step	Action
	Pre-Condition	The User is logged in to the system..
	1 User selects "View Schedule". 2 System fetches available slots from the clinic's schedule. 3 System displays available appointment dates and times.	
	Post-condition	The available appointment schedule is displayed.
Alternate Flow	Pre-Condition	No available appointment
	1. If no slots are available, system shows a message: "No appointments available."	
	Post-condition	
Robust Flow	Pre-Condition	
	-	
	Post-condition	

Table 13 Use case statement - View Schedule

Use Case ID	UC-009	
Use Case Name	Book Appointment	
Description	This use case allows the User to request an appointment based on available slots.	
Primary Actor	User	
Include use cases	Login	
Pre-Condition	The User is logged in and has viewed the available schedule.	
Post-condition	The appointment request is submitted and pending approval by the clinic	
Main Flow	Step	Action
	Pre-Condition	The User is logged in to the system.
	1 User selects a preferred date and time from the schedule. 2 User submits the booking request. 3 System saves the request and notifies the clinic.	
	Post-condition	The appointment request is submitted and pending approval by the clinic
Alternate Flow	Pre-Condition	No available appointment
	1. If the selected time slot becomes unavailable during submission, system notifies the user and asks to select another time.	
	Post-condition	
Robust Flow	Pre-Condition	
	-	
	Post-condition	

Table 14 Use case statement - Book appointment

Use Case ID	UC-010	
Use Case Name	Set Reminder	
Description	This use case allows the User to set a calendar reminder for their approved appointment.	
Primary Actor	User	
Include use cases	Book Appointment	
Pre-Condition	The appointment has been approved by the Clinic	
Post-condition	A reminder is added to the User's calenda	
Main Flow	Step	Action
	Pre-Condition	The appointment has been approved by the Clinic
	1 User views the approved appointment. 2 User selects “Set Reminder”. 3 System asks for preferred reminder time (e.g., 1 day before). 4 Reminder is added to the user's system calendar or notified internally.	
	Post-condition	A reminder is added to the User's calendar.
Alternate Flow	Pre-Condition	No available appointment
	Post-condition	
Robust Flow	Pre-Condition	
	-	
	Post-condition	

Table 15 Use case statement - Set reminder

Use Case ID	UC-011	
Use Case Name	Cancel/Reschedule Appointment	
Description	This use case allows the User to cancel or reschedule an existing appointment.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	The User has an existing appointment.	
Post-condition	Appointment is canceled or rescheduled successfully	
Main Flow	Step	Action
	Pre-Condition	The User has an existing appointment.
	1 User selects “Cancel/Reschedule Appointment”. 2 System displays upcoming appointments. 3 User chooses to either cancel or pick a new slot. 4 If rescheduled, system sends new request to the Clinic. 5 If canceled, system removes the appointment.	
	Post-condition	Appointment is canceled or rescheduled successfully
Alternate Flow	Pre-Condition	No available appointment slot
	If rescheduling and new slot is unavailable, system notifies user to try again later.	
	Post-condition	Notify user to try later
Robust Flow	Pre-Condition	
	-	
	Post-condition	

Table 16 Use case statement - Cancel/reschedule Appointment

Use Case ID	UC-012	
Use Case Name	Approve Appointment	
Description	This use case allows the Clinic to view and approve or reject appointment requests made by users. Approval confirms the appointment and notifies the user.	
Primary Actor	Clinic	
Include use cases	-	
Pre-Condition	The Clinic is logged into the system and there are pending appointment requests.	
Post-condition	Appointment is either approved (confirmed) or rejected (removed), and the user is notified accordingly.	
Main Flow	Step	Action
	Pre-Condition	pending appointment requests.
	1 Clinic navigates to "Pending Appointments" section. 2 System displays all pending requests. 3 Clinic selects an appointment request. 4 Clinic approves the appointment. 5 System marks the appointment as confirmed and notifies the user.	
	Post-condition	Appointment approved
Alternate Flow	Pre-Condition	pending appointment requests
	1Clinic rejects the appointment request. 2 System deletes or archives the request and notifies the user of rejection.	
	Post-condition	Appointment rejected
Robust Flow	Pre-Condition	No pending request
	If no pending appointments are available, system displays “No pending requests.”	
	Post-condition	Display messages

Table 17 Use case statement - Approve Appointment

Use Case ID	UC-013	
Use Case Name	Manage Clinic Schedule	
Description	This use case allows the Clinic to view and approve or reject appointment requests made by users. Approval confirms the appointment and notifies the user.	
Primary Actor	Clinic	
Include use cases	-	
Pre-Condition	Clinic is authenticated and has access to the schedule management interface.	
Post-condition	Clinic schedule is updated and reflected in the user-facing booking system.	
Main Flow	Step	Action
	Pre-Condition	Clinic is authenticated and has access to the schedule management interface.
	1 Clinic navigates to the "Manage Schedule" section. 2 Clinic views the current schedule. 3 Clinic adds, updates, or deletes time slots. 4 System saves the changes and updates the available schedule for users.	
	Post-condition	Clinic schedule is updated and reflected in the user-facing booking system.
Alternate Flow	Pre-Condition	A pending appointment is schedule on the slot
	1 Clinic attempts to modify a time slot with an existing confirmed appointment. 2 System prevents the change and prompts the clinic to handle existing appointments before editing.	
	Post-condition	
Robust Flow	Pre-Condition	Server error
	If the system is unable to retrieve the schedule due to a server error, it displays an error message and logs the issue.	
	Post-condition	Display error message

Table 18 Use case statement - Manage clinic schedule

4.1.4 use case: Manage user

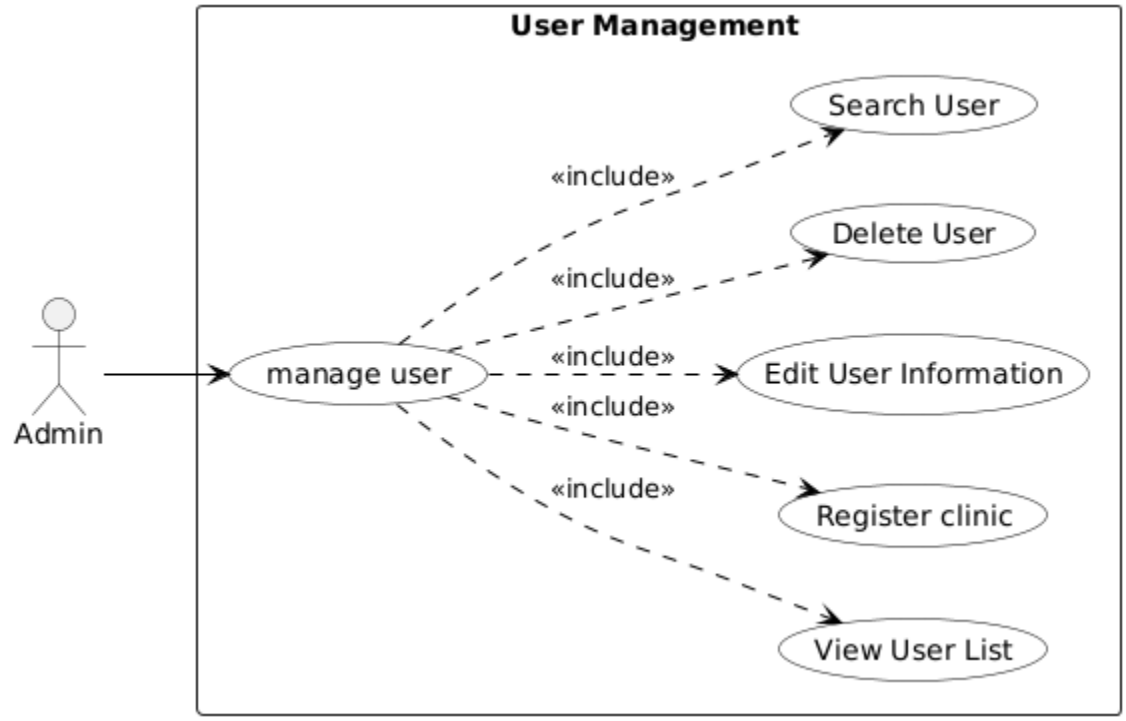


Figure 7 Use case: Manage user

Use Case ID	UC-014	
Use Case Name	Manage User	
Description	Allows the admin to manage registered users in the system, including viewing, deleting, or resetting their passwords.	
Primary Actor	Admin	
Include use cases	View User List, Delete User, search user, edit user information, register clinic.	
Pre-Condition	Admin is logged into the system.	
Post-condition	User information is accessed or updated based on admin action.	
Main Flow	Step	Action
	Pre-Condition	Admin is logged into the system.
	1 Admin navigates to the Manage Users section on the dashboard. 2 System displays available user management actions. 3 Admin chooses to view, delete, or reset password of a user. 4 Corresponding included use case is triggered.	
	Post-condition	User information is accessed or updated based on admin action
Alternate Flow	Pre-Condition	No users exist in the system
	1 System displays "No users found" message.	
	Post-condition	Admin is notified of the issue; no changes to user data occur.
Robust Flow	Pre-Condition	System fails to load the user list due to server error
	System displays an error and suggests retrying later.	
	Post-condition	No updates are performed; system state remains unchanged.access based on security protocol.

Table 19 Use case statement - Manage user

Use Case ID	UC-015
Use Case Name	View User List
Description	

	Allows the Admin to view the list of all registered users.	
Primary Actor	Admin	
Include use cases	View User List, Delete User, Reset User Password	
Pre-Condition	Admin is authenticated and authorized..	
Post-condition	System displays user list.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin clicks on “View User List.” 2 System fetches and displays all users in a table format.	
	Post-condition	System displays user list.
Alternate Flow	Pre-Condition	No users exist in the system
	1 System displays "No users found" message.	
	Post-condition	System still remains functional; no list is shown
Robust Flow	Pre-Condition	database is unreachable
	1 If database is unreachable, system shows error and retry option.	
	Post-condition	No data shown; admin remains on the same page.

Table 20 Use case statement - view user list

Use Case ID	UC-016	
Use Case Name	Delete User	
Description	Allows the Admin to remove a user account from the system.	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	Admin is authenticated and user exists in the system.	
Post-condition	User account is deleted and removed from user list.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin selects a user from the list. 2 Admin clicks "Delete." 3 System confirms action and deletes the user.	
	Post-condition	System displays user list.
Alternate Flow	Pre-Condition	Admin cancels the deletion
	1 Admin cancels the deletion at the confirmation prompt..	
	Post-condition	No action is taken; user remains in the system.
Robust Flow	Pre-Condition	database is unreachable
	1 If database is unreachable, system shows error and retry option.	
	Post-condition	No data shown; admin remains on the same page.

Table 21 Use case statement - Delete user

Use Case ID	UC-017	
Use Case Name	Edit user information	
Description	Allows the Admin to update user details such as name, contact information, or role.	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	User account exists and admin is authenticated	
Post-condition	The updated user information is saved and reflected in the system.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin selects a user from the user list. 2 Admin clicks "Edit." 3 System displays editable user information form. 4 Admin updates the desired fields. 5 Admin submits the changes. 6 System validates and saves the updated information.	
	Post-condition	System displays a success message and updated user list.
Alternate Flow	Pre-Condition	Admin cancels the update process.
	1 Admin clicks "Cancel" or navigates away before saving.	
	Post-condition	No changes are made; user information remains unchanged.
Robust Flow	Pre-Condition	System encounters a validation or database error.
	1 If there is a validation error (e.g., invalid email), system prompts for correction.	

	2	If database is unreachable, system shows an error and allows retry or cancel.
	Post-condition	User data is not updated; admin remains on the edit page until resolved.

Table 22 Use case statement - Edit user information

Use Case ID	UC-018	
Use Case Name	Search User	
Description	Allows the Admin to search for a specific user in the system using keywords like name or email.	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	Admin is authenticated and the system has existing users.	
Post-condition	Search results are displayed based on the entered criteria.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin enters a search keyword (e.g., name, email) in the search bar. 2 Admin clicks "Search" or presses Enter. 3 System processes the request and queries the database. 4 System displays a list of matching users.	
	Post-condition	System displays the filtered user list based on the search criteria.
Alternate Flow	Pre-Condition	Search input does not match any user.
	1. System displays "No users found" message.	
	Post-condition	System returns to the search page; no users are displayed..
Robust Flow	Pre-Condition	System encounters input error or database issue.
	1 If search input is invalid (e.g., too short), system displays validation error. 2 If the database is unreachable, system shows error and retry option.	
	Post-condition	No results displayed; admin is notified and remains on the search page.

Table 23 Use case statement - Search user

Use Case ID	UC-019	
Use Case Name	Register Clinic	
Description	Allows the Admin to register a new clinic account in the system by entering clinic details and creating associated login credentials.	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	Admin is authenticated and has permission to register clinics.	
Post-condition	A new clinic account is successfully created and added to the system.	
Main Flow	Step	Action
	Pre-Condition	Admin is authenticated and authorized.
	1 Admin clicks on "Register Clinic" button. 2 System displays clinic registration form. 3 Admin enters required clinic details (e.g., clinic name, address, email, phone number, login credentials). 4 Admin submits the form. 5 System validates the input and creates the clinic account. 6 System displays confirmation message and adds the clinic to the list.	
	Post-condition	Clinic is saved in the system and appears in the clinic list.
Alternate Flow	Pre-Condition	Admin cancels the registration process.
	1. Admin clicks "Cancel" or navigates away before submitting the form.	
	Post-condition	No clinic is registered; form is discarded
Robust Flow	Pre-Condition	System encounters input or database error.
	1 If required fields are missing or invalid, system prompts for correction. 2 If email already exists, system shows duplication error. 3 If the database is unreachable, system shows error and retry/cancel option.	

	Post-condition	Clinic account is not created until errors are resolved.

Table 24 Use case statement - Register Clinic

4.1.5 use case: manage profile

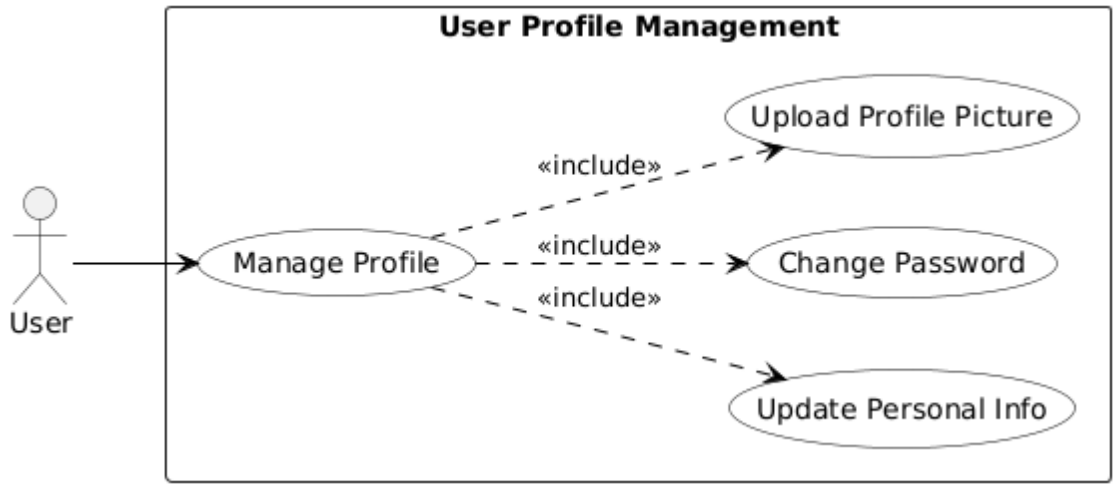


Figure 8 Use case: Manage Profile

Use Case ID	UC-020	
Use Case Name	Update Personal Info	
Description	Allows the user to edit their personal information such as full name, email, phone number, or address.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User is authenticated and has a profile.	
Post-condition	Updated personal info is saved and displayed in the user's profile.	
Main Flow	Step	Action
	Pre-Condition	User is logged into the system.
	1 User navigates to the profile section. 2 User clicks on “Edit Personal Info.” 3 System displays editable fields.	

	4 User updates the desired fields. 5 User clicks "Save." 6 System validates and saves the changes.	
	Post-condition	Updated information is shown in the user profile.
Alternate Flow	Pre-Condition	User cancels the update.
	1. User clicks "Cancel" before saving. 1.	
	Post-condition	No changes are made
Robust Flow	Pre-Condition	Invalid input or server/database issue.
	1 System displays error messages for invalid fields. 2 If the database is unreachable, system shows an error and retry option.	
	Post-condition	Changes are not saved until the issue is resolved.

Table 25 Use case statement - Update personal info

Use Case ID	UC-021	
Use Case Name	Change Password	
Description	This use case allows the user to securely change their password.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User is authenticated.	
Post-condition	Password is updated in the system.	
Main Flow	Step	Action
	Pre-Condition	User is logged into the system.
	1 User goes to the "Change Password" section. 2 User enters current password. 3 User enters and confirms new password. 4 System validates input. 5 System updates the password and shows confirmation.	

	Post-condition	New password is saved and used for future logins..
Alternate Flow	Pre-Condition	User cancels password change.
	1 User clicks "Cancel" before saving.	
	Post-condition	Password remains unchanged.
Robust Flow	Pre-Condition	Input errors or backend failure.
	1 If current password is incorrect, system displays an error. 2 If new password doesn't meet criteria, prompt user to fix. 3 If system error occurs, show retry/cancel options.	
	Post-condition	Password is not changed until errors are resolved.

Table 26 Use case statement - Change password

Use Case ID	UC-022	
Use Case Name	Upload Profile Picture	
Description	Allows the user to upload or change their profile picture.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User is authenticated.	
Post-condition	New profile picture is saved and shown in the user profile.	
Main Flow	Step	Action
	Pre-Condition	User is logged into the system.
	1 User clicks “Upload Profile Picture.” 2 System prompts user to choose an image file. 3 User selects an image and clicks “Upload.” 4 System validates file format and size. 5 System uploads the picture and updates the profile.	
	Post-condition	New profile picture is displayed in the system.
Alternate Flow	Pre-Condition	User cancels the upload
	1 User clicks "Cancel" before saving.	
	Post-condition	No image is uploaded or changed..
Robust Flow	Pre-Condition	Invalid file
	1 If file is not an image or too large, system shows error. 2 If upload fails due to network or server issue, system shows error and retry option.	
	Post-condition	No changes made until a valid image is successfully uploaded.

Table 27 Use case statement - upload profile picture

4.1.6 use case manage feedback

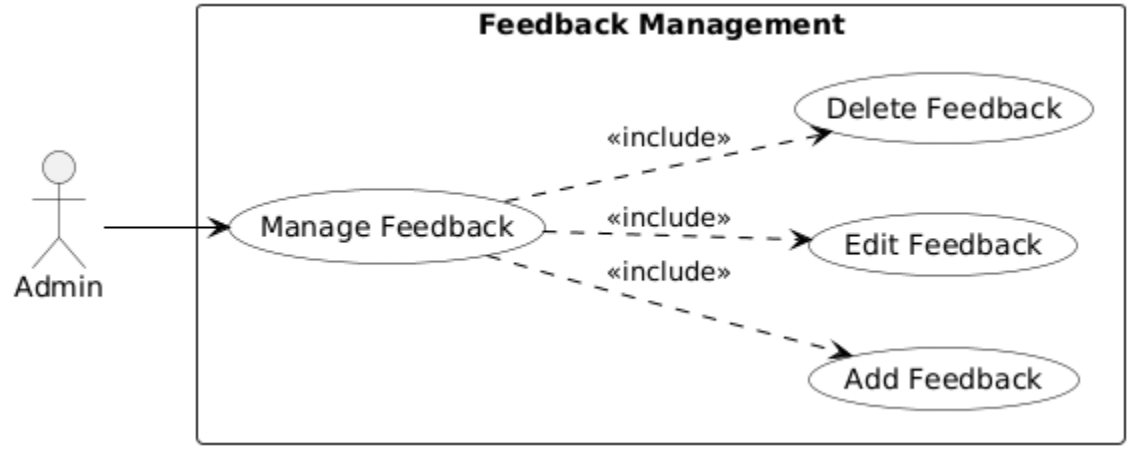


Figure 9 Use case: Manage Feedback

Use Case ID	UC-023	
Use Case Name	Add Feedback	
Description	Allows the Admin to manually add feedback into the system	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	Admin is authenticated.	
Post-condition	New feedback is stored in the system	
Main Flow	Step	Action
	Pre-Condition	User is logged into the system.
	1 Admin clicks "Add Feedback." 2 Admin enters details such as feedback text 3 Admin submits the feedback. 4 System validates and stores the data.	
	Post-condition	System displays success message and shows the added feedback.
Alternate Flow	Pre-Condition	Missing Fields
	1 Admin submits incomplete data. 2 System shows validation errors.	
	Post-condition	No feedback is stored.

Robust Flow	Pre-Condition	System Error
	1 If the system fails to store data, an error message and retry option are shown.	
	Post-condition	Admin remains on the form.

Table 28 Use case statement - Add feedback

Use Case ID	UC-024	
Use Case Name	Edit Feedback	
Description	Allows the Admin to edit existing feedback records linked to object detection results.	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	Admin is authenticated and feedback exists.	
Post-condition	Feedback is updated in the system.	
Main Flow	Step	Action
	Pre-Condition	Admin accesses the feedback list.
	1 Admin selects feedback to edit. 2 Admin updates fields (e.g., text, label, comment). 3 Admin saves the changes. 4 System updates the record.	
	Post-condition	Updated feedback is displayed in the list.
Alternate Flow	Pre-Condition	Cancel Editing
	1. Admin exit without saving	
	Post-condition	No changes made.
Robust Flow	Pre-Condition	Update Error
	1 If saving fails, system shows an error message.	
	Post-condition	Original data remains unchanged.

Table 29 Use case statement - edit feedback

Use Case ID	UC-025
Use Case Name	Delete Feedback

Description	Allows the Admin to remove stored feedback entries that are no longer needed.	
Primary Actor	Admin	
Include use cases	-	
Pre-Condition	Admin is authenticated and feedback exists.	
Post-condition	Feedback entry is deleted from the system.	
Main Flow	Step	Action
	Pre-Condition	Admin is logged in and viewing feedback list.
	1 Admin selects a feedback entry. 2 Admin clicks "Delete." 3 System asks for confirmation. 4 Admin confirms deletion. 5 System removes the feedback.	
	Post-condition	Feedback is no longer visible in the list.
Alternate Flow	Pre-Condition	Cancel Deletion
	1. Admin cancels the confirmation prompt.	
	Post-condition	Feedback remains unchanged.
Robust Flow	Pre-Condition	Deletion Failure
	1 If deletion fails, system displays an error.	
	Post-condition	Feedback remains in the system.

Table 30 Use case statement - Delete feedback

4.1.7 use case progress tracking

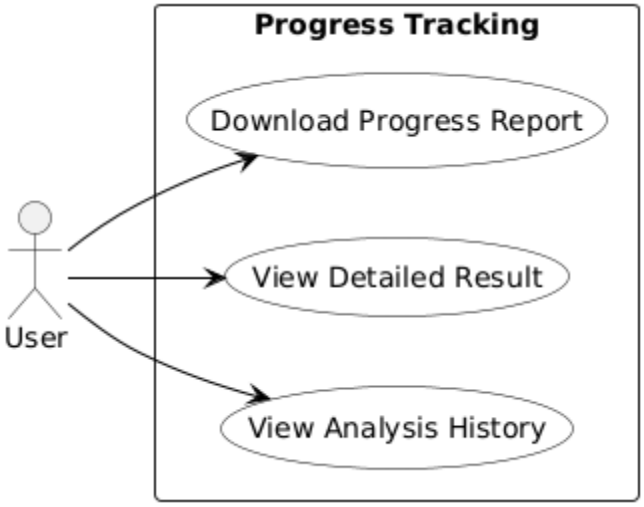


Table 31 Use case: Track Progress

Use Case ID	UC-026	
Use Case Name	View Analysis History	
Description	Allows the user to view a list of previously analyzed images along with the result summary.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User is authenticated and has at least one analysis history.	
Post-condition	A list of past analyses is displayed.	
Main Flow	Step	Action
	Pre-Condition	User is logged in.
	1 User navigates to "Progress" page. 2 System retrieves analysis history. 3 System displays each record (date, result, image thumbnail).	
	Post-condition	User sees a list of past analyses.
Alternate Flow	Pre-Condition	**No history exists**
	System shows “No history available” message.	
	Post-condition	User remains on same page.

Robust Flow	Pre-Condition	**Database is unreachable**
	System shows error message.	
	Post-condition	User stays on page with retry option .

Table 32 Use case statement - view analysis history

Use Case ID	UC-027	
Use Case Name	View Detailed Result	
Description	Allows the user to click on an individual analysis entry to view full detection results and feedback.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User has selected an analysis entry.	
Post-condition	Full result and image are shown.	
Main Flow	Step	Action
	Pre-Condition	User is logged in.
	1 User selects an entry from the history. 2 System fetches detailed result (condition, severity, feedback). 3 System displays detailed result and associated image.	
	Post-condition	Detailed data is shown.
Alternate Flow	Pre-Condition	**Selected entry is missing or corrupted**
	1 System notifies user of issue.	
	Post-condition	User remains on same page.
Robust Flow	Pre-Condition	**Server error occurs**

	1 System shows error with retry button.	
	Post-condition	User remains on previous page.

Table 33 Use case statement - View detailed history

Use Case ID	UC-028	
Use Case Name	Download Progress Report	
Description	Allows the user to download a PDF report of their past dental analyses.	
Primary Actor	User	
Include use cases	-	
Pre-Condition	User has at least one analysis history.	
Post-condition	File is downloaded successfully.	
Main Flow	Step	Action
	Pre-Condition	User has at least one analysis history.
	1 User clicks on "Download Report" button. 2 System generates a PDF summary of analysis results. 3 System downloads the file to user's device.	
	Post-condition	User receives the file.
Alternate Flow	Pre-Condition	**No history available**
	1 System disables button or shows "No data" alert.	
	Post-condition	User remains on same page.
Robust Flow	Pre-Condition	

		PDF generation fails
	1 System shows error with retry button.	
	Post-condition	User remains on page with retry option.

Table 34 Use case statement - Download Progress Report

4.2 Test Case Specification

4.2.1 Test case : Register Management

Test Cases for UC-001: Register User

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		

Test Case Number	TC001.1			
Test Case Name	Functionality Test Case for User Registration			
Requirement Traceability	REQ_UR101			
Test Case Description	To verify that a new user can successfully register using valid details and is redirected to the login page upon successful registration			
Item(s) to be tested				
1	Registration form fields			
2	Form Validation logic (email unique , matching passwords)			
3	Database entry creation			
4	Successful messages and redirection to login page			
Pre-condition	1.User does not already have an account. 2.User is on the registration page.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User Inputs valid registration details	Full Name : Ammar Syahansyah Email : ammar@gmail.com Password : @TEst123 Confirm Password : @TEst123	System validates the input create a new user record, display a successful message and redirects to login page		

Post Condition	1.A new user account is successfully created and stored in the database 2.User is redirected to the login page.

Table 35 Test Case Specification - Functionality Test Case for user Registration

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC001.2		
Test Case Name	UI Validation Test Case for Required Fields		
Requirement Traceability	REQ_UR101		
Test Case Description	To ensure the registration form does not allow submission if any mandatory field is left blank.		
Item(s) to be tested			
1	Required field validation		
2	Error message display on incomplete submission		
Pre-condition	1. User is on the registration page 2. User has not filled all required fields		
Test Procedure Specification			

Input	Test Data	Expected Output/Result	Actual Result	Pass/Fail
User leaves email blank	Full Name : Ammar Syahansyah Email : Password : @TEst123 Confirm Password : @TEst123	System highlights the email field and displays :"Email is required"		
User leaves password blank	Full Name : Ammar Syahansyah Email : ammar@gmail.com Password : Confirm Password :	System highlights the password field and displays: "Pass word is required"		
Post Condition	1.Form is not submitted 2.User is prompted to fill all mandatory fields			

Table 36 Test Case Specification - UI validation Test Case for Required Fields

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	5 (Highest)		
Test Case Number	TC001.3		
Test Case Name	SQL Injection Input Validation Test		

Requirement Traceability	REQ_UR101			
Test Case Description	To test whether the registration form properly sanitizes malicious SQL input in the email or username field.			
Item(s) to be tested				
1	SQL injection protection			
2	Input sanitation mechanisms			
Pre-condition	1. User is on registration page 2. User attempts malicious input			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User enters SQL in email field	Full Name : Ammar Syahansyah Email : '1' = '1' Password : @TEst123 Confirm Password : @TEst123	System treats input as plain text, displays "Invalid email format" , and prevents registration		
Post Condition	1.System remains secure and no database operations are altered 2.Malicious code is not executed			

Table 37 Test Case Specification - input validation test

Test Cases for UC-002 : Login

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		
Test Case Number	TC002.1		
Test Case Name	Functionality Test Case for User/Admin Login		
Requirement Traceability	REQ_UR201		
Test Case Description	To verify that users and admins can log in using valid credentials and are redirected to the appropriate main page based on their role.		

Item(s) to be tested	
1	Login form fields (username and password
2	Credential validation logic
3	Role based redirection (User/Admin)
4	Error handling for invalid login
5	Password recovery process

Pre-condition	1.User does already have an account. 2.User is on the login page.

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

Registered user enters valid login details	Email : ammar@gmail.com Password : @TEst123	System authenticates user and redirects to user dashboard		
Admin enters valid login details	Email : Admin101 Password : @ADmin123	System authenticates admin and redirects to admin dashboard		
Post Condition	1.The user is authenticated 2.Redirected to their respective main page			

Table 38 Test Case Specification - Functionality Test Case for User/Admin Login

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Medium)		
Test Case Number	TC002.2		
Test Case Name	Forgot Password Functionality Test		
Requirement Traceability	REQ_UR201		
Test Case Description	To verify the password recovery process via email.		
Item(s) to be tested			

1	Forgot password link
2	Email input validation
3	Reset link delivery
4	New password setup

Pre-condition	1. User is on login page and selects “Forgot Password”.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User clicks “Forgot Password” and enters registered email	Email : ammar@gmail.com	System sends a reset link to the email		
User clicks “Forgot Password” and enters unregistered email	Email : Axxx@gmail.com	System displays: “Email not found in our records.”		

Post Condition	1. Password reset link is sent only to registered users. 2. New password setup works without compromising security.

Table 39 Test Case specification - Forgot Password Functionality Test

Test case for UC-003:Logout

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		

Risk Number		1 (Lowest)		
Test Case Number		TC003.1		
Test Case Name		Functionality Test Case for User Logout		
Requirement Traceability		REQ_UR101		
Test Case Description		To verify that a logged-in user can successfully log out and is redirected to the login page		
Item(s) to be tested				
1	Logout button functionality			
2	Session termination			
3	Redirection to the login page			
4	Restriction of access to authenticated pages after logout			
Pre-condition	1. User is already logged in to the system 2. User is on the main dashboard or any authenticated page			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User clicks the Logout button	Click on “Logout” on the navigation bar	System ends user session, displays a logout confirmation or redirects to login page, and prevents access		

		to authenticated pages		
Post Condition	1 User session is terminated 2 User is redirected to the login page 3 Access to authenticated pages is blocked unless the user logs in again			

Table 40 Test Case Specification - Functionality Test Case for User Logout

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		
Test Case Number	TC003.2		
Test Case Name	Alternate Flow Test Case for Logout after Session Expiry		
Requirement Traceability	REQ_UR101		
Test Case Description	To verify that the system automatically redirects the user to the login page when the session has expired and a protected page or logout function is accessed		
Item(s) to be tested			
1	Session expiration handling		
2	Automatic redirection to login page		

3	Restriction of access to protected pages without a valid session
Pre-condition	1. User was previously logged in 2. Session has expired due to inactivity 3. User attempts to access a protected page or click the logout button

Test Procedure Specification

Input	Test Data	Expected Output/Result	Actual Result	Pass/Fail
User accesses a protected page or clicks Logout after session has expired	Simulate session timeout due to inactivity, then click a protected route or Logout	System detects expired session, securely terminates it, and redirects to login page with optional timeout message		

Post Condition	1 User is securely logged out 2 User is redirected to the login page 3 Access to authenticated pages remains blocked until re-authentication

Table 41 Test Case Specification - Alternate Flow Test Case for Logout after Session Expiry

4.2.2 Test case : Analyze Image

Test Cases for UC-003 : Upload image

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		
Test Case Number	TC003.1		
Test Case Name	Functionality Test Case for Image Upload		
Requirement Traceability	REQ_UR301		
Test Case Description	To verify that a logged-in user can upload a valid intraoral image, and the system processes it correctly for AI analysis.		

Item(s) to be tested	
1	File selection and upload
2	File validation logic (type/size)
3	Upload to storage (cloud/local)
4	AI model processing trigger
5	Success notification

Pre-condition	1. User is logged in 2. User is on the upload image page
Test Procedure Specification	

Input	Test Data	Expected Output/Result	Actual Result	Pass/Fail
User uploads valid image	File: teeth_scan.jpg (JPEG, 1.8 MB)	System accepts the image, stores it, triggers AI analysis, and displays a success message		
Post Condition	1. Image is stored 2. Analysis is triggered 3. Feedback/result generated			

Table 42Test Case Specification - Functionality Test Case for Image Upload

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC003.2		
Test Case Name	Validation for Invalid File Types		
Requirement Traceability	REQ_UR302		
Test Case Description	To verify that the system blocks invalid file types and prompts re-upload.		
Item(s) to be tested			

1	File format validation
2	Error messaging for invalid formats

Pre-condition	1. User is logged in 2. User is on the upload image page

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User uploads PDF	File: report.pdf	System displays: "Invalid file type. Please upload an image (JPG, PNG)."		

Post Condition	1. No image is stored, and the user is prompted to upload a valid image.

Table 43 Test Case Specification - Validation for Invalid File Types

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Medium)		
Test Case Number	TC003.3		
Test Case Name	Oversized Image Upload Rejection		

Requirement Traceability	REQ_UR303			
Test Case Description	To ensure that the system restricts upload of large image files (e.g., >5MB).			
Item(s) to be tested				
1	File size validation			
2	Error handling for oversize uploads			
Pre-condition	1. User is logged in 2. User is on the upload image page			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Upload large image	File: ultra_hd_scan.png (6.2 MB)	System displays: "File size exceeds limit (Max 5MB). Please upload a smaller image."		
Post Condition	1. Upload rejected, and user must upload a smaller file.			

Table 44 Test Case Specification - Oversized Image Upload Rejection

Table 45 Test Case Specification - Upload Failure Due to Server/Network Error

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		

Test Design Technique	Use Case Testing			
Risk Number	4 (High)			
Test Case Number	TC003.4			
Test Case Name	Upload Failure Due to Server/Network Error			
Requirement Traceability	REQ_UR304			
Test Case Description	To test system resilience in the event of an upload failure due to backend or network issues.			
Item(s) to be tested				
1	Backend communication			
2	Graceful error messages			
3	Retry functionality			
Pre-condition	1. User is logged in and tries to upload during a backend failure or slow network.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Upload with backend error	File: scan.jpg (normal size)	System displays: "Upload failed. Please try again later."		
Post Condition	1. No data is stored, user is notified, and can retry or contact support.			

Table 46 Test Case Specification - Upload Failure Due to Server/Network Error

Table 47 Test Case Specification - Functional Test Case for Image Analysis

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC004.1		
Test Case Name	Functional Test Case for Image Analysis		
Requirement Traceability	REQ_UR401		
Test Case Description	To verify that the system correctly processes a valid intraoral image using the AI model and displays the detected dental conditions to the user.		

Item(s) to be tested	
1	Image handoff to AI model
2	AI inference process
3	Result generation with labels and confidence
4	Result storage in database
5	Display of feedback to user

Pre-condition	1. A valid intraoral image has been uploaded and linked to a user account

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

Valid image is processed by AI	File: teeth_01.jpg	System returns results (e.g., "Cavity: 85%", "Gum Inflammation: 78%"), stores in DB, and displays to user		
Post Condition	1. AI result is stored and linked to the user; feedback is shown in UI.			

Table XX Test Case Specification – Functional Test Case for Image Analysis

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		
Test Case Number	TC004.2		
Test Case Name	Analysis Result – No Condition Detected		
Requirement Traceability	REQ_UR401		
Test Case Description	To ensure the system can gracefully handle cases where no dental issues are detected in the image.		
Item(s) to be tested			
1	AI result interpretation logic		
2	User notification with neutral results		
Pre-condition	1. A valid image is submitted and analyzed		

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Image with no issues	File: teeth_01.jpg	System displays: "No dental conditions detected."		
Post Condition	1. Message is displayed to the user, indicating a clean scan.			

Table 48 Test Case Specification - Analysis Result – No Condition Detected

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	4 (High)		
Test Case Number	TC004.3		
Test Case Name	AI Analysis Failure Handling		
Requirement Traceability	REQ_UR401		
Test Case Description	To verify that the system handles AI model failures or image corruption gracefully without crashing.		
Item(s) to be tested			
1	Error detection in model pipeline		

2	Logging and user feedback
3	Retry options for the user

Pre-condition	1. An uploaded image is corrupted or the AI model encounters a failure

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Corrupted image or server issue	File: teeth_x.jpg	System displays: "Analysis failed. Please re-upload the image or try again later." Logs error.		

Post Condition	1. System does not crash, logs the issue, and user is allowed to retry.

Table 49 Test Case Specification - AI Analysis Failure Handling

Test Cases for UC-005 : View Feedback

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		
Test Case Number	TC005.1		
Test Case Name	Functional Test Case for Viewing Feedback		

Requirement Traceability	REQ_UR501			
Test Case Description	To verify that a user can successfully view feedback generated from previous image analyses.			
Item(s) to be tested				
1	Feedback section navigation			
2	Retrieval of feedback records			
3	Display of feedback details (text, date, suggestions)			
4	Database-to-UI mapping			
Pre-condition	1. User is logged in 2. User has completed at least one image analysis with available feedback			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User accesses Feedback section	Logged-in as: ammar@gmail.com	System retrieves and displays: • Cavity: 85% • Suggested Treatment: Filling • Date: 2025-05-10		
Post Condition	1. Feedback is visible and accessible to the user at any time.			

Table 50 Test Case Specification - Functional Test Case for Viewing Feedback

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC005.2		
Test Case Name	No Feedback Available Case		
Requirement Traceability	REQ_UR501		
Test Case Description	To ensure the system gracefully informs the user when no feedback records are available.		

Item(s) to be tested	
1	Feedback availability check
2	Empty state messaging

Pre-condition	1. User has not completed any image analysis yet.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Access Feedback with no prior analysis	Logged-in as: newuser@gmail.com	System displays message: "No feedback available yet."		

Post Condition	1. User is prompted to upload an image or return to dashboard.

Table 51 Test Case Name Test Case Specification -No Feedback Available Case

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Medium)		
Test Case Number	TC005.3		
Test Case Name	Feedback Retrieval Failure Handling		
Requirement Traceability	REQ_UR501		
Test Case Description	To verify the system's behavior when a database or system error occurs while fetching feedback.		

Item(s) to be tested	
1	Error handling during database fetch
2	System error logging
3	User feedback and retry option

Pre-condition	1. User has completed analysis 2. A database issue or connectivity failure occurs

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

Attempt to access feedback during DB failure	Existing user with records	System displays: "Error retrieving feedback. Please try again later." Logs error internally		
Post Condition	1. User is given a retry option and system remains stable and secure.			

Table 52 Test Case Specification - Feedback Retrieval Failure Handling

Test Cases for UC-006 : Generate Feedback

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		
Test Case Number	TC006.1		
Test Case Name	Functional Test Case for Feedback Generation		
Requirement Traceability	REQ_UR601		
Test Case Description	To verify that the system successfully generates feedback based on AI image analysis and stores it in the database.		
Item(s) to be tested			
1	Triggering AI image analysis		
2	Feedback generation based on diagnosis		

3	Storing feedback in the database
4	Linking feedback to user profile

Pre-condition	1. User has uploaded a valid intraoral image 2. Image analysis completes successfully

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User uploads a valid image for analysis	Image: img_tooth_01.jpg	System generates feedback: • Condition: "Cavity detected (85%)" • Recommendation: "Filling required" • Feedback saved and accessible		

Post Condition	1. Feedback is generated, stored in the system, and linked to the user's profile

Table 53 Test Case Specification - Functional Test Case for Feedback Generation

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		

Risk Number	3 (Medium)			
Test Case Number	TC006.2			
Test Case Name	Handling Invalid Image and AI Model Errors			
Requirement Traceability	REQ_UR601			
Test Case Description	To ensure the system handles invalid images or internal AI processing errors without crashing.			
Item(s) to be tested				
1	File format and clarity validation			
2	AI model runtime exception handling			
3	Error messaging and retry flow			
Pre-condition	1. User uploads an invalid or corrupted image			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Upload unsupported file	Image: img_tooth_01.pdf	System throws: "Unsupported image format. Please upload a valid intraoral image."		
AI model fails internally	Image causes unexpected AI error	System logs error and shows: "Feedback generation failed. Please try again later."		

Post Condition	1. User is prompted to retry. System remains stable and logs the failure for administrative review.

Table 54 Test Case Specification - Handling Invalid Image and AI Model Errors

Table XX Test Case Specification – Handling Invalid Image and AI Model Errors

4.2.3 Test case : Booking Appointment

Test Cases for UC-007 : View Schedule

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		
Test Case Number	TC007.1		
Test Case Name	Functionality Test Case for Viewing Appointment Schedule		
Requirement Traceability	REQ_UR701		
Test Case Description	To verify that a logged-in user can view available appointment slots provided by the clinic.		

Item(s) to be tested				
1	Appointment schedule access			
2	Data retrieval from the schedule database			
3	Display of available time slots			
Pre-condition	1. User is logged into the system 2. The clinic has available appointment slots in the system			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User clicks "View Schedule"	Logged-in user	System retrieves and displays available slots: e.g., "15 May 2025, 10:00 AM – Dr. Farah"		
Post Condition	1. The user sees a list of available appointment times and can proceed to book.			

Table 55 Test Case Specification - Functionality Test Case for Viewing Appointment Schedule

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC007.2		

Test Case Name	No Available Appointment Slots Handling			
Requirement Traceability	REQ_UR701			
Test Case Description	To verify that the system handles situations where no appointment slots are available and provides proper user feedback.			
Item(s) to be tested				
1	Schedule lookup when empty			
2	Message display logic when no slots are found			
Pre-condition	1. User is logged into the system 2. The clinic has no available appointment slots in the system			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User clicks "View Schedule"	Schedule table is empty	System shows message: "No appointments available."		
Post Condition	1. User is informed of the unavailability and may try again later.			

Table 56 Test Case Specification - No Available Appointment Slots Handling

Test Cases for UC-008 : Book Appointment

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		

Test Case Number	TC008.1			
Test Case Name	Functionality Test Case for Booking Appointment			
Requirement Traceability	REQ_UR801			
Test Case Description	To verify that a logged-in user can book an appointment using an available time slot and that the system properly logs and notifies the clinic.			
Item(s) to be tested				
1	Date/time selection logic			
2	Booking request submission			
3	Database update and clinic notification			
Pre-condition	1. User is logged into the system 2. User has already viewed the available schedule			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User selects an available slot and clicks "Book"	Date: 20 May 2025, Time: 2:00 PM	System confirms booking, sets status to "Pending Approval", notifies clinic		
Post Condition	1. Appointment is stored in the database with a status of "Pending Approval", and the clinic receives the request.			

Table 57 Test Case Specification - Test Case Name Functionality Test Case for Booking Appointment

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC008.2		
Test Case Name	Handling Booking Conflict for Time Slot		
Requirement Traceability	REQ_UR801		
Test Case Description	To ensure the system properly handles booking conflicts when the selected time slot becomes unavailable before submission is completed.		

Item(s) to be tested	
1	Real-time validation of slot availability
2	Error message display
3	Prompt to reselect slot

Pre-condition	1. User is logged into the system 2. Time slot becomes unavailable due to another user booking the same slot

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User submits a slot already booked by another user	Date: 20 May 2025, Time: 2:00 PM	System shows error: "Selected time slot is no longer available.		

		Please choose another."		
Post Condition	1. User is prompted to select a different available time slot.			

Table 58 Test Case Specification - Handling Booking Conflict for Time Slot

Test Cases for UC-009 : Set Reminder

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Lowest)		
Test Case Number	TC009.1		
Test Case Name	Functionality Test Case for Setting Appointment Reminder		
Requirement Traceability	REQ_UR901		
Test Case Description	To verify that the user can set a reminder for an approved appointment and that the system saves or syncs the reminder correctly.		
Item(s) to be tested			
1	Viewing approved appointments		
2	Reminder time selection		
3	Reminder calendar creation/sync		
Pre-condition	1. User is logged into the system 2. User has at least one approved appointment		

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User selects "Set Reminder" for an approved appointment	Date: 20 May 2025, Time: 2:00 PM	Reminder is added to calendar or app's notification system		
Post Condition	1. A calendar or system reminder is created based on user preference.			

Table 59 Test Case Specification - Functionality Test Case for Setting Appointment Reminder

Table XX Test Case Specification – Handling Booking Conflict for Time Slot

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC009.2		
Test Case Name	Handling Reminder Setup with No Approved Appointment		
Requirement Traceability	REQ_UR901		
Test Case Description	To ensure the system does not allow reminder setup if no appointments have been approved yet.		
Item(s) to be tested			
1	Validation for reminder eligibility		
2	System notification on unapproved status		

Pre-condition	1. User is logged into the system 2. No approved appointments available			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User attempts to set a reminder without any approved appointment		System displays: "No approved appointments found. Cannot set reminder."		
Post Condition	1. System prevents invalid reminder setup and suggests booking or waiting for approval.			

Table 60 Test Case Specification - Test Case Name Handling Reminder Setup with No Approved Appointment

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Medium)		
Test Case Number	TC009.3		
Test Case Name	Reminder Setup Failure Due to Calendar/Sync Error		
Requirement Traceability	REQ_UR901		
Test Case Description	To verify how the system handles errors when it cannot create or sync the reminder due to technical issues.		
Item(s) to be tested			

1	Calendar integration failure handling
2	User error messaging

Pre-condition	1. Appointment is approved 2. Calendar sync service fails

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User tries to set reminder but calendar sync fails	Appointment Date: 23 May 2025	System shows: "Failed to set reminder. Please try again later or contact support."		

Post Condition	1. System logs error and allows retry or alternate notification method.

Table 61 Test Case Specification - Reminder Setup Failure Due to Calendar/Sync Error

Test Cases for UC-010 : Cancel/Reschedule Appointment

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC010.1		
Test Case Name	Cancel Appointment Functionality		
Requirement Traceability	REQ_UR1001		

Test Case Description	To verify that the user can cancel an existing appointment and the system removes it successfully from the schedule.			
Item(s) to be tested				
1	Display of upcoming appointments			
2	Appointment cancellation			
3	Appointment removal confirmation			
Pre-condition	1. User is logged in 2. User has at least one existing appointment			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User selects "Cancel Appointment"	Appointment Date: 22 May 2025	System removes the appointment and shows: "Appointment successfully canceled."		
Post Condition	1. Appointment is removed and not listed under upcoming appointments.			

Table 62 Test Case Specification - Cancel Appointment Functionality

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		

Test Case Number	TC010.2			
Test Case Name	Reschedule Appointment Functionality			
Requirement Traceability	REQ_UR1001			
Test Case Description	To verify that the user can reschedule an existing appointment to a new available time slot.			
Item(s) to be tested				
1	Selection of new time slot			
2	Validation of new slot availability			
3	Rescheduling request handling			
Pre-condition	1. User has an existing appointment 2. New appointment slot is available			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User selects "Reschedule" and picks a new slot	New Date: 25 May 2025	System updates the appointment and shows: "Appointment successfully rescheduled."		
Post Condition	1. The appointment is rescheduled to the new time and stored accordingly.			

Table 63 Test Case Specification - Reschedule Appointment Functionality

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing

Test Type	Black Box Testing – Functional Testing			
Test Design Technique	Use Case Testing			
Risk Number	3 (Medium)			
Test Case Number	TC010.3			
Test Case Name	Reschedule Attempt with No Available Slots			
Requirement Traceability	REQ_UR1001			
Test Case Description	To verify the system's behavior when no available slots exist during a reschedule attempt.			
Item(s) to be tested				
1	Slot availability validation			
2	Error messaging			
Pre-condition	1. User is logged in 2. No rescheduling slots are available			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User tries to reschedule appointment	Selected Date: Fully Booked	System shows message: "No available time slots. Please try again later."		
Post Condition	1. User remains on reschedule screen with option to try again later.			

Table 64 Test Case Specification - Reschedule Attempt with No Available Slots

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	4 (High)		
Test Case Number	TC010.4		
Test Case Name	Error Handling During Appointment Update		
Requirement Traceability	REQ_UR1001		
Test Case Description	To ensure system handles server or network failures gracefully during cancel or reschedule operations.		

Item(s) to be tested	
1	Server/network error handling
2	User-facing error notification
3	Retry mechanism

Pre-condition	1. User initiates a cancel/reschedule 2. Network/server error occurs

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User cancels or reschedules appointment	Simulated network timeout	System displays: "Failed to update appointment. Please try again later."		

Post Condition	1. User is notified of the error and may retry the operation later.

Table 65 Test Case Specification - Error Handling During Appointment Update

Test Cases for UC-011 : Approve Appointment

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing	
Test Type	Black Box Testing – Functional Testing			
Test Design Technique	Use Case Testing			
Risk Number	2 (Low)			
Test Case Number	TC011.1			
Test Case Name	Approve Appointment Functionality			
Requirement Traceability	REQ_UR1101			
Test Case Description	To verify that the clinic can approve a pending appointment and the user is notified.			
Item(s) to be tested				
1	Appointment listing in "Pending Appointments"			
2	Approval mechanism			
3	User notification of approval			
Pre-condition	1. Clinic staff is logged in 2. At least one pending appointment exists			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

Clinic selects and approves appointment request	Request ID: APT-0081	System marks the appointment as "Confirmed" and notifies the user		
Post Condition	1. Appointment is confirmed and shown under approved/active appointments.			

Table 66 Test Case Specification - Approve Appointment Functionality

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Medium)		
Test Case Number	TC011.2		
Test Case Name	Reject Appointment Functionality		
Requirement Traceability	REQ_UR1101		
Test Case Description	To verify that the clinic can reject an appointment request and the user is properly notified.		
Item(s) to be tested			
1	Appointment rejection process		
2	Notification system for user		
Pre-condition	1. At least one pending appointment exists 2. Clinic staff is logged in		

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Clinic selects and rejects appointment	Request ID: APT-0082	System deletes or archives the appointment and notifies user of rejection		
Post Condition	1. The request is no longer pending or visible; user is informed.			

Table 67 Test Case Specification - Reject Appointment Functionality

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC011.3		
Test Case Name	Empty Pending Appointments View		
Requirement Traceability	REQ_UR1101		
Test Case Description	To verify that the system handles cases with no pending appointments and notifies the clinic appropriately.		
Item(s) to be tested			
1	Pending request view		
2	Error/empty messaging		
Pre-condition	1. Clinic is logged in		

	2. No pending requests exist			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Clinic opens "Pending Appointments"	No pending data	System displays: “No pending requests.”		
Post Condition	1. The clinic is informed that no actions are needed.			

Table 68 Test Case Specification - Empty Pending Appointments View

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	4 (High)		
Test Case Number	TC011.4		
Test Case Name	Error Handling During Appointment Approval		
Requirement Traceability	REQ_UR1101		
Test Case Description	To verify that the system properly handles failure scenarios during the approval/rejection process.		
Item(s) to be tested			
1	Server error handling		
2	Error message displayed to clinic staff		
3	Logging and retry mechanism		

Pre-condition	1. A request is selected 2. Backend failure simulated

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Clinic tries to approve request	Simulate DB connection failure	System shows: "Error approving appointment. Please try again."		

Post Condition	1. System logs the error and allows retry.

Table 69 Test Case Specification - Error Handling During Appointment Approval

Test Cases for UC-012 : Manage Clinic Schedule

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Medium)		
Test Case Number	TC012.1		
Test Case Name	Modify Clinic Schedule		
Requirement Traceability	REQ_UR1201		
Test Case Description	To verify that the clinic can view and modify their schedule (add/update/delete slots) and changes reflect in the user-facing system.		

Item(s) to be tested				
1	Access to schedule management interface			
2	Schedule modification (add, edit, delete)			
3	Real-time update in user system			
Pre-condition	1. Clinic staff is logged in 2. Has access to schedule management panel			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Add new slot	Date: 20-May-2025, Time: 10:00 AM	Slot is added and visible to users		
Edit existing slot	Change 9:00 AM to 10:30 AM	Time is updated in user booking view		
Delete slot	Remove 2:00 PM slot	Slot no longer appears to users		
Post Condition	1. Appointment is confirmed and shown under approved/active appointments.			

Table 70 Test Case Specification - Modify Clinic Schedule

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		

Risk Number	4 (High)			
Test Case Number	TC012.2			
Test Case Name	Attempt to Modify Booked Slot			
Requirement Traceability	REQ_UR1201			
Test Case Description	To verify that the system prevents schedule changes to slots with existing appointments and prompts the clinic for manual intervention.			
Item(s) to be tested				
1	Booking conflict detection			
2	Prompting with appropriate messages			
Pre-condition	1. A confirmed appointment exists for the selected slot			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Edit slot with appointment	Date: 22-May-2025, Time: 3:00 PM	System prevents update, shows "Slot has active appointment"		
Post Condition	1. Clinic is prompted to handle appointment before slot modification.			

Table 71 Test Case Specification - Attempt to Modify Booked Slot

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		

Test Design Technique	Use Case Testing			
Risk Number	5 (Highest)			
Test Case Number	TC012.3			
Test Case Name	Server Error on Load			
Requirement Traceability	REQ_UR1201			
Test Case Description	To verify system behavior when the schedule page fails to load due to a backend/server issue.			
Item(s) to be tested				
1	Error handling during server failure			
2	Logging and messaging			
Pre-condition	1. Clinic accesses schedule panel 2. Simulated backend failure occurs			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Load schedule page	Simulated server downtime	System displays: "Unable to load schedule. Please try again later."		
Post Condition	1. System logs the error and gracefully informs the user.			

Table 72 Test Case Specification - Server Error on Load

4.2.4 Test case : Manage user

Test Cases for UC-013 : Login

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC013.1		
Test Case Name	Successful Admin Login		
Requirement Traceability	REQ_UR1301		
Test Case Description	Verify that an admin with valid credentials can successfully log in and access the dashboard.		

Item(s) to be tested	
1	Admin login form
2	Credential verification
3	Dashboard redirection

Pre-condition	1. Admin must be pre-registered with a valid username and password.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Username	admin001	Username is accepted		
Password	Admin001	Password is verified, dashboard loads		

Post Condition	Admin is authenticated and granted access to the admin dashboard.

Table XX Test Case Specification – Successful Admin Login

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Medium)		
Test Case Number	TC013.2		
Test Case Name	Login with Invalid Credentials		
Requirement Traceability	REQ_UR1301		
Test Case Description	To verify that the system rejects invalid credentials and prompts the admin to try again or reset password.		

Item(s) to be tested	
1	Login input validation
2	Error handling for incorrect login

Pre-condition	1. Admin account exists but incorrect credentials are entered.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Username and password	Username : admin001 Password : WrongAdmin	"Invalid username or password" message is displayed		

Post Condition	1. Admin is prompted to re-enter credentials or initiate password reset.

Table XX Test Case Specification – Login with Invalid Credentials

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	4 (High)		
Test Case Number	TC013.3		
Test Case Name	Lockout After Failed Logins		
Requirement Traceability	REQ_UR1301		
Test Case Description	To verify that the system locks the admin account after multiple failed login attempts and sends a security alert.		

Item(s) to be tested

1	Login failure counter
2	Lockout mechanism
3	Email notification

Pre-condition	1. Admin fails to log in more than 5 times consecutively.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
6x incorrect passwords	Username : admin001 Password : WrongAdmin	System locks account and sends email notification		

Post Condition	1. Admin remains locked out and access is restricted temporarily as per security protocol.

Table XX Test Case Specification – Lockout After Failed Logins

Test Cases for UC-014 : Manage User

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing	
Test Type	Black Box Testing – Functional Testing			
Test Design Technique	Use Case Testing			
Risk Number	3 (Medium)			
Test Case Number	TC014.1			
Test Case Name	View All Users			
Requirement Traceability	REQ_UR1301			
Test Case Description	Verify that the admin can successfully view a list of all registered users after logging in.			
Item(s) to be tested				
1	Access to "Manage Users" section			
2	Display of registered users list			
Pre-condition	1. Admin is logged into the system.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Admin Action	Navigate to Manage Users	User list is displayed with management options		
Post Condition	1. Admin sees a list of registered users.			

Table 73 Test Case Specification - Test Case Name View All Users

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	4 (High)		
Test Case Number	TC014.2		
Test Case Name	Delete User		
Requirement Traceability	REQ_UR1301		
Test Case Description	Verify that an admin can delete a user account from the system.		

Item(s) to be tested	
1	User selection
2	Delete operation
3	Confirmation message

Pre-condition	1. Admin is logged in and user list is available.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Admin selects user “user123”	Clicks "Delete"	System removes user and confirms deletion		

Post Condition	1. User account is deleted from the system.

Table 74 Test Case Specification - Delete User

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	4 (High)		
Test Case Number	TC014.3		
Test Case Name	Reset User Password		
Requirement Traceability	REQ_UR1301		
Test Case Description	Verify that the admin can reset a user's password and notify the user.		

Item(s) to be tested	
1	Password reset functionality
2	Notification system
3	Confirmation message

Pre-condition	1. Admin is logged in and user list is available.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Admin selects user "user123"	Click "Reset Password"	Password is reset and user is notified		

Post Condition	1. User's password is reset and an email/SMS is sent to notify them.

Table XX Test Case Specification – Reset User Password

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Negative Test Case		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC014.4		
Test Case Name	No Users Found		
Requirement Traceability	REQ_UR1301		
Test Case Description	Verify the system's response when there are no registered users.		

Item(s) to be tested	
1	User list retrieval
2	Empty state message

Pre-condition	1. Admin is logged in and user list is available.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Navigate to Manage Users		System displays "No users found."		

Post Condition	1. System notifies admin that there are no users. No action taken.

Table XX Test Case Specification – Reset User Password

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing

Test Type	Negative Test Case			
Test Design Technique	Use Case Testing			
Risk Number	5 (High)			
Test Case Number	TC014.5			
Test Case Name	Server Error Handling			
Requirement Traceability	REQ_UR1301			
Test Case Description	Ensure system displays an error if the user list cannot be retrieved due to a server issue.			
Item(s) to be tested				
1	Error handling			
2	Retry mechanism			
Pre-condition	1. System experiences server-side error when fetching user data.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Navigate to Manage Users	Simulate backend error	System shows error message and retry suggestion		
Post Condition	1. No user management actions are performed; admin is advised to try later.			

Table XX Test Case Specification – Server Error Handling

Test Cases for UC-015 : View User List

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Medium)		
Test Case Number	TC015.1		
Test Case Name	Display All Registered Users		
Requirement Traceability	REQ_UR1501		
Test Case Description	Verify that an authenticated admin can view a list of all registered users.		

Item(s) to be tested	
1	Navigation to "View User List"
2	User list display

Pre-condition	1. Admin is logged in and authorized.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Click "View User List"	Authenticated session	System displays all registered users in a table format		

Post Condition	User list is visible to admin.

Table 75 Test Case Specification - Display All Registered Users

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Negative Test Case		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC015.2		
Test Case Name	No Registered Users		
Requirement Traceability	REQ_UR1501		
Test Case Description	Verify system behavior when no users are registered.		

Item(s) to be tested	
1	User list retrieval
2	Empty state handling

Pre-condition	1. System has no users registered.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Click "View User List"	Authenticated session	System shows "No users found" message		

Post Condition	1. Admin is informed that no users exist. No list is shown.

Table 76 Test Case Specification - No Registered Users

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing

Test Type	Negative Test Case			
Test Design Technique	Use Case Testing			
Risk Number	5 (Highest)			
Test Case Number	TC015.3			
Test Case Name	Database Unreachable			
Requirement Traceability	REQ_UR1501			
Test Case Description	Verify the system response when the database is unreachable during user list retrieval.			
Item(s) to be tested				
1	Error message handling			
2	Retry mechanism			
Pre-condition	1. Database is temporarily down.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Click "View User List"	Simulate DB failure	System shows error and retry option		
Post Condition	1. Admin remains on the same page with an error prompt; no data is shown.			

Table 77 Test Case Specification - Database Unreachable

Test Cases for UC-016 : Delete User

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing

Test Type	Black Box Testing – Functional Testing			
Test Design Technique	Use Case Testing			
Risk Number	4 (High)			
Test Case Number	TC016.1			
Test Case Name	Delete Registered User			
Requirement Traceability	REQ_UR1601			
Test Case Description	Verify that an admin can successfully delete an existing user from the system.			
Item(s) to be tested				
1	Delete button functionality			
2	Confirmation prompt			
3	User list update			
Pre-condition	1. Admin is logged into the system.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Select user + Click "Delete"	Valid user ID selected	System prompts for confirmation, deletes user, updates list		
Post Condition	1. User is removed from the list and database.			

Table 78 Test Case Specification - Delete Registered User

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Negative Test Case		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC016.2		
Test Case Name	Cancel User Deletion		
Requirement Traceability	REQ_UR1601		
Test Case Description	Verify the system behavior when the admin cancels the delete action.		

Item(s) to be tested	
1	Confirmation prompt
2	Cancel action

Pre-condition	1. Admin selects a user but cancels the deletion in the confirmation prompt.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Select user + Cancel deletion	Valid user ID selected	User remains in list, no deletion occurs		

Post Condition	1. No user is deleted and system state remains unchanged.

Table 79 Test Case Specification - Cancel User Deletion

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Negative Test Case		
Risk Number	5 (Highest)		
Test Case Number	TC016.3		
Test Case Name	Deletion Fails Due to DB Error		
Requirement Traceability	REQ_UR1601		
Test Case Description	Verify the system response if a database connection issue occurs during deletion.		

Item(s) to be tested	
1	Error handling
2	Retry option

Pre-condition	1. Database is down or unreachable.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Attempt delete action	Valid user ID	System shows an error and provides retry option		

Post Condition	1. No deletion occurs and admin remains on the same page with error prompt.

Table 80 Test Case Specification - Deletion Fails Due to DB Error

Test Cases for UC-017 : Edit user information

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC017.1		
Test Case Name	Successful Edit of User Information		
Requirement Traceability	REQ_UR1701		
Test Case Description	Verify that an admin can successfully edit and save user information.		

Item(s) to be tested	
1	User list
2	Edit form
3	Data validation
4	Save functionality
5	Success message

Pre-condition	1. Admin is logged in and authorized. 2. User account exist

Test Procedure Specification

Input	Test Data	Expected Output/Result	Actual Result	Pass/Fail
User field	User002	User selected,name field updated,		

Name field Role Action	John doe User Click save	role updated and success message shown		
Post Condition	The user's updated information is saved and displayed in the system.			

Table 81 Test Case Specification - Successful Edit of User Information

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Negative Test Case		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC017.2		
Test Case Name	Cancel Edit Operation		
Requirement Traceability	REQ_UR1701		
Test Case Description	Verify that canceling an edit operation does not change user data.		

Item(s) to be tested	
1	User list
2	Edit form
3	Cancel button

Pre-condition	1. Admin is authenticated 2. User account exists

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

Select user	User 002 John doe Click “cancel”	Edit canceled, no change in user list		
Post Condition	User data remains unchanged in the system.			

Table 82 Test Case Specification - Cancel Edit Operation

Tested By	Ammar Syahansyah	**Test Level**	User Acceptance Testing
Test Type	Negative Test Case		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC017.3		
Test Case Name	Edit User – Validation Error		
Requirement Traceability	REQ_UR1701		
Test Case Description	Verify that system handles input validation errors properly during edit.		
Item(s) to be tested			
1	1. Edit form		
2	2. Input validation		
3	3. Error messages		
Pre-condition	1. Admin is authenticated 2. User account exists		

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Email Action	johndoe@invalid Click “Save”	System show “invalid email format” data not saved		
Post Condition	User information is not saved; admin remains on the edit page to fix input.			

Table 83 Test Case Specification - Edit User – Validation Error

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	User Acceptance Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC018.1		
Test Case Name	Successful User Search		
Requirement Traceability	REQ_UR1801		
Test Case Description	Verify that the admin can successfully search for a user using a valid keyword.		
Item(s) to be tested			
1	Search bar		
2	Input field		

3	Search button
2	Search results display

Pre-condition	1. Admin is authenticated 2. System contains user data

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Keyword Action	john@example.com Click "Search"	System returns matching user(s) with name/email containing john@example.com. Search results displayed in user list		

Post Condition	System displays filtered list of users matching the search keyword.

Table 84 Test Case Specification - Successful User Search

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	User Acceptance Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC018.2		
Test Case Name	No Matching User Found		
Requirement Traceability			

	REQ_UR1802			
Test Case Description	Verify that the system displays an appropriate message when no users match the sear			
Item(s) to be tested				
1	Search bar			
2	Search message handling			
Pre-condition	1. Admin is authenticated 2. System contains user data			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Keyword Action	no_match_123 Click “Search”	"No users found" message displayed No users shown		
Post Condition	System returns to search page; no users are displayed.			

Table 85 Test Case Specification - No Matching User Found

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	User Acceptance Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		

Test Case Number	TC018.3			
Test Case Name	Search – Validation or System Error			
Requirement Traceability	REQ_UR1803			
Test Case Description	Verify system handles invalid search inputs and server/database errors correctly.			
Item(s) to be tested				
1	Input validation			
2	Error message display			
3	Retry mechanism			
Pre-condition	1. Admin is authenticated			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Keyword Action	valid term	Retry or error message shown: "Unable to fetch data, please try again"		
Post Condition	User remains on search page with appropriate error feedback; no data is changed.			

Table 86 Test Case Specification - Search – Validation or System Error

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	User Acceptance Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC019.1		
Test Case Name	Successful Clinic Registration		
Requirement Traceability	REQ_UR1901		
Test Case Description	Verify that an authenticated admin can successfully register a clinic with valid input data.		

Item(s) to be tested	
1	Register Clinic button
2	Registration form
3	Input validation

Pre-condition	Admin is authenticated and has permission to register clinics.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Admin click submit button	Sunshine Dental Care, 123 Main St, KL,sunshine@clinic.com, 0123456789, sunshineclinic,	System creates clinic account and shows confirmation message		

	Sunshine@123			
Post Condition	Clinic is added to the system and appears in the clinic list.			

Table 87 Test Case Specification - Successful Clinic Registration

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	User Acceptance Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC019.2		
Test Case Name	Cancel Clinic Registration		
Requirement Traceability	REQ_UR1902		
Test Case Description	Verify that no clinic is created if the admin cancels the registration before submitting.		

Item(s) to be tested	
1	Cancel button
2	Form discard behavior
Pre-condition	Admin is authenticated and accesses the registration form.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Admin click cancel button	Sunshine dental			

		Form closes/discards data and no clinic is added		
Post Condition	No changes are made; the system returns to the previous page without saving clinic data.			

Table 88 Test Case Specification - Cancel Clinic Registration

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	User Acceptance Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC01903		
Test Case Name	Clinic Registration – Validation and System Error		
Requirement Traceability	REQ_UR1903		
Test Case Description	Verify the system handles input validation errors and database issues properly.		

Item(s) to be tested	
1	Form input validation
2	Duplicate detection
3	System/database error handling

Pre-condition	Admin is authenticated and accesses the registration form.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

Admin click cancel button	Clinic Name (Leave blank)	Validation error: "Clinic name is required"		
Post Condition	Clinic is not registered until all errors are corrected and system is stable.			

Table 89Test Case Specification - Clinic Registration – Validation and System Error

4.2.5 Test case: Manage profile

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC020.1		
Test Case Name	Successful Update of Personal Information		
Requirement Traceability	REQ_UR2001		
Test Case Description	Verify that a user can successfully update their personal information with valid inputs.		
Item(s) to be tested			
1	Edit Personal Info button		
2	Editable form fields		
3	Data update logic		
Pre-condition	User is authenticated and has a complete profile.		
Test Procedure Specification			

Input	Test Data	Expected Output/Result	Actual Result	Pass/Fail
User click "save" button	Ammar S. Ahmad,ammar@example.com,0123456789, No.10 Jalan Meranti	System saves and reflects changes on profile with confirmation message		
Post Condition	Updated information is saved and shown in the user profile.			

Table 90 Test Case Specification - Successful Update of Personal Information

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC020.2		
Test Case Name	Cancel Update of Personal Info		
Requirement Traceability	REQ_UR2002		
Test Case Description	Verify that no data is saved if the user cancels the update process.		
Item(s) to be tested			
1	Cancel button		
2	Discard changes logic		
Pre-condition	User is authenticated and has accessed the edit profile page.		
Test Procedure Specification			

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Input full name and click cancel	Ammar S. Ahmad,	Form closes, no changes made, profile remains unchanged		
Post Condition	No changes saved; user profile remains as before.			

Table 91 Test Case Specification - Cancel Update of Personal Info

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC020.3		
Test Case Name	Successful Update of Personal Information		
Requirement Traceability	REQ_UR2001		
Test Case Description	Update Personal Info – Input Validation		
Item(s) to be tested			
1	Field validation		
2	Error message handling		
Pre-condition	User is authenticated and editing profile information.		
Test Procedure Specification			

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

User input invalid value and click save	invalidemail.com, “abcdefg” in phone number field	Validation error: "Please enter a valid email address", Validation error: "Phone number must be numeric"		
Post Condition	No changes saved until valid input is given			

Table 92 Test Case Specification - Successful Update of Personal Information

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC021.1		
Test Case Name	Successful Password Change		
Requirement Traceability	REQ_UR2101		
Test Case Description	Verify that the user can change their password successfully using the correct credentials.		
Item(s) to be tested			
1	Change Password form		
2	Password validation		
3	Password update logic		
Pre-condition	User is authenticated and logged in.		
Test Procedure Specification			

Input	Test Data	Expected Output/Result	Actual Result	Pass/Fail
User input current password, new password and confirm new password and click "save"	Current PasswordOldPass123 New Password: NewSecurePass456 Confirm New Password: NewSecurePass456	Password is updated and confirmation message is shown		
Post Condition	New password is stored and used for subsequent logins.			

Table 93 Test Case Specification - Test Case Name

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC021.2		
Test Case Name	Cancel Password Change		
Requirement Traceability	REQ_UR2102		
Test Case Description	Verify that no changes are made if the user cancels the password change process.		
Item(s) to be tested			

1	Cancel button
2	State preservation

Pre-condition	User is on the Change Password screen.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User input current password, new password and confirm new password and click "cancel"	Current PasswordOldPass123 New Password: NewSecurePass456 Confirm New Password: NewSecurePass456	Password remains unchanged		

Post Condition	Password remains as it was before.

Table 94 Test Case Specification - Cancel Password Change

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC021.3		
Test Case Name			

	Password Change – Input Validation and System Error			
Requirement Traceability	REQ_UR2103			
Test Case Description	Ensure proper error messages are shown for incorrect input or server failures during password change.			
Item(s) to be tested				
1	Input validation			
2	Error messages			
3	Retry/cancel options			
Pre-condition	User is authenticated and attempting to change password.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User input wrong current password, invalid format new password. And system error logic handling	Current WrongPass999 New Password: 123 Simulate DB failure	Error: "Current password is incorrect" Error: "Password must be at least 8 characters with letters and numbers" Error: "Unable to update password. Please try again."		
Post Condition	Password is not changed until issues are corrected.			

Table 95 Test Case Specification - Password Change – Input Validation and System Error

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing

Test Type	Black Box Testing – Functional Testing			
Test Design Technique	Use Case Testing			
Risk Number	2 (Low)			
Test Case Number	TC022.1			
Test Case Name	Successful Profile Picture Upload			
Requirement Traceability	REQ_UR2201			
Test Case Description	Verify that the user can successfully upload a valid image as a profile picture.			
Item(s) to be tested				
1	Upload button			
2	File validation			
3	Profile picture update			
Pre-condition	User is logged in and on the profile page.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
User select file and click "upload profile picture"	File selected: profile.jpg(size<5mb)	Profile picture is uploaded and updated		
Post Condition	New profile picture is shown on the user's profile page.			

Table 96 Test Case Specification - Successful Profile Picture Upload

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC022.2		
Test Case Name	Cancel Profile Picture Upload		
Requirement Traceability	REQ_UR2201		
Test Case Description	Ensure that no changes are made if the user cancels the upload before completion.		

Item(s) to be tested	
1	Cancel button
2	No update to profile picture

Pre-condition	User is on the Upload Profile Picture screen.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
File Selected Click "Cancel"	Any image file	Image previewed but not uploaded Upload is aborted; existing profile picture remains		

Post Condition	No image is uploaded or changed.

Table 97 Test Case Specification - Cancel Profile Picture Upload

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC022.2		
Test Case Name	Cancel Profile Picture Upload		
Requirement Traceability	REQ_UR2202		
Test Case Description	Ensure that no changes are made if the user cancels the upload before completion.		

Item(s) to be tested	
1	Cancel button
2	No update to profile picture

Pre-condition	User is on the Upload Profile Picture screen.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
File Selected Click "Cancel"	Any image file	Image previewed but not uploaded Upload is aborted; existing profile picture remains		

Post Condition	No image is uploaded or changed.

Table 98 Test Case Specification - Cancel Profile Picture Upload

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC022.3		
Test Case Name	Invalid File Format or Upload Error		
Requirement Traceability	REQ_UR2202		
Test Case Description	Ensure that no changes are made if the user cancels the upload before completion.		

Item(s) to be tested	
1	File format/size validation
2	Error handling
3	Retry mechanism

Pre-condition	User is logged in and uploads an image.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
File Selected	profile.pdf	Error: "Invalid file type. Please upload an image."		

Post Condition	No changes made until a valid image is successfully uploaded.

Table 99 Test Case Specification - Invalid File Format or Upload Error

4.2.6 Test case: Manage feedback

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing	
Test Type	Black Box Testing – Functional Testing			
Test Design Technique	Use Case Testing			
Risk Number	3 (Low)			
Test Case Number	TC023.1			
Test Case Name	Successfully Add Feedback			
Requirement Traceability	REQ_UR2301			
Test Case Description	Verify that admin can successfully add new feedback into the system.			
Item(s) to be tested				
1	1. Feedback form			
2	2. Data validation			
3	3. Feedback list update			
Pre-condition	Admin is logged in and navigates to the "Add Feedback" form.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Feedback Text	"Patient showed signs of plaque."	System accepts input Feedback is validated, stored, and confirmation is shown		
Post Condition	Feedback is added and displayed in the feedback list.			

Table 100 Test Case Specification - Successfully Add Feedback

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Negative Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Low)		
Test Case Number	TC023.2		
Test Case Name	Attempt to Add Feedback with Missing Data		
Requirement Traceability	REQ_UR2302		
Test Case Description	Verify that the system shows validation errors when feedback text is left empty.		

Item(s) to be tested	
1	1. Validation mechanism
2	2. Error messages
Pre-condition	Admin is on the feedback form screen.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Feedback Text	(leave blank)	System displays: "Feedback text is required."		
Post Condition	No feedback is saved; form prompts for complete data.			

Table 101 Test Case Specification - Attempt to Add Feedback with Missing Data

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing

Test Type		Black Box Testing – Error Handling		
Test Design Technique		Use Case Testing		
Risk Number		4 (Moderate)		
Test Case Number		TC023.3		
Test Case Name		System Error During Feedback Submission		
Requirement Traceability		REQ_UR2303		
Test Case Description		Verify system response when a backend/database error occurs during submission.		
Item(s) to be tested				
1	1. Error handling			
2	2. Retry option			
3	Form retention			
Pre-condition	Admin submits valid data, but system error is simulated (e.g., DB disconnected			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Feedback Text	"Tooth decay detected on molar." Click "Retry"	System fails to store data and displays: "Error saving data. Try again." System attempts to resubmit the data		
Post Condition	Feedback is only stored after error is resolved.			

Table 102 Test Case Specification - System Error During Feedback Submission

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Low)		
Test Case Number	TC024.1		
Test Case Name	Successfully Edit Feedback		
Requirement Traceability	REQ_UR2401		
Test Case Description	Verify that an Admin can successfully edit existing feedback and that updates are reflected.		

Item(s) to be tested	
1	1. Feedback list
2	2. Edit form
3	3. Data update logic
Pre-condition	Admin is logged in and feedback items exist in the system.

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Feedback Text	"Updated comment for better clarity	System updates record and displays success message		
Post Condition	Feedback is updated and displayed in the list with new content.			

Table 103 Test Case Specification - Successfully Edit Feedback

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing

Test Type	Black Box Testing – Negative Testing			
Test Design Technique	Use Case Testing			
Risk Number	2 (Low)			
Test Case Number	TC024.2			
Test Case Name	Cancel Editing Feedback			
Requirement Traceability	REQ_UR2402			
Test Case Description	Verify that no changes are saved if Admin cancels during feedback editing.			
Item(s) to be tested				
1	1. Cancel button			
2	2. Data persistence checks			
Pre-condition	Admin is on the Edit Feedback screen.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Action	Click "Cancel" or close	System returns to feedback list without saving changes		
Post Condition	Original feedback remains unchanged.			

Table 104 Test Case Specification - Cancel Editing Feedback

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Error Handling		
Test Design Technique	Use Case Testing		

Risk Number	4 (Moderate)			
Test Case Number	TC024.3			
Test Case Name	Save Failure During Feedback Edit			
Requirement Traceability	REQ_UR2403			
Test Case Description	Verify the system handles failure to save edited feedback due to server or DB issues.			
Item(s) to be tested				
1	Error handling			
2	2. Retry option			
Pre-condition	Admin modifies existing feedback; system failure (e.g., server down) is simulated.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Action	Click "Save"	System displays: "Failed to save. Please retry or cancel."		
Post Condition	Feedback is not changed; original version is preserved until successful save.			

Table 105 Test Case Specification - Save Failure During Feedback Edit

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Moderate)		

Test Case Number	TC025.1			
Test Case Name	Successfully Delete Feedback			
Requirement Traceability	REQ_UR2501			
Test Case Description	Verify that an Admin can successfully delete a feedback entry from the system.			
Item(s) to be tested				
1	Delete confirmation			
2	Deletion logic			
Pre-condition	Admin is logged in and feedback exists.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Action	Click "Delete" on a feedback Click "Confirm" on prompt	System prompts for confirmation System deletes the feedback and shows success message		
Post Condition	Feedback is not changed; original version is preserved until successful save.			

Table 106 Test Case Specification - Successfully Delete Feedback

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Negative Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC025.2		

Test Case Name	Cancel Feedback Deletion			
Requirement Traceability	REQ_UR2502			
Test Case Description	Verify that feedback is not deleted when Admin cancels the deletion confirmation.			
Item(s) to be tested				
1	1. Delete confirmation prompt			
2	2. Cancel action			
Pre-condition	Admin is viewing feedback and selects one for deletion.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Action	Click "Delete" on a feedback Click "Confirm" on prompt	System prompts for confirmation System deletes the feedback and shows success message		
Post Condition	Feedback is not changed; original version is preserved until successful save.			

Table 107 Test Case Specification - Cancel Feedback Deletion

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Error Handling		
Test Design Technique	Use Case Testing		
Risk Number	4 (High)		
Test Case Number	TC025.3		
Test Case Name	Deletion Failure Handling		

Requirement Traceability	REQ_UR2503			
Test Case Description	Ensure system handles feedback deletion failures (e.g., database or server error).			
Item(s) to be tested				
1	1. Error message			
2	2. Retry/Cancel logic			
Pre-condition	Simulate backend/database error during deletion.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Click "Delete" and Confirm	Simulated error (e.g., timeout)	System displays error: "Unable to delete. Please try again."		
Post Condition	Feedback entry is not deleted and remains intact until successful operation.			

Table 108 Test Case Specification - Deletion Failure Handling

4.2.7 Test case: tracking progress

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC026.1		
Test Case Name	Successfully View Analysis History		
Requirement Traceability	REQ_UR2601		

Test Case Description		Ensure that the user can view their past analysis history records with date, result, and image.		
Item(s) to be tested				
1	Navigation to Progress page			
2	History retrieval			
Pre-condition	User is logged in and has at least one analysis record.			
Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Navigation	Click on "Progress"	System retrieves and displays past analysis records with summary & thumbnails		
Post Condition	User sees list of past analyses with all details displayed.			

Table 109 Test Case Specification - Successfully View Analysis History

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Negative Testing		
Test Design Technique	Use Case Testing		
Risk Number	1 (Very Low)		
Test Case Number	TC026.2		
Test Case Name	No Analysis History		
Requirement Traceability	REQ_UR2602		
Test Case Description	Verify that an appropriate message is shown when there is no analysis history available.		

Item(s) to be tested					
1	Message display logic				
Pre-condition	User is logged in but has no analysis records.				
Test Procedure Specification					
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**	
Navigation	Click "Progress"	System displays "No history available"			
Post Condition	User sees a blank history page with appropriate message.				

Table 110Test Case Specification - No Analysis History

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Error Handling		
Test Design Technique	Use Case Testing		
Risk Number	3 (Moderate)		
Test Case Number	TC026.3		
Test Case Name	Database Unreachable during History Load		
Requirement Traceability	REQ_UR2603		
Test Case Description	Validate that the system displays an error and retry option when database retrieval fails.		
Item(s) to be tested			
1	Error message		
2	Retry logic		
Pre-condition	Simulated DB/server error preventing data fetch.		

Test Procedure Specification				
Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Navigation	Click “Progress”	System shows error message and retry button		
Post Condition	User is informed of the error and can retry fetching the history.			

Table 111Test Case Specification - Database Unreachable during History Load

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Functional Testing		
Test Design Technique	Use Case Testing		
Risk Number	3 (Moderate)		
Test Case Number	TC027.1		
Test Case Name	Successfully View Detailed Result		
Requirement Traceability	REQ_UR2701		
Test Case Description	Ensure that the user can view full detection results, severity, and feedback from a history entry.		
Item(s) to be tested			
1	History list selection		
2	Detailed data retrieval		
3	Display interface		
Pre-condition	User is logged in and selects a valid analysis record.		
Test Procedure Specification			

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**
Analysis Entry	Click a valid history record	System fetches and displays the condition, severity, feedback, and image		
Post Condition	Full detailed result is shown correctly.			

Table 112 Test Case Specification - View Detailed Result

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Negative Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Low)		
Test Case Number	TC027.2		
Test Case Name	Missing or Corrupted Entry		
Requirement Traceability	REQ_UR2702		
Test Case Description	Ensure the system informs the user when the selected history entry is not found or corrupted.		
Item(s) to be tested			
1	Entry validation		
2	Error message display		
Pre-condition	User selects an entry that is missing or corrupted.		
Test Procedure Specification			

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

Analysis Entry	Click invalid/corrupted entry	System shows notification: "Unable to load result."		
Post Condition	User stays on the same page with informative feedback.			

Table 113 Test Case Specification - Missing or Corrupted Entry

Tested By	Amir Keizrul	**Test Level**	User Acceptance Testing
Test Type	Black Box Testing – Error Handling		
Test Design Technique	Use Case Testing		
Risk Number	4 (High)		
Test Case Number	TC027.3		
Test Case Name	Server Error During Detail Retrieval		
Requirement Traceability	REQ_UR2703		
Test Case Description	Validate that system handles backend/server errors gracefully when fetching details.		

Item(s) to be tested	
1	Backend connection
2	Retry mechanism
3	Error UI
Pre-condition	Server/database error occurs during detailed result fetch.

Test Procedure Specification

Input	**Test Data**	**Expected Output/Result**	**Actual Result**	**Pass/Fail**

Analysis Entry	Click valid record	System shows "Server error. Try again." with retry button		
Post Condition	User is not redirected and can retry loading data.			

Table 114 Test Case Specification - Server Error During Detail Retrieval

5. TEST ENVIROMENT

This chapter section outlines the required resources that are important and part of the test environment in order to execute the testing. These resources are the hardware, software, facilities, distinctive test tools and other necessary special requirement necessary for the AI Dental Assessment System.

5.1 Test Site

5.1.1 Developer Site

Testing will take place in a specialized test lab environment where testers execute comprehensive testing from dedicated workstations equipped with AI model testing capabilities and dental image processing tools.

5.1.2 Clinical Testing Site

A secondary testing site will be established in partnership with dental clinics to conduct real-world validation testing with actual dental professionals and patient scenarios under controlled conditions.

5.2 Facilities Required

To facilitate effective testing of the AI dental assessment in the dedicated lab environment, specific facilities are essential.

5.2.1 Lab Space

A testing lab with an approximate area of 600 sqft will be designated for testing purposes. Within this space, there will be allocated desktop areas covering 250 sqft to accommodate workstations for tester, with additional space reserved for AI model training servers and dental imaging equipment testing stations.

5.2.2 Furniture

The lab will be equipped with adjustable tables and chairs to provide comfortable workstations for the testing team during extended AI model testing sessions. The arrangement of furniture will be designed to facilitate effective communication and collaboration among team members. Tables and chairs will be positioned to create collaborative zones for discussing AI model performance result and testing outcome.

5.2.3 Power Infrastructure

Multiple high-capacity plug points will be strategically placed near workstations to ensure uninterrupted access to power for:

- High-performance testing devices and AI processing devices.
- Dental imaging simulation devices.
- Multiple monitor setup for image analysis testing.
- Server equipment for AI model training and validation.

5.2.4 Communication Facilities

To enhance collaboration and communication among the testing team, the lab layout will be designed to encourage face-to-face interaction. This setup aims to facilitate efficient discussions regarding AI model accuracy, dental condition detection results, and clinical validation findings during the testing process

5.2.5 Network Connectivity

A highspeed and secure network connection with minimum 500 Mbps bandwidth will be established within the testing lab to ensure:

- Seamless communication and access to cloud-based AI model repositories
- Rapid upload and processing of large dental image datasets

- Real-time collaboration with remote clinical testing sites
- Secure access to medical database

5.2.6 Specialized Testing Equipment

Additional specialized equipment required for comprehensive AI dental system Testing:

5.2.6.1 Image Testing Equipment

- High resolution dental image samples and datasets
- Various image format testing file
- Image quality assessment tools
- Dental photography simulation equipment

5.2.6.2 AI Model Testing Infrastructure

- GPU enabled workstations for AI model performance testing
- Dedicated servers for concurrent user load testing
- Model accuracy validation datasets
- Performance monitoring and logging system

5.3 Required hardware and software specifications

5.3.1 Hardware Requirement

Category	Technologies	
	Amir Keizrul Bin Mudzir	Ammar Syahansyah Bin Mohd Reza
Device	PC	Laptop

Operating System	Windows 11	Windows 11
Browser	Google Chrome	Mozilla Firefox
Memory(RAM)	16GB	16GB
Processor	Intel Core i7-14700f	Intel Core i5-9300h
Processor Speed	Min : 2.1 GHz Max : 5.4 GHz	Min : 2.4 GHz Max : 4.1 GHz
System Type	64-bit	64-Bit
Internal Storage	2 TB Hard disk 512 GB nvme SSD	1 TB SSD

Table 115 Hardware Requirement

5.3.2 Software Requirements

Category	Technologies
Database	• Flask • Firebase
Design Tools	• Figma • Canva
Development Tools	• Visual Studio Code • Microsoft 365
Programming Language	• CSS

	• JavaScript • JavaScript XML • Python (Development Backend)

Table 116 Software Requirements

5.4 Testing Group

Test Sequence	Feature	Tester
01	Register New User (Register Management)	Amir Keizrul Bin Mudzir
02	Login (Register Management)	Amir Keizrul Bin Mudzir
03	Logout (Register Management)	Amir Keizrul Bin Mudzir
04	Upload Image (Image Analysis)	Amir Keizrul Bin Mudzir
05	Analyze Image (Image Analysis)	Amir Keizrul Bin Mudzir
06	Generate Feedback (Feedback Module)	Amir Keizrul Bin Mudzir
07	View Feedback (Feedback Module)	Amir Keizrul Bin Mudzir
08	View Schedule (Appointment Booking)	Amir Keizrul Bin Mudzir
09	Book Appointment (Appointment Booking)	Amir Keizrul Bin Mudzir
10	Set Reminder (Appointment Booking)	Ammar Syahansyah Bin Mohd Reza
11	Cancel/Reschedule Appointment	Ammar Syahansyah Bin Mohd Reza

12	Approve Appointment (Clinic Admin)	Ammar Syahansyah Bin Mohd Reza
13	Manage Clinic Schedule (Clinic Admin)	Ammar Syahansyah Bin Mohd Reza
14	Manage Users (Admin Panel)	Ammar Syahansyah Bin Mohd Reza
15	View User List (Admin Panel)	Ammar Syahansyah Bin Mohd Reza

Table 117 Testing Group

5.5 Preparation and training required of the test team

Roles	Responsibilities
Test Planner / Controller • Ammar Syahansyah Bin Mohd Reza • Amir Keizrul Bin Mudzir	• Ensure Phase 1 is delivered to schedule and quality. • Produce high level and detailed test conditions for AI model validation. • Produce expected results for dental condition detection accuracy. • Report progress at regular status reporting meetings. • Co-ordinate review and signoff of test conditions.

	• Manage individual test cycles and resolve tester queries/problems. • Oversee AI model performance testing and clinical validation phases.
Lead Tester • Ammar Syahansyah Bin Mohd Reza • Amir Keizrul Bin Mudzi	• • Identify test data including dental image datasets and clinical scenarios. • Execute test conditions and mark-off results for AI diagnostic accuracy. • Prepare software error reports and AI model performance issues. • Administrate error measurement system and model accuracy tracking. • Ensure test systems outages/problems are reported immediately & followed up. • Ensure entrance criteria are achieved prior to the start of the system test. • Ensure exit criteria are achieved prior to system test signoff. • Validate AI model predictions against clinical ground truth data. • Coordinate with dental professionals for clinical validation testing.

Test Designer • Ammar Syahansyah Bin Mohd Reza • Amir Keizrul Bin Mudzir	• Design test components such as test case, test data, test sequence, test procedures, test scenario, etc. • Determine appropriate techniques, tools, and guidelines to implement AI model testing. • Create comprehensive test scenarios for dental condition detection algorithms. • Design user interface testing protocols for healthcare professionals. • Develop performance testing strategies for image processing workflows. • Establish AI model accuracy benchmarks and validation criteria. • Design cross-platform compatibility testing for mobile and web interfaces.

Table 118 Preparation and training required of the test team

6. TEST SCHEDULE

This section presents a comprehensive overview of the project testing timeline, highlighting key academic development phases and quality assurance milestones throughout the Final Year Project (FYP) duration. The test schedule is strategically aligned with the overall project timeline to ensure systematic validation and verification of project deliverables.

Academic Milestrone	Scope of Workc / Activity	Date	Milestone	Quality Assurance Activities
FYP Briefing	**Planning & Project Feasibility** Planning is the process of determining the general scope of the AI dental assessment project and how to carry it out. Research on AI model architectures, dental image analysis techniques, and web application frameworks. Approach supervisor and explain the AI dental assessment concept and its feasibility for detecting gingivitis, ulcers, caries, and tooth discoloration. Finally, write a comprehensive	1/3/25 – 14/3/25	• Feasibility Studies • Agreed Proposal • Concept Approval • Supervisor Agreement	For test development, this is where test plan knowledge is obtained and researched on how to write a test plan for AI-powered dental assessment system. Initial testing strategy formulation for deep learning model validation and web application functionality testing.

	proposal and get supervisor approval for the AI dental assessment idea.			
Defend Proposal	Analysis & Requirement Study After proposal defence, next is a process of gathering, organizing, documenting and analyzing all the user and technical requirements for the AI dental assessment system. Define functional requirements for image upload, AI analysis, user management, appointment booking, and admin functionalities.	15/3/25 – 31/3/25	• Software Requirement Specification • Software Test Plan	For test development, this phase outlines the test plan structure covering AI model accuracy testing, web application functionality testing, and user experience validation. Test plan development focusing on AI model performance metrics, image processing accuracy, and

				web application reliability testing.
Complete Documentation	**Design & Setup IT Infrastructure** **This activity focuses on the process of transforming user requirements into software artifact and developing technical detailed design of system architecture. Design AI model architecture using CNN for dental image analysis, database schema for user data and dental records, and web application interface design.**	1/4/25 – 14/4/25	• UML Diagram • ERD Diagram • Redefined SRS	For test development, writing test plan is almost finalized to be shown to supervisor including AI model testing procedures and web application test cases. Test environment setup planning for AI model training validation, database testing, and web application performance testing.

Progress Demo to Supervisor and Assessor	**Develop & Setup Application** Development activity where it emphasizes on coding, integrate testing, and implementing the component into a whole web application system. Develop Flask-based web application, implement user authentication, image upload functionality, AI model integration, and appointment booking system.	15/4/25 – 30/4/25	• User Interface • Prototype Web App • Database • AI Model Training	For test development, test plan version 1.0 is completed and shown to supervisor covering comprehensive testing for all system components. Unit testing implementation for individual modules, AI model accuracy validation with dental image datasets, and database functionality testing.

Continue Development Progress	**Continue App Development & AI Model Training** Continue development of AI dental assessment web application. Implement deep learning model training using dental image datasets from Roboflow, integrate trained model with web application, develop user progress tracking, and implement feedback management system.	1/5/25 – 14/5/25	• Enchanced Prototype • Trained AI Model • Database Integration • User Interface Refinement	Still in software development. For test development, test environment preparation for comprehensive system testing including AI model performance evaluation. Integration testing for AI model with web application, performance testing for image processing speed, and user interface testing for different browsers and devices.

Testing	**System Testing & Acceptance Testing** Comprehensive testing phase including AI model accuracy testing with diverse dental images, web application functionality testing, user acceptance testing with dental professionals and patients, performance testing under various load conditions.	15/5/25 – 31/5/25	• Test Log • Test Summary Report • Updated STP • AI Model Validation Report	For test development, this is the execution phase of planned testing for acceptance testing of the AI dental assessment system. Execute all planned test cases, document AI model accuracy rates for different dental conditions, validate web application performance, and conduct user experience testing. Also, document the execution phase and organize it

				well for developer to refer later.
FYP2 Presentation	**Deployment & Final Documentation** This activity process emphasizes on deployment of the AI dental assessment system so that it can be deployed smoothly on web servers. Prepare final documentation, user guides, and deployment procedures for the complete system.	1/6/25 – 15/6/25	• Product Release • Final Documentation • Deployment Guide • User Manual	For test development, prepare all necessary testing documents including final test report, AI model performance analysis, and system validation certificates before project submission. Final system validation, performance benchmarking, security testing, and preparation of comprehensive test documentation

				for project submission.

Table 119 TEST SCHEDULE

7. DEFINITIONS, ACRONYMS AND ABREVIATION

7.1 Definitions

Term	Definition
Coding	The process of translating system design and functional requirements into executable source code, such as implementing system features like image analysis or user registration as outlined in the SRS.
Defect	A deviation from expected behavior in the system, such as incorrect image analysis results or failure to register a user, identified during testing and documented for correction.
Functional Testing	A type of black-box testing focused on verifying that the system behaves according to its functional requirements from the Software Requirements Specification (SRS), such as user login or dental condition feedback generation.
Risk Analysis	The process of evaluating potential system failures (e.g., AI model misdiagnosis, failed image uploads) by calculating impact and likelihood to prioritize testing of high-risk features.
Software Requirements Specification (SRS)	A key deliverable that outlines the system's functional and non-functional requirements, such as modules for feedback generation, appointment booking, and user authentication

Test Item	An individual component or feature to be tested, such as the login form, appointment scheduler, or AI-based dental analysis module.
Tester	A person skilled in executing test scenarios and cases to validate whether the application meets its requirements and to document any observed defects for resolution.

Table 120 Definitions

7.2 Abbreviations

Abbreviation	Meaning
ATP	Acceptance Test Plan
SRS	Software Requirements Specification
IEEE	Institute of Electrical and Electronics Engineeers
ISO	International Organization for Standardization
ISTQB	International Software Testing Qualifications Board
STP	Software Test Plang
UAT	User Acceptance Testing
SDLC	Software Development Life Cycle
RAD	Rapid Application Development
CNN	Convolutional Neural Network
AI	Artificial Intelligence
YOLO	You Only Look Once (object detection algorithm

UI	User Interface
UC	Use Case
ID	Identifier

Table 121 Abbreviations


AI-Powered Dental Assessment System
Project Gantt Chart | March 2025 - June 2025
Students: Amir Keizrul Bin Mudzir & Ammar Syahansyah Bin Mohd Reza
Task / Milestone
Mar 2025
Apr 2025
May 2025
Jun 2025
PHASE 1: FINAL DEVELOPMENT & INTEGRATION
Flask Web Application Completion
AI Model Optimization & Fine-tuning
Appointment Booking System Integration
User Management System Finalization
Database Integration & Optimization
Unit Testing Completion
System Integration Milestone
PHASE 2: COMPREHENSIVE TESTING
AI Model Accuracy Testing
System Integration Testing
Performance & Load Testing
Security Testing
Cross-Browser Compatibility Testing
User Acceptance Testing
Testing Completion Milestone
PHASE 3: DEPLOYMENT & FINAL PREPARATION
Production Environment Setup
System Deployment
Post-Deployment Testing
Bug Fixes & System Optimization
User Training & System Demo
System Go-Live Milestone
PHASE 4: DOCUMENTATION & PROJECT COMPLETION
Final Project Report Writing
Technical Documentation Completion
Final Test Report
User Manual & Installation Guide
Presentation Preparation
Project Demonstration Video
FYP2 Final Presentation
ADDITIONAL QUALITY ASSURANCE
Code Review & Refactoring
Automated Testing Setup
Performance Benchmarking
Backup & Recovery Testing
System Maintenance Plan
Planning & Research
Analysis & Requirements
Design & Architecture
Development & Implementation
Testing & Quality Assurance
Deployment & Documentation
Academic Milestones

Figure 10 Gantt chart